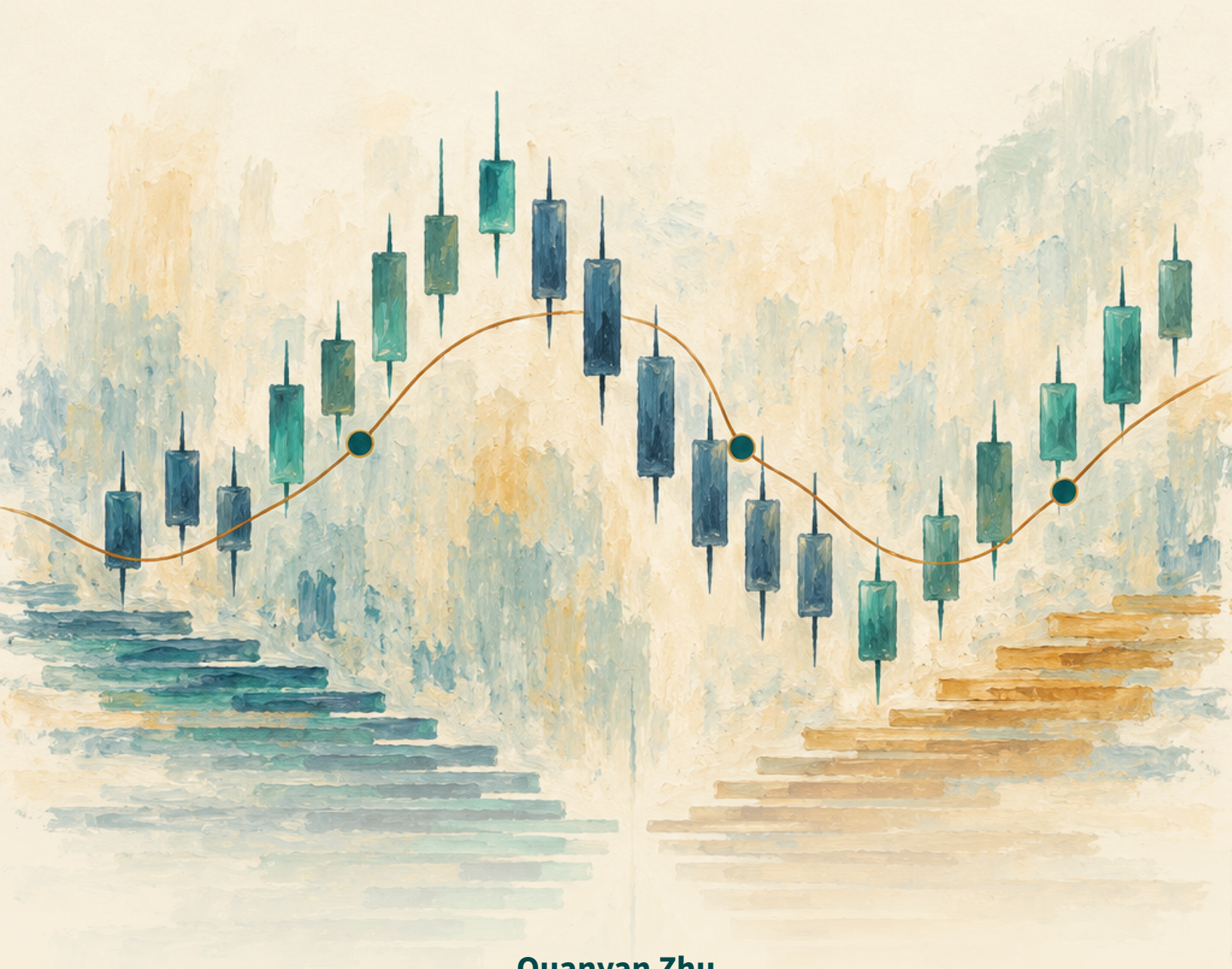


Financial Markets and Trading for Engineers

A primer on market microstructure and optimal trading



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Draft edition, October 3, 2026.

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How to read this book

This book is written for a reader who is unusual in one specific way: you already know the mathematics, and you do not know the finance. You can state the dynamic programming principle, you know what a viscosity solution is, you have solved semidefinite programs, you can recite the KKT conditions in your sleep, and yet if someone says “we got run over by adverse selection at the top of book and had to widen quotes,” none of that machinery helps you.

Most introductions to trading are written for the opposite reader: someone comfortable with the institutions who needs the mathematics kept out of sight. Those books spend three pages explaining what a share is and then, at the crucial moment, say “the price is assumed to follow a random walk” and move on. For you that is exactly backwards. You do not need random walks explained. You need to know *which* objects in a market are states, which are controls, which are constraints, where the noise enters, what is observable, and (most importantly) which of the modelling assumptions are load-bearing and which are decoration.

The organizing claim

A financial market is a controlled dynamical system with three unusual features: (i) the noise is endogenous, because it is generated by other controllers; (ii) your control moves the state you are trying to profit from, so the “plant” reacts to you; and (iii) the objective is almost always a risk-adjusted terminal cash position, which makes the problem a stochastic control problem with a convex running penalty. Everything else (the vocabulary, the institutions, the acronyms) is detail layered on top of that skeleton.

What this book covers

Part I builds the objects. We start by placing a market inside the standard control template (Chapter 1). Chapter 2 is a visual dictionary: a picture for each of the two dozen words that the rest of the book uses without explanation, from *tick* and *spread* to *slippage* and *margin*. Chapter 3 then gets the accounting exactly right, an unglamorous chapter that will save you from sign errors for the rest of your career. Chapter 4 dissects the limit order book, the data structure that is the modern market, and shows why it is naturally a measure-valued state with a deterministic matching map. Chapter 5 explains where price models come from and why the absence of arbitrage is, quite literally, a linear programming duality theorem. Chapter 6 answers the question that separates finance from applied probability: why do prices move at all, and why does your own trading move them? Chapter 7 makes “liquidity” precise enough to appear in a constraint set.

Part II is where the machinery you already have takes over. Optimal execution (Chapter 8) is a finite-horizon linear-quadratic control problem with a liquidation constraint. Market making (Chapter 9) is impulse-and-intensity control of a queueing system. Hedging (Chapter 10) is a stochastic target problem, exactly solvable in the complete-market fantasy and a projection problem otherwise. Portfolio management (Chapter 11) runs from a quadratic program to a singular control problem with a free boundary.

Part III deals with everything that makes practice harder than theory: risk measures and the constraints that bind in reality (Chapter 12), the statistics of estimating any of this from data (Chapter 13), the technology regime in which speed rather than prediction decides who profits (Chapter 14), what changes when the controller is an autonomous agent rather than a function (Chapter 15, which also sets out the three-layer architecture that separates economic intent from exchange messages), and a survey of what remains open (Chapter 16).

Conventions

Five recurring boxes carry most of the translation work.

Control-theoretic dictionary

Appears when a piece of financial vocabulary maps onto an object you already know. *Inventory* is a state; *quoting* is a control; *fill probability* is a jump intensity; *slippage* is a running cost.

How it actually works

Appears when the mathematics is clean but the institution is messy, and the mess matters: tick sizes, fee schedules, priority rules, settlement, and the other facts that decide whether a model is usable.

Where the model breaks

Appears when a standard model is about to be pushed past the regime where it is true. Every model in this book is wrong somewhere; these boxes say where.

In plain terms

Appears at the head of every chapter, and wherever a formal statement needs one. It says what is going on without using any mathematics, for a reader who wants the idea before the notation.

Problem n : the canonical statement

Numbered problem boxes state each decision problem once, formally and completely: state, control, dynamics, admissible set, objective. If you read nothing else in a chapter, read its problem box.

Margin notes carry cross-references, translations of terminology, and asides that would break the flow of an argument. Figures are computed, not sketched: unless a caption says otherwise, every curve in this book comes from a simulation whose parameters are printed in the caption, so you can reproduce it.

What this book is not

It is not a course in mathematical finance: there is no measure-theoretic development of stochastic integration here, and the derivative-pricing content is confined to what you need to understand hedging as a control problem. It is not an empirical survey; the stylized facts are stated and illustrated, not established. And it is emphatically not investment advice. Every *figure* in this book is produced by a simulator, and the

empirical magnitudes quoted in the text are calibrations taken from the literature rather than measurements made here.

Notation

Notation is fixed once, here, and used consistently for the rest of the book. Where the finance literature is internally inconsistent (and it very often is) we choose the convention that keeps the control-theoretic correspondence clean and flag the alternatives in a margin note.

Probability and analysis

$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$	filtered probability space; $\mathcal{F}_t$ is the information available at time t
$\mathcal{G}_t \subseteq \mathcal{F}_t$	the <i>observation</i> filtration of a particular agent (strictly smaller in any realistic model)
$\mathbb{Q}$	risk-neutral / pricing measure; equivalent martingale measure
W, B	standard Brownian motions
N, N^b, N^a	counting processes; superscripts index the bid and ask side
$\lambda(\cdot)$	intensity of a counting process
$\mathbb{E}_t[\cdot] := \mathbb{E}[\cdot \mathcal{F}_t]$	conditional expectation
$\mathcal{A}$	set of admissible controls
$a \wedge b, a \vee b$	$\min(a, b), \max(a, b)$
$x^+ := x \vee 0$	positive part
$\mathbf{1}_{\{A\}}$	indicator of the event A

Market state

S_t	<i>latent</i> efficient (fundamental) price of the traded asset; not observable
p_t^b, p_t^a	best bid and best ask price; $p_t^b < p_t^a$
$M_t = \frac{1}{2}(p_t^b + p_t^a)$	observed midprice; a computed quantity, and not the same object as S_t
$S_t = p_t^a - p_t^b$	quoted spread
Δ	tick size, the price grid spacing
L_t	full limit order book state: a signed measure on the price grid
q_t^b, q_t^a	displayed size at the best bid / best ask
I_t	order-flow (queue) imbalance, $(q_t^b - q_t^a)/(q_t^b + q_t^a)$
ν_t	trading <i>rate</i> (control). In the general ledger of Chapter 3 , $dQ_t = \nu_t dt$, so $\nu_t > 0$ is buying; in the liquidation problems of §1.5 and Chapter 8 , where the position only falls, $\nu_t \geq 0$ is the <i>sell</i> rate and $dQ_t = -\nu_t dt$
δ^b, δ^a	quoted half-spreads of a market maker, relative to S

Agent state and objective

Q_t	inventory (position) in shares/contracts; may be negative
X_t	cash account
$V_t = X_t + Q_t S_t$	mark-to-market wealth
$g(\cdot)$	permanent price-impact function
$h(\cdot)$	temporary (instantaneous) impact / execution cost function
ϕ	running inventory penalty (urgency)
α	terminal liquidation penalty
γ	risk aversion
σ	volatility of the efficient price
T	terminal time / trading horizon
$H(t, x, q, p, M)$	Hamiltonian; p, M are the adjoint and Hessian slots
$v(t, \cdot)$	value function

The master dictionary

An engineer says

A trader says

state	position, inventory, book, exposure
control	order, quote, hedge ratio, weight
control constraint	position limit, risk limit, margin
plant / dynamics	price process, order-book dynamics
disturbance	order flow, news, “the market”
running cost	transaction cost, slippage, financing, borrow
terminal cost	liquidation cost, closing at the bell
value function	expected P&L to go
observation noise	microstructure noise, the bid-ask bounce
filtering	signal extraction, alpha estimation
model uncertainty	model risk
feedback law	strategy, algo, execution schedule
adjoint / costate	marginal value of inventory
jump intensity	fill rate, arrival rate, trade intensity
LP dual variable	state price, risk-neutral probability
reference command	parent order, target weight, hedge target
actuator input	the exchange message a broker will actually accept
supervisory loop	the tactical layer that adjusts how an order is executed
safety filter	pre-trade risk check, certified execution kernel

PART I

Objects and Mechanisms

Seven chapters that build the objects: a visual dictionary of the vocabulary, then what a position is, what a price is, what the order book is, why prices move, and what liquidity means once you insist on writing it into a constraint set. Nothing here is optimized. The goal is to make the models of Part II statements about something real.

The Market as a Controlled System

Before any vocabulary, here is the claim in its most compressed form. A financial market is a *queueing system with prices*, operated by a deterministic matching rule, driven by the orders of competing agents, each of whom is solving a stochastic control problem whose disturbance process is the aggregate of everyone else's controls. If you hold that sentence in mind, the rest of this book is a series of specializations of it.

This chapter installs the frame. We describe the physical system (§1.1), place it in the standard control template (§1.2), isolate the four features that make markets genuinely different from the systems you are used to (§1.3), and then state the four canonical decision problems that Part II solves in detail (§1.4). We close with a two-period toy market that you can solve in your head, and which already exhibits price impact, adverse selection, and inventory risk (§1.5).

One warning about how to take this chapter. It uses perhaps thirty words of market vocabulary that it does not stop to define: bid, ask, spread, inventory, leverage, fill, slippage and the rest. That is deliberate, because the frame is easier to see before the vocabulary than after it, and because every one of those words gets a picture and a definition in [Chapter 2](#), which follows immediately. Read this chapter for the shape of the argument and let the words wash over you; nothing here depends on their precise meanings, and everything here is repeated once you have them.

1.1 The institutional environment

Strip away the institutions and a modern electronic market is a single computer program (the **matching engine**) that maintains a sorted data structure of resting orders and applies a fixed rule to incoming messages. Everything else is plumbing around that program.

Four facts about [Fig. 1.1](#) deserve emphasis, because each one breaks an assumption that is usually harmless in engineering applications.

- (1) **The plant is discrete and deterministic.** The matching engine contains no randomness. Given the book and the incoming message stream, the sequence of trades is a deterministic function. All the stochasticity lives in *when* messages arrive and *who* sends them.
- (2) **The disturbance is other agents' controls.** What you model as exogenous order flow is the output of other optimizers. This is why stationarity fails: change your behaviour enough and the disturbance distribution changes.
- (3) **Observation is lagged and partial.** You see the public book, not the parent orders behind it, and you see it τ seconds late. Two agents at different latencies literally solve different filtering problems.
- (4) **Actions are not free.** Sending a marketable order consumes resting liquidity and moves the price against you. The control appears inside the state equation with a sign that hurts.

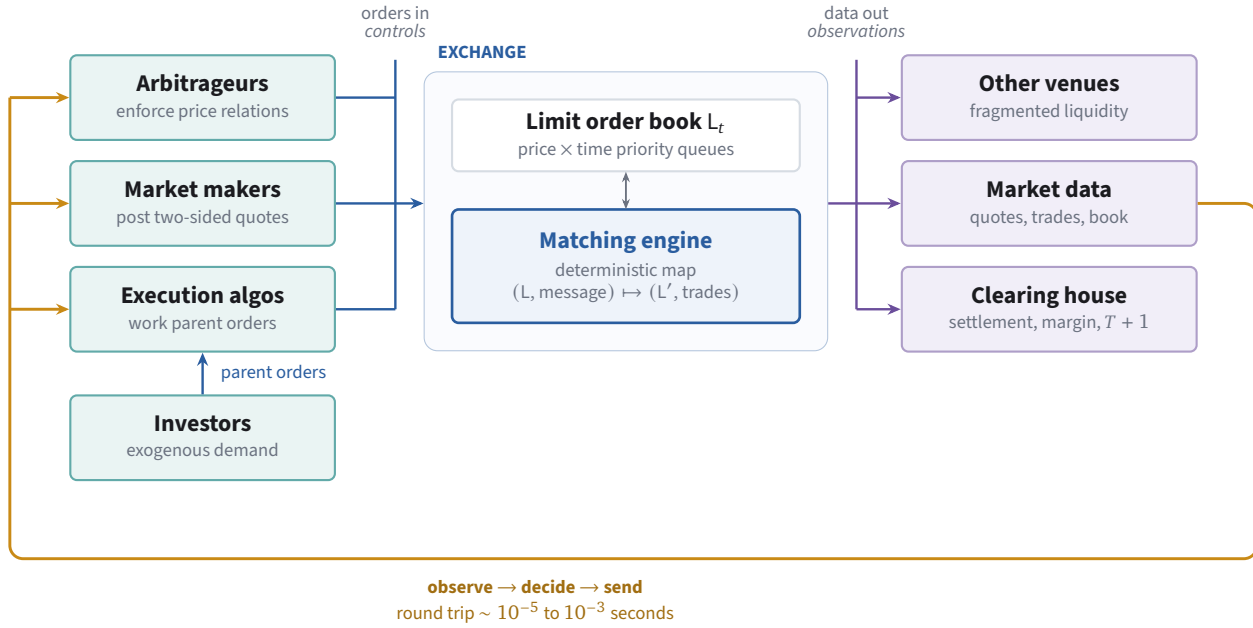


Figure 1.1. The ecosystem. Participants send messages (limit orders, market orders, cancellations) into a matching engine that maintains the book L_t and applies a deterministic priority rule. The engine emits trades and quote updates, which reach every participant as market data after a latency. Investors do not usually trade directly; they hand parent orders to execution algorithms, which is why Chapter 8 exists. The amber path is the only genuinely dynamic element: your own control re-enters the state that generates your observations, which is why price impact is unavoidable rather than a modelling choice.

How it actually works

Real venues add layers that models usually suppress but that decide profitability: a *fee schedule* (typically a rebate for posting liquidity and a fee for taking it, of order 0.1 to 0.3 basis points, a basis point being one hundredth of one percent), *tick size* rules that quantize the price grid, *order types* beyond the plain limit/market pair (pegged, hidden, iceberg, immediate-or-cancel), opening and closing *auctions* that concentrate a large fraction of daily volume into two instants, and *circuit breakers* that halt trading after large moves. A model that ignores fees can get the sign of a market-making strategy's P&L wrong; Chapter 4 treats these carefully.

1.2 The standard control template

Here is the frame into which every problem in this book will be poured. Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ satisfying the usual conditions. An agent chooses a control $\nu = (\nu_t)_{t \in [0, T]}$, adapted to an observation filtration $\mathcal{G}_t \subseteq \mathcal{F}_t$, from an admissible set $\mathcal{A}$. The state $Z_t = (X_t, Q_t, S_t, L_t, \dots)$ evolves as

$$dZ_t = b(t, Z_t, \nu_t) dt + \sigma(t, Z_t) dW_t + \int_E \beta(t, Z_{t-}, e) \tilde{N}(dt, de), \quad (1.1)$$

and the agent maximizes

$$J(\nu) = \mathbb{E} \left[\int_0^T \ell(t, Z_t, \nu_t) dt + \Psi(Z_T) \right], \quad \nu(t, z) = \sup_{\nu \in \mathcal{A}} \mathbb{E}_{t, z} [\cdot]. \quad (1.2)$$

Symbol	Control theory	What it is in a market
Q_t	state component	inventory : how many shares you own. Signed; negative means <i>short</i> , i.e. you sold something you borrowed.
X_t	accounting state	cash : the running integral of everything you have paid and received. Its dynamics are induced by trades, fees and financing rather than by a state equation of its own.
S_t	state component	price : the latent efficient value. Modelled as an exogenous martingale <i>plus your own impact</i> , so it is controlled, not exogenous.
L_t	state component	the book : the queue of resting orders. Infinite-dimensional in principle; usually projected onto a few statistics.
ν_t	control	trading : either a rate (shares per second) or a discrete order (buy 500 now) or a quote (offer at $S + \delta^a$).
$\mathcal{A}$	admissible set	mandate and risk limits : no shorting, gross exposure caps, maximum order size, capital. Usually convex.
ℓ	running cost	transaction costs and risk : the price paid for trading fast, plus a penalty on carrying inventory. Convex in ν .
Ψ	terminal reward	mark-to-market : the cash you would have if you liquidated at T , less the cost of that liquidation.
$\mathcal{G}_t$	observation	your data : public feed plus your own fills. Strictly coarser than $\mathcal{F}_t$, which contains other agents' intentions.

Table 1.1. The template twice over. Every problem in Part II is (1.1) to (1.2) with specific choices of b , ℓ , Ψ and $\mathcal{A}$. The chapters differ in which component of the state is the interesting one.

Nothing here is new to you. What is new is the *interpretation* of each symbol, and Table 1.1 gives it.

The dynamic programming principle applies unchanged, and when the state is Markov and the value function is smooth enough, v solves the Hamilton-Jacobi-Bellman equation

$$\partial_t v(t, z) + \sup_{u \in U} \left\{ \ell(t, z, u) + \mathcal{L}^u v(t, z) \right\} = 0, \quad v(T, z) = \Psi(z), \quad (1.3)$$

with $\mathcal{L}^u$ the generator of (1.1) under the constant control u . The financial specification enters through all of the state dynamics b and σ and the jump term β , the observation filtration $\mathcal{G}$, the admissible set $\mathcal{A}$, the running reward ℓ and the terminal criterion Ψ ; in microstructure the last three are often more economically consequential than the drift. The mathematical content is in whether (1.3) is solvable in closed form, which for a surprising number of the canonical models it is.

Appendix B collects the verification theorem, the standing regularity assumptions, and the BSDE representation, in the form used in the later chapters.

Control-theoretic dictionary

The single most useful translation in this book: *the marginal value of inventory*, $\partial_q v(t, z)$, is what a trader calls the **fair transfer price** of one more share. It is not the market price S_t . It equals S_t plus a correction for the risk and the liquidation cost of carrying the extra share to the horizon. Almost every trading rule in Part II is an instruction to trade whenever the market price differs from $\partial_q v$ by more than the cost of trading.

1.3 Four distinguishing features of markets

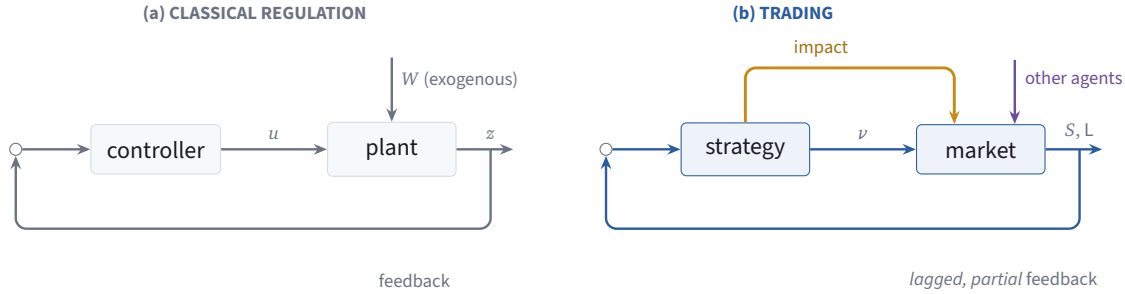


Figure 1.2. Two feedback loops. In (a) the disturbance W is exogenous and the control affects only the state trajectory. In (b) the control enters the plant twice, once as an order and once as impact on the price, the “disturbance” is the aggregate control of other optimizers, and the feedback path is both delayed and lossy. The complications of Part II are variations on these three arrows.

1.3.1 Price impact and state-dependent control

In a temperature-control problem, the thermostat does not change the laws of thermodynamics. In a market, buying raises the price. The mechanism is mechanical, not psychological: a buy order removes shares from the ask queue, and when the queue at the best ask is exhausted the best ask moves up.

Formally, the price dynamics carry a control-dependent drift,

$$dS_t = \underbrace{g(\nu_t) dt}_{\text{permanent impact}} + \sigma dW_t, \quad \text{and you transact at } \hat{S}_t = S_t + \underbrace{h(\nu_t)}_{\text{temporary impact}}, \quad (1.4)$$

with g, h increasing and $g(0) = h(0) = 0$. The distinction matters enormously and is the subject of §6.5: g changes the price for everyone forever, h is a cost you alone pay, on this trade only.

1.3.2 Strategic disturbances

Figure 1.2(b) shows other agents entering as an exogenous input. This is a modelling convenience with real content: it is exactly the assumption that you are *small*. When you are not small (when your order is a visible fraction of the day’s volume, or when your counterparty is better informed) the honest formulation is a stochastic game, and the solution concept changes from optimality to equilibrium.

Where the model breaks

The “small trader” assumption is doing enormous work. It justifies (i) treating the price process as unaffected by anything except your own mechanical impact, (ii) treating fill probabilities as exogenous functions of your quoted distance, and (iii) ignoring the possibility that other agents detect and exploit your pattern. In practice, execution algorithms are routinely reverse-engineered from their footprints. Chapter 16 discusses what changes under a game-theoretic formulation, where the relevant object is a mean-field or Nash equilibrium rather than a single HJB.

1.3.3 Partial observation

The efficient price S_t is a modelling fiction: what you observe are transaction prices and quotes, which equal S_t plus a microstructure distortion. Formally, the observation is $Y_t = S_t + \eta_t$ with η a bounded, strongly mean-reverting noise generated by the discreteness of the price grid and the bid-ask bounce. Estimating S from Y is a

filtering problem, and (as Chapter 13 shows) naïvely estimating volatility from Y at high frequency does not converge to the volatility of S ; it diverges.

1.3.4 The role of the risk criterion

A pure expected-profit objective produces absurd strategies: infinite leverage, unbounded inventory, and a positive probability of ruin that the expectation does not see. Every usable objective therefore contains a risk term, and the choice of that term (variance, exponential utility, a running inventory penalty, a coherent risk measure) changes both the optimal policy and the tractability of the problem. Table 1.2 previews the four choices we use.

Risk term	Where used	Consequence
$-\phi \int_0^T Q_t^2 dt$	execution, Ch. 8	keeps the problem linear-quadratic; closed-form tanh solution
$-\frac{\gamma}{2} \text{Var}[\cdot]$	mean-variance, Ch. 11	a quadratic program; time-inconsistent in dynamic settings
$-\frac{1}{\gamma} \log \mathbb{E}[e^{-\gamma \cdot}]$	market making, Ch. 9	entropic; linearizes the HJB via a Hopf-Cole transform
CVaR_α	risk limits, Ch. 12	coherent; convex in general, and a <i>linear</i> program for finitely many scenarios with affine losses (Rockafellar-Uryasev)

Table 1.2. Four risk criteria and what each buys you. The last row is the one that will feel most familiar: with finitely many scenarios and losses affine in the position, minimizing conditional value-at-risk is a linear program.

1.4 Four canonical decision problems

Figure 1.3 is the map of Part II. Each panel is the same template with a different component of the state promoted to the leading role.

1.5 A two-period illustration

The following toy model is deliberately trivial, and yet it already contains price impact, inventory risk, and the trade-off that drives execution algorithms in practice. We will refer back to it repeatedly.

Problem 0: two-period liquidation

An agent must sell $q_0 > 0$ shares over two dates, $t = 1$ and $t = 2$. Let $\nu_1, \nu_2 \geq 0$ be the quantities sold, with $\nu_1 + \nu_2 = q_0$. The unaffected price is a random walk,

$$S_1 = S_0, \quad S_2 = S_1 - \kappa \nu_1 + \sigma \varepsilon, \quad \varepsilon \sim (0, 1),$$

where $\kappa > 0$ measures *permanent* impact. Selling ν at date t walks the price down by $\kappa \nu$ as the order executes, so the average price realized along the way is $S_t - \frac{\kappa}{2} \nu$; on top of that the seller pays a temporary concession $\frac{\lambda}{2} \nu$ that is gone by the next date. The realized price is therefore $S_t - \frac{\kappa}{2} \nu - \frac{\lambda}{2} \nu$, with $\lambda > 0$ the *temporary* impact, and the agent maximizes mean-variance revenue

$$\max_{\nu_1 + \nu_2 = q_0} \mathbb{E}[R] - \frac{\gamma}{2} \text{Var}[R], \quad R = \sum_{t=1,2} \nu_t \left(S_t - \frac{\kappa}{2} \nu_t - \frac{\lambda}{2} \nu_t \right).$$

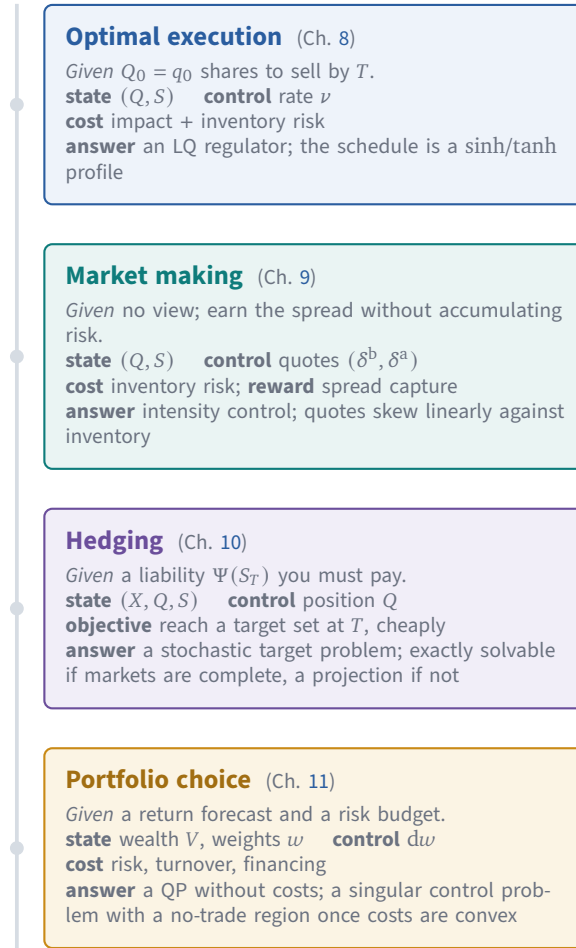


Figure 1.3. Part II in one figure. The four problems differ in which state variable is exogenous and which is chosen. In execution the terminal inventory is *constrained*; in market making it is *penalized*; in hedging it is *determined* by a liability; in portfolio choice it is the decision variable.

Charging the impact as it is incurred, rather than applying it only after the order has printed, is what makes the permanent cost below schedule-independent and matches the continuous-time convention of [Chapter 8](#); the exercises examine what goes wrong under the other choice. Every symbol has a job. q_0 is the position to be liquidated and ν_1, ν_2 the two decisions; κ is how much of your selling pressure sticks to the price permanently and λ how much of it you pay only on the shares you are selling at that instant; σ is the volatility of the price over one period and ε the shock that realizes it; and $\gamma \geq 0$ is the agent's risk aversion, the exchange rate at which variance of revenue is traded against its mean. Note that λ here is a *temporary*-impact coefficient, written k in [Chapter 8](#); it is not Kyle's λ of [Chapter 6](#), which measures permanent impact. Setting $\gamma = 0$ makes the agent indifferent to risk, and $\gamma \rightarrow \infty$ makes it liquidate immediately.

Substituting $\nu_2 = q_0 - \nu_1$ and writing $q_1 = q_0 - \nu_1$ for the inventory carried into the second period,

$$\mathbb{E}[R] = q_0 S_0 - \frac{\kappa}{2} q_0^2 - \frac{\lambda}{2} (\nu_1^2 + q_1^2), \quad (1.5)$$

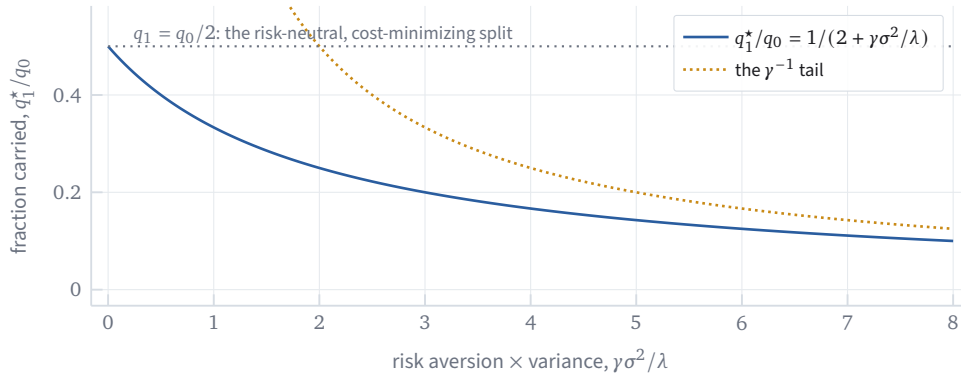


Figure 1.4. Urgency is a ratio. The fraction of the order carried into the second period, from (1.8). At $\gamma = 0$ the agent splits evenly, minimizing quadratic execution cost. As risk aversion or volatility grows, the agent front-loads: $q_1^* \rightarrow 0$ like γ^{-1} . Permanent impact does not appear: it costs $\frac{\kappa}{2}q_0^2$ whatever the schedule, so it changes the bill and not the decision. A single dimensionless group, $\gamma\sigma^2/\lambda$, therefore indexes the whole family.

$$\text{Var}[R] = \sigma^2 q_1^2. \quad (1.6)$$

Two remarks before we optimize. First, the permanent-impact cost collapses to $\frac{\kappa}{2}q_0^2$, which depends only on the *total* quantity and not on the schedule: whatever the split, the price is walked down by κq_0 in total and the seller realizes the average along the way. Because it is a constant, κ drops out of the optimization entirely. Second, the variance depends only on q_1 : risk is the cost of *holding*, not of trading.

Dropping constants, the agent solves

$$\min_{q_1} \frac{\lambda}{2} [(q_0 - q_1)^2 + q_1^2] + \frac{\gamma\sigma^2}{2} q_1^2, \quad (1.7)$$

a one-dimensional strictly convex program whenever $\lambda > 0$, with solution

$$q_1^* = \frac{\lambda}{2\lambda + \gamma\sigma^2} q_0, \quad \nu_1^* = q_0 - q_1^*. \quad (1.8)$$

(1.8) is worth staring at, because the same structure survives all the way to the continuous-time solution of Chapter 8:

- **Risk-neutral limit** $\gamma \rightarrow 0$: $q_1^* = q_0/2$. With no risk aversion, the only cost is the convex temporary impact, and convexity says spread the trading evenly. This is the **TWAP** (time-weighted average price) schedule.
- **Risk-dominant limit** $\gamma\sigma^2 \rightarrow \infty$: $q_1^* \rightarrow 0$. Sell everything immediately, paying the impact to eliminate the risk. This is a **market order**.
- **Interior**: the optimum trades off a λ -weighted cost of speed against a $\gamma\sigma^2$ -weighted cost of exposure. The ratio $\gamma\sigma^2/\lambda$ is the only thing that matters: an *urgency* parameter with units of inverse time in the continuous-time version.

The efficient frontier of execution

Sweeping γ from 0 to ∞ traces a curve in the (expected cost, variance of cost) plane. That curve is the **efficient frontier of execution**: no strategy lies below it, and every point on it is optimal for *some* risk aversion. The entire practical

Schedule-independence needs two things: that impact be *linear*, and that it be charged as it is incurred. It is the reason Chapter 8 can ignore its permanent-impact coefficient when optimizing, and it fails as soon as impact is nonlinear or transient.

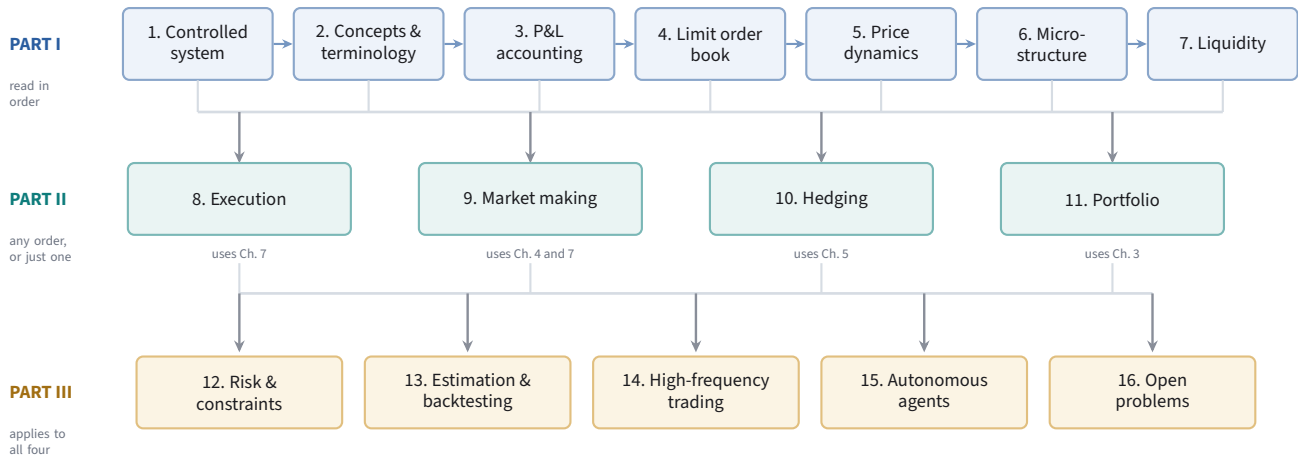


Figure 1.5. How the book fits together. Part I is a genuine chain: each chapter assumes the one before it, so read it in order. Part II is not a chain at all. Its four problems share the Part I material and nothing else, so they may be taken in any order or singly, and the small notes say which Part I chapters each leans on hardest. Part III applies to all four. A reader in a hurry can take Chapters 1, 2, 3, 4 and 7, then go directly to whichever problem they came for.

content of “choose an execution algorithm” is choosing a point on this curve. We compute it exactly in [Section 8.5](#).

1.6 Organization of the book

[Figure 1.5](#) shows the dependency structure. The chapters of Part I are genuinely prerequisite: Part II’s models are only meaningful once you know what a fill is and where impact comes from. Within Part II the chapters are independent, and a reader in a hurry can go straight from Chapters 4 and 7 to whichever of Chapters 8 to 11 matches their interest.

Exercises

Exercise 1.1 (Three-period toy). Extend Problem 0 to three dates with $\nu_1 + \nu_2 + \nu_3 = q_0$. Show that the optimizer solves a tridiagonal linear system, and that the inventory path $q_0, q_1, q_2, 0$ is convex decreasing. (This is the discrete Almgren-Chriss problem; [Section 8.4](#) does the general case.)

Exercise 1.2 (When impact is concave). Replace the temporary impact $\frac{\lambda}{2}\nu$ by $\lambda \nu^{p-1}/p$ with $p \in (1, 2)$, so that the cost of trading ν is $\lambda \nu^p/p$: empirically $p \approx 3/2$. Show that the risk-neutral optimum is still the even split, but that the problem is no longer quadratic and the risky case has no closed form. What happens as $p \rightarrow 1^+$?

Exercise 1.3 (Why the timing convention matters). Problem 0 charges permanent impact as it is incurred. Suppose instead that the whole of ν_t printed at the pre-trade price S_t , with the impact $\kappa \nu_t$ applied only afterwards, so that $R = \sum_t \nu_t (S_t - \frac{\lambda}{2}\nu_t)$. Show that (i) $\mathbb{E}[R]$ then carries a cross term $-\kappa \nu_1 q_1$ and is no longer schedule-independent; (ii) the optimizer is (1.8) with λ replaced by $\lambda - \kappa$, so it requires $\lambda > \kappa$ to be well posed; and (iii) a round trip selling ν at the first date and buying it back at the second earns $(\kappa - \lambda)\nu^2$ in expectation, so the model admits price manipulation when $\kappa > \lambda$. Reconcile (iii) with [Theorem 6.6](#), which says linear permanent impact never permits a profitable round trip, and conclude that the manipulation is an artifact of charging impact asymmetrically in time rather than a property of the impact being permanent.

Exercise 1.4 (Observation lag). Suppose the agent in Fig. 1.1 observes the price with a delay $\tau > 0$, so controls must be $\mathcal{G}_t = \mathcal{F}_{t-\tau}$ -measurable. In the two-period model, does the optimal ν_1^* change? Does the achieved value change? Explain the difference between the two answers.

Fundamental Concepts and Terminology

This chapter contains almost no mathematics. Its job is to attach a picture to each of the two dozen words that the rest of the book uses without explanation: market, asset, tick, bid, ask, spread, mid, limit order, market order, fill, slippage, return, volatility, long, short, leverage, margin, queue, diversification, derivative, hedge, arbitrage.

Every one of these ideas is simple. Taken together they are the reason a reader who is entirely comfortable with viscosity solutions can still be unable to follow a sentence about a trading desk. Finance has an unusually large vocabulary for an unusually small number of underlying concepts, and almost none of it is defined anywhere in the mathematical literature, because everyone writing that literature already knew it.

So read this chapter slowly and once. Each figure is a definition. The caption states the term precisely and says where in the book the idea is formalized, so you can come back to it later as a reference rather than a narrative.

2.1 The economic function of a market

Begin at the beginning, because nothing later makes sense without it.

An **asset** is an economic resource or contractual claim that carries value across time and across states of the world. A share of a company is a claim on its future profits and a bond a claim on a fixed schedule of payments; a barrel of oil is nobody's promise at all, but it stores value and can be sold later, which is all the definition asks. What all assets have in common is that they are worth something to hold, that different people value them differently, and that those valuations change as people learn things.

Because valuations differ, there are gains from trade. If you value a share at \$101 and I value it at \$99, then any transfer between us at a price between \$99 and \$101 makes us both better off. A **market** is a mechanism for finding those transfers and agreeing the price. That is the entire economic purpose of the machinery in this book.

The difficulty is that we are not in the same room. You want to buy today; the person who wants to sell arrives tomorrow. Somebody has to bridge that gap in time, and bridging it is risky, because the price may move in between. The institution that emerged to do this is an **exchange**: a continuously running auction in which anyone may leave a standing offer, and anyone may accept one.

In plain terms

Three roles recur throughout the book and it is worth fixing them now. An **investor** trades because they want the asset, or want to be rid of it, for reasons outside the market. A **market maker** does not want the asset at all; they buy and sell continuously and try to end each day owning nothing, earning the small difference between the two prices. An **arbitrageur** looks for two places

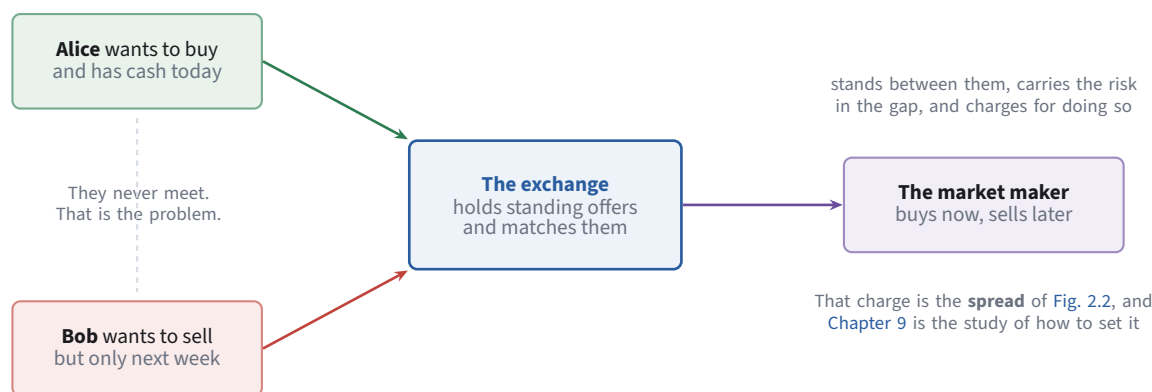


Figure 2.1. Asset, market, exchange, market maker. Two people who value the same **asset** differently would both gain from trading, but they are not present at the same time. An **exchange** lets each leave a standing offer so the two sides can meet asynchronously. When even that is not enough, a **market maker** bridges the remaining gap by buying from the seller now and selling to the buyer later, carrying the risk of the price moving in between. Everything else in this book is a consequence of that last sentence: somebody is holding a position they do not want, for a while, and being paid for it.

where the same thing has two prices. [Chapter 8](#) is the investor's problem and [Chapter 9](#) the market maker's; [Chapter 10](#) and [Chapter 11](#) are variations on both.

2.2 Quoted prices and the bid-ask spread

Start with the object that causes the most confusion. In a model, “the price” is a real number S_t . In a market, there is no such number. There is a price at which you may sell immediately, a different and higher price at which you may buy immediately, and a grid of allowed prices in between at which nobody is currently offering anything at all.

The reason is the asynchrony of [Fig. 2.1](#). Since buyers and sellers do not arrive together, each of them leaves behind a standing instruction: *I will buy up to this much at this price, and I will wait*. The exchange keeps all of these instructions in a list, sorted by price. That list is the **order book**, and it is the single most important object in this book; [Chapter 4](#) takes it apart properly. For now we only need to look at its two edges.

Naturally, everyone who is waiting to buy would like to pay as little as possible, and everyone waiting to sell would like to receive as much as possible. So the buy orders pile up below, the sell orders pile up above, and in between there is a gap where nobody is willing to stand. The gap must exist: if a buyer were ever willing to pay what a seller was asking, the exchange would match them immediately and both orders would vanish. What is left is the picture in [Fig. 2.2](#).

Four numbers summarize that picture and you will meet all four again.

- The **tick** is the resolution of the price grid. Prices are not real numbers; they are multiples of a fixed increment, typically one cent on a share worth tens of dollars. This sounds like a triviality and is not: a coarse grid changes what traders compete over, as [Fig. 2.17](#) shows.
- The **best bid** and **best ask** are the two edges of the gap. If you want to sell right now, the best bid is what you get. If you want to buy right now, the best ask is what you pay. Note the asymmetry that trips up newcomers: *you buy at the ask*

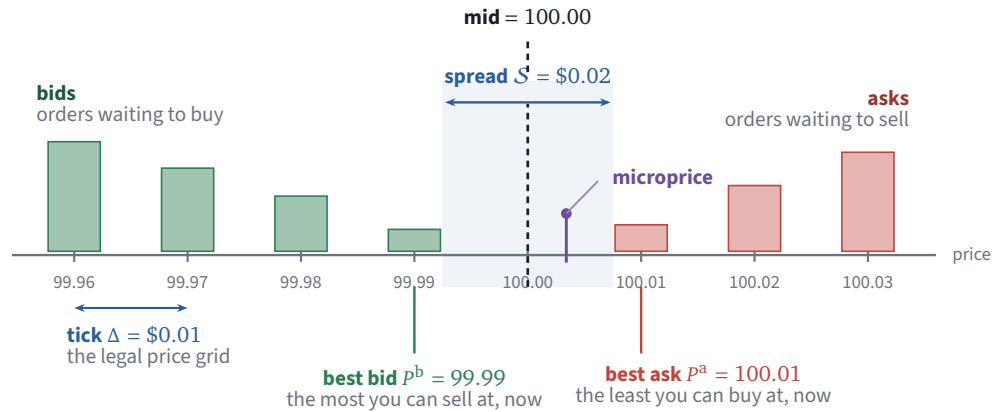


Figure 2.2. Tick, bid, ask, spread, mid, microprice. The **tick** Δ is the spacing of the legal price grid. The **best bid** is the highest price anyone is currently willing to pay and the **best ask** the lowest price anyone will accept; the gap between them is the **spread**. It is empty by definition rather than by accident: an order posted inside the gap immediately becomes the new best bid or best ask, and one posted across it trades at once, so no resting order can survive between the two edges. The **mid** is their average, a convenient fiction that is not a price at which anyone will trade. The **microprice** corrects the mid toward whichever side holds less volume, on the reasoning that the thin side will be consumed first. Formalized in Section 4.3.

and sell at the bid, never the other way around, so a round trip loses money even if the price never moves.

- The **spread** is the width of the gap, and by the previous sentence it is the cost of changing your mind immediately.
- The **mid** is the average of the two edges. It is what people mean by “the price” when they say the price. It is a convenience, not a transactable quantity, and forgetting that is the source of a great many optimistic backtests.

Why the spread is not a fee

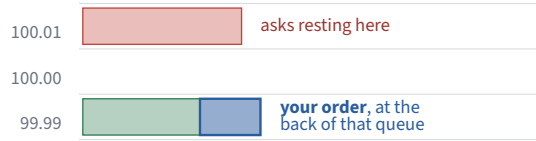
It is tempting to read the spread as a transaction cost, like a commission. It is better read as a *price of immediacy*. Two counterparties with opposite needs and no urgency could meet at the mid and both do better. The spread exists because one of them wants to trade *now*, and the other, who is willing to wait, must be paid for the risk of waiting. Chapter 6 shows that most of that payment is not compensation for waiting at all, but for the risk that the person in a hurry knows something.

2.3 Limit orders and market orders

Everything you can do in a market is a combination of two primitives. The choice between them is the fundamental trade-off of trading, and it has the shape of the engineering trade-offs you already know: you may fix the price or fix the time, but not both.

2.4 Execution benchmarks and slippage

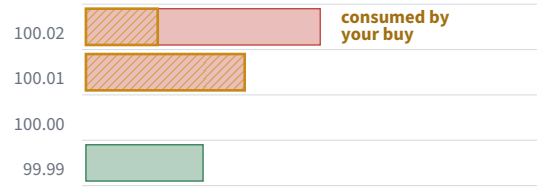
Slippage is the single most important cost in this book, and the word is used for at least three different quantities. They differ only in the **benchmark** they measure

(a) LIMIT ORDER: name your price, wait

price certain: 99.99, or nothing

execution uncertain

You may wait a long time. You may never fill.
And you will fill soonest precisely when the
price is about to fall through you.

(b) MARKET ORDER: name your size, pay

execution certain, immediate

price uncertain, and worse than the touch

You pay 100.01 for the first lots and 100.02 for
the rest. The average is your **fill price**; the gap
to 100.01 is **slippage**, and it grows with size.

Figure 2.3. Limit order, market order, fill, slippage. A **limit order** names a price and joins a queue; a **market order** names a quantity and takes whatever prices are available, **walking the book** upward until it is filled. A **fill** is an execution, whole or partial. The party who posts is the **maker** and the party who takes is the **taker**; many electronic venues pay the first and charge the second, though inverted and flat schedules also exist. Choosing between the two is choosing which of price and time you are willing to leave uncertain, and it is the decision that Chapters 8 and 9 formalize.

against, that is, in the reference price the realized fills are compared with. Figure 2.4 separates them on one timeline.

Two benchmarks matter here. The **arrival price** is the mid at the instant the decision to trade was taken; comparing your average fill price with it gives the **implementation shortfall**, the full economic cost of having wanted the position. The **VWAP**, or volume-weighted average price, is the average price at which the whole market traded over the same window, each trade weighted by its size,

$$\text{VWAP} = \frac{\sum_j v_j p_j}{\sum_j v_j}, \quad (2.1)$$

the sum running over every trade j in the window at price p_j and size v_j . Comparing against it asks a narrower question: not what the decision cost, but whether you traded better or worse than everyone else who was trading at the same time. The two can point in opposite directions, and the difference between them is the subject of Section 8.2.

2.5 Returns and volatility

Prices are quoted in dollars, but almost nothing in finance is expressed in dollars, because a dollar move means something different for a \$10 stock and a \$1000 stock. Everything is expressed in **returns**: the fractional change over some period. If the price goes from 100 to 101, the return is 1%.

Returns in this subject are usually small, so they are quoted in **basis points**. One basis point, written 1 bp, is one hundredth of one percent, that is 10^{-4} . A one-cent spread on a \$40 share is $0.01/40 = 2.5$ bp; a strategy earning 8% a year earns 800 bp; a typical market-making margin is a fraction of a basis point. The unit exists because the interesting quantities in trading are three or four orders of magnitude smaller than the prices they are computed from, and this book uses it throughout without further comment.

There are two conventions for the return itself and they differ in ways that matter later. The **arithmetic return** is $(S_t - S_{t-1})/S_{t-1}$; the **log return** is $\log(S_t/S_{t-1})$. They agree to first order for small moves. Log returns add up across time, which makes them the right choice for describing growth; arithmetic returns add up across assets,

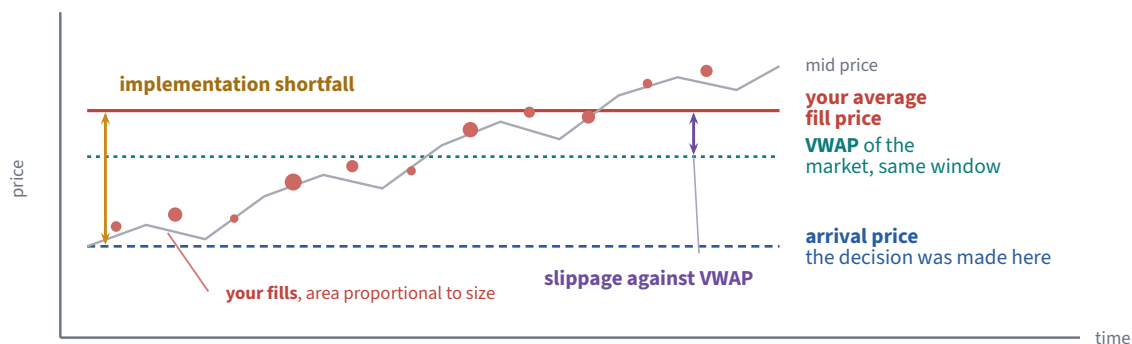


Figure 2.4. Slippage three ways. A buy order arrives at the left and is executed toward the right; the dots are the individual fills. Two different things happen at once and separating them is the whole job of transaction cost analysis. Measured against the **arrival price**, the mid at the moment the decision was made, the total gap is the **implementation shortfall**: it contains both the cost you caused and the drift you suffered while working. Measured against the market's VWAP over the same window, it isolates how well you traded relative to everyone else, which is the benchmark a desk prefers to be judged on and the one that tells an investor least. Trading faster shrinks the variance of the first gap and widens its mean; that trade-off is [Chapter 8](#).

which makes them the right choice for describing a portfolio. You cannot have both, and [Section 3.3](#) explains the consequences.

Volatility is the standard deviation of returns, quoted per year by convention. It is the standard second-moment measure of return uncertainty and the scale parameter that appears in every model in Part II. It is not the whole of risk: skewness, jumps and tail dependence are invisible to it, which is the objection [Chapter 12](#) develops at length. As a rule of thumb a large-cap share has a volatility of 15 to 30% per year; in a crisis this can triple.

Return and volatility are almost never quoted alone, because either can be made to look good at the expense of the other. What is compared instead is their ratio. Write r for the return an investment earns in excess of cash, the **excess return**, and σ for its volatility. Then

$$SR = \frac{r}{\sigma} \quad (2.2)$$

is the **Sharpe ratio**: reward per unit of risk, quoted per year like its two ingredients.

In plain terms

The Sharpe ratio is the closest thing the subject has to a universal score, and it is worth calibrating your expectations against it now, because numbers will be quoted against it for the rest of the book. What follows are order-of-magnitude calibrations rather than constants; they move with the period, the asset class and the measurement convention. A broad equity index earns roughly 0.4 over long periods. A good systematic strategy runs at 1. A sustained 2 is rare and a sustained 3 is rarer still, and both should prompt the question that [Chapter 13](#) is about, which is how many candidates were examined before this one was reported.

Two properties explain why it is used. It is scale-free, so a levered version of a strategy has the same Sharpe ratio as the unlevered one, which means it measures skill rather than size. And under the standard approximation that returns are serially independent it scales with the square root of the horizon, exactly as volatility does, so a daily Sharpe ratio of s annualizes to $s\sqrt{252}$; autocorrelation, overlapping positions and smoothed marks all break

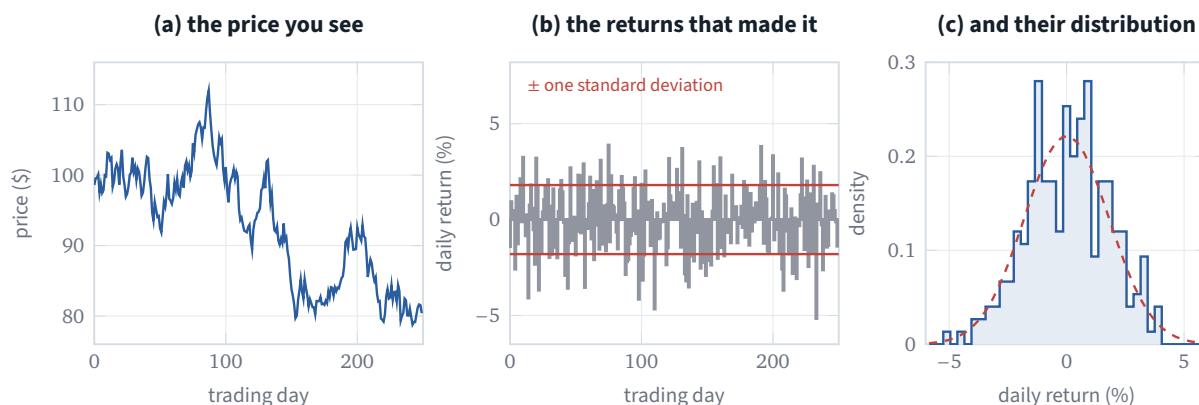


Figure 2.5. Return, volatility, random walk. The same simulated year seen three ways. (a) The price wanders. (b) Its daily **returns** are what actually gets modelled: they have no visible pattern, which is the empirical content of the statement that prices are close to a random walk. (c) Their spread is the **volatility**, here 1.8% per day, which by the square-root-of-time rule is about $1.8\sqrt{252} \approx 28\%$ per year. The dashed curve is the matched normal distribution. Real returns have fatter tails than this, and Fig. 5.3 shows how much fatter.

that scaling, generally in the flattering direction. What it does not do is see the shape of the tail, which is the objection Chapter 12 develops at length.

In plain terms

Two rules of thumb do most of the work in practice. First, volatility scales with the *square root* of time: if daily volatility is σ_d , then volatility over n days is $\sigma_d\sqrt{n}$. Second, a 30% annual volatility means a typical day moves about 2%, because $0.30/\sqrt{252} \approx 0.019$. Both follow from returns being roughly independent across time, which is the one empirical fact about prices that almost every model assumes and that is very nearly true.

2.6 Diversification

Here is the closest thing to a free lunch that finance has. It deserves a section rather than a sentence, because the reason it works is not the reason most people think.

2.6.1 The case for combining assets

Suppose you have identified the asset with the highest expected return. Why hold anything else? Adding a worse asset must lower your expected return, so surely it can only hurt.

It does lower the expected return, exactly in proportion. What it also does is lower the *risk*, and by more than in proportion, so long as the two assets do not move in lockstep. The expected return of a mixture is the weighted average of the two expected returns, and averages are linear. The variance of a mixture is not the average of the two variances, and that is the whole story.

Write it out for two assets with weights w and $1 - w$:

$$\sigma_{\text{portfolio}}^2 = w^2\sigma_A^2 + (1 - w)^2\sigma_B^2 + 2w(1 - w)\rho\sigma_A\sigma_B. \quad (2.3)$$

The first two terms are the diversifiable part: squaring the weights shrinks them. The third term is the interaction, and its size is governed by the **correlation** ρ , which

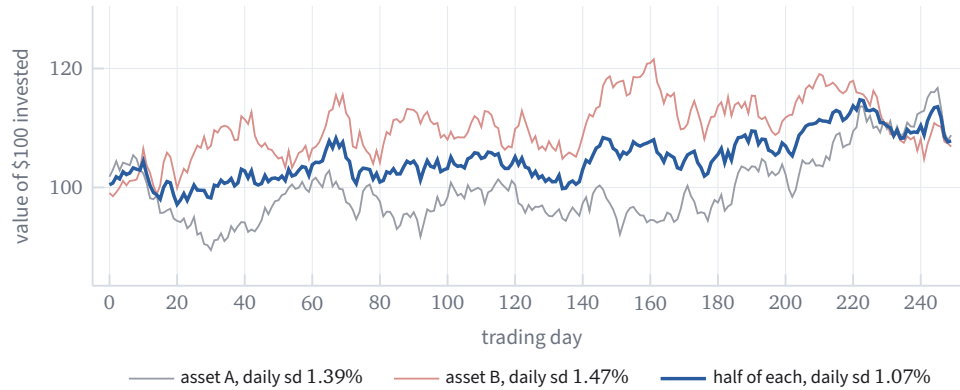


Figure 2.6. The same journey with a smoother ride. Two simulated assets with correlation 0.15, and the portfolio holding half of each. All three end in much the same place, so nothing was gained in return. But the blue line is visibly less jagged than either ingredient: its daily volatility is a quarter lower, and that reduction cost nothing, required no forecast, and involved no skill.

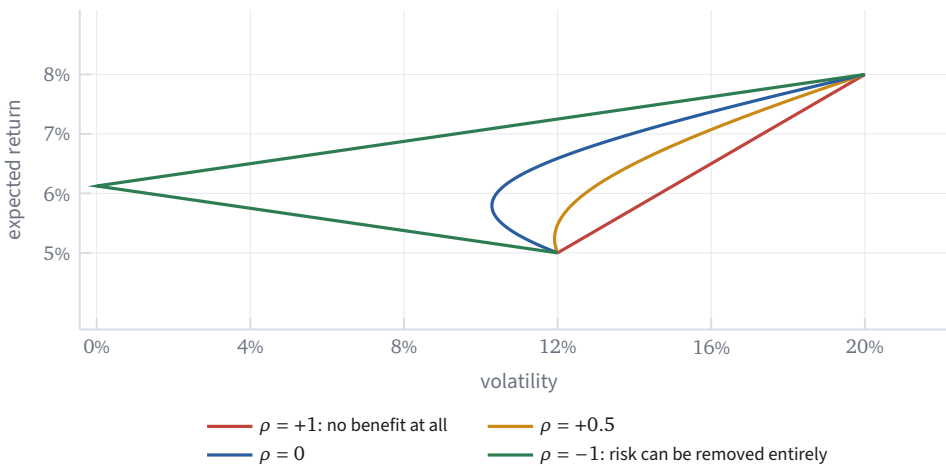


Figure 2.7. Correlation is the dial that controls how much diversification is available. The same two assets, A with 8% return and 20% volatility and B with 5% and 12%, mixed in every proportion, for four different correlations. When $\rho = +1$ the set of mixtures is a straight line and combining them achieves nothing. As ρ falls the curve bows left, and every point of that bow is risk that has been destroyed rather than transferred. At $\rho = -1$ the bow reaches the vertical axis: some mixture of two risky assets is completely riskless. Real correlations between broad asset classes typically sit between 0 and 0.6, which is why the effect is large but not miraculous.

runs from -1 (the two always move oppositely) through 0 (unrelated) to $+1$ (always together).

For equal volatilities σ and $w = \frac{1}{2}$, (2.3) collapses to a formula worth memorizing:

$$\sigma_{\text{portfolio}} = \sigma \sqrt{\frac{1+\rho}{2}}. \quad (2.4)$$

At $\rho = 1$ you get σ back and nothing has been gained. At $\rho = 0$ you get $\sigma/\sqrt{2}$, a 29% reduction, for free. At $\rho = -1$ you get zero: the two risks cancel exactly, which is the mathematical extreme that hedging in §2.7 tries to approach on purpose.

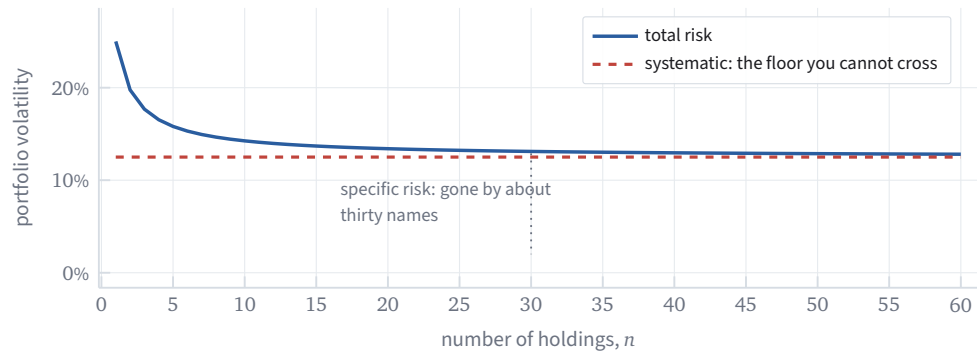


Figure 2.8. Diversification works quickly and then stops. Equally weighted portfolios of n assets, each with 25% volatility and each pair correlated at 0.25, from (2.5). The first ten holdings do most of the work, cutting risk from 25% to 14.3%; the next fifty take it only to 12.8%, against a floor of $\sigma\sqrt{\rho} = 12.5\%$ that no amount of further diversification can breach. This is why a portfolio of thirty well-chosen names behaves much like a portfolio of five hundred, and why the residual common risk is the thing that has to be priced, hedged or simply borne.

2.6.2 The many-asset limit

Two assets show the mechanism; many assets show its limit, and the limit is the more important lesson.

Hold n assets in equal weights, each with volatility σ and each pair correlated by the same ρ . Then (2.3) generalizes to

$$\sigma_{\text{portfolio}}^2 = \frac{\sigma^2}{n} + \left(1 - \frac{1}{n}\right)\rho\sigma^2 \xrightarrow{n \rightarrow \infty} \rho\sigma^2. \quad (2.5)$$

The first term is the part that averages away. The second does not: no matter how many names you own, you cannot get below $\sigma\sqrt{\rho}$.

That split has a name and it is one of the most-used ideas in the subject. The part that vanishes is **specific** or **idiosyncratic** risk, the thing that happens to one company and not the others. The part that survives is **systematic** or **market** risk, the thing that happens to everything at once. Diversification is free protection against the first and no protection whatsoever against the second.

Where the model breaks

The catch is in the word “same” in (2.5). Correlations are not constants; they are estimated, they drift, and they have a strong tendency to rise toward one precisely during the episodes that diversification was supposed to protect against. A portfolio whose historical correlations imply a 12% volatility can behave like a single asset for a week. This is not a subtlety at the edges: it is the standard way that diversified portfolios lose money, and Fig. 7.3 shows the liquidity half of the same phenomenon.

2.7 Derivatives and hedging

A **derivative** is a contract whose payoff is a function of something else. The something else is called the **underlying**. That definition is so general as to be useless on its own, so start instead with the question of why anyone would want such a thing.

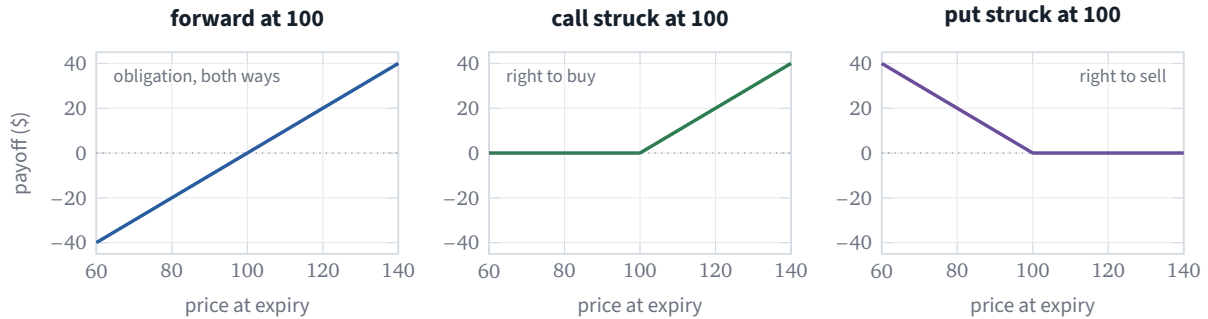


Figure 2.9. Forward, call, put. A forward is a straight line: symmetric, free to enter, and an obligation. The two options are kinked: asymmetric, and therefore not free, because a right that you exercise only when it suits you is worth something. Any payoff that depends only on the terminal price can be assembled from these three shapes plus cash, which is the observation behind static replication and the reason positions in them add in a way that positions in the underlying do not.

2.7.1 The economic rationale for derivatives

An airline knows it will buy jet fuel every month for the next three years. It is good at running an airline and has no view whatever on the oil market, but its profits swing violently with the oil price all the same. A farmer will harvest wheat in September and would like to know in June what it will fetch. A European manufacturer sells into the United States and is paid in dollars, so its earnings depend on an exchange rate it neither forecasts nor controls.

In each case somebody is carrying a risk that is incidental to their actual business. A derivative is the instrument that lets them hand it to somebody else. The airline buys a contract that pays out when fuel is expensive; the farmer sells one that fixes their September price; the manufacturer sells dollars forward. None of them has made a bet. Each has *removed* one.

That is the economic function, and it is worth holding on to, because the mathematics that follows is easy to mistake for a machine for speculation. The speculation exists, and it is what makes the market liquid enough for the airline to trade in, but the reason the instruments were invented is insurance.

2.7.2 Forwards, calls and puts

Almost everything traded is assembled from three shapes.

A **forward** is an agreement to trade at a fixed price on a fixed future date, with no money changing hands today. Its payoff is a straight line: you gain a dollar for every dollar the price ends above the agreed level and lose one for every dollar below. This is the farmer's instrument, and (2.6) will show that its fair price is forced by arbitrage.

A **call option** is the *right*, not the obligation, to buy at a fixed **strike** price K on a fixed date. If the price ends above K you exercise and collect the difference; if below, you walk away and lose only what you paid for the option. Its payoff is $\max(S_T - K, 0)$: flat, then a line with a kink at the strike. This is the airline's instrument.

A **put option** is the mirror image, the right to *sell* at K , worth $\max(K - S_T, 0)$. It is insurance against a fall.

2.7.3 Asymmetry and the option premium

A forward is free to enter. An option is not, and the reason is exactly the asymmetry. The holder of a call participates fully when the price rises and not at all when it falls, so the seller is taking a strictly one-sided risk and must be paid a **premium** for it.

That premium is what Chapter 10 computes, and it also produces the feature that draws speculators. A call struck at \$100 on a \$100 share might cost \$5. If the share

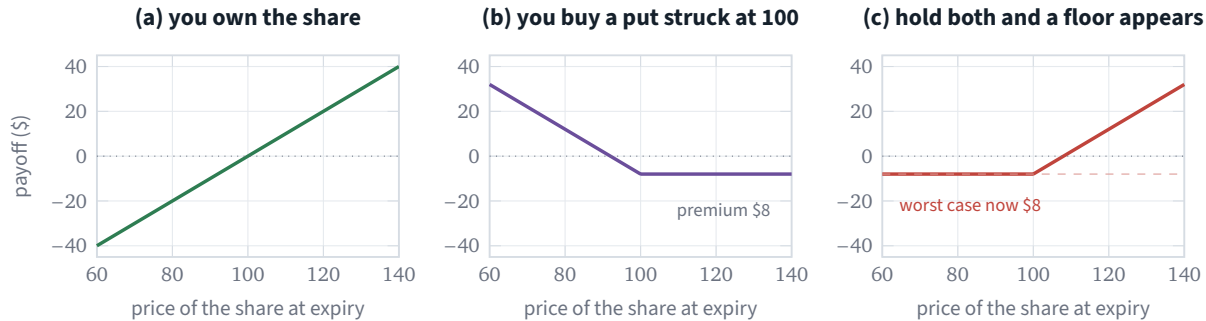


Figure 2.10. Hedging is addition. (a) Owning a share loses as the price falls, down to the whole of the initial outlay but no further: with limited liability the price cannot go below zero. (b) A put pays only when the price falls, and costs a premium up front. (c) Add the two payoff curves pointwise and the losses below \$100 cancel exactly, leaving a position that keeps all the upside and whose worst case is the premium paid. This is what an insurance policy looks like when drawn as a function.

risers to \$110 the option is worth \$10, so a 10% move in the underlying has doubled your money, while the most you could have lost was the \$5. Options embed leverage without borrowing, which is useful and dangerous in the usual proportions.

2.7.4 Hedging as the addition of payoffs

Payoffs are additive. If you hold two positions, the payoff of the pair is the sum of the two payoffs, and this is the fact that makes the whole subject tractable, because it turns the design of a desired exposure into a *linear* problem before any probability appears.

Hedging exploits exactly that. Take a second position whose payoff curve bends the opposite way to the one you already have, and the sum is flatter than either. [Figure 2.10](#) does this in the simplest case: own a share, buy a put, and the loss below the strike is cancelled, leaving a payoff with a floor.

The question this book actually asks

[Figure 2.10](#) assumes somebody will sell you the put. The interesting question, and the one [Chapter 10](#) is about, is what the person selling it should charge and what they should do afterwards, since they now hold the risk the buyer shed.

The remarkable answer is that they need not hold it at all. By trading the underlying share continuously in the right pattern, they can manufacture the put's payoff themselves out of shares and cash. If they can do that exactly, then the fair premium is simply what that trading program costs, and it can be computed without any opinion about where the share is going. [Section 10.1](#) does this in a two-state world in half a page, and it is the single most surprising calculation in finance.

2.8 Arbitrage

Arbitrage is the technical word for free money, and it deserves more space than it usually gets, because it does something no other idea in finance does: it lets you work out what things must be worth without having any opinion at all about the future.

Definition 2.1 (Arbitrage, informally). A trading strategy is an arbitrage if all three of the following hold.

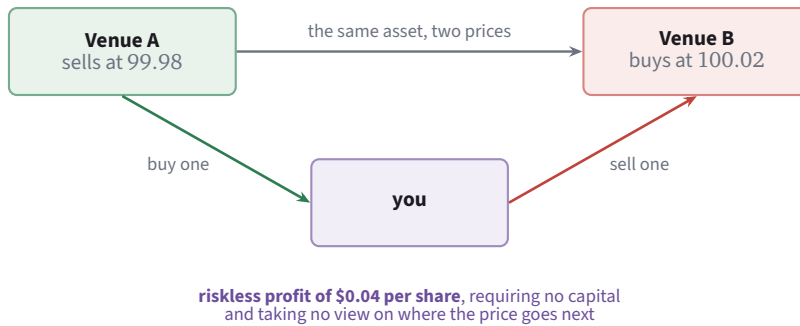


Figure 2.11. The simplest arbitrage. Two venues quoting the same instrument at different prices. You buy on one and sell on the other in the same instant, so you never hold a position and never carry a risk. Note that you did not need to believe anything about the company, the economy or tomorrow. That is the defining feature.

- (1) It costs nothing to set up today, or better, it pays you something.
- (2) Under *no* future scenario whatsoever does it lose money.
- (3) Under *at least one* scenario it makes money.

Consider those three conditions again, because the strength of the idea lies in the word “no” in the second one. This is not a good bet, or a bet with favourable odds. It is not a bet at all. Nothing has to go right.

Such things should not exist, and mostly they do not. The interesting move is to turn that around and use their non-existence as an axiom: *assume* that no arbitrage is available, and see what that assumption forces prices to be. It turns out to force a great deal. Three examples, in increasing order of usefulness.

2.8.1 Arbitrage across venues

The same asset, two venues, two prices. Buy where it is cheap, sell where it is dear, and the difference is yours immediately, with no position left over and no exposure to anything.

This version is real but rare and short-lived, which is the point: someone does it within microseconds, their buying lifts venue A and their selling depresses venue B, and the gap closes. Arbitrage is self-destroying, which is exactly why assuming its absence is a reasonable modelling premise.

2.8.2 Triangular arbitrage

Now a version where nothing is obviously mispriced, because three separate prices are individually reasonable and only jointly impossible.

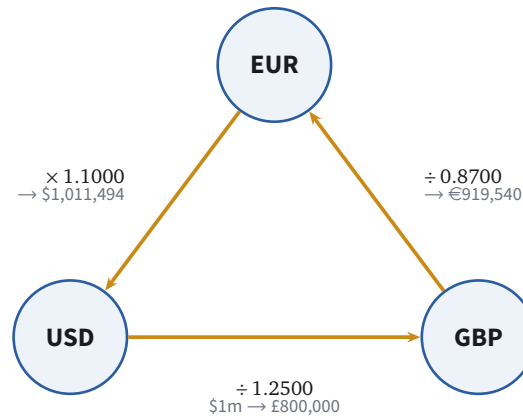
Three currencies. The market quotes

EUR/USD = 1.1000	one euro buys \$1.10
GBP/USD = 1.2500	one pound buys \$1.25
EUR/GBP = 0.8700	one euro buys 0.87 pounds

Nothing about any single quote looks wrong. But the first two together already determine the third: a euro is worth \$1.10, and \$1.10 buys $1.10/1.25 = 0.8800$ pounds. The third quote says 0.8700, so euros are cheap when bought with pounds.

Take a million dollars around the triangle.

\$1,000,000	→	\$ 800,000	at 1.2500
	→	€ 919,540	at 0.8700
	→	\$1,011,494	at 1.1000



\$11,494 richer for going round in a circle. The three quotes are individually plausible and jointly impossible: any two of them determine the third.

Figure 2.12. Triangular arbitrage. No single price here is obviously wrong, which is what makes the example instructive. Arbitrage is a statement about the *consistency* of a set of prices, not about whether any one of them is too high. Real mispricings of this kind are a tiny fraction of the size shown and are gone in microseconds, which is precisely the evidence that somebody is enforcing the relationship.

You end the loop with \$11,494 more than you started, having held each currency for a fraction of a second and having taken no view on any of them.

2.8.3 Cash-and-carry arbitrage

The previous two examples removed inconsistencies. This one manufactures a price out of thin air, and it is the template for the whole of [Chapter 10](#).

Gold costs \$2,000 an ounce today. Borrowing costs 5% for a year, and storing gold is free. What should it cost to agree *today* to buy an ounce of gold in one year, with payment on delivery? Such an agreement is a **forward contract**, and its price is called the forward price.

You might think the answer depends on where you expect gold to be in a year. It does not, and here is why. Suppose the forward price were \$2,150.

- Borrow \$2,000 and buy an ounce of gold today. Put it in a vault.
- Simultaneously agree to sell an ounce forward at \$2,150.
- A year later: deliver the gold you already own, collect \$2,150, and repay the loan, which has grown to $2,000 \times 1.05 = \$2,100$.

You are left with \$50 and no position. Gold could have gone to \$5,000 or to \$500; you would still have exactly \$50, because you never had any gold exposure. That is an arbitrage, so \$2,150 cannot be the price.

Run the same argument with the forward at \$2,050: borrow the gold, sell it, invest the proceeds at 5%, and buy it back through the forward, again ending with \$50 and no risk. So \$2,050 is impossible too.

The only price that admits neither trade is

$$F = 2,000 \times 1.05 = \$2,100, \quad (2.6)$$

and it was forced on us by two facts, the spot price and the interest rate, neither of which is a forecast.

What arbitrage actually buys you

(2.6) contains no probabilities, no expected return on gold, and no risk aversion. It did not require anyone to agree about the future. It required only that a specific pair of trades not be free money.

That is the move that makes mathematical finance possible. Instead of asking “what is this worth?”, which needs a forecast and a utility function and which two reasonable people will answer differently, we ask “what must this cost, *given* the prices of things that already trade?”, which often has a single answer that everybody is forced to accept. [Chapter 5](#) shows that this question is a linear program, and [Chapter 10](#) shows that its solution is a trading strategy rather than merely a number.

Where the model breaks

Real arbitrage is not quite free. You need capital to post as collateral, the two legs may not execute at the same instant, borrowing the asset to short it may be expensive or impossible, and a position that is riskless at maturity can lose a great deal in between and force you out before you get there. The academic phrase is *limits to arbitrage*, and it is why the relationships above hold approximately rather than exactly, and why the people who enforce them are paid.

2.9 The bid-ask bounce and microstructure noise

One last picture, because it explains a phenomenon that will otherwise look like a bug in your data.

Transactions alternate between buyer-initiated and seller-initiated more or less at random. Buyer-initiated trades print at the ask; seller-initiated trades print at the bid. So even when the underlying value of the asset is perfectly constant, the sequence of *traded prices* oscillates up and down by the width of the spread.

This has a name, **microstructure noise**, and two consequences worth absorbing now.

The first is that the price you can measure is not the price you want. Somewhere behind the quotes there is a number representing what the asset is currently worth, and every model in Part II is written about that number. It is not observable. What is observable is that number plus an error which is small, bounded by roughly half the spread, and violently mean-reverting, because a trade at the ask tends to be followed by a trade at the bid.

The second is that this particular kind of error is unusually destructive to statistics. Ordinary measurement noise makes an estimate less precise, and you recover by collecting more data. Here the opposite happens. Suppose you estimate volatility by adding up squared price changes. Over a long interval the genuine price movement is large and the bounce is negligible; over a very short interval the genuine movement is tiny and almost everything you are measuring is the bounce. Sampling more finely therefore adds more noise than signal, and the estimate does not converge to the truth but grows without bound. [Proposition 5.4](#) proves this and [Fig. 5.4](#) shows it happening.

The practical upshot is a rule you will meet repeatedly: at high frequency more data is not automatically better, and the sampling interval is a parameter to be chosen rather than a nuisance to be minimized.



Figure 2.13. The bid-ask bounce as microstructure noise. The dashed line is the **efficient price**, the value the asset would have if it could be observed directly. It is not observable. What you observe are transactions, and those alternate between the bid and the ask according to who happened to be in a hurry, adding a large, rapidly mean-reverting error to every measurement. The consequence is severe: the observed price sequence looks far more volatile than the thing it measures, and [Proposition 5.4](#) shows that naïve volatility estimates do not merely lose accuracy at high frequency, they diverge.

2.10 Long and short positions

Your **position**, also called your **inventory**, is simply how much of the asset you are holding. If you own a hundred shares you are **long** a hundred; if you own none you are **flat**. Modelling this is easy, and Part II will treat it as a single signed number Q_t .

The complication is that the number is allowed to go negative. Being **short** means you have sold something you did not own, which sounds impossible and is in fact a piece of financial engineering. You borrow the share from somebody who owns it and has no immediate plans for it, you sell the borrowed share, and you now owe a share rather than owning one. Later you buy one back and return it. If the price fell in between, you buy back cheaper than you sold and keep the difference.

Why bother? Two reasons, and both matter for this book. First, it is the direct way to express a negative view: without shorting, a trader who thinks something is overpriced can only decline to own it, or buy a put from somebody who will short it for them. Second, and far more important here, shorting is what makes *hedging* possible: to cancel the risk of something you hold, you generally need to be short something else. Half the positions in [Chapters 10](#) and [11](#) are negative for exactly this reason.

The trouble is that the mathematics is symmetric and the institution is not, in three ways that a model writing $Q_t \in \mathbb{R}$ will silently miss. A long position can lose at most what it cost, because the price cannot fall below zero, whereas a short position's loss is unbounded, because the price can rise without limit. A long position is free to hold, whereas a short pays a **borrow fee** that ranges from negligible to punitive. And a long position is yours to close when you choose, whereas a short can be **recalled** by the lender at a moment of their choosing, forcing you to buy back at whatever price is then available.

2.11 Leverage and margin

Trading is done with borrowed money far more often than not, and the amount borrowed is what turns a modest strategy into an interesting one, or into a catastrophe.

Leverage is the ratio of the total value of what you control to the capital that is actually yours. If you have \$100 of your own, borrow \$400, and buy \$500 of an asset,

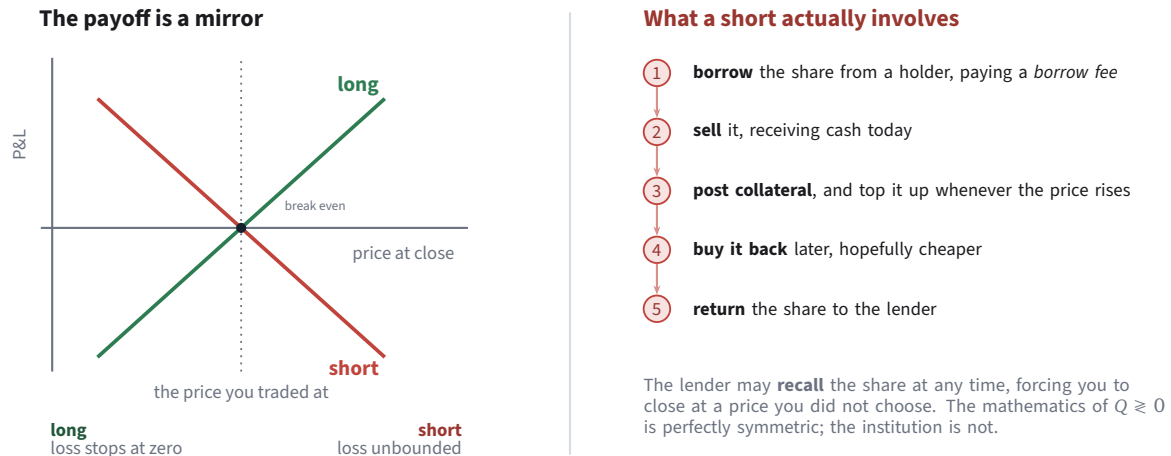


Figure 2.14. Long, short, borrow, recall. A **long** position profits when the price rises and cannot lose more than it cost. A **short** position is created by borrowing the asset and selling it, so it profits when the price falls, and its loss is unbounded above. Both appear in Part II as a signed inventory Q_t , which hides the asymmetry entirely, so (3.8) restores it as an explicit borrow cost.

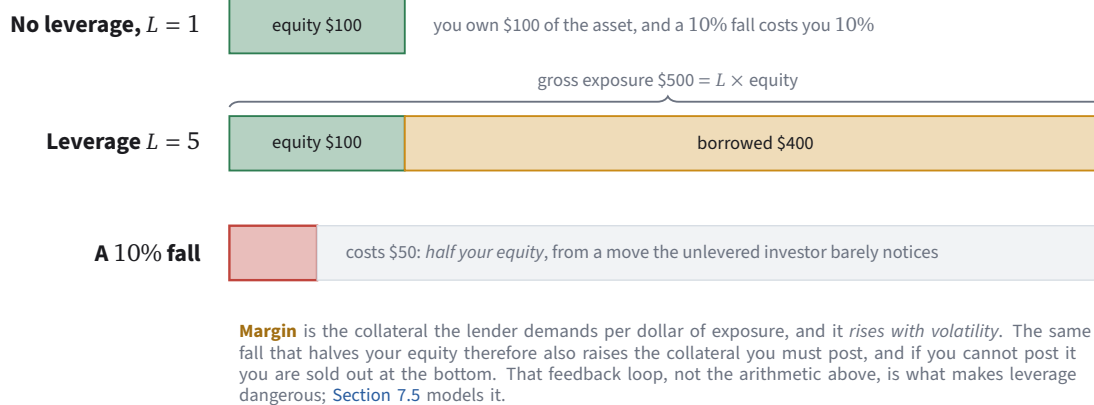


Figure 2.15. Leverage, margin, and the margin call. Leverage L is gross exposure divided by equity. It multiplies gains and losses equally, which is the part everyone knows. The part that matters is that the **margin** requirement is not a constant: it tightens exactly when prices fall and volatility rises, so the constraint binds at the worst possible moment.

your leverage is five. The arithmetic is obvious: every percentage move in the asset is now worth five times as much to you, in both directions.

The part that is not obvious, and that causes the trouble, is the **margin**. The lender does not simply hand over the \$400; they require collateral, and they mark the position to market continuously. If the value falls, your equity falls with it, and the lender demands more collateral: a **margin call**. If you cannot meet it, they liquidate the position for you, at whatever price is available, at the worst possible moment.

Worse, the required margin per dollar of exposure is not a constant. It is set from a risk model, and risk models raise their estimates when volatility rises. So the same shock that reduces your equity also raises the collateral demanded against it, and the constraint tightens from both sides at once. Section 7.5 models this feedback explicitly, and Fig. 7.3 shows the empirical shape of it.

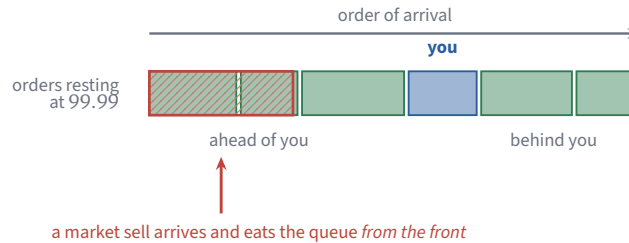


Figure 2.16. Queue position and price-time priority. At a given price, resting orders are filled in the order they arrived. Your **queue position** is the volume ahead of you, and it decides whether you trade at all. This is why microseconds matter even to strategies whose horizon is minutes: the contest is not to predict the price, it is to be early in the queue when a price level opens. [Section 4.5](#) turns this picture into a closed-form fill probability.

2.12 Queue position and priority

Return to the limit order of [Fig. 2.3](#). You have named a price and joined a queue. Whether you actually trade now depends on a detail that no continuous-time model contains: how many other people named the same price before you did.

Exchanges resolve competing orders by **price-time priority**. Better prices go first, which is what makes the book sorted; among orders at the *same* price, earlier arrivals go first. The volume sitting in front of you is your **queue position**, and it is the difference between trading and watching.

The consequence is easy to state and surprisingly large. When a price level first opens, everybody wants to be at the front of it, and the only way to get there is to arrive first, measured in microseconds. This is the origin of the latency competition that dominates modern market structure, and it is why a strategy whose ideas play out over minutes may still need hardware measured in nanoseconds. [Section 4.5](#) converts the picture below into a fill probability, and finds that being twenty lots back rather than at the front can cost fifteen to thirty percentage points of the chance of ever trading.

2.13 Tick size regimes

The last piece of vocabulary is the one most likely to seem like a detail and least likely to behave like one.

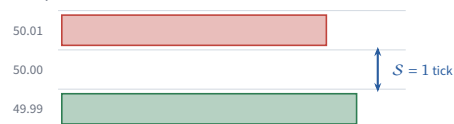
Prices live on a grid, and the spacing of that grid, the **tick**, is set by the exchange rather than by the market. Whether the tick is large or small is not an absolute question but a relative one: large or small compared with how much the price typically moves between consecutive trades. That single comparison determines which of two quite different games the participants are playing.

If the tick is large, the spread is pinned at one tick and cannot narrow. There is then no way to compete by offering a better price, because the next better price is a long way away and would cost more than the trade is worth. Competition moves to *time*: queues at the touch grow very long, and the scarce resource is priority, which is why [Section 4.5](#) matters so much in this regime.

If the tick is small, a quote can be improved by a trivial amount. Anyone can step in front of anyone else for a hundredth of a cent, so queues never grow, priority is worth almost nothing, and competition happens in *price* instead. Regulators change tick sizes deliberately, and the same instrument can move between these two regimes when they do.

Large tick

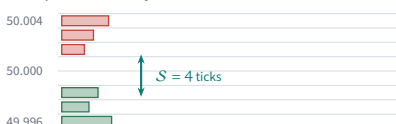
the spread cannot narrow below one tick



You cannot compete on price, so you compete on *time*. Queues are long, and **priority is the scarce resource**.

Small tick

the spread is many ticks wide



Undercutting costs almost nothing, so you compete on *price*. Queues are short, and **priority is cheap to buy**.

Figure 2.17. The tick size decides which game is being played. When Δ is large relative to the price movement between consecutive trades, the spread sits on its floor of one tick and cannot narrow, so competition moves into the queue and the model of Section 4.5 is the relevant one. When Δ is small, a quote can be improved for a negligible amount, queues stay short, and competition happens in price, which is the regime that the market-making model of Chapter 9 assumes. The same instrument, listed with different tick sizes, needs genuinely different models.

A one-page summary

The whole vocabulary in order

A **market** is a queue of **limit orders** at each price on a grid of **ticks**. The best buy order is the **bid**, the best sell order the **ask**, their gap the **spread**, and their average the **mid**. You may join a queue and wait, or send a **market order** and take. Taking is certain but expensive: you **walk the book** and pay **slippage**. Your holding is your **inventory** or **position**, positive for **long**, negative for **short**. Its size relative to your capital is **leverage**, and the collateral you must post against it is **margin**. Everything else in this book is either a measurement of these quantities or a decision about them.

Positions and Profit and Loss Accounting

P&L, spoken as “pee and ell”, is short for *profit and loss*: the money a position has made or lost over some period. It is the quantity that every objective function in this book is ultimately written in, and this chapter pins down exactly how it is computed.

That makes it the chapter nobody wants and everybody needs. It contains no optimization. Its purpose is to make the state equation of a trading problem *exactly* right, because most of the sign errors you will make in this subject are made here, and a sign error in the state equation produces a model that is not merely inaccurate but backwards.

The payoff is a single identity (the self-financing condition, (3.5)) from which the objective functions of Chapters 8 to 11 all descend.

In plain terms

This chapter is bookkeeping. If you buy something and later sell it, how much did you make? The answer looks trivial and is not, because three different things are changing at once: the number of units you hold, the cash in your account, and the price at which the world would currently value what you hold. Keeping those three straight is what lets us say, precisely, that *you earn on what you held and you pay on what you changed*. Every objective function in Part II is built from that sentence.

3.1 Three primitive quantities

A position in one asset is completely described by two numbers and one convention.

Together, cash and holdings form what accountants call a **ledger**: the running record of what you own and what you owe, updated every time something happens. This chapter is about keeping that record correctly, and the two numbers below are all it contains.

Definition 3.1 (Position). At each time t the agent holds $Q_t \in \mathbb{R}$ units of a risky asset and $X_t \in \mathbb{R}$ units of cash. We allow both to be negative: $Q_t < 0$ is a **short** position, $X_t < 0$ is borrowing. The **mark-to-market wealth** is

$$V_t := X_t + Q_t S_t, \quad (3.1)$$

where S_t is the **mark**: a single price at which the position is notionally valued.

Where the model breaks

(3.1) is a convention, not a fact. There is no single price at which you can trade Q_t units: you can sell at the bid, buy at the ask, and neither is available in unlimited size. Marking at the mid $S_t = \frac{1}{2}(P_t^b + P_t^a)$ systematically overstates the

value of a position you would have to liquidate. Chapter 7 makes the correction explicit as a *liquidity-adjusted* mark $V_t^{\text{liq}} = X_t + \int_0^{Q_t} p_t^b(u) du$, which is concave in Q_t , and it is that concavity, not (3.1), which makes large positions genuinely risky.

Control-theoretic dictionary

Q is the state you control; X is an accounting state whose dynamics are induced by trading rather than given by an independent state equation; S is the latent price, exogenous only in the absence of impact. Wealth V is an *output*, not a state: writing dynamics for V directly is possible but hides the fact that only Q is controlled.

3.2 The self-financing condition

3.2.1 The discrete-time ledger

Trade only at $t_0 < t_1 < \dots < t_n$. Let $\Delta Q_k = Q_k - Q_{k-1}$ be the quantity bought at time t_k (negative if sold), and suppose it executes at price $\hat{S}_k$, which need not equal the mark S_k . Cash changes by the negative of what you paid:

$$X_k = X_{k-1} - \hat{S}_k \Delta Q_k. \quad (3.2)$$

No money enters or leaves the account from outside, which is precisely what **self-financing** means. Wealth then satisfies

$$\begin{aligned} V_k - V_{k-1} &= (X_k + Q_k S_k) - (X_{k-1} + Q_{k-1} S_{k-1}) \\ &= \underbrace{Q_{k-1} (S_k - S_{k-1})}_{\text{market P\&L}} - \underbrace{(\hat{S}_k - S_k) \Delta Q_k}_{\text{execution cost}}. \end{aligned} \quad (3.3)$$

(3.3) is the whole of trading P&L, and it says something worth internalizing: *you earn on the position you held, and you pay on the position you changed*. The first term rewards being right about direction; the second punishes being impatient. Everything in Part II is a trade-off between these two terms.

3.2.2 The continuous-time limit

Let the agent trade at a rate $\nu_t = \dot{Q}_t$ (shares per unit time), and let the *execution price* carry a temporary impact $\hat{S}_t = S_t + h(\nu_t)$ as in (1.4). Then (3.2) becomes

$$dX_t = -(S_t + h(\nu_t)) \nu_t dt, \quad dQ_t = \nu_t dt, \quad (3.4)$$

and Itô's formula applied to (3.1) gives the continuous-time self-financing identity

$$dV_t = Q_t dS_t - h(\nu_t) \nu_t dt. \quad (3.5)$$

In words: *wealth changes by position times price change, minus the cost of trading*. In the frictionless limit $h \equiv 0$ this is the familiar $dV_t = Q_t dS_t$, i.e. wealth is a stochastic integral against the price, which is exactly why the martingale property of S under a pricing measure will turn “no arbitrage” into “no free lunch” in Section 5.2.

Remark 3.2. The term $-h(\nu)\nu$ is the *only* place friction enters, and it is nonpositive whenever h has the same sign as ν : i.e. whenever buying pushes the price up. If $h(\nu) = \lambda\nu$ then the running cost is $\lambda\nu^2$, quadratic and strictly convex: the mathematical reason execution problems are LQ problems.

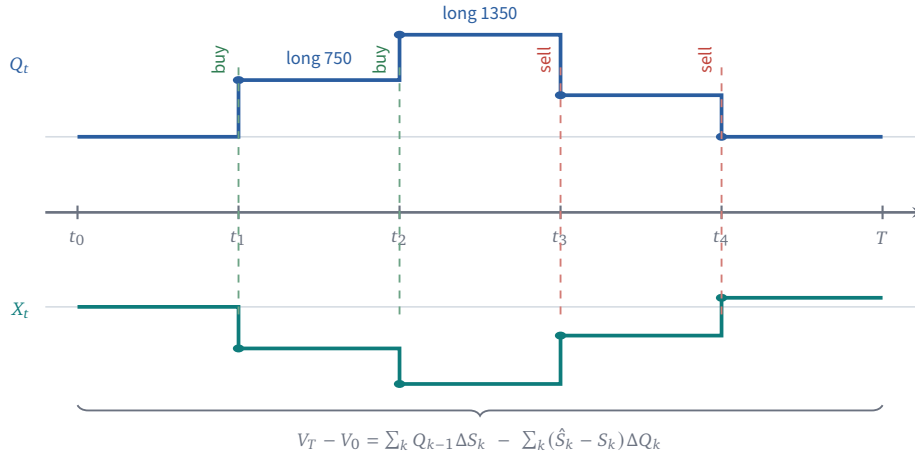


Figure 3.1. A round trip in two ledgers. Inventory Q (blue) is piecewise constant and jumps at trade times; cash X (teal) jumps in the opposite direction by the executed value. The position returns to flat at t_4 , so the final wealth is pure realized P&L. Note that the cash ladder does not return to its starting level: the difference is the profit, net of the execution costs paid at each of the four jumps.

3.3 Arithmetic and logarithmic returns

Practitioners quote **returns**, not prices. There are two definitions and the difference is not pedantic.

$$R_t^{\text{arith}} = \frac{S_t - S_{t-1}}{S_{t-1}}, \quad R_t^{\text{log}} = \log \frac{S_t}{S_{t-1}}. \quad (3.6)$$

Log returns add across time; arithmetic returns add across assets. You cannot have both, and the choice of which to use is dictated by which sum you need. Portfolio construction (Chapter 11) needs $R_{\text{portfolio}}^{\text{arith}} = \sum_i w_i R_i^{\text{arith}}$ and so uses arithmetic returns; multi-period growth analysis uses log returns.

The gap between them is the **volatility drag**. If $dS_t = \mu S_t dt + \sigma S_t dW_t$, then

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \quad \text{but} \quad S_t = S_0 \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t\right], \quad (3.7)$$

so the *typical* path grows at $\mu - \sigma^2/2$ while the *average* path grows at μ . For $\sigma = 35\%$ the drag is 6.1% per year, which is larger than most equity **risk premia**: the risk premium of an asset is the return it is expected to earn in excess of cash, and it is the compensation for holding something uncertain rather than something safe.

This is the same $-\sigma^2/2$ that appears in the Black-Scholes drift change and in the Kelly criterion. It is not a finance fact; it is Jensen's inequality.

3.4 Financing and borrowing costs

How it actually works

Going *long* is simple: you pay cash, you own the asset. Going *short* is not. You must *borrow* the asset from someone who owns it, sell it, and stand ready to buy it back and return it. You pay a *borrow fee* (a few basis points annually for liquid large caps, tens of percent for genuinely hard-to-borrow names), you post collateral, and the lender may *recall* the shares at any time and force you to close. The symmetric mathematics of $Q \geq 0$ conceals a deeply asymmetric institution.

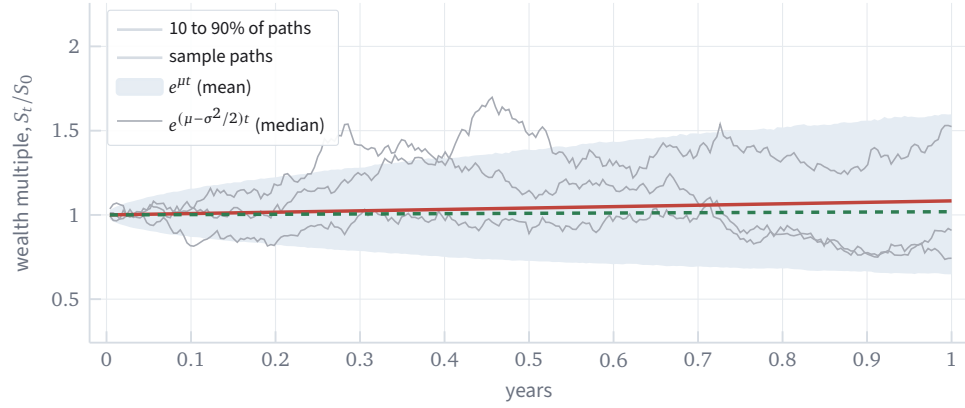


Figure 3.2. The average is not the typical. Geometric Brownian motion with $\mu = 8\%$, $\sigma = 35\%$, 4000 paths. The mean (red) grows at μ ; the median (dashed green) grows at $\mu - \sigma^2/2$ and is *below* the starting level for much of the year. An investor experiences the median, a naive expectation calculation reports the mean, and the difference is 6.1% per annum. Simulation: scripts/generate_data.py.

Introduce a financing rate r on cash and a borrow spread $b \geq 0$ charged on short positions. (3.5) becomes

$$dV_t = Q_t dS_t + \underbrace{rX_t dt}_{\text{interest on cash}} - \underbrace{b(Q_t)^- S_t dt}_{\text{borrow cost}} - h(\nu_t)\nu_t dt, \quad (3.8)$$

where $(x)^- = \max(-x, 0)$. The accumulation of these frictions is what makes a *market-neutral* strategy (with Q long in some assets and short in others, so that $\sum_i Q_i S_i \approx 0$) expensive to run even when it has no market exposure.

Leverage is the ratio of gross exposure to equity,

$$L_t = \frac{\sum_i |Q_t^i| S_t^i}{V_t}, \quad (3.9)$$

and it is the constraint that binds first in practice. A strategy with a Sharpe ratio of 2 and a volatility of 1% is only interesting at $L = 10$; at $L = 10$ a 10% adverse move is a total loss. The relationship between leverage, the position limit in $\mathcal{A}$, and the margin requirement imposed by the clearing house is developed in Chapter 12.

3.5 Profit and loss attribution

Given (3.3), a desk decomposes each day's P&L into interpretable pieces. For a portfolio of derivatives with value $\Pi(t, S, \sigma)$, the second-order Taylor expansion gives the standard attribution

$$\Delta \Pi \approx \underbrace{\frac{\partial \Pi}{\partial S} \Delta S}_{\text{delta}} + \underbrace{\frac{1}{2} \frac{\partial^2 \Pi}{\partial S^2} (\Delta S)^2}_{\text{gamma}} + \underbrace{\frac{\partial \Pi}{\partial \sigma} \Delta \sigma}_{\text{vega}} + \underbrace{\frac{\partial \Pi}{\partial t} \Delta t}_{\text{theta}} + \text{carry} - \text{costs}. \quad (3.10)$$

The four named terms are the **Greeks**, and they are simply the partial derivatives above given traders' names. **Delta** is the sensitivity to the underlying price, **gamma** the sensitivity of delta to the same price, **vega** the sensitivity to volatility, and **theta** the loss to the mere passage of time. Chapter 10 derives them and explains why the second is the one that causes trouble; here they are only labels for the terms of a Taylor expansion, and the point is that a day's P&L can be attributed to them exactly.

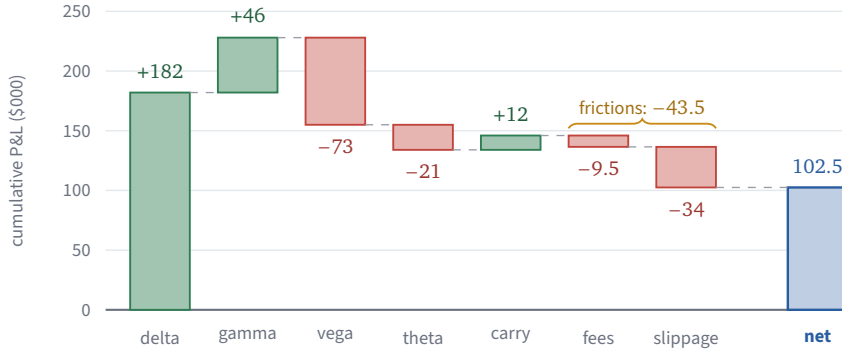


Figure 3.3. Where the money came from. A stylized one-day attribution for an options desk, in thousands of dollars. The directional bet (delta) made money and the convexity (gamma) helped, but a fall in implied volatility (vega) gave most of it back, and the frictions (fees plus slippage) cost \$43.5k. Attribution is the diagnostic that tells you whether a strategy is earning the risk premium it claims to earn, or something else entirely.

Attribution as a diagnostic

Attribution is the empirical counterpart of the value-function decomposition. If your model says the strategy earns a premium for bearing inventory risk, then the realized P&L should load on the inventory term and *not* on, say, vega. When it does not, either the model is misspecified or the implementation is not the policy you designed. In control language: attribution is how you detect that the plant is not the plant you modelled.

3.6 The multi-asset ledger

With d assets, $Q_t \in \mathbb{R}^d$, $S_t \in \mathbb{R}^d$, and (3.1) becomes $V_t = X_t + Q_t^\top S_t$. It is usually more convenient to work in **weights**

$$w_t^i = \frac{Q_t^i S_t^i}{V_t}, \quad \sum_{i=1}^d w_t^i + w_t^{\text{cash}} = 1, \quad (3.11)$$

because the frictionless wealth dynamics then become scale-free,

$$\frac{dV_t}{V_t} = w_t^\top \frac{dS_t}{S_t} + (1 - w_t^\top \mathbf{1})r dt, \quad (3.12)$$

which is the state equation of the Merton problem (Section 11.4). Note the change of control variable: in execution problems the control is the trading rate ν ; in portfolio problems it is the position w itself. The two are related by $\nu dt = dQ$, and the choice matters enormously as soon as transaction costs are present, because costs are a function of dw , not w .

Where the model breaks

(3.12) silently assumes you can rebalance to any w instantaneously and for free. Every real portfolio problem has a cost term $c(dw)$, and the moment you add it the problem stops being a pointwise optimization over w and becomes a genuinely dynamic problem with memory: the optimal w_t depends on where

you already are. This is the single biggest gap between textbook portfolio theory and practice, and [Section 11.5](#) is devoted to it.

3.7 Instruments requiring a modified ledger

Everything so far assumed the simplest possible instrument: you pay cash, you own a thing, the thing has a price. A great deal of what trades does not work that way, and in each case the failure is in the *ledger* rather than in the mathematics. Three cases matter enough to state explicitly, because each one breaks (3.1) in a different place.

3.7.1 Futures and variation margin

A **futures contract** is an agreement to trade at a fixed price on a future date. Crucially, entering one costs nothing: no cash changes hands at inception, because nothing has been bought yet.

That immediately breaks the ledger of (3.1), which wants to record your holding times its price. What is the holding worth? At inception, nothing, because the agreed price is by construction the fair one. A day later the market has moved and the contract has acquired a value, but rather than letting that value accumulate the exchange settles it *daily* in cash, a process called **variation margin**. If the contract has gained a dollar, a dollar is credited to your account that evening; if it has lost, a dollar is taken.

The correct dynamics are therefore

$$dX_t = Q_t dF_t, \quad V_t = X_t, \quad (3.13)$$

where F is the futures price. The position has zero mark-to-market value at all times, and all the P&L arrives as a stream of cash rather than as an appreciating asset.

Two consequences follow. First, futures are the natural hedging instrument in [Chapter 10](#), because a hedge you never have to fund is much easier to carry than one you do. Second, the daily settlement means you can be forced to find cash on a day when you have lost money, even though your view has not changed and the position has not been closed. That is a liquidity risk which the elegant (3.13) conceals entirely, and it has bankrupted firms whose eventual view turned out to be correct.

3.7.2 Nonlinear instruments

An option (§2.7) has a value that is a nonlinear function of the underlying price, so the term $Q dS$ in (3.5) is only the first-order piece of the story. Expanding one order further,

$$d\Pi \approx \frac{\partial \Pi}{\partial S} dS + \frac{1}{2} \frac{\partial^2 \Pi}{\partial S^2} (dS)^2 + \frac{\partial \Pi}{\partial t} dt, \quad (3.14)$$

and now the ledger has to track not just how much you hold but how that holding *changes* as the price moves. An option position that looks market-neutral this morning is long the market by lunchtime simply because the price moved.

This is not an accounting inconvenience, it is the entire business of [Chapter 10](#). The first derivative is the hedge ratio; the second is the rate at which the hedge must be revised; and the third is what you are paid or charged for holding the position while nothing happens.

3.7.3 Swaps and repurchase agreements

An interest rate **swap** exchanges a stream of fixed payments for a stream of floating ones over several years. A **repo** is a sale of a security combined with an agreement to buy it back later, which is economically a secured loan.

Neither has a “quantity held” in the sense the ledger expects. What you hold is a *schedule* of future cash flows, and its value depends on the whole interest rate curve rather than on a single price. The natural state is therefore not a scalar Q but a vector of sensitivities, one for each point on the curve, and the risk of the position is a vector too.

The mathematics is not harder, it is just higher-dimensional: $Q \in \mathbb{R}^d$ where d indexes maturities, and everything in Chapters 11 and 12 that was written for a vector of assets applies directly with the assets reinterpreted as points on a curve.

In plain terms

The common thread is worth extracting. In each case the object being held is not “ q units of a thing at price S ”, so the two-column ledger of (3.1) needs modifying: futures move the value into the cash column, options make the holding’s value curved, and swaps replace the single holding with a vector. Once the ledger is right, everything downstream, the self-financing condition, the objective functions, the optimal policies, goes through unchanged. That is exactly why this chapter is worth the trouble.

Exercises

Exercise 3.1 (Deriving (3.5)). Verify (3.5) by applying Itô’s product rule to $V = X + QS$ with Q of finite variation. Where exactly does the absence of a $d\langle Q, S \rangle_t$ term come from, and what would change if the agent were allowed to trade with unbounded variation?

Exercise 3.2 (Drag and the Kelly criterion). An investor puts a fraction f of wealth in the risky asset of Fig. 3.2 and the rest in cash at rate r . Show that the long-run growth rate of wealth is $g(f) = r + f(\mu - r) - \frac{1}{2}f^2\sigma^2$, and that it is maximized at $f^* = (\mu - r)/\sigma^2$. Explain why $g(2f^*) = r$: doubling the Kelly fraction gives away the entire risk premium.

Exercise 3.3 (Short-selling asymmetry). Show that a long position has P&L bounded below by $-Q_0S_0$ and unbounded above, while a short position has the reverse. Using (3.8), compute the break-even holding period for a short position in an asset with borrow spread $b = 8\%$ that you expect to fall 5%.

Exercise 3.4 (Attribution as a projection). Formalize (3.10) as a least-squares projection of realized daily P&L onto a set of risk-factor increments. Under what conditions on the factor covariance is the attribution unique? What is the financial interpretation of a large residual?

Electronic Markets and the Limit Order Book

The limit order book is the single most important object in this book, and it is not usually presented to mathematicians in a form they can use. It is described either as a picture of two staircases, or as a list of institutional rules. Neither tells you what kind of mathematical object it is.

Here is the answer, stated first and justified afterwards: *the book is a finite signed measure on a discrete price grid, and the exchange applies a deterministic, piecewise-linear, order-preserving map to it every time a message arrives*. All the probability lives in the arrival process of messages. All the economics lives in who sends them and why. This chapter develops the measure, the map, and the arrival process, in that order.

In plain terms

Chapter 2 showed the two edges of the order book. This chapter opens it up. The book is nothing more than a list of unexecuted orders sorted by price, together with a completely mechanical rule for what happens when a new order arrives. There is no economics in the rule and no randomness in it: the exchange is a computer program that anyone could write. All the uncertainty lies in *which* orders arrive and *when*. Understanding this cleanly matters because it is the source of the cost of trading: a large order eats through the list and pays worse and worse prices as it goes.

4.1 The order book as a measure

Fix a **tick size** $\Delta > 0$; prices live on the grid $\Delta\mathbb{Z} = \{\dots, -\Delta, 0, \Delta, 2\Delta, \dots\}$. A **limit order** is a commitment: *I will buy (or sell) up to v units at price p , and I will wait*. A **market order** is the opposite: *I will buy (or sell) v units now, at whatever prices are available*. The book is the record of all outstanding limit orders.

Definition 4.1 (Limit order book). The book at time t is a function $L_t : \Delta\mathbb{Z} \rightarrow \mathbb{R}$ with finite support, where

$$L_t(p) = \begin{cases} + (\text{volume resting at price } p \text{ on the buy side}), & p \leq P_t^b, \\ - (\text{volume resting at price } p \text{ on the sell side}), & p \geq P_t^a, \\ 0, & P_t^b < p < P_t^a, \end{cases}$$

and the **best bid** and **best ask** are

$$P_t^b = \max\{p : L_t(p) > 0\}, \quad P_t^a = \min\{p : L_t(p) < 0\}.$$

The defining constraint is $P_t^b < P_t^a$: the book never crosses, because crossing orders are executed on arrival.

Control-theoretic dictionary

Think of $\mathcal{L}_t$ as the state of a spatially extended queueing system: one FIFO queue per price level, coupled through the boundary condition $P^b < P^a$. The state space is $\mathcal{M} = \{\mathcal{L} \in \ell^1(\Delta\mathbb{Z}; \mathbb{Z}) : \mathcal{L} \text{ satisfies Definition 4.1}\}$, which is countable but very large; a book with 20 active levels on each side and depths up to 500 lots has of order 500^{40} states. This is why every practical model projects $\mathcal{L}$ onto a handful of statistics. §4.3 tells you which ones, and §4.7 says when the projection is defensible.

4.2 The matching map

The exchange is a deterministic transition system. Let $\mathcal{E}$ be the message alphabet: a message is a tuple $m = (\text{side}, \text{type}, p, v, \text{id})$ with $\text{type} \in \{\text{limit}, \text{market}, \text{cancel}\}$. The matching engine implements

$$\Phi : \mathcal{M} \times \mathcal{E} \longrightarrow \mathcal{M} \times \mathcal{T}, \quad (\mathcal{L}, m) \longmapsto (\mathcal{L}', \tau), \quad (4.1)$$

where $\tau \in \mathcal{T}$ is the (possibly empty) list of resulting trades. The rule is **price-time priority**: among resting orders, better prices execute first, and at equal price, earlier arrivals execute first.

The matching map precisely

For an incoming buy order of size v at limit price p (with $p = +\infty$ encoding a market order), define the cumulative available volume $A(p) = \sum_{p' \leq p} |\mathcal{L}(p')| \mathbf{1}_{\{\mathcal{L}(p') < 0\}}$. Then:

- (1) *Executed volume* is $v \wedge A(p)$, filled greedily from P^a upward.
- (2) *Residual* $(v - A(p))^+$ joins the buy queue at price p , at the *back*.
- (3) *Cash paid* is $\int_0^{v \wedge A(p)} P^a(u) du$, where $P^a(u)$ is the **marginal ask price** for the u -th unit: a nondecreasing step function. Hence the cost is *convex* in v .

The convexity in item 3 is the single most consequential structural fact in this book: it is why every execution problem has a convex running cost, and therefore why the optimizers of Part II are unique and computable.

Proposition 4.2 (Execution cost is convex). *Let $C(v) = \int_0^v P^a(u) du$ be the cash paid for a market buy of size v against a fixed book. Then C is convex, nondecreasing, piecewise linear; $C(0) = 0$, and $C'(0^+) = P^a$. The **effective price** $C(v)/v$ is nondecreasing in v , and the **slippage***

$$\text{slip}(v) = \frac{C(v)}{v} - P^a$$

is nonnegative and nondecreasing. Symmetric statements hold for sells with $P^b(\cdot)$ nonincreasing.

Proof. $P^a(\cdot)$ is nondecreasing by price priority, so its integral is convex, and a convex function through the origin has nondecreasing average $C(v)/v$. $\square$

Where the model breaks

Proposition 4.2 holds *against a fixed book*. In reality, by the time your second slice arrives the book has partially refilled (the **resilience** of Section 7.1.3)

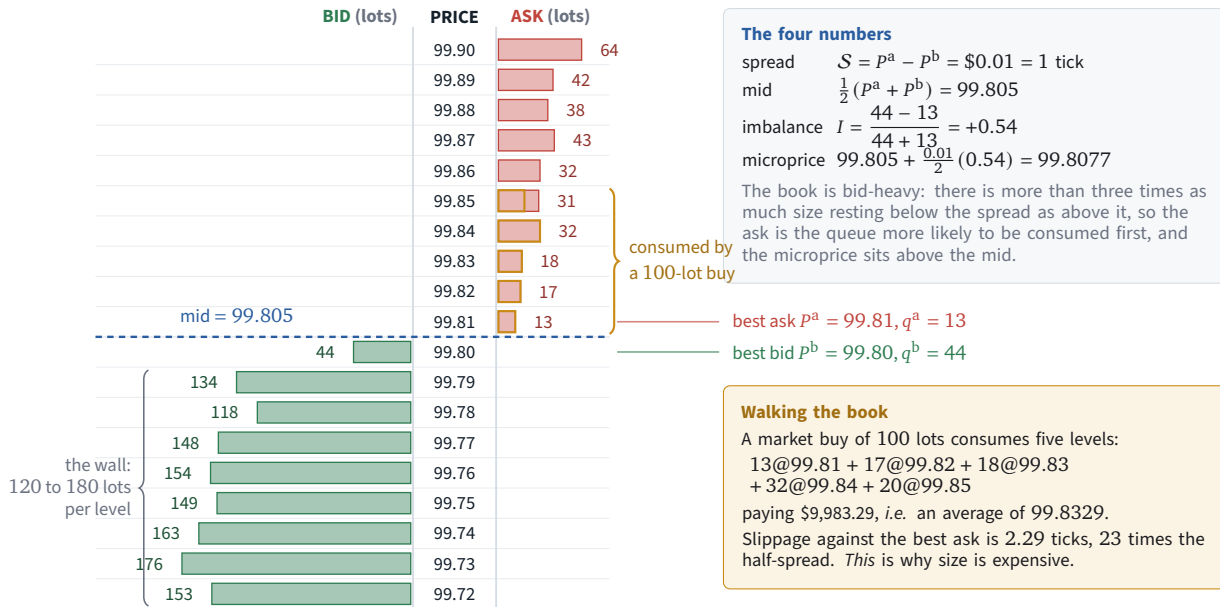


Figure 4.1. The book as a trader sees it. One snapshot from the zero-intelligence simulation of §4.4, with tick size $\Delta = \$0.01$. Two features recur throughout this book. First, the *touch is thin*: 44 and 13 lots at the best quotes against 120 to 180 deeper in, so the marginal price is far more fragile than the average price. Second, a market order of any real size consumes the queues in price order (**walking the book**) and pays a volume-weighted average price strictly worse than the best quote. That gap is the mechanical origin of the temporary impact term h in (1.4), and its convexity is Proposition 4.2.

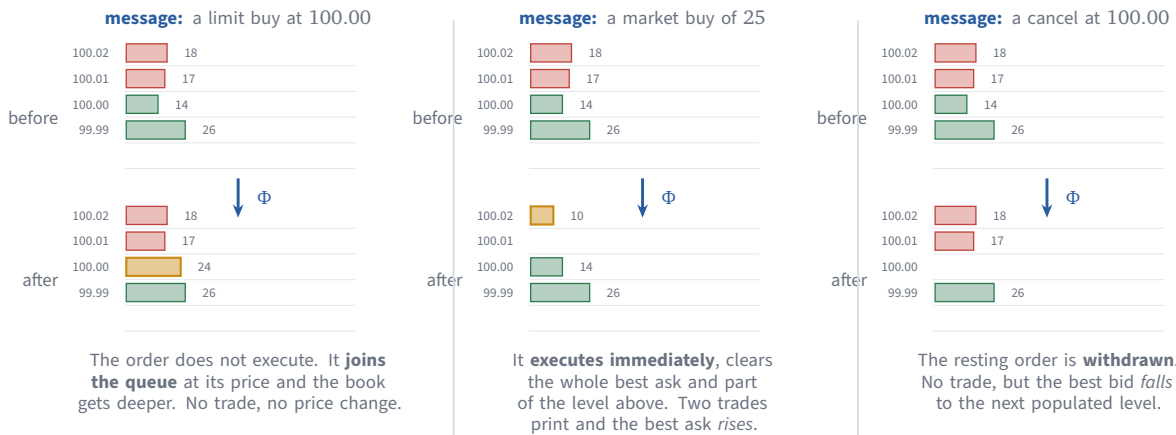


Figure 4.2. Three messages and three different outcomes. The same book receives each of the three message types, and the deterministic map Φ of (4.1) returns a new book and a list of trades. Only the middle column produces a trade. Notice that the price moved in the second and third columns for completely different reasons: in one case liquidity was *consumed*, in the other it was merely *withdrawn*, and from the outside these are indistinguishable in the quote feed. Distinguishing them is one of the recurring practical difficulties of working with this data.

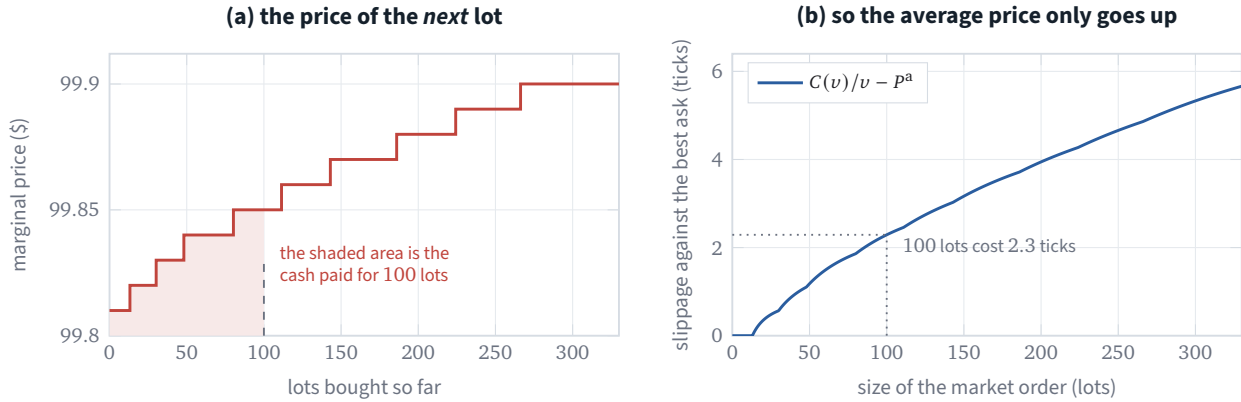


Figure 4.3. Why trading costs are convex. (a) As a market buy eats through the book of Fig. 4.1, each successive lot is bought at a price at least as high as the last, so the marginal price $P^a(u)$ is a nondecreasing staircase. The cash you pay for v lots is the area under it, which makes $C(v) = \int_0^v P^a(u) du$ convex: this is the whole content of Proposition 4.2. (b) The same fact expressed the way a trader experiences it. The average price paid rises with size, from nothing at all for a single lot to more than five ticks for the whole visible book. Convexity here is the reason every execution problem in Part II has a well-posed answer, and the reason that, absent a forecast, the answer is to spread the order out.

so the cost of trading v in one shot is not $C(v)$ but something between $C(v)$ and $v \cdot P^a$ depending on how long you take. Modelling that interpolation is exactly the content of transient-impact models (Section 6.4), and getting it wrong is the most common way to build an execution algorithm that backtests beautifully and loses money live.

4.2.1 Economic controls and exchange messages

One consequence of (4.1) deserves to be stated now, because it is invisible in the models of Part II and decides the shape of any system built from them. The alphabet $\mathcal{E}$ is the *only* thing the market accepts. Every decision made anywhere above it must eventually be compiled into a finite sequence of tuples (side, type, p , v , id).

That is not what the models produce. Chapter 8 produces a trading rate in shares per unit time, Chapter 9 a pair of distances from the mid, Chapter 10 a desired number of shares, and Chapter 11 a vector of weights. None of these is a message. A rate of five thousand shares a minute cannot be submitted anywhere; it is a description of how quickly a position should disappear, not an instruction any exchange can execute.

Control-theoretic dictionary

Call a variable an **economic control** when it appears naturally in an optimization model and summarizes what the trader wants to accomplish: a liquidation rate ν_t , a quoted distance δ , a hedge target Q_t^* , a weight vector w^* . Call a message $m \in \mathcal{E}$ a **market action**, since it is what actually changes the state through Φ .

The relationship between the two is the one a control engineer already knows from cascaded loops. The economic control is a *reference command* issued by an outer loop; the market action is the *actuator input* of the inner one. The outer loop decides where the state should go, the inner loop decides what signal to apply, and confusing the two produces a controller with no actuator.

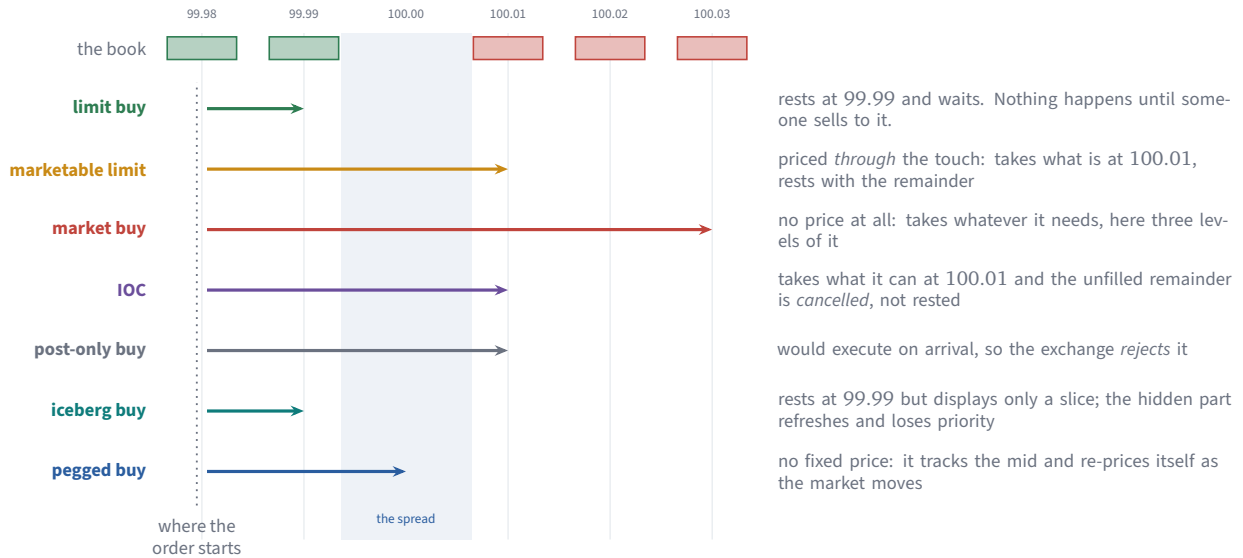


Figure 4.4. Order types are answers to one question. All of them start on the buy side and reach to the right; the length of the arrow is how far the order is willing to pay. A **limit** order stops short of the spread and waits. A **market** order reaches as far as necessary. Between those extremes sit the variants that differ only in what happens to the unfilled remainder: rest it, cancel it, or refuse the order entirely. Nothing here is conceptually new, which is exactly why the vocabulary is worth compressing into one picture.

The gap between them varies by problem, and it is worth ranking the four canonical problems by how wide it is. Market making is closest to the book, since a quoted distance almost determines a message and needs only a size and a queue to join. Execution is one step further away, because a rate has to be turned into discrete child orders. Hedging is further still, since its output is a desired *state* rather than any action at all. Portfolio choice is furthest, because a target weight vector says nothing about the order in which a multi-asset rebalance should be sequenced. [Section 8.6](#), [§9.5.1](#), [§10.6](#) and [§11.7](#) each pick this up where it arises.

Where the model breaks

The gap is not an argument against rate models, quote rules or portfolio optimization. It is an argument about *scope*. A rate model can be an entirely correct fluid-scale description of how fast inventory should disappear while saying nothing about how orders are placed, cancelled, repriced and filled, and both statements can be true at once. The error is to read the output of a Part II model as though it were an instruction to the exchange, because that interpretation quietly assumes the missing layer is trivial. It is not: [Section 4.5](#) shows that the choice of *which* order to send at a given price can change the fill probability by tens of percentage points, which is larger than most of the effects the outer model is optimizing.

4.2.2 Order types

Essentially every order type in use is a variation on one question: how far are you willing to reach, and what should happen to the part that does not execute? [Figure 4.4](#) places the common answers on a single price axis.

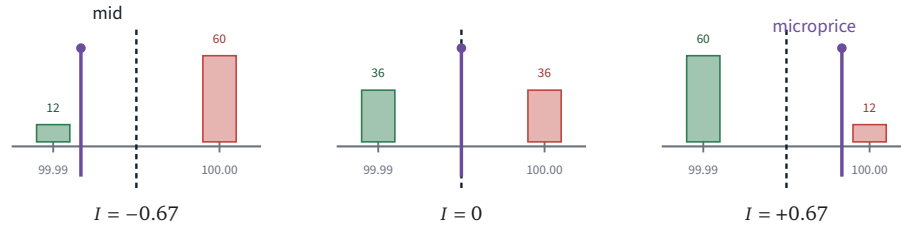


Figure 4.5. Imbalance moves the price before the price moves. Three books with identical quotes and different resting sizes. The **mid** (dashed) is the same in all three, because it depends only on the two prices. The **microprice** (solid) slides toward the *thin* side, because that is the queue which will be exhausted first and therefore the direction the quote is most likely to move next. Nothing here involves information or beliefs; it is a statement about which of two queues empties sooner.

How it actually works

Limit	Rest at a price; you supply liquidity, earn the spread, and bear the risk of being filled exactly when you least want to be.
Market	Take whatever is there. Certain execution, uncertain price.
Marketable limit	A limit order priced through the touch: takes what it can, rests with the remainder. Bounds your worst price.
IOC / FOK	Immediate-or-cancel and fill-or-kill: take now, do not rest. The building block of most execution algorithms.
Iceberg	Displays only a slice of its true size; the hidden part refreshes on execution but loses time priority.
Pegged	Price tracks a reference (e.g. the mid) automatically, saving the round trip to your own machines.
Post-only	Rejected if it would execute on arrival. Guarantees you earn the maker rebate rather than paying the taker fee.

4.3 Summary statistics of the book

Practical models never carry L_t ; they carry a projection. The four statistics below account for the overwhelming majority of the predictive content of the book at short horizons.

$$\text{spread} \quad S_t = P_t^a - P_t^b \quad (\text{cost of immediacy}) \quad (4.2)$$

$$\text{mid} \quad M_t = \frac{1}{2}(P_t^a + P_t^b) \quad (\text{the usual, biased, reference}) \quad (4.3)$$

$$\text{imbalance} \quad I_t = \frac{q_t^b - q_t^a}{q_t^b + q_t^a} \in [-1, 1] \quad (\text{the short-horizon predictor}) \quad (4.4)$$

$$\text{microprice} \quad S_t^{\text{micro}} = \frac{q_t^a P_t^b + q_t^b P_t^a}{q_t^b + q_t^a} \quad (\text{the imbalance-weighted mid}) \quad (4.5)$$

The microprice deserves comment, because it is the first genuinely nontrivial object in this chapter. Note the crossed weights: a *large* bid queue pulls the microprice *up*, toward the ask. The reason is a queueing argument, not an economic one: if the bid

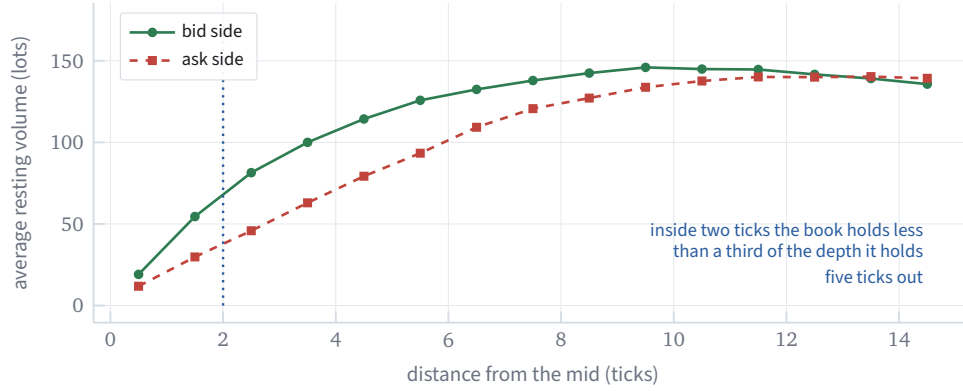


Figure 4.6. The shape of the book is a queueing phenomenon. Time-averaged depth from 1200 simulated seconds of the zero-intelligence model of §4.4: limit orders arrive uniformly in a band, cancel at a constant per-order rate, and market orders eat the touch. No agent in this simulation has beliefs, information, or an objective, and yet the characteristic concave-then-flat profile of real markets emerges. The touch is thin because it is the only place that market orders consume.

queue is long and the ask queue is short, the ask will be consumed first, so the next price change is more likely to be upward. Equivalently

$$S_t^{\text{micro}} = M_t + \frac{S_t}{2} I_t, \quad (4.6)$$

so the microprice is the mid, shifted within the spread by the imbalance. Empirically (Stoikov 2018), $\mathbb{E}[S_{t+\delta}^{\text{mid}} | \mathcal{F}_t]$ is much closer to S_t^{micro} than to M_t for δ of order seconds: a fact we return to in Section 6.4.

4.4 The book as a queueing system

The cleanest generative model of a book treats it as a system of coupled birth-death queues. Fix a band of $2K$ price levels around the mid. Then:

- *limit orders* arrive at each level p on the passive side as a Poisson process of rate $\lambda(p)$, adding one unit;
- *cancellations* remove each resting unit at rate ν independently, so a queue of length q depletes at rate νq ;
- *market orders* arrive at rate μ on each side and consume from the touch, walking the book if the order exceeds the touch depth.

This is the **zero-intelligence** model of Smith et al. (2003). Its generator acting on functions of the book is

$$\begin{aligned} \mathcal{A}f(L) &= \sum_p \lambda(p) [f(L + e_p) - f(L)] && \text{(limit orders)} \\ &+ \sum_p \nu |L(p)| [f(L - e_p) - f(L)] && \text{(cancellations)} \\ &+ \mu \sum_{\pm} [f(\Phi(L, m^{\pm})) - f(L)], && \text{(market orders)} \end{aligned} \quad (4.7)$$

with e_p the unit mass at p and $m^{\pm}$ the buy/sell market order. The model contains no optimization whatever, and yet reproduces the depth profile of Fig. 4.6, a diffusive

mid price at long horizons with strong mean reversion at short ones, and a spread distribution concentrated on one to three ticks.

A market with no agents in it

The zero-intelligence book is the correct *null hypothesis*. Any stylized fact it reproduces (the shape of the book, diffusive prices, the distribution of the spread) needs no economic explanation, because a mechanical queue produces it. Only deviations from the zero-intelligence baseline are evidence of strategic behaviour. A great deal of the microstructure literature consists of establishing exactly which deviations are real, and [Chapter 6](#) is organized around them.

4.5 Queue position and fill probability

Here is the decision a market maker faces thousands of times a day, and an execution algorithm faces on every child order: post a limit order and wait, or cross the spread and pay. Everything in [Chapter 9](#) is downstream of it.

4.5.1 Definition and observability

When you post a limit buy at a price, you join a queue. Under price-time priority (§4.2) the orders already resting at that price will be filled before yours, in the order they arrived. Your **queue position** is the total volume ahead of you at the moment you join.

Two facts about it are worth stating plainly because they surprise people.

First, *you cannot observe it directly*. The public feed tells you the total size at each price; it does not tell you where within that total your own order sits, because it does not identify orders. You know the depth when you arrived, so you know your initial position, and after that you must infer: trades at your price consume from the front, and cancellations remove orders you cannot see and whose position relative to yours you can only guess. Order by order data (§13.2) makes this exact and most participants do not have it.

Second, *the queue only shortens from the front, but it can vanish from the middle*. Executions consume the front. Cancellations remove orders from anywhere, and a cancellation ahead of you is pure good news: you advance for free. This is why the analysis below treats the two mechanisms separately.

4.5.2 Derivation of the fill probability

Problem 1: fill probability

Your order rests at the best bid with k lots ahead of it. Volume ahead of you disappears in two ways: market sells arrive at rate μ and consume one lot from the front, and each lot ahead of you is cancelled independently at rate ν . Meanwhile, at rate ρ , the opposite side is lifted and the whole market moves up, after which your quote is stale and you will not trade at this price. What is $\mathbb{P}(\text{fill})$?

The four symbols are the whole model. k is your queue position, which is what you control by deciding when to post. The rate μ is the flow that helps you, since every market sell eats one lot from the front. The rate ν also helps you, because a cancellation ahead of you advances you for free, and it is why queue position improves even when nothing trades. The rate ρ is the one that hurts: it is the chance that the market moves away before your turn arrives, in which

case you never trade at this price. A *fill* means your own lot is reached before that happens, so the question is a race between k departures and one adverse event.

The derivation is a race between exponential clocks and takes three lines. Suppose j lots remain ahead of you. Two things can happen next that matter. The ahead-queue can shrink by one, which happens at rate $\mu + \nu j$, since either a market order arrives or one of the j lots cancels. Or the price can run away, which happens at rate ρ . These are competing exponentials, so the probability that the queue shrinks before the price moves is

$$\frac{\mu + \nu j}{\mu + \nu j + \rho}. \quad (4.8)$$

To be filled you must win this race k times in a row, at $j = k$, then $j = k - 1$, and so on down to $j = 1$. The events are independent because exponential clocks are memoryless, so

$$\mathbb{P}(\text{fill} \mid k \text{ ahead}) = \prod_{j=1}^k \frac{\mu + \nu j}{\mu + \nu j + \rho}. \quad (4.9)$$

In plain terms

Consider the factors. Each one is close to one when the queue in front of you is melting fast relative to how quickly the price moves, and close to zero when it is not. The νj term is why the first few lots of a long queue disappear quickly: with many orders ahead, many of them are cancelling at once. The product is why being far back is so damaging: you need every single one of those races to go your way.

4.5.3 The posting decision

Now use it. You must buy one lot within the next few seconds. Posting at the bid costs p^b and works with probability p ; crossing costs P^a and works for certain. If posting fails you must cross anyway, by which time the price may have moved against you by Δ on average, since the reason you were not filled is usually that the market went up.

Expected cost of crossing now is P^a . Expected cost of posting is

$$p P^b + (1 - p)(P^a + \Delta), \quad (4.10)$$

so posting is better exactly when

$$p S > (1 - p) \Delta, \quad \text{that is} \quad p > \frac{\Delta}{S + \Delta}. \quad (4.11)$$

The threshold is illuminating. If failing to be filled costs you nothing ($\Delta = 0$) you should always post. If the adverse move on failure is as large as the spread you win, you need better than even odds. And since p is determined by queue position through (4.9), the whole decision reduces to *where in the queue you would land*, which is why so much engineering effort goes into landing near the front.

Why microseconds matter to a slow strategy

Suppose a price level has just opened, so the queue is empty. The first participant to post is at position zero and enjoys the top of Fig. 4.8; the tenth arrival is a few hundred lots back and may as well not have bothered. The gap in expected

value between those two is realized in the microseconds it takes to detect the new level and respond.

This is why a strategy whose ideas play out over minutes may still need hardware measured in nanoseconds. The **alpha**, which is the trade jargon for whatever predictive edge a strategy believes it has, lives at the horizon of minutes; the *execution* of it is a race, and losing the race converts a good strategy into one that pays the spread.

Remark 4.3 (Adverse selection lurks here). (4.9) computes the probability of a fill, not the value of one. The fills you get are not a random sample of the times you were quoting: you are filled precisely when someone chose to trade against you, and their reasons are correlated with what happens next. Conditional on being filled at the bid, the price is more likely to fall. This is **adverse selection**, it is the central cost of market making, and Section 9.6 quantifies it.

Where the model breaks

(4.9) assumes the queue ahead of you never grows, that cancellations are uniform across positions, and that μ , ν and ρ are constants. All three are false. Queues do grow; cancellations concentrate at the back, where the orders are newest and least committed; and the rates vary by orders of magnitude across the trading day. The formula is a good qualitative guide and a poor quantitative one, and any desk relying on it in production replaces it with an empirical fill model estimated from its own executions.

4.6 Institutional features and their consequences

Everything so far has treated the exchange as a clean abstraction. It is not, and the details below are not trivia: each one changes what the optimal policy looks like, and a model that ignores them can be wrong by more than the quantity it is trying to estimate.

4.6.1 Tick size

The grid spacing Δ is set by the exchange or the regulator, not by the market, and whether it is “large” or “small” is a relative question: large or small compared with how far the price typically moves between consecutive trades. Figure 2.17 drew the two regimes; here is why they matter.

In a large-tick market the spread is pinned at one tick and cannot narrow. No participant can offer a better price, because the next better price is a long way away and would cost more than the trade is worth. Competition therefore moves entirely into *time*: queues at the touch grow to hundreds of lots, queue position becomes the scarce resource, and (4.9) is the governing equation. Market making in such a name is a queueing problem.

In a small-tick market a quote can be improved for a negligible amount. Priority is cheap to buy, so queues stay short, nobody bothers waiting, and competition happens in price. Here (4.9) is nearly irrelevant and the models of Chapter 9, which choose a continuous quoted distance, apply directly.

The dimensionless quantity that separates the regimes is $\Delta/(\sigma\sqrt{\text{trade duration}})$, the tick measured in units of the price move per trade. Regulators change tick sizes deliberately and occasionally by large factors, and when they do the same instrument moves between the two regimes and the appropriate model changes with it.

4.6.2 Fees and the maker-taker model

Most venues do not charge both sides equally. The common arrangement pays a **rebate** to whoever posted the resting order and charges a **fee** to whoever took it, on the theory that liquidity provision is a service worth subsidizing. Typical magnitudes are a rebate of about 0.2 basis points and a fee of about 0.3.

Those numbers look small and are not. A market maker capturing a one-tick spread on a \$100 stock earns one basis point gross. The fee schedule is therefore a third of the gross margin, and the sign of a strategy's profitability can depend on it entirely: the same quoting policy can be profitable on a maker-taker venue and loss-making on one with the inverted schedule.

Two behavioural consequences follow, and both are visible in real data. Orders migrate toward venues with the best rebate for the side being provided, which is part of why liquidity is fragmented. And the *post-only* order type exists specifically so that an order which would accidentally take liquidity, and thereby pay the fee instead of earning the rebate, is rejected rather than executed.

4.6.3 Fragmentation and routing

The same instrument trades simultaneously on many venues, each with its own book. There is no single order book for a US equity; there are a dozen, plus several venues that do not display quotes at all.

"The" best bid is therefore a computed object, an aggregate across venues, and in the US it has a regulatory definition and a name, the national best bid and offer. Deciding where to send an order becomes an optimization in its own right, trading off displayed size, fee schedule, expected queue position, and the latency to each venue. It also creates the arbitrage of Fig. 2.11 as a routine occurrence rather than an anomaly: keeping a dozen books consistent is somebody's full-time business.

4.6.4 Auctions

Opening and closing auctions abandon continuous matching altogether. Orders accumulate for several minutes without trading, and then everything clears at once, at the single price that maximizes the volume that can change hands. Formally the exchange solves $\max_p \min\{D(p), S(p)\}$ over the cumulative demand and supply schedules, a one-dimensional problem whose solution is where the two curves cross.

Three features make an auction a genuinely different mechanism from the continuous book, and worth separate treatment. There is no spread, because everyone transacts at the same price. There is no queue and no time priority, so none of §4.5 applies. And the clearing price is determined by the whole schedule rather than by the marginal order, which makes it much harder to move with a single large order and correspondingly attractive to anyone trading size.

These two instants carry a large fraction of the day's volume, and for an investor whose benchmark is the closing price the closing auction is not part of the strategy, it is the strategy.

4.6.5 Priority rules

Equities use price-time priority, which is what makes §4.5 meaningful. Many futures markets do not. Under **pro rata** allocation an incoming marketable order is split across all resting orders at the best price in proportion to their size, with no reference to who arrived first.

This single change rewrites the strategic problem. Time priority is worthless, so there is no race to be first. Instead, since your fill is proportional to your share of the displayed size, you are rewarded for quoting more, and the equilibrium involves quoting substantially more than you actually wish to trade. That is only stable because

overquoting carries a risk: a large incoming order can fill you in full, and having quoted five times your target size you are then five times too long.

The mathematics of the two regimes is genuinely different, and a market-making model built for one is not merely miscalibrated on the other, it is solving the wrong problem.

4.6.6 Circuit breakers and risk controls

Finally, a set of mechanisms that exist to stop things rather than to price them. Exchanges halt trading in a name after a sufficiently large move, reject orders priced too far from the current market, and cap message rates. Brokers enforce pre-trade limits between a strategy and the exchange: maximum order size, maximum position, and a kill switch.

For a modelling exercise these look like a nuisance. For an operating desk they are the dominant tail risk control, because the tail risk of an algorithmic trading operation is dominated by software failure rather than by market moves, and no objective function in Part II contains a term for that.

4.7 State-space reduction

We can now be precise about the state space of the problems in Part II, and about the fact that choosing it is the most consequential modelling decision in the whole exercise.

4.7.1 The dimensionality problem

The full state is (L_t, Q_t, X_t) . The last two are scalars. The first is not: it is a function on the price grid, and even truncated to twenty active levels on each side with depths up to five hundred lots it has of order 500^{40} configurations. No dynamic programming recursion is going to run on that, and no amount of computing power changes the exponent.

So every usable model projects L_t onto a handful of numbers. The question is which handful, and the honest answer is that this is chosen by judgement rather than derived, because there is no theory of sufficiency for order books (§16.3.2 returns to this as an open problem).

4.7.2 Three standard reductions

- (1) **Price only.** Keep S_t and summarize the entire book by one or two impact coefficients: $L_t \rightsquigarrow S_t$, with a diffusion for the price and a cost function for trading against it.

This is the state space of [Chapter 8](#), and it is right for horizons of minutes to days, where the book refills many times over during the life of the decision and only its average shape matters. What it cannot express is anything about *when* to trade within the next few seconds, because the state contains nothing that varies on that timescale.

- (2) **Price and touch.** Keep (S_t, q_t^b, q_t^a) , which gives you the spread, the imbalance of [Fig. 4.5](#) and hence a short-horizon drift.

This is the state of most market-making models ([Chapter 9](#)) and it is right for horizons of seconds. It is enough to express “lean against the imbalance” and “widen when the book is thin”. It is not enough to express anything about queue position, because two books with identical touch sizes can leave your own order in very different places.

Decision	Horizon	State needed	Where
how fast to liquidate	hours	S, Q	Chapter 8
where to quote	seconds	S, Q, q^b, q^a	Chapter 9
post or cross	seconds	the above plus queue position	§4.5
which price to post at	sub-second	the full book	simulation only
which venue to use	sub-second	books across venues, fees	§4.6

Table 4.1. The state you need is determined by the decision you are making, not by the data you happen to have. Down the table, each row adds state and loses tractability; the top two rows have closed-form solutions and the bottom two do not.

- (3) **The full book.** Keep L_t , with dynamics along the lines of (4.7). Necessary for any decision about the *placement* of orders rather than their sizing: which price to post at, whether to cancel and re-post, whether the queue you are in is worth keeping.

Almost never analytically tractable. This is the domain of simulation and, increasingly, of reinforcement learning (Section 16.2), with all the difficulties that section describes.

4.7.3 Aggregation and latency

Even after choosing a reduction, the state you actually have differs from the state you modelled in two ways.

It is aggregated. The feed reports total size at each price. It does not say whether the two hundred lots at the bid are one determined institution or forty small orders that will evaporate at the first sign of trouble, and those two books behave completely differently. Participants spend real effort inferring this from the pattern of individual message sizes.

It is stale. Between the exchange writing a quote update and your strategy acting on it there is a delay of tens of microseconds at best. Your state is a lagged version of the true one, so formally the object you control is $\mathcal{G}_t = \mathcal{F}_{t-\tau}$ rather than $\mathcal{F}_t$, and the fastest participant is solving a different, easier problem than you are. This is not a detail that averages out: it is the entire economic basis of the latency industry.

The reduction is a modelling decision

Choosing the state space is choosing which trade-offs your controller can represent. A model whose state is S_t alone cannot express “wait, I am third in the queue and the imbalance is favourable”, because none of those words correspond to anything in its state. It is therefore not a market-making model, however carefully you solve it, and no amount of mathematical care will repair the omission.

The practical instruction is to match the reduction to the decision. Ask what the policy needs to be able to condition on, then include exactly that and nothing more. Every extra state variable costs you tractability; every missing one costs you a policy you cannot express.

Exercises

Exercise 4.1 (The matching map is monotone). Show that Φ of (4.1) is monotone in the incoming size: if $v_1 \leq v_2$ then the executed volume and the cash paid are both nondecreasing. Deduce Proposition 4.2 from monotonicity of the marginal price alone.

Exercise 4.2 (Microprice as a conditional expectation). Assume the next price move is $\pm\Delta/2$ from the mid, up with probability π and down with probability $1 - \pi$, and that $\pi = q^b/(q^b + q^a)$. Show that the expected next mid equals the microprice (4.5). Then explain the empirical failure of this model: real imbalance is autocorrelated, so a single-step calculation understates the drift.

Exercise 4.3 (Stationary depth in the ZI model). Consider a single price level far from the touch, receiving limit orders at rate λ and cancelling each resting unit at rate ν , with no market orders. Show that the stationary queue length is Poisson with mean λ/ν . Explain why the observed profile in Fig. 4.6 flattens out at exactly this value away from the touch, and is depressed near it.

Exercise 4.4 (Optimal placement, one shot). You must buy one lot within the next T seconds. You may post at the bid (filled with probability $p(T)$ from (4.9), price P^b) or cross the spread now (price P^a , certain). If unfilled at T you must cross anyway, at a price that has moved by $\pm\Delta$ with equal probability. Write down the expected cost of each policy and find the threshold in $p(T)$ at which posting becomes optimal. Where does the spread enter, and where does the volatility?

Exercise 4.5 (Pro rata changes everything). Under pro-rata allocation, a marketable order of size v against resting orders $q_1, \dots, q_m$ fills order i by $v q_i / \sum_j q_j$. Show that a risk-neutral trader who wishes to trade q^* should quote strictly more than q^* , and that if all traders reason this way the total quoted size is not an equilibrium unless there is a cost to overquoting. What plays the role of that cost in a real market?

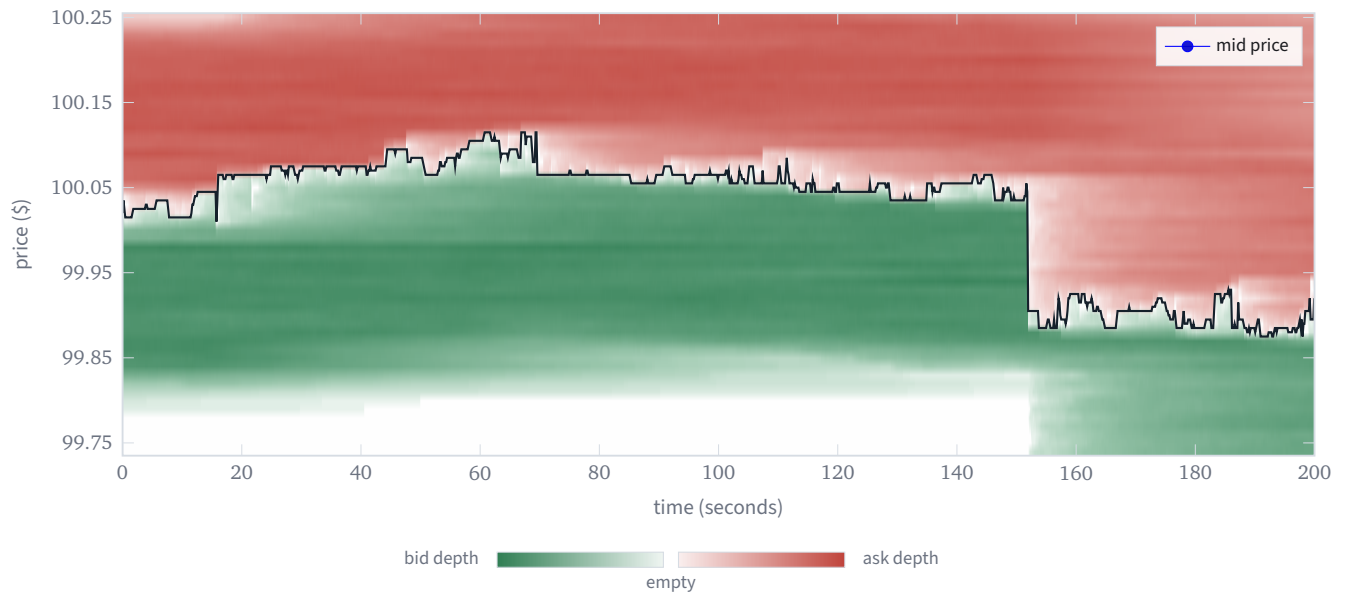


Figure 4.7. Two hundred seconds of a limit order book. Each vertical slice is one snapshot of L_t on a fixed price grid, coloured by resting volume: green for bids, red for asks, white where the book is empty. Three things are worth seeing. First, the white corridor is the spread, and it is *always* narrow: the book is two dense walls with a thin, constantly refilled channel between them. Second, the walls move: the mid price (black) is not an exogenous process but the boundary between two queueing systems. Third, the wall is thinnest exactly at the boundary, so the marginal lot is always the fragile one. Colour uses a $\gamma = 0.45$ power scale so that the ten-lot queues at the touch remain visible against the hundred-lot queues behind them.

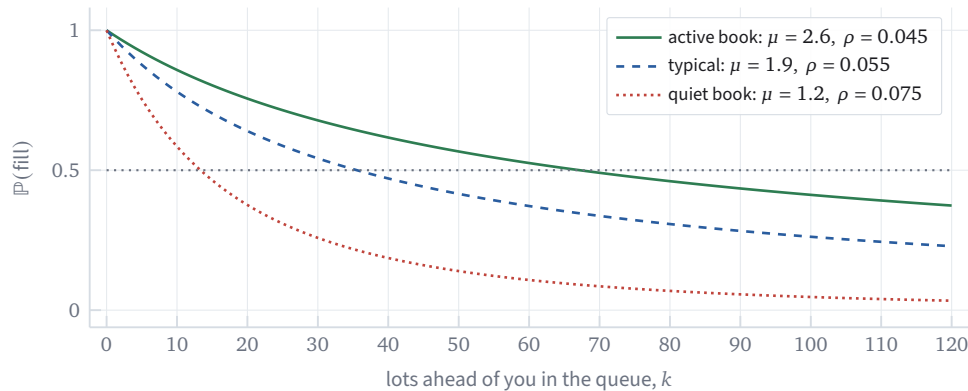


Figure 4.8. Queue position is destiny. (4.9) for three market regimes. Being 20 lots back rather than at the front costs 15 to 30 percentage points of fill probability, and the curves flatten only once the queue is so long that arrival is hopeless. Note how much the regime matters: the same queue position is worth far more in a busy market, because the queue in front melts faster relative to the rate at which the price runs away.

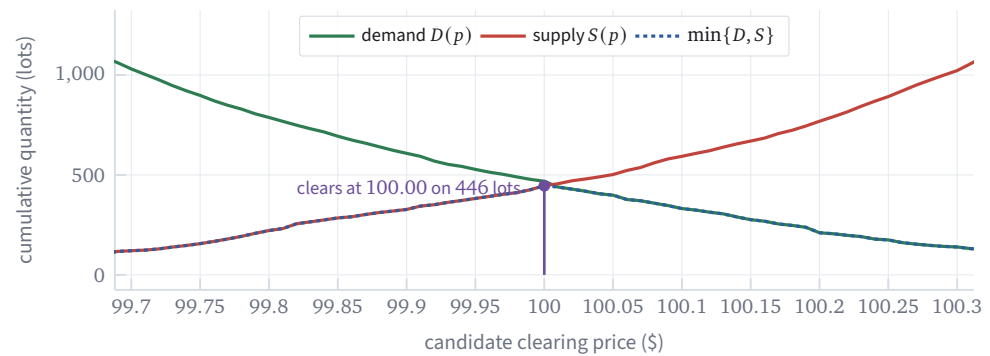


Figure 4.9. An auction is a one-dimensional optimization. Every buy order is a promise to take a quantity at or below some price, so cumulative demand slopes down; every sell order slopes it up. The exchange picks the price that maximizes the volume able to change hands, which is the peak of the dashed curve and hence the crossing point. Everyone who bought pays the same price regardless of what they were willing to pay, which makes an auction a fundamentally different mechanism from the continuous book of the rest of this chapter: no queue, no priority, and no spread.

Price Dynamics and No-Arbitrage Duality

There are two entirely different questions hiding inside “what are the dynamics of the price?”. The first is normative: what *must* a price process satisfy for the market not to contain a money pump? The answer is a theorem in linear programming, and §5.2 states it in exactly those terms. The second is descriptive: what do prices actually do? The answer is a list of robust empirical regularities that no simple model reproduces all of, surveyed in §5.5.

Confusing the two questions is the most common error in this subject. No arbitrage is a constraint on the *pricing measure* $\mathbb{Q}$; the stylized facts are properties of the *physical measure* $\mathbb{P}$. A model can be perfect for one and useless for the other.

In plain terms

Two questions get confused constantly and this chapter separates them. The first is *what must be true* of prices for the market not to contain free money. That is a logical question, it has a complete answer, and the answer turns out to be a theorem about linear programs. The second is *what prices actually do*, which is an empirical question with a messy answer: returns are unpredictable but their size is not, big moves are far more common than a bell curve allows, and the market has a distinct rhythm over the trading day. Models that are excellent for the first question can be useless for the second.

5.1 A three-state example

The general theorem is short and abstract, so do the arithmetic first. It takes one page and contains every idea in the chapter.

Tomorrow a share will be worth 120, 100 or 80, and nothing else. Call these states 1, 2 and 3. Today the share costs 100, and there is also a bond that pays 1 in every state and costs 1, so interest rates are zero. Nobody has told us how likely the three states are, and it will turn out that nobody needs to.

State prices. Ask a strange-sounding question: what would it cost today to buy a contract paying exactly \$1 if state i occurs and nothing otherwise? Call that price ψ_i . If such contracts existed, every other asset would just be a bundle of them, so its price would have to be the bundle’s cost. That gives one equation per traded asset:

$$\text{bond: } \psi_1 + \psi_2 + \psi_3 = 1, \quad (5.1)$$

$$\text{share: } 120\psi_1 + 100\psi_2 + 80\psi_3 = 100. \quad (5.2)$$

Two equations, three unknowns. Solving, $\psi_3 = \psi_1$, so the whole family of solutions is

$$\psi = (t, 1 - 2t, t), \quad \text{and positivity forces } 0 < t < \frac{1}{2}. \quad (5.3)$$

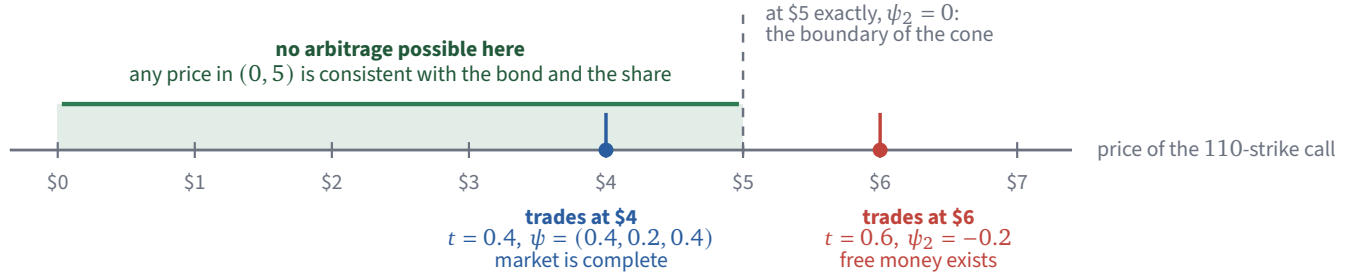


Figure 5.1. No arbitrage is a positivity condition with a picture. Given only the bond and the share, the state prices form the one-parameter family (5.3), and positivity confines the 110-strike call to the shaded interval. Any quoted price inside it is consistent with some strictly positive state-price vector, which by Theorem 5.2 means no arbitrage exists. Any price outside it forces a state price negative, and a negative state price is not an abstraction: it is a specific portfolio, written out above, that pays you today and can never lose. The whole of arbitrage pricing is the observation that this interval is a projection of a convex cone.

The market does not pin down ψ ; it pins down a *segment* of possible ψ 's. That is not a defect of the calculation, it is the honest answer: two instruments cannot determine three unknowns.

What that costs you. Now try to price a call option struck at 100, which pays 20 in state 1 and nothing otherwise. Its price must be $20\psi_1 = 20t$, and since t ranges over $(0, \frac{1}{2})$ the price ranges over $(0, 10)$. The model tells us almost nothing. This is **incompleteness**, and Fig. 5.1 shows it as an interval rather than a point.

Adding an instrument. Suppose a second option, struck at 110 and paying 10 in state 1, actually trades, at \$4. That is a third equation, $10t = 4$, so $t = 0.4$, which lies inside the permitted range. Now everything is determined:

$$\psi = (0.4, 0.2, 0.4), \quad \text{and the 100-strike call is worth } 20(0.4) = \$8. \quad (5.4)$$

Three instruments, three states, one answer. The market has become **complete**, and completeness is precisely the rank condition of Proposition 5.3.

And if the price is wrong. Suppose instead that the 110-call traded at \$6. Then $t = 0.6$, so $\psi_2 = 1 - 1.2 = -0.2$: a *negative* state price, meaning somebody is paying you to receive a dollar in state 2. That cannot survive, and we can construct the free money explicitly. Look for a portfolio paying $(0, 1, 0)$:

$$\underbrace{-4}_{\text{bonds}} \times (1, 1, 1) + \underbrace{0.05}_{\text{shares}} \times (120, 100, 80) + \underbrace{-0.2}_{\text{calls}} \times (10, 0, 0) = (0, 1, 0), \quad (5.5)$$

which you may verify line by line. Its cost today is $-4(1) + 0.05(100) - 0.2(6) = -\0.20 . You are *paid* twenty cents to hold a position that can never lose and sometimes pays a dollar. That is an arbitrage, and its existence is exactly the statement $\psi_2 < 0$.

What the example already contains

Everything in the rest of this chapter is visible above. *State prices exist if and only if there is no arbitrage* is the positivity of ψ , and it is Theorem 5.2. *Prices are a range, not a number, when the market is incomplete* is the interval $(0, 10)$ for the 100-strike call, and it is the gap between the two sides of the

linear program (5.7). Risk-neutral probabilities are normalized state prices is $q = (0.4, 0.2, 0.4)$, which sums to one and under which the share's expected value is $120(0.4) + 100(0.2) + 80(0.4) = 100$, its current price. The share is a martingale under q and under nothing else.

5.2 The absence of arbitrage and Farkas' lemma

Start in the simplest possible setting, because the whole theory is visible there. One period, two dates $t = 0$ and $t = 1$, finitely many states of the world $\omega_1, \dots, \omega_m$, and n traded assets. Asset j costs p_j today and pays A_{ij} in state ω_i . Collect these into $p \in \mathbb{R}^n$ and $A \in \mathbb{R}^{m \times n}$.

A **portfolio** is a vector $\theta \in \mathbb{R}^n$ (short positions allowed). It costs $p^\top \theta$ today and pays $A\theta \in \mathbb{R}^m$ tomorrow.

Definition 5.1 (Arbitrage). An arbitrage, defined informally in Section 2.8, is a portfolio θ with

$$p^\top \theta \leq 0, \quad A\theta \geq 0, \quad (p^\top \theta, A\theta) \neq 0.$$

In words: it costs nothing (or less), never loses, and somewhere strictly gains.

Theorem 5.2 (First fundamental theorem, finite version; Harrison and Kreps 1979; Harrison and Pliska 1981). *There is no arbitrage if and only if there exists a strictly positive state-price vector $\psi \in \mathbb{R}_{++}^m$ with*

$$p = A^\top \psi. \quad (5.6)$$

Proof. This is Stiemke's lemma, a variant of Farkas'. The set $K = \{(-p^\top \theta, A\theta) : \theta \in \mathbb{R}^n\}$ is a linear subspace of $\mathbb{R}^{1+m}$; "no arbitrage" says $K \cap \mathbb{R}_+^{1+m} = \{0\}$. A subspace meeting the nonnegative orthant only at the origin can be separated strictly from the simplex, which produces a strictly positive vector (ψ_0, ψ) orthogonal to K ; normalizing $\psi_0 = 1$ gives (5.6). $\square$

Pricing as a linear program

Consider the **superreplication** problem: what is the cheapest portfolio that dominates a payoff $c \in \mathbb{R}^m$?

$$\begin{array}{ll} \text{(P)} \quad \min_{\theta \in \mathbb{R}^n} p^\top \theta & \text{(D)} \quad \max_{\psi \geq 0} c^\top \psi \\ \text{s.t. } A\theta \geq c & \text{s.t. } A^\top \psi = p. \end{array} \quad (5.7)$$

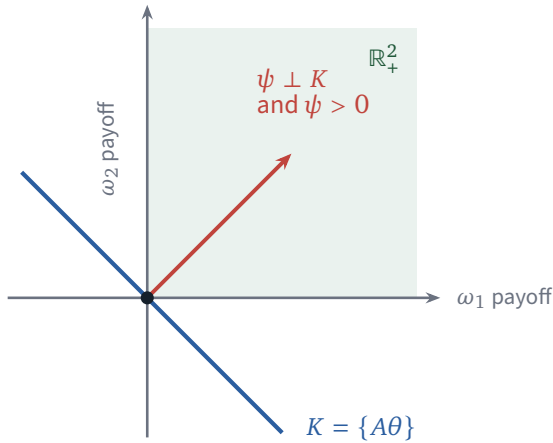
(P) is the seller's hedging problem: buy something that pays at least the liability in every state, as cheaply as possible. (D) is the search over state-price vectors consistent with observed prices, seeking the one that values the payoff most highly. Strong duality gives $\text{val(P)} = \text{val(D)}$ whenever either is feasible and bounded, which by Theorem 5.2 is exactly the no-arbitrage condition. *The upper price of a claim is an LP value, and the risk-neutral measure is its dual solution.*

5.2.1 From state prices to a probability measure

Suppose one asset (the **numeraire**, say a bank account) pays 1 in every state and costs B_0 . Then (5.6) gives $B_0 = \sum_i \psi_i$, so

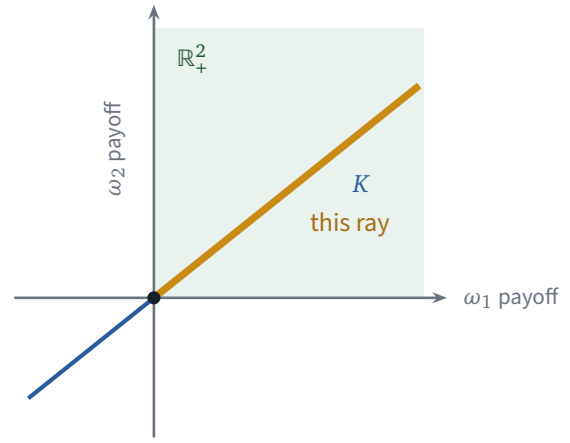
$$q_i := \frac{\psi_i}{\sum_k \psi_k} \quad (5.8)$$

(a) NO ARBITRAGE



K touches the orthant only at the origin, so a normal exists with *both* coordinates positive. Those coordinates are the state prices.

(b) ARBITRAGE EXISTS



Every point on the amber ray is a portfolio that costs nothing and pays nonnegative amounts in both states. No positive normal can exist.

Figure 5.2. No arbitrage is a separation statement. The attainable payoffs form a subspace K ; here it is a line because there are two states and one non-trivial direction to trade in. In (a) K meets the nonnegative orthant only at the origin, and a hyperplane can be slipped between them whose normal ψ has strictly positive coordinates: those coordinates are the state prices of (5.6). In (b) K cuts through the orthant, so there are portfolios costing nothing that pay nonnegative amounts in every state and something in at least one, and no strictly positive normal exists. [Theorem 5.2](#) says these two situations are the only possibilities.

is a probability vector, and (5.6) reads

$$p_j = B_0 \mathbb{E}^{\mathbb{Q}}[A_{\cdot,j}] \iff \frac{p_j}{B_0} \text{ is a } \mathbb{Q}\text{-martingale.} \quad (5.9)$$

Control-theoretic dictionary

$\mathbb{Q}$ is not a forecast. It is a normalized dual variable. Its “probabilities” are marginal prices of insurance against states, and they differ from real-world probabilities by exactly the risk premium the market charges for that state. Asking “is $\mathbb{Q}$ the true probability?” is like asking whether a Lagrange multiplier is a physical quantity: it is a shadow price, and it is meaningful, but it is not the primal object.

Where the model breaks

[Theorem 5.2](#) is a finite-dimensional statement. In continuous time with infinitely many states the correct version (Delbaen and Schachermayer 1994) replaces “no arbitrage” with the strictly stronger *no free lunch with vanishing risk* and requires a closure argument in a space of semimartingales. The economics is unchanged; the functional analysis is not. Note also that (5.7) produces a *range* of prices $[\underline{p}, \bar{p}]$ when markets are incomplete (the LP has a non-unique dual solution) and the width of that range is the honest statement of how much a model can tell you. [Section 10.4](#) returns to this.

5.2.2 Completeness as a rank condition

Proposition 5.3. *The market is **complete** (every payoff $c \in \mathbb{R}^m$ is attainable) if and only if $\text{rank } A = m$. In that case the state-price vector is unique, the LP (5.7) has a unique dual solution, and $\underline{p} = \bar{p}$ for every claim.*

Completeness is thus a statement about how many independent risks the traded assets span, and it is generically false: there are far more states of the world than liquid instruments. Chapter 10 is organized around what to do when $\text{rank } A < m$, which is to say, always.

5.3 Prediction markets

The state-contingent claims of §5.1 were introduced as a device: contracts paying \$1 in one state and nothing otherwise, useful for the proof and not obviously traded anywhere. They are traded, and the venues that trade them are called **prediction markets** (Wolfers and Zitzewitz 2004).

A prediction market lists a contract that pays \$1 if a stated event occurs and \$0 if it does not. If the contract trades at \$0.37 then, by (5.6) with zero interest rates, $\psi = 0.37$ is the state price of that event, and by (5.9) the corresponding risk-neutral probability is 0.37. The price is a probability, which is what makes these markets interesting well beyond finance: they are a mechanism for aggregating dispersed beliefs into a single number that anyone can read.

In plain terms

The mechanism is worth appreciating. Nobody surveys anybody. Anyone who thinks the market's number is too low can buy, anyone who thinks it too high can sell, and both are paid for being right and charged for being wrong. What emerges is a consensus weighted by conviction and by capital, updated continuously, and with a built-in incentive to be honest that no survey has.

5.3.1 When the price is not the probability

Three corrections stand between the quoted price and an honest probability, and each one is an application of something earlier in this book.

Risk premia. (5.9) gives the price as a *risk-neutral* expectation, not a physical one. If the event is correlated with the rest of a trader's wealth, they will pay more or less than its true probability for insurance reasons, exactly as this chapter has been saying throughout. For most prediction market questions the correlation is negligible and the correction is small, which is why the identification usually works.

Costs. The spread and the fees of Chapter 4 put a band around the price inside which no trade is worth making, so a contract quoted 0.35/0.39 identifies the probability only to within four points.

Thinness. Prediction markets have no natural hedgers. A wheat farmer must sell wheat forward; nobody *must* trade a contract on an election, so the only participants are speculators and the market maker. The deep book of Fig. 4.6 has no analogue here: the depth is small, and a moderate order moves the price by an amount that has nothing to do with information.

That last point interacts badly with the first two, and it is the standard explanation for the observed **longshot bias**: contracts trading at very low prices are systematically overpriced relative to their frequency of paying out, because a few cents of spread is an enormous relative cost on a three-cent contract.

5.3.2 Automated market making and a convex potential

Thinness creates a chicken-and-egg problem: nobody trades because there is no liquidity, and there is no liquidity because nobody trades. The standard solution is an **automated market maker**, an algorithm that quotes both sides unconditionally, and the standard algorithm has an elegant structure that will look familiar.

Let $q \in \mathbb{R}^n$ be the vector of contracts the market maker has sold on each of n mutually exclusive outcomes. Define a **cost function**

$$C(q) = b \log \sum_{i=1}^n e^{q_i/b}, \quad (5.10)$$

and let the cost of moving the outstanding vector from q to q' be $C(q') - C(q)$. This is the logarithmic market scoring rule of Hanson (2003) and Hanson (2007). Then the instantaneous price of outcome i is

$$p_i = \frac{\partial C}{\partial q_i} = \frac{e^{q_i/b}}{\sum_j e^{q_j/b}}, \quad (5.11)$$

which is a softmax and therefore automatically a probability vector: positive and summing to one, whatever anybody trades.

The market maker as a convex potential

(5.10) is the log-sum-exp function, the smooth convex conjugate of the negative entropy on the simplex. Three properties follow immediately and they are exactly the properties one wants.

Prices are *always* a valid probability distribution, by construction, so no arbitrage of the kind in Fig. 5.2 can ever exist inside the mechanism.

The maker's worst-case loss is bounded by $b \log n$, because C is bounded between $\max_i q_i$ and $\max_i q_i + b \log n$. The parameter b therefore buys liquidity: larger b means prices move less per contract, and the subsidy the operator is prepared to lose is the price of that depth.

And the whole mechanism is the gradient of a convex potential, which is the same structure as a proper scoring rule. Quoting honestly is optimal for the same reason that reporting your true belief maximizes a logarithmic scoring rule: the mechanism was constructed as the market version of that rule.

Control-theoretic dictionary

For a control-theoretic reader the point of (5.10) is that a market maker's price schedule is a *gradient field of a convex function*. That is not special to prediction markets: the convexity of the execution cost in Proposition 4.2 is the same statement about the order book, since $C(v) = \int_0^v P^a(u) du$ is convex and its derivative is the price. An automated market maker simply chooses that convex function explicitly rather than letting it emerge from a queue.

Choosing it explicitly is what makes the design tractable: you pick the potential, and liquidity, bounded loss and no-arbitrage all follow from its curvature. Abernethy et al. (2013) develop this correspondence in general, characterizing exactly which convex potentials give well-behaved market makers and connecting the resulting mechanism to regret bounds in online learning.

Where the model breaks

Prediction markets are a good mechanism and a poor oracle for anything rare, anything far in the future, or anything whose resolution is disputable. The first two make the capital cost of holding a position dominate the information, the third makes the contract unenforceable, and all three are why the successful applications cluster on frequent, short-dated, objectively resolvable questions.

5.4 The continuous-time formulation

Let S be a positive semimartingale and $B_t = e^{rt}$. The continuous-time analogue of (5.9) is: there is a measure $\mathbb{Q} \sim \mathbb{P}$ under which $e^{-rt}S_t$ is a martingale. If

$$dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}, \quad (5.12)$$

then Girsanov's theorem says $W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \frac{\mu-r}{\sigma}t$ is a $\mathbb{Q}$ -Brownian motion, and under $\mathbb{Q}$

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}. \quad (5.13)$$

The quantity $\varsigma = (\mu - r)/\sigma$ is the **market price of risk**, the Sharpe ratio of the asset, and it is exactly the Radon-Nikodym exponent. Two consequences are worth stating plainly because they surprise people:

- (1) **The drift μ does not appear in the price of a derivative.** Hedging removes it. This is the content of the Black-Scholes argument, and it is why option pricing is a PDE problem rather than a forecasting problem.
- (2) **The drift μ is everything for a portfolio.** The Merton problem of [Section 11.4](#) depends on μ through ς and on almost nothing else.

The same process, two questions, opposite answers about which parameter matters. Keep the distinction: *pricing lives under $\mathbb{Q}$, decisions live under $\mathbb{P}$.*

5.5 Empirical properties of returns

The models above are analytically convenient and empirically false in specific, well-documented ways. [Figure 5.3](#) shows the three most important discrepancies, computed from a simulation calibrated to look like an equity index.

5.6 Microstructure noise in observed prices

Everything above concerns an *efficient price* S_t that is not observable. What you observe is a transaction price or a quote, and the two differ by an amount that is small, bounded, and violently mean-reverting:

$$Y_t = S_t + \eta_t, \quad \eta_t \text{ bounded, } \mathbb{E}[\eta_t] \approx 0, \quad \text{Cov}(\eta_t, \eta_{t+h}) \approx 0 \text{ for } h > 0. \quad (5.14)$$

The noise η comes from the bid-ask bounce (trades alternate between buyer- and seller-initiated), from price discreteness, and from transient inventory effects. It is tiny (often less than a tick) and it is catastrophic for high-frequency statistics.

The natural estimator of how much a price moved over a day is to chop the day into intervals, square each price change and add them up. That sum is the **realized variance**, and it is what every practitioner computes. The following says that on high-frequency data it does not work.

Model	Tails	Clustering	Closed form	Use it when
Bachelier	no	no	yes	horizons are short and prices can go negative (spreads, rates)
Black-Scholes	no	no	yes	you need a baseline, or a quoting convention
Heston	mild	yes	semi	you need a smile and can tolerate two extra parameters
Jump diffusion	yes	no	semi	gaps matter (earnings, credit)
Rough volatility	yes	yes	no	you care about the term structure of volatility-of-volatility
Hawkes / point process	yes	yes	rarely	you are modelling at the tick scale, Chapter 6

Table 5.1. Choosing a price model is choosing which stylized fact you refuse to give up. For the control problems of Part II the choice matters less than you would expect: execution and market-making solutions depend on volatility mainly through σ^2 over the trading horizon, and are remarkably robust to the shape of the return distribution.

Proposition 5.4 (Realized variance diverges). Let $Y_{i\delta} = S_{i\delta} + \eta_{i\delta}$ with η i.i.d. of variance ω^2 , independent of S , and let $\text{RV}(\delta) = \sum_{i=1}^{\lfloor T/\delta \rfloor} (Y_{i\delta} - Y_{(i-1)\delta})^2$. Then

$$\mathbb{E}[\text{RV}(\delta)] = \underbrace{\int_0^T \sigma_s^2 ds}_{\text{what you want}} + \underbrace{\frac{2T\omega^2}{\delta}}_{\text{noise, and it blows up}}. \quad (5.15)$$

Sampling more finely makes the estimator worse, without bound.

Proof. Each increment contributes $\mathbb{E}[(\Delta S)^2] + \mathbb{E}[(\Delta \eta)^2] = \int \sigma^2 + 2\omega^2$, and there are T/δ of them. $\square$

Volatility estimation as filtering

(5.14) is a state-space model: a latent semimartingale observed in additive noise. Everything you know about filtering applies, and the modern high-frequency volatility estimators are, in effect, smoothers exploiting the spectral separation between a diffusive signal (variance $\propto \delta$) and i.i.d. noise (variance constant). Once you see it that way, the “choose a sampling frequency” folklore becomes a bandwidth selection problem with a standard bias-variance trade-off.

5.7 The choice of time scale

A last, easily overlooked, modelling choice. Prices are usually written as functions of calendar time, but almost every microstructure quantity is better behaved in a **business clock**:

- *tick time*, indexed by transactions;
- *volume time*, indexed by cumulative traded volume;
- *intrinsic time*, indexed by realized variance.

Under a volume clock the intraday U-shape of [Fig. 5.3\(d\)](#) largely disappears, returns are closer to Gaussian, and the arrival of information looks stationary. Formally

we are writing $S_t = \tilde{S}_{\Lambda_t}$ with Λ an increasing time change, which is exactly the Dambis-Dubins-Schwarz representation of a continuous martingale as time-changed Brownian motion. Execution algorithms exploit this directly: a schedule that is linear in *volume* time is the VWAP strategy of [Section 8.2](#), and it is far more robust than one linear in clock time.

Exercises

Exercise 5.1 (Duality by hand). Take $m = 3$ states, $n = 2$ assets, $A = \begin{pmatrix} 1 & 3 \\ 1 & 1 \\ 1 & 0 \end{pmatrix}$, $p = (1, 1.2)$. Find the set of state-price vectors, verify it is a one-parameter family, and compute the upper and lower prices of the digital claim $c = (1, 0, 0)$ by solving (5.7) explicitly. Confirm that the range collapses when a third asset spanning the states is added.

Exercise 5.2 (The dual is a measure). Show that in (5.7) the optimal dual variable, normalized by the price of the numeraire, is a probability measure supported on the states where the optimal superreplicating portfolio binds ($A\theta^* = c$). Interpret this: the hedge is exact exactly where the pricing measure puts mass.

Exercise 5.3 (Signature plot slope). Using (5.15), show that a log-log plot of $\mathbb{E}[\text{RV}(\delta)]$ against δ has slope -1 at high frequency and 0 at low frequency, and that the crossover occurs at $\delta^* = 2\omega^2/\sigma^2$. Estimate ω from [Fig. 5.4](#), given $\sigma = 30\%$ annualized.

Exercise 5.4 (Time change). Let Λ_t be the cumulative volume traded by time t and suppose $S_t = \tilde{S}_{\Lambda_t}$ with $\tilde{S}$ a Brownian motion of unit variance rate. Show that returns sampled at equal volume increments are i.i.d. Gaussian even though returns sampled at equal clock increments are not. What does this say about the origin of the fat tails in [Fig. 5.3\(a\)](#)?

Exercise 5.5 (Incompleteness has a width). In a one-period model with $m = 4$ states and $n = 2$ assets, characterize the set of state-price vectors as a polytope. Show that the price interval $[\underline{p}, \bar{p}]$ of a claim is the image of that polytope under a linear map, and that its width is zero if and only if the claim lies in the span of the traded payoffs.

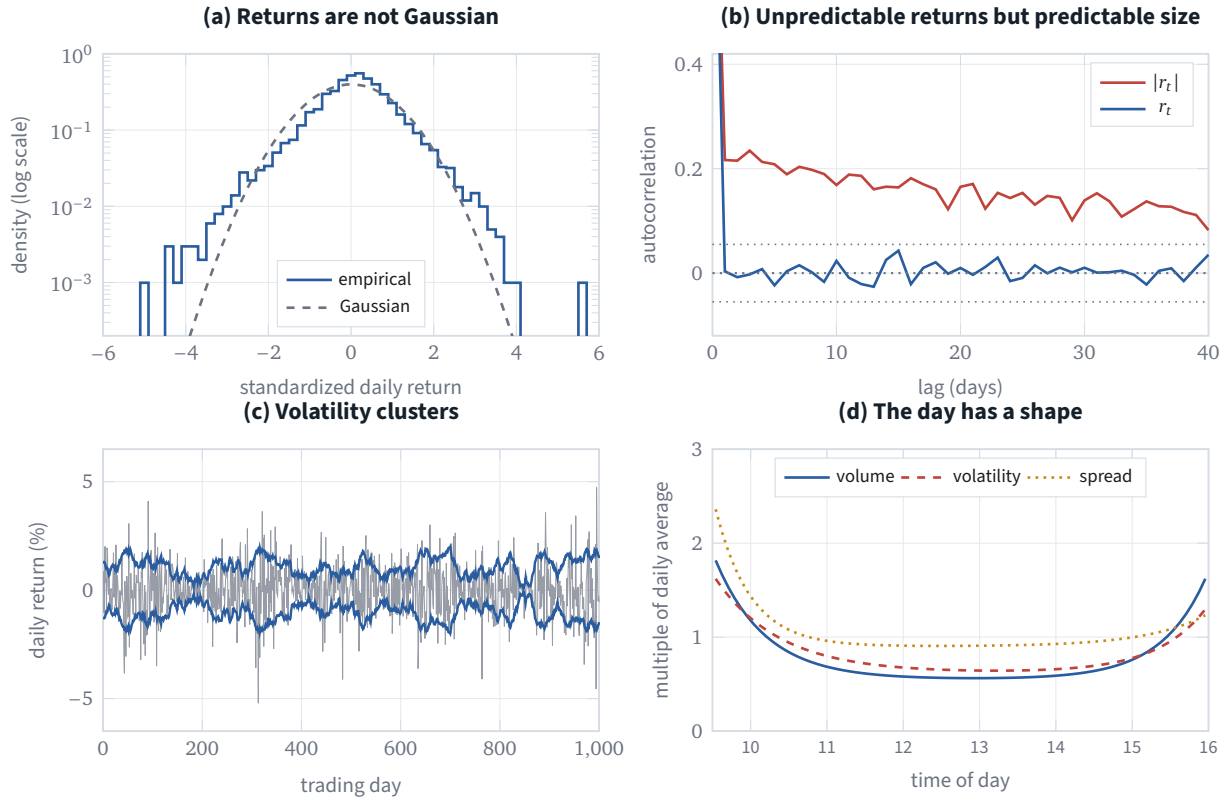


Figure 5.3. Three facts every price model is measured against, in four panels. (a) Standardized daily returns have far more mass beyond four standard deviations than a Gaussian (here roughly forty times as much) so variance is not a sufficient statistic for risk. (b) Returns themselves are essentially uncorrelated, which is a property of $\mathbb{P}$ and not something (5.9) implies, but their absolute values are strongly and persistently correlated: volatility is predictable even though direction is not. (c) The same fact seen as a path; the blue envelope is the latent volatility. (d) Volume, volatility and spread all follow a pronounced intraday pattern; ignoring it makes every execution schedule in Chapter 8 wrong at the open and the close. Panels (a) to (c): Heston dynamics with jumps, 5040 days (20 years); panel (d): a stylized profile.

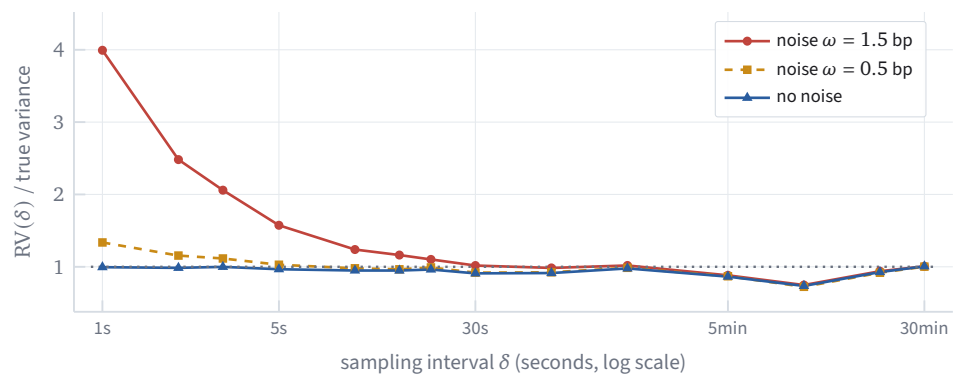


Figure 5.4. The volatility signature plot. Realized variance from a single simulated day, as a function of sampling frequency, relative to the true integrated variance. With realistic noise, one-second sampling overstates variance fourfold, exactly as (5.15) predicts; the bias vanishes only when δ is large enough that $2\omega^2/\delta \ll \sigma^2$, at the cost of throwing away data. The practitioner's fix (sample every five minutes) is a crude bias-variance trade-off; the modern fix is a kernel or pre-averaging estimator (Zhang et al. 2005; Barndorff-Nielsen et al. 2008) (Section 13.3).

Market Microstructure and Price Formation

Chapter 5 told you what a price process must look like to a pricer. This chapter asks the harder question: what makes it move? The answer is not “news”. News is far too rare. Prices move continuously all day on days when nothing happens, and the overwhelming majority of price variance is contemporaneous with *trades*, not with announcements.

Two mechanisms compete for the explanation, and both are correct:

- (1) **Information.** Some traders know something. The market maker cannot tell who, so every trade is evidence, and quotes must move against the incoming flow. This produces a spread even with zero costs and zero risk aversion (§6.1), and it makes the price a martingale that *learns* (§6.2).
- (2) **Mechanics.** Buying consumes the ask queue. Even a trade carrying no information moves the price, because the price is the boundary of a queueing system (Chapter 4). This produces impact that is partly transient and decays as the book refills (§6.5).

The intellectual history of the field is largely a fight over the relative weight of these two, and the current consensus (that they are empirically almost indistinguishable at the level of a single trade) is itself interesting, and is stated precisely in §6.6.

In plain terms

Prices move much more than news arrives. Why? This chapter gives the two competing answers and shows that both are partly right. The first is *information*: some of the people trading know something, the market maker cannot tell which ones, and so must charge everybody a little to cover the losses to the informed. That charge is the spread. The second is *mechanics*: buying consumes the orders that were waiting to sell, and when they run out the price simply moves. That is price impact, and it is the reason a large order is expensive quite apart from anybody knowing anything.

6.1 The Glosten-Milgrom model of adverse selection

Strip the market to its bones. A single asset is worth $V \in \{0, 1\}$, revealed at some future date. A market maker quotes a bid b and an ask a , is risk-neutral, and competes so hard that expected profit is zero. Traders arrive one at a time and trade one unit; with probability α the trader is **informed** (knows V), otherwise **uninformed** and buys or sells with probability $\frac{1}{2}$ each.

Problem 2: quoting against unknown information

Let $\pi_t = \mathbb{P}(V = 1 \mid \mathcal{F}_t)$ be the market maker's belief before the t -th trade. Zero expected profit on each side requires that the quotes equal the conditional expectation of V given the direction of the order:

$$a_t = \mathbb{E}[V \mid \mathcal{F}_t, \text{ buy}], \quad b_t = \mathbb{E}[V \mid \mathcal{F}_t, \text{ sell}].$$

Here $V \in \{0, 1\}$ is the asset's unknown value, $\mathcal{F}_t$ everything the market maker has seen before the t -th trade (the sequence of buys and sells, nothing more), π_t the belief that summarizes it, and a_t and b_t the ask and the bid. The conditioning on the direction of the order is the whole content of the problem: the maker must quote a price that is already fair *after* taking into account what the arrival of that order reveals.

Bayes' rule gives, writing $\mathbb{P}(\text{buy}) = \alpha\pi_t + \frac{1-\alpha}{2}$,

$$a_t = \frac{\alpha\pi_t + \frac{1-\alpha}{2}\pi_t}{\alpha\pi_t + \frac{1-\alpha}{2}}, \quad b_t = \frac{\frac{1-\alpha}{2}\pi_t}{\alpha(1-\pi_t) + \frac{1-\alpha}{2}}, \quad (6.1)$$

and the belief updates to $\pi_{t+1} = a_t$ after a buy, b_t after a sell.

Proposition 6.1 (The spread is pure adverse selection). Write $u = \frac{1-\alpha}{2}$, so that $\alpha + 2u = 1$. At $\pi_t = \frac{1}{2}$ the quotes of (6.1) are

$$a_t = \frac{1+\alpha}{2}, \quad b_t = \frac{1-\alpha}{2}, \quad \text{hence} \quad S = a_t - b_t = \alpha.$$

The spread is exactly the probability that the next counterparty is informed. It is strictly positive for $\alpha > 0$ and vanishes as $\alpha \rightarrow 0$. There are no inventory costs, no order-processing costs, and no risk aversion anywhere in this model: the entire spread is compensation for trading against people who know more than you do.

Proof. At $\pi = \frac{1}{2}$ both $\mathbb{P}(\text{buy})$ and $\mathbb{P}(\text{sell})$ equal $\frac{1}{2}$. Then $a = \frac{1}{2}(\alpha + u)/\frac{1}{2} = \alpha + u$ and $b = \frac{1}{2}u/\frac{1}{2} = u$, and $a - b = \alpha$ follows from $\alpha + 2u = 1$. $\square$

The martingale property survives

Note that $\mathbb{E}[\pi_{t+1} \mid \mathcal{F}_t] = \mathbb{P}(\text{buy})a_t + \mathbb{P}(\text{sell})b_t = \pi_t$: the mid is a martingale, as (5.9) demands. Adverse selection does *not* break the efficient-market property. What it does is make the martingale *informative*: the price is a running conditional expectation, and the order flow is the observation process. Glosten-Milgrom is a filtering problem whose state is V and whose observations are trade signs.

6.2 Kyle's model of strategic information

Glosten-Milgrom has an informed trader who trades whenever they can. Kyle's model asks what an informed trader who is *strategic* would do, and the answer introduces the single most used number in all of market microstructure.

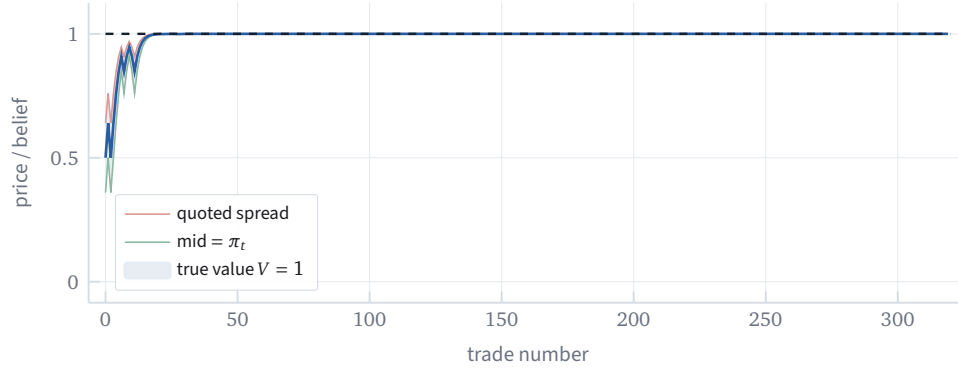


Figure 6.1. Learning from order flow. A single realization of (6.1) with $\alpha = 0.28$ and $V = 1$. The mid is the market maker's posterior; it is a bounded martingale and therefore converges, here to the truth. The shaded band is the quoted spread, which starts wide (the maker knows nothing and fears everything) and collapses as the posterior concentrates. Every price move in this figure is caused by a trade, and no news ever arrives.

Problem 3: Kyle's single-period equilibrium

The asset is worth $v \sim \mathcal{N}(p_0, \Sigma_0)$. Three players:

- an *informed trader* who observes v and submits a quantity $x = X(v)$, maximizing $\mathbb{E}[(v - p)x]$;
- *noise traders* who submit $u \sim \mathcal{N}(0, \sigma_u^2)$, independent of v ;
- a *market maker* who observes only the total order flow $y = x + u$ and sets a price $p = P(y)$ that is competitive, i.e. $P(y) = \mathbb{E}[v | y]$.

An equilibrium is a pair (X, P) such that each is optimal given the other.

The symbols divide into a prior, a signal and two decisions. p_0 and Σ_0 are the mean and variance of what the market believes about v before trading; σ_u^2 is the size of the noise trading that the informed trader hides behind; x is the informed trader's order, u the noise, and y the only thing the market maker ever sees; and p is the single price at which everything clears. The two unknowns are the functions X and P , not numbers.

Theorem 6.2 (Kyle). *There is a unique linear equilibrium,*

$$X(v) = \beta (v - p_0), \quad P(y) = p_0 + \lambda y, \quad \beta = \sqrt{\frac{\sigma_u^2}{\Sigma_0}}, \quad \lambda = \frac{1}{2} \sqrt{\frac{\Sigma_0}{\sigma_u^2}}. \quad (6.2)$$

In equilibrium the informed trader's expected profit is $\frac{1}{2} \sqrt{\Sigma_0 \sigma_u^2}$, and exactly half of the private information is revealed by the price: $\text{Var}(v | y) = \frac{1}{2} \Sigma_0$.

Sketch. Guess $P(y) = p_0 + \lambda y$. The informed trader maximizes $\mathbb{E}[(v - p_0 - \lambda(x + u))x] = (v - p_0)x - \lambda x^2$, giving $x = (v - p_0)/(2\lambda)$, so $\beta = 1/(2\lambda)$. Given that, $y = \beta(v - p_0) + u$ is jointly normal with v , and the projection theorem gives $\mathbb{E}[v | y] = p_0 + \frac{\beta \Sigma_0}{\beta^2 \Sigma_0 + \sigma_u^2} y$, so $\lambda = \beta \Sigma_0 / (\beta^2 \Sigma_0 + \sigma_u^2)$. Solving the two equations simultaneously yields (6.2). $\square$

λ is **Kyle's lambda**, and it is the inverse of market depth: the price moves λ dollars per unit of net order flow. It is measured empirically by regressing price changes on signed volume, and it is the single number that most execution models use to summarize the entire order book. In the notation of (1.4), λ is the slope of the permanent impact function g at the origin.

The formula $\lambda = \frac{1}{2}\sqrt{\Sigma_0/\sigma_u^2}$ says something you should find intuitive: *depth is the ratio of noise to information*. Markets are deep where there is a lot of uninformed trading to hide behind, and thin where every order is suspicious.

Where the model breaks

Kyle's equilibrium is linear because everything in it is Gaussian and the objective is quadratic: it is, in effect, an LQG problem with an information-revelation twist. That elegance is also its limitation: it gives *linear* permanent impact, whereas the empirical impact of §6.5 is strongly *concave*. Reconciling the two occupies much of the modern literature; the leading explanation is that the concavity comes from order splitting and the latent, unrevealed nature of supply and demand, not from the shape of the book on any given day.

6.3 Decomposition of the spread

Three distinct forces widen the spread, and they respond to different controls:

$$\underbrace{S}_{\text{quoted}} = \underbrace{c_{\text{proc}}}_{\text{fees, technology}} + \underbrace{c_{\text{inv}}}_{\text{risk of holding}} + \underbrace{c_{\text{adv}}}_{\text{losing to the informed}}. \quad (6.3)$$

Only the third is present in Glosten-Milgrom; only the second appears in the market-making models of Chapter 9; the first is an institutional constant. Distinguishing them empirically is a classical problem, and the classical answer is elegant enough to state.

Proposition 6.3 (Roll's estimator). *Suppose the efficient price is a random walk and observed transaction prices are $Y_t = S_t + \frac{S}{2}\epsilon_t$ with $\epsilon_t = \pm 1$ i.i.d. trade signs independent of S . Then*

$$\text{Cov}(\Delta Y_t, \Delta Y_{t-1}) = -\frac{S^2}{4}, \quad \text{so} \quad \widehat{S} = 2\sqrt{-\widehat{\text{Cov}}(\Delta Y_t, \Delta Y_{t-1})}. \quad (6.4)$$

Proof. $\Delta Y_t = \Delta S_t + \frac{S}{2}(\epsilon_t - \epsilon_{t-1})$. The only surviving cross term at lag one is $-\frac{S^2}{4}\mathbb{E}[\epsilon_{t-1}^2]$. $\square$

Remark 6.4. Roll's estimator is a beautiful piece of reasoning that fails on real data, because the assumption “ ϵ_t i.i.d.” is spectacularly false, as §6.4 shows immediately below. The sample covariance is often positive, and the estimator returns the square root of a negative number. The failure is more instructive than the formula.

6.4 The long memory of order flow

The sign of successive trades is one of the most strongly autocorrelated series in all of empirical finance. The autocorrelation of trade signs decays as a power law $\text{Cov}(\epsilon_t, \epsilon_{t+\ell}) \sim \ell^{-\gamma}$ with $\gamma \approx 0.5$, remaining significantly positive over thousands of trades, which is hours or days of activity.

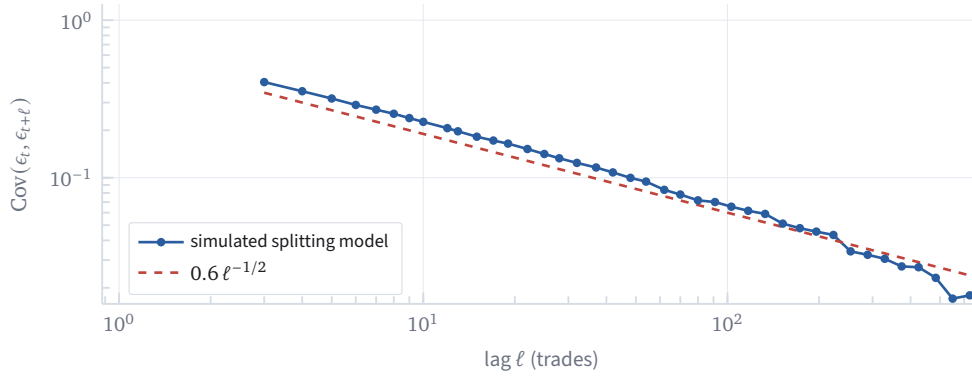


Figure 6.2. Long memory from splitting alone. Autocorrelation of trade signs from 400,000 simulated trades in which metaorder sizes are Pareto with tail exponent $\zeta = 1.5$ and each metaorder is executed as a run of same-signed children. No agent in the simulation has information, yet the sign autocorrelation is 0.6 at lag one and still positive after hundreds of trades, matching the empirical power law $\ell^{-(\zeta-1)}$. The downward bend at large lags is a finite-sample effect.

This looks paradoxical. If buys predict buys, and buys push the price up, why is the price not predictable? The resolution is one of the most satisfying results in the field, and it has two parts.

Long memory from order splitting

Large investors do not trade their full size at once; they split a **metaorder** into hundreds of child orders executed over hours. A single decision therefore produces a long run of same-signed trades. If metaorder sizes have a Pareto tail with exponent ζ , the resulting sign process has autocorrelation decaying like $\ell^{-(\zeta-1)}$. With $\zeta \approx 1.5$, we get $\gamma \approx 0.5$: the observed value, produced by a model containing no information at all. The symbol ζ here is unrelated to the informed fraction α of §6.1; it is the tail index of the distribution of *order sizes*.

Transient impact restores efficiency

If order flow is predictable and impact were permanent, prices would be predictable, and (5.9) would fail. The market resolves this by making impact *decay*: the price reverts after each trade by exactly enough to offset the predictability of the next one. Formally, in the **propagator** model

$$S_t = S_{-\infty} + \sum_{s < t} G(t-s) \epsilon_s f(v_s) + \text{noise}, \quad (6.5)$$

the kernel G must decay like $\ell^{-(1-\gamma)/2}$ for S to be a martingale. The long memory of the flow and the decay of impact are two views of a single fact.

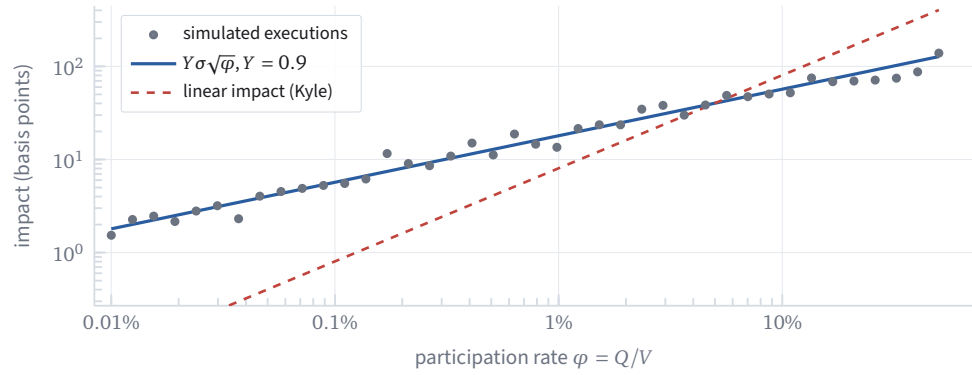


Figure 6.3. Impact is dramatically concave. On log-log axes the square-root law is a line of slope $\frac{1}{2}$; linear impact would have slope 1. Calibrated to daily volatility $\sigma = 2\%$: a trade of 1% of daily volume costs about 18 basis points, while a trade a hundred times smaller costs about 1.8, not 0.18. Concavity is why large orders are executed at all, and why the marginal cost of the first shares is so much higher than the average.

6.5 The shape of price impact

We can now state the empirical facts that Chapter 8 takes as given. Let a metaorder of size Q be executed over a period during which total market volume is V , in an asset with volatility σ over the same period. The ratio

$$\varphi = \frac{Q}{V} \quad (6.6)$$

is the **participation rate**: the fraction of everything that traded which was you. It is the natural size variable throughout this book, because impact depends on how much of the market you were rather than on how many shares you bought, and §7.3 turns it into the most common risk limit in the industry.

The square-root law

Across markets, asset classes, decades, and execution styles, the average price move from the start to the end of a metaorder is

$$\mathbb{E}[\text{impact}] \approx Y \sigma \sqrt{\varphi}, \quad Y \in [0.5, 1.5], \quad (6.7)$$

with Y remarkably stable. The exponent is close to $1/2$ (empirical estimates cluster in $[0.4, 0.7]$) and is emphatically *not* 1. This is the most robust quantitative regularity in market microstructure and it has no fully accepted theoretical derivation.

6.5.1 Permanent, temporary and transient components

The distinction promised in (1.4) can now be drawn properly. Watch the price around a metaorder (Fig. 6.4):

- during execution, the price drifts against you, concavely, reaching a peak at completion;
- after you stop, it *reverts*, typically by about one third, leaving a permanent component of roughly $\frac{2}{3}$ of the peak;
- the reversion is not instantaneous but decays over a timescale comparable to the execution horizon itself.

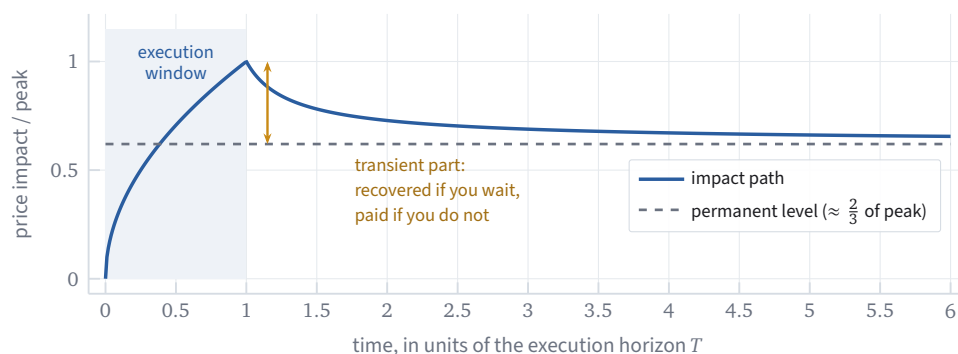


Figure 6.4. The impact of a metaorder before and after. Impact accumulates concavely (roughly as $\sqrt{t}$) while you are trading, peaks at completion, and then relaxes toward a permanent level near two thirds of the peak. The gap is the transient component. An execution algorithm that measures its performance at the moment it finishes trading will systematically overstate its costs; one that measures a day later will understate them.

Only the permanent part is a genuine transfer of wealth; the transient part is the cost of demanding liquidity faster than it is replenished, and it can be reduced by trading more slowly. This is precisely the trade-off that [Chapter 8](#) optimizes.

6.6 No-arbitrage constraints on impact

Here is a result that will appeal to you specifically, because it turns an empirical modelling question into a theorem about convex functionals.

Definition 6.5 (Price manipulation). A round trip is a trading schedule with $Q_0 = Q_T = 0$. A model of impact admits **price manipulation** if some round trip has strictly positive expected profit.

Theorem 6.6 (Impact must be linear, or transient in a particular way). Consider dynamics $dS_t = g(\nu_t) dt + \sigma dW_t$ with execution at $S_t + h(\nu_t)$. If g is nonlinear then price manipulation exists. If g is linear, $g(\nu) = \kappa\nu$, then no round trip profits, and the total permanent cost of any schedule is $\frac{\kappa}{2}q_0^2$, independent of the schedule.

Sketch. The permanent-impact cost of a schedule is $\int_0^T Q_t g(\nu_t) dt$; integrating by parts with $Q_0 = Q_T = 0$ shows this vanishes for linear g regardless of the path. For strictly convex or concave g one constructs a two-speed round trip (trade fast in one direction, slowly in the other) whose cost is strictly negative. The full statement, including the transient case, is Gatheral's no-dynamic-arbitrage principle: the propagator G in (6.5) and the instantaneous cost f must satisfy a joint condition, and e.g. G exponentially decaying with f nonlinear is inadmissible. $\square$

Where the model breaks

Theorem 6.6 is why the models of [Chapter 8](#) use *linear permanent* impact and put all the nonlinearity into the *temporary* term, even though the measured aggregate impact (6.7) is concave. It is a genuine tension in the field: the schedule-level empirical law and the trade-level no-arbitrage requirement point in different directions, and every practical model chooses a compromise. Be suspicious of any execution model whose optimal solution depends delicately on the curvature of g .

Exercises

Exercise 6.1 (Glosten-Milgrom converges). Show that π_t in (6.1) is a bounded martingale and hence converges a.s. Show further that the limit is V (the truth) whenever $\alpha > 0$, and give the rate: how does $\mathbb{E}[\pi_t(1 - \pi_t)]$ decay?

Exercise 6.2 (Kyle with n periods). Extend Theorem 6.2 to n trading rounds with the informed trader maximizing total profit. Show that the equilibrium is again linear, that λ_k is constant across rounds, and that information is revealed at a constant rate: the informed trader spreads their trading exactly evenly. Compare with the risk-neutral execution schedule of Section 8.4, and explain why the two problems have the same answer for different reasons.

Exercise 6.3 (Roll fails). Simulate the splitting model of Fig. 6.2, add a bid-ask bounce of one tick, and compute Roll's estimator of Proposition 6.3. Show that it returns a negative variance whenever the sign autocorrelation at lag one exceeds a threshold, and find that threshold.

Exercise 6.4 (Manipulation with concave impact). Take $g(\nu) = \kappa \operatorname{sgn}(\nu) |\nu|^{1/2}$ and construct an explicit round trip with positive expected profit, thereby proving the first half of Theorem 6.6 by counterexample. Interpret your construction: in what sense are you exploiting the fact that a fast trade “costs less per share” in permanent terms?

Exercise 6.5 (Propagator and efficiency). In (6.5) with $f \equiv 1$ and $\operatorname{Cov}(\epsilon_t, \epsilon_{t+\ell}) = c \ell^{-\gamma}$, show that requiring $\mathbb{E}[S_{t+1} - S_t | \mathcal{F}_t] = 0$ forces $G(\ell) \sim \ell^{-\beta}$ with $\beta = (1 - \gamma)/2$. Verify that $\gamma = 1/2$ gives $\beta = 1/4$, and comment on how slowly that decays.

Liquidity and Its Measurement

Start with a question that sounds like it has an obvious answer. You hold two million shares of a company whose stock last traded at \$40. What is your position worth?

The accountant says \$80 million, and writes it down. But suppose you had to sell it this afternoon. The stock trades four million shares on a normal day, so you are holding half a day's volume, and Fig. 4.3 says that selling that much at once will not happen at \$40. Suppose instead you sold it over a week: better prices, but the stock might move a long way in a week. Suppose the market is in the middle of a panic: then everything above gets worse at once, and by more than you expected.

Every one of those complications is called “liquidity”, which is the most overloaded word in finance. It is used for at least four distinct properties. They are empirically correlated, which is why one word survives for all of them, but they are mathematically different objects and they enter an optimization problem in different places. A model that conflates them will misprice exactly the situations you care about.

This chapter separates them, attaches a measurable quantity to each, and shows where each one lands in the problems of Part II: as a cost coefficient, as a constraint, or as an extra state variable.

The four faces of liquidity

Face	Question it answers	Where it enters
<i>Tightness</i>	What does a round trip of one lot cost?	the spread S : a fixed cost per trade
<i>Depth</i>	How much can I trade before the price moves?	the impact coefficients λ, κ
<i>Resiliency</i>	How fast does the book come back?	the decay rate ρ of transient impact; a <i>state</i>
<i>Immediacy</i>	How long must I wait for a counterparty?	fill intensities, and the horizon T itself

In plain terms

Everyone in finance uses the word “liquidity” and almost nobody defines it. An asset is called liquid when you can trade a lot of it, quickly, without moving the price much, and get out again if you change your mind. Those are four separate properties, they are measured differently, and they enter an optimization problem in different places: one as a fixed cost, one as a cost that grows with size, one as a state that recovers over time, and one as a constraint. The chapter separates them and then makes the uncomfortable observation that all four deteriorate together, exactly when you need them.

7.1 Measuring the four dimensions

Each of the four has a standard measurement, and each measurement has a standard way of misleading you.

7.1.1 Tightness

Tightness is the cost of changing your mind about one share. Buy at the ask, sell immediately at the bid, and you are out the spread: on a \$40 stock with a one-cent spread that is 2.5 basis points for the round trip, before fees.

The **quoted spread** $S = p^a - p^b$ is the obvious measure and it is systematically wrong, for two reasons. Trades often execute inside the quotes, against hidden or mid-point liquidity, so you may not pay the full spread. And the mid moves after you trade, so the price you compare against matters. Three refinements are standard. For a trade at price $\hat{S}$ with sign $\epsilon = \pm 1$ at time t :

$$\text{effective spread} \quad 2\epsilon(\hat{S} - M_t) \quad \text{what the taker paid} \quad (7.1)$$

$$\text{realized spread} \quad 2\epsilon(\hat{S} - M_{t+\delta}) \quad \text{what the maker kept} \quad (7.2)$$

$$\text{price impact} \quad 2\epsilon(M_{t+\delta} - M_t) \quad \text{the difference} \quad (7.3)$$

In plain terms

Work through one trade. The mid is \$40.00 and you buy at the ask, \$40.005. Your *effective* spread is $2 \times 0.005 = 1$ cent, and that is your cost. A minute later the mid has drifted to \$40.002, because your buy conveyed a little information. The market maker who sold to you can now buy back at \$40.002 rather than \$40.00, so what they actually *kept* is $2 \times (40.005 - 40.002) = 0.6$ cents, the *realized* spread. The missing 0.4 cents did not go to them; it went into the price, and it is the *impact*.

The identity $\text{effective} = \text{realized} + \text{impact}$ holds by construction, and it is the empirical version of (6.3). When the realized spread goes negative, the maker is losing money on average: the adverse selection of Section 6.1 exceeds the spread they charged, and they must widen or stop.

7.1.2 Depth

Depth is how much you can trade before the price moves against you, and it is the face that enters Part II most directly, as the impact coefficient.

The canonical measure is Kyle's λ (Theorem 6.2), estimated by regressing mid changes on signed volume over some interval: $\Delta S = \lambda v + \text{noise}$. Its units are dollars per share traded, so $1/\lambda$ is a depth: the number of shares that moves the price by one dollar. A liquid large-cap might have λ such that a million dollars of net buying moves the price a few basis points; an illiquid small cap can be a hundred times worse.

Two warnings about estimating it, both of which Chapter 13 develops. The regression is biased upward, because news moves the price and provokes the flow simultaneously, so the flow gets credit for a move it did not cause. And λ is not a constant: it varies by time of day, by regime, and with the size of the trade being measured, which is exactly what (6.7) says.

When volume data is coarse or unavailable there is a cruder scale-free cousin (Amihud 2002), the **Amihud illiquidity** ratio

$$\text{ILLIQ} = \mathbb{E} \left[\frac{|r_t|}{\text{volume}_t} \right], \quad (7.4)$$

the average absolute return per dollar traded. Its virtue is that it can be computed from daily closing data for every asset that has ever traded anywhere, which is why

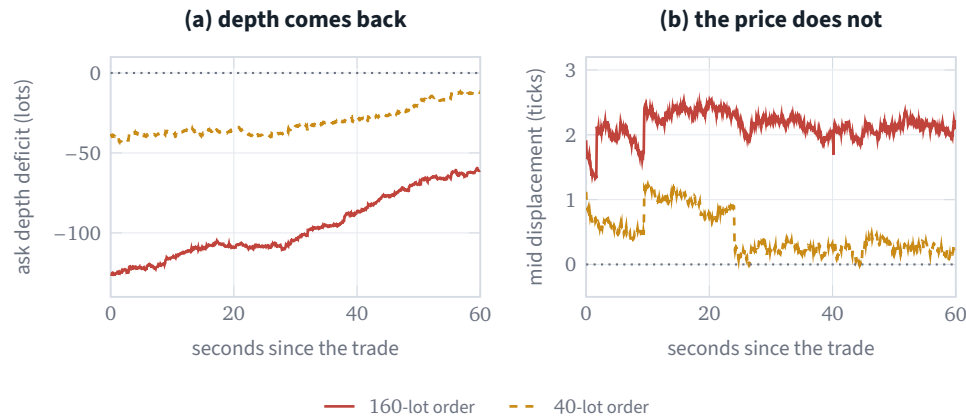


Figure 7.1. The impulse response of a book. A market buy is fired into the zero-intelligence book of §4.4 and the result is compared with a matched control run driven by the identical random stream, so the difference isolates the metaorder’s own effect. (a) Depth on the affected price levels recovers on a timescale set by the limit-order arrival rate, here about a minute because the model refills at three lots per second per level. (b) The mid, once displaced, stays displaced.

it dominates the empirical asset-pricing literature. Its defect is that it is a ratio of two noisy quantities and is therefore extremely noisy itself.

7.1.3 Resiliency

Resiliency is how fast the book comes back after you have eaten it, and it is the only one of the four that is genuinely *dynamic*. That makes it the only one that adds a state variable to your control problem rather than a coefficient.

Why it matters is easiest to see at the extremes. If the book refilled instantly, then breaking a large order into pieces would cost nothing extra, because each piece would meet a full book: the optimal execution problem of Chapter 8 would be trivial and you would trade as slowly as your risk appetite allowed. If the book never refilled, then the total cost would depend only on the total quantity and not at all on the schedule, and the problem would be trivial in the other direction. Everything interesting happens in between, and *how fast* it refills is the parameter that decides where in between you are.

It is measured by an impulse response: hit the book with a metaorder and watch what comes back.

Two things in that figure deserve comment, and the second is the more interesting.

The depth recovery in panel (a) has a characteristic timescale, and that timescale is the parameter an execution algorithm cares about. Trade again before the book has recovered and you pay the deficit twice; wait longer than the recovery time and you have gained nothing further by waiting. The optimal spacing of child orders is essentially this timescale, which is why Section 8.7.2 produces a schedule with a characteristic rate rather than a smooth curve.

Panel (b) is the surprise. In a book with no strategic participant, the price does *not* come back. Nothing in the zero-intelligence model knows where the price “ought” to be, so once a queue has been consumed and the quote has moved, the refilling happens at the new level rather than the old one. Impact in that model is permanent.

Resiliency is economic rather than mechanical

That result is worth dwelling on, because the transient impact of Fig. 6.4 is real and empirically robust: after a metaorder finishes, the price does typically revert by about a third. Panel (b) says queueing alone cannot produce that reversion.

So the reversion must come from somewhere else, and the only candidate is participants who have a view about value and who lean against a move they believe is temporary. Resiliency is therefore not a property of the matching engine, it is evidence of somebody deciding that the price has gone too far. Which means it can disappear: exactly when everyone agrees the move was justified, or when the people who would have leaned against it have run out of capital, the market stops bouncing back. Section 7.5 is about the second of those.

7.1.4 Immediacy

The last face is the one that is hardest to measure and easiest to feel: how long you must wait for a counterparty who wants the other side.

For a single small order this is the fill-intensity story of Section 4.5: post, and you are filled with a probability that depends on the queue. For a large position it is a different question, because you are not waiting for one counterparty but for many, arriving over hours.

The practical summary is **average daily volume**, universally ADV, and the practical quantity derived from it is the participation rate (6.6), the fraction of market volume that is you. A position of half a day's ADV cannot be sold in an hour at any sensible price, because the natural counterparties for it simply have not arrived yet.

That reasoning turns into the single most common risk limit in the industry: *never be more than about ten per cent of volume*. It is a crude rule and everybody knows it is crude, but it encodes something real, which is that beyond roughly that level you stop being a participant in the market and start being the market. Section 7.3 writes it as a constraint.

7.2 The liquidity-adjusted mark

Chapter 3 warned that $V = X + QS$ is a convention. Here is the correction. Selling Q lots immediately into the book yields

$$L(Q) = \int_0^Q p^b(u) du, \quad (7.5)$$

with $p^b(\cdot)$ the nonincreasing marginal bid price of Proposition 4.2. Hence L is concave, $L(Q) \leq QS$, and the gap grows superlinearly.

Control-theoretic dictionary

Concavity of L is the same fact as convexity of the execution cost C from Proposition 4.2: $L(Q) = QS - C(Q)$ with C convex. In the language of Part II, $-L$ is the terminal cost Ψ of a liquidation problem, and its convexity is what guarantees that the value functions of Chapters 8 and 9 are concave in inventory, which is in turn what makes the optimal policies monotone in Q . Every qualitative feature of the solutions in Part II can be traced back to this one picture.

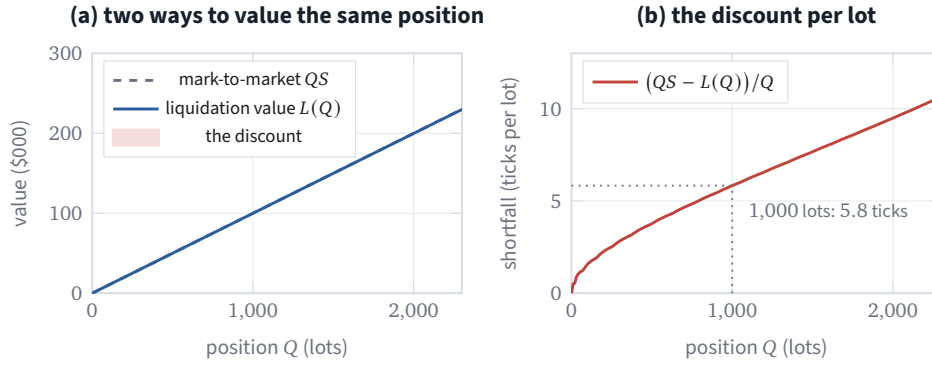


Figure 7.2. Marking a position honestly. Computed from the time-averaged book of Fig. 4.6. In (a) the straight dashed line is the accounting convention and the curve below it is what you would actually receive by selling into the book at once; the two are close enough that the shortfall is easy to overlook, which is exactly the problem. Panel (b) plots that shortfall per lot, and it grows without bound: six ticks a lot at a thousand lots, more than ten at two thousand. Because L is concave, variance-based risk measures applied to QS systematically understate the risk of a large position, and the correct object is a concave function of Q .

7.3 Liquidity as a constraint set

For the optimization problems of Part II the most useful representation of liquidity is often not a cost at all but a *set*. Costs are traded off against other costs; constraints are absolute, and in practice it is the constraints that stop desks rather than the objective.

The admissible controls $\mathcal{A}$ are typically cut out by four conditions.

$$|\nu_t| \leq \varphi_{\max} \mathcal{V}_t \quad \text{participation cap} \quad (7.6)$$

$$|Q_t| \leq \bar{q} \quad \text{position limit} \quad (7.7)$$

$$\sum_i |Q_t^i| S_t^i \leq L_{\max} V_t \quad \text{gross leverage} \quad (7.8)$$

$$Q_T = 0 \quad \text{terminal liquidation} \quad (7.9)$$

Take them in turn, because each encodes a different one of the four faces.

The *participation cap* (7.6) is immediacy written as algebra: trade at most $\varphi_{\max}$ of the market's volume $\mathcal{V}_t$ (written $\mathcal{V}$ to keep it distinct from wealth V), with $\varphi_{\max}$ typically between five and twenty per cent. It exists because impact models are unreliable at high participation and because being a large fraction of volume makes you visible.

The *position limit* is depth written as algebra. It caps how much inventory you can be holding when something goes wrong, and it is usually set so that the position could be unwound within a stated number of days at the participation cap, which ties the two constraints together.

The *leverage constraint* is the funding face, and §7.5 is about what happens when it binds.

The *terminal condition* is what makes an execution problem an execution problem rather than a portfolio problem: you have promised to be flat, and that promise is not negotiable.

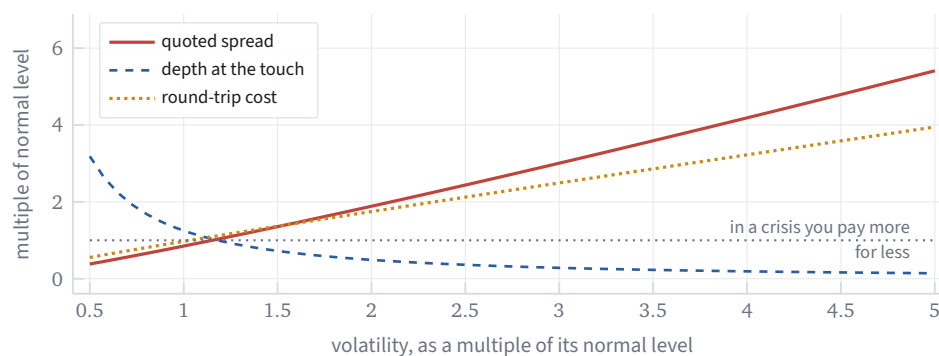


Figure 7.3. The liquidity spiral in one picture. Stylized elasticities calibrated to the commonly reported values: spread $\propto \sigma^{1.15}$, depth $\propto \sigma^{-1.35}$. A market that is three times as volatile as usual costs roughly three and a half times as much to trade in. Because margin requirements also rise with volatility, forced sellers arrive at precisely this moment: the feedback loop between funding liquidity and market liquidity (Brunnermeier and Pedersen 2009) that Section 12.6 treats.

Why these are all convex

Each of the four is either a norm bound or a linear equality, so $\mathcal{A}$ is a convex set and the problems of Part II remain convex programs. That is not a lucky accident of exposition.

Risk managers write limits as intersections of half-spaces because those are the limits that can be checked in real time, on every order, by a piece of software that must answer in microseconds. A limit that required solving an optimization to evaluate would not be usable. So the shape of the constraint set is dictated by the latency budget of the compliance system, and it happens to be exactly the shape that keeps the mathematics tractable.

Where the model breaks

The participation cap has a defect that is easy to miss: its right-hand side $\mathcal{V}_t$ is a *random, forward-looking* quantity. You do not know today's volume until today is over, so in practice it is replaced by a forecast and the constraint becomes soft.

The resulting problem is a chance constraint or a robust constraint rather than a hard one, and treating the forecast as a certainty is one of the two or three most common sources of blow-ups in systematic trading. On a quiet day your ten per cent turns out to have been twenty-five, and the impact you paid was priced for the wrong regime. Section 11.8 gives the robust version.

7.4 The correlation of liquidity with volatility

The final and most important empirical fact about liquidity is that it is strongly, nonlinearly, and *adversely* correlated with volatility. Spreads widen roughly linearly in volatility, depth falls faster than one-for-one, and the round-trip cost of a position therefore rises steeply exactly in the states where you most want to reduce it.

What this breaks in Part II

Every model in Part II treats the liquidity parameters λ, S, σ as constants, or at best as deterministic functions of time. The figure above says they are stochastic, and correlated with the very shocks the model is supposed to protect you from. Three responses, in increasing order of ambition:

- (1) *Regime-condition*. Solve the model separately in each liquidity regime and switch. Crude, and by far the most common.
- (2) *Add a state*. Let λ_t be a mean-reverting process and solve the resulting higher-dimensional HJB. Tractable in special cases (Section 16.5).
- (3) *Be robust*. Optimize the worst case over a set of liquidity parameters. Convex, conservative, and increasingly standard (Section 11.8).

7.5 Funding liquidity and the liquidity spiral

There is a second kind of liquidity that has nothing to do with order books: your ability to *finance* a position at all.

A leveraged position is funded by borrowing against collateral, and the lender applies a **haircut**: they will lend, say, ninety cents against a dollar of stock, so you must fund the remaining ten cents yourself. Write m_t for that margin requirement per unit of exposure. The constraint is

$$m_t \sum_i |Q_t^i| S_t^i \leq V_t, \quad (7.10)$$

and the crucial fact is that m_t is not a constant. It is set from a risk model, and risk models raise their estimates when volatility rises. So m_t is increasing in σ_t .

Now trace what happens when volatility rises, one step at a time.

- (1) Volatility rises, so the lender's risk model raises m_t .
- (2) The left-hand side of (7.10) grows while the right-hand side, your equity, has if anything shrunk. The constraint binds.
- (3) You must reduce $\sum_i |Q_t^i| S_t^i$, which means selling, and selling now rather than at your convenience.
- (4) Everybody subject to the same constraint is selling at the same time, for the same reason, into a market whose depth has already fallen (Fig. 7.3).
- (5) That selling moves prices and *raises* realized volatility.
- (6) Return to step 1.

This is the **liquidity spiral**, and it is the mechanism behind essentially every rapid deleveraging event on record. Notice what makes it vicious: no participant in it is behaving irrationally, or panicking, or making an error. Each is complying with a contract. The loop is a property of the system rather than of anybody's judgement.

Control-theoretic dictionary

Mathematically this is a constrained control problem in which the *feasible set depends on the state*, and the state is driven by the aggregate of all agents' controls. That is a *mean-field game* with a state-dependent constraint: a game

with so many identical small players that each optimizes against the average behaviour of the population rather than against any individual, with the requirement that the average be consistent with what the individuals then do. [Section 16.1](#) develops it.

Both features are individually hard and together they are close to the frontier. Standard mean-field theory assumes the constraint set is fixed; standard constrained control assumes the constraint does not depend on what everybody else does. This is why crises are hard to model, why the models that exist are qualitative rather than predictive, and why [Chapter 16](#) lists this among the open problems rather than the solved ones.

In plain terms

The practical lesson does not require the mathematics. Any risk calculation that treats your ability to reduce a position as independent of the reason you want to reduce it is wrong in the same direction every time. The states in which you most want to sell are precisely the states in which selling is most expensive and most crowded, and a model with constant liquidity parameters cannot represent that at all.

Exercises

Exercise 7.1 (Effective, realized, impact). Verify the identity $\text{effective} = \text{realized} + \text{impact}$ directly from the definitions. Then show that under the Glosten-Milgrom model of [Section 6.1](#) the realized spread is exactly zero at every trade, and explain what that means about the market maker's business.

Exercise 7.2 (Concavity of the liquidation value). Prove that L in (7.5) is concave, and that for a book with constant depth q lots per tick, $L(Q) = QS - \frac{\Delta}{2q}Q^2$ exactly. Deduce that constant depth is precisely the case of *linear* impact, and identify Kyle's λ in terms of Δ and q .

Exercise 7.3 (Resiliency as a state variable). Model the deviation of the book from its steady state by $dD_t = -\rho D_t dt + \eta \nu_t dt$, so that the execution price is $S_t + D_t$. Show that this is equivalent to a transient impact model with an exponential propagator $G(\ell) = \eta e^{-\rho \ell}$, and that as $\rho \rightarrow \infty$ it degenerates to the purely temporary impact of (1.4). This is the model of Obizhaeva and Wang (2013); you will solve it in [Chapter 8](#).

Exercise 7.4 (A liquidity-adjusted risk measure). Let $R(Q) = \text{Var}[L(Q)]$ with L from (7.5) and a stochastic book. Show that R grows faster than quadratically in Q , and that the mean-variance portfolio problem of [Chapter 11](#) therefore has a strictly smaller optimal position than the frictionless version. Where does the extra term enter the KKT conditions?

Exercise 7.5 (The spiral has a fixed point). Consider n identical leveraged agents with margin $m(\sigma)$ increasing and volatility σ (net selling) increasing. Write down the map from aggregate leverage to itself and give a condition on the elasticities under which it has a unique stable fixed point. Interpret the failure of that condition.

PART II

Decision Problems

Four decision problems, each stated once as a stochastic control problem and then solved as far as closed form allows: execution, market making, hedging, and portfolio choice. These chapters are independent of one another and can be read in any order.

Optimal Execution

A portfolio manager decides to sell five hundred thousand shares. That decision is not this chapter. This chapter is what happens next: the order is handed to a trading desk with an instruction like “work it over the day,” and someone (increasingly, some algorithm) must decide how fast to trade.

Trade too fast and you pay the convex cost of [Proposition 4.2](#). Trade too slowly and you carry inventory through a random price for longer. That is the entire problem, and it is a finite-horizon linear-quadratic control problem with a terminal constraint, the execution analogue of a regulator. You already know how to solve it; what follows is mostly a matter of getting the model right and then interpreting the answer carefully.

In plain terms

Somebody has decided to sell half a million shares. That decision is not this chapter. This chapter is the question of *how fast*. Sell it all at once and you eat through the order book and get a terrible average price. Sell it slowly over the day and you get better prices, but you are still holding an unsold position while the market wanders, and it may wander against you. So there is a genuine trade-off between the cost of trading quickly and the risk of trading slowly, it has a precise mathematical answer, and the answer is a curve of possible schedules from which you pick one according to how much uncertainty you are willing to bear.

8.1 The trade-off in quantitative terms

Put concrete figures on the problem before formalizing it, because the whole chapter turns on the size of two competing effects.

A fund decides to sell 500,000 shares of a stock trading at \$100. The stock trades about two million shares on a normal day, so this order is a quarter of a day’s volume: large, but not absurd. Its daily volatility is 2%, or \$2 a share.

Option one: sell it all now. You send the whole order to the market. It walks through the book exactly as in [Fig. 4.3](#), and empirically an order of this size moves the price by something like $0.9 \sigma \sqrt{0.25} \approx 0.9\%$, or roughly \$0.90 a share. On 500,000 shares that is \$450,000, about 90 basis points of the value of the order. You are, however, completely finished, and nothing that happens tomorrow can hurt you.

Option two: sell evenly over the day. Now each individual slice is tiny and barely moves the price. Working through the same impact model, the total cost falls to around 30 basis points, a third of the immediate execution. But you are now holding an unsold position for hours while the price wanders, and the wandering has a standard deviation of about $2\%/\sqrt{3} \approx 1.2\%$, or 116 basis points of the order. Your expected cost improved by 60 basis points and your uncertainty went from nothing to more than a hundred.

So which is it? Neither, in general. The answer depends on how much uncertainty you are willing to accept in exchange for a lower expected cost, and *that* is a preference, not a fact. What the mathematics can do, and what the rest of this chapter does, is compute the entire menu of efficient compromises between the two extremes, so that the only thing left to choose is a point on it. That menu is Fig. 8.2, and the two options above are its endpoints.

In plain terms

Every execution problem has this shape. Trading fast is expensive because you consume liquidity that has not had time to replenish. Trading slowly is risky because you are still holding the position while the market moves. The cost of haste is roughly predictable; the cost of patience is not, which is why the comparison is between a number and a distribution. There is no schedule that is best under all circumstances, only a family of schedules indexed by how much risk you will tolerate.

8.2 Execution benchmarks

Before optimizing, fix the objective. Execution desks are measured against benchmarks, and the choice of benchmark changes the optimal strategy completely.

How it actually works

Implementation shortfall (arrival price). Compare the average execution price with the mid at the moment the order arrived. This is the economically correct benchmark, since it measures the full cost of the decision, and it is the one we optimize below. It is also the one with the most variance, because it is exposed to the price path.

VWAP. Compare with the volume-weighted average price over the execution window. Beating it requires only that you trade in proportion to market volume, so it is nearly risk-free to track and correspondingly uninformative. A desk measured on VWAP will trade a fixed participation rate and ignore everything else.

Closing price. Common for index funds. Concentrates everything into the closing auction, and creates the enormous, predictable order imbalances observed at the close.

The lesson is general: a benchmark is a *terminal cost*, and specifying one is specifying Ψ in (1.2). Choose it carelessly and you will get exactly what you asked for.

8.3 The model

Problem 4: optimal liquidation

Sell $q_0 > 0$ shares over $[0, T]$. The control is the trading rate $\nu_t \geq 0$ (shares per unit time), inventory and cash evolve as

$$dQ_t = -\nu_t dt, \quad dX_t = (S_t - k \nu_t) \nu_t dt,$$

and the efficient price carries permanent impact,

$$dS_t = -b \nu_t dt + \sigma dW_t.$$

The agent maximizes terminal cash plus the value of any remaining inventory, penalized for the risk carried along the way:

$$\sup_{\nu \in \mathcal{A}} \mathbb{E} \left[X_T + Q_T (S_T - \alpha Q_T) - \phi \int_0^T Q_t^2 dt \right]. \quad (8.1)$$

Every symbol has a job. q_0 is the position to be sold and T the deadline; Q_t is what is left at time t and X_t the cash raised so far; S_t is the price the rest of the market sees, moved by volatility σ through the Brownian motion W ; and $\mathcal{A}$ is the set of trading rates you are allowed to choose, meaning adapted and square integrable. The four parameters are the model. $k > 0$ is *temporary* impact, the concession you pay on the shares you are selling right now and which vanishes the moment you stop; $b \geq 0$ is *permanent* impact, the part of that concession that stays in the price for everybody afterwards; $\phi \geq 0$ is the running inventory penalty, the **urgency**, which charges you for every second the unsold position is still exposed; and $\alpha > 0$ is the terminal penalty charged on whatever is left at T , with $\alpha \rightarrow \infty$ turning the soft penalty into the hard constraint $Q_T = 0$.

Control-theoretic dictionary

Everything is where you expect: Q is the state, ν the control, $k\nu^2$ the control cost, ϕQ^2 the state cost, αQ_T^2 the terminal cost. It is an LQ problem, so the value function is quadratic in Q and the optimal control is linear feedback. The only unusual feature is that the “noise” σdW enters the objective only through the inventory it multiplies, which is why the risk term is ϕQ^2 rather than a control-dependent variance.

8.3.1 Two simplifications

First, *permanent impact drops out*. Applying Itô to $Q_t S_t$ and using $dQ = -\nu dt$,

$$X_T + Q_T S_T = q_0 S_0 - \underbrace{\frac{b}{2}(q_0^2 - Q_T^2)}_{\text{path-independent}} - k \int_0^T \nu_t^2 dt + \sigma \int_0^T Q_t dW_t. \quad (8.2)$$

This is [Theorem 6.6](#) in action: with linear permanent impact the total permanent cost depends on the endpoints alone and not on the path between them. Under the hard constraint $Q_T = 0$ it is $\frac{b}{2}q_0^2$, a constant, and b may be ignored when optimizing. For finite α the term $+\frac{b}{2}Q_T^2$ survives and offsets part of the stated terminal penalty, which is where the combination $\alpha - \frac{b}{2}$ in [Theorem 8.1](#) comes from.

Second, *an optimal schedule may be taken deterministic*. This is not the same as saying the objective is deterministic, since ν is an adapted control and Q_t , ν_t^2 and Q_t^2 are random in general. It holds because the model carries no predictive state and the reduced cost is convex in the inventory path: by Jensen, replacing an adapted Q by its mean path weakly improves every term, and that mean path is itself admissible. Taking expectations in (8.2) and dropping constants, (8.1) therefore reduces to

$$\min_{Q(\cdot)} \int_0^T \left[k \dot{Q}_t^2 + \phi Q_t^2 \right] dt + \left(\alpha - \frac{b}{2} \right) Q_T^2, \quad Q_0 = q_0, \quad (8.3)$$

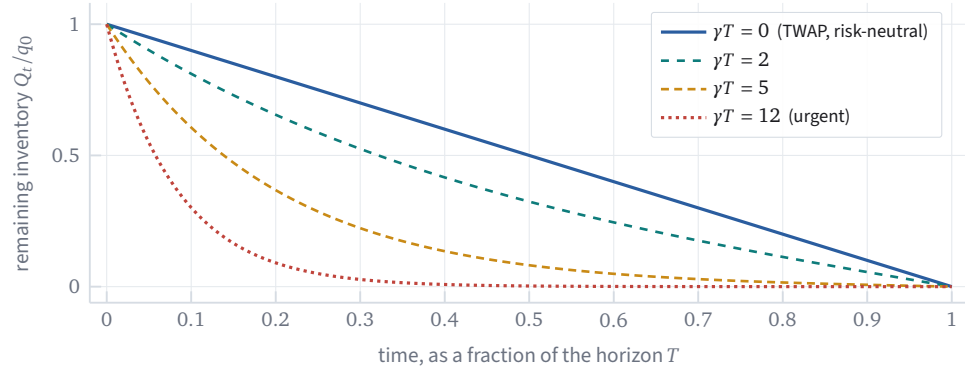


Figure 8.1. One family of curves from one parameter. The Almgren-Chriss inventory paths of (8.4). Risk neutrality ($\phi = 0$) gives the straight line: trade at a constant rate, the TWAP schedule. As urgency $\gamma = \sqrt{\phi/k}$ grows the path becomes front-loaded, approaching an immediate market order. Note that γ has units of inverse time; the only thing that matters is the dimensionless product γT , which compares the horizon with the natural timescale of the risk-cost trade-off.

a deterministic calculus-of-variations problem, convex and well posed when $\alpha > \frac{b}{2}$. The randomness has been absorbed into the penalty ϕ , which is why the answer is a *schedule* and not a feedback law: a fact we revisit critically in §8.8.

8.4 Solution of the model

Theorem 8.1 (Almgren and Chriss 2001). Let $\gamma = \sqrt{\phi/k}$. The minimizer of (8.3) with $\alpha = \infty$ (full liquidation) is

$$Q_t^* = q_0 \frac{\sinh(\gamma(T-t))}{\sinh(\gamma T)}, \quad v_t^* = q_0 \gamma \frac{\cosh(\gamma(T-t))}{\sinh(\gamma T)}. \quad (8.4)$$

For finite α the same formula holds with $\sinh$ replaced by a ζ -weighted combination of exponentials, $Q_t^* \propto \zeta e^{\gamma(T-t)} - e^{-\gamma(T-t)}$ with $\zeta = (\alpha - \frac{b}{2} + \sqrt{k\phi})/(\alpha - \frac{b}{2} - \sqrt{k\phi})$.

Proof. The Euler-Lagrange equation of (8.3) is $k\ddot{Q} = \phi Q$, i.e. $\ddot{Q} = \gamma^2 Q$, with general solution $Ae^{\gamma t} + Be^{-\gamma t}$. Imposing $Q_0 = q_0$ and $Q_T = 0$ gives (8.4). Convexity of the integrand in $(Q, \dot{Q})$ makes the stationary point the global minimum. $\square$

Equivalently, via dynamic programming: the ansatz $v(t, x, q, S) = x + qS + h(t)q^2$ reduces the HJB (1.3) to the Riccati equation $\dot{h} = \phi - h^2/k$ with $h(T) = -\alpha$, whose solution is $-\sqrt{k\phi} \coth(\gamma(T-t) + \text{const})$, and the optimal feedback is $v_t^* = -h(t)Q_t/k$. That form is the one to remember: *trade at a rate proportional to your inventory, with a time-dependent gain that blows up at the horizon.*

Three interpretations of the answer

Three interpretations of (8.4), each useful:

- (1) As a *filter*. $\gamma^{-1} = \sqrt{k/\phi}$ is a time constant. If $T \ll \gamma^{-1}$ there is no time for risk to matter and the schedule is linear; if $T \gg \gamma^{-1}$ the trader liquidates in about γ^{-1} and idles.
- (2) As a *scaling law*. γ does not involve q_0 at all, so the shape of the optimal schedule does not depend on the size of the order. Only the vertical scale

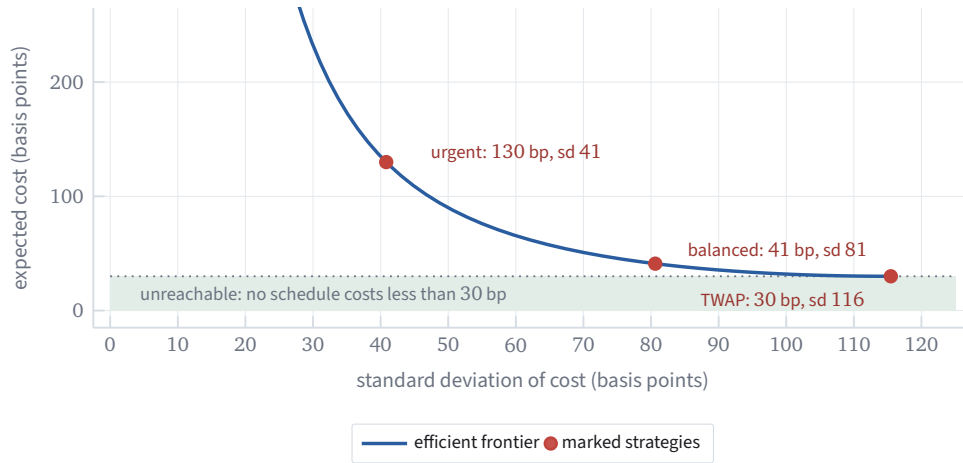


Figure 8.2. Choosing an algorithm is choosing a point. Frontier for a 500,000-share order in a \$100 stock with 2% daily volatility, 25% of average daily volume, executed over one day. Moving from TWAP to the “balanced” schedule buys a 30% reduction in risk, from 116 to 81 basis points, for 11 basis points of expected cost. Moving further, to the urgent schedule, costs another 89 basis points and buys only another 40 basis points of risk: the frontier is steep, and the marginal price of certainty rises fast. The shaded region is unreachable because no schedule has a lower expected cost than the slowest one, which pays 20 basis points of temporary impact on top of the 10 of permanent impact that no schedule can avoid.

does. This is an artifact of quadratic costs and it is false empirically (see [Section 8.8](#)).

- (3) *As a feedback law.* $\nu^* = \gamma \coth(\gamma(T-t)) Q_t$. The gain diverges as $t \rightarrow T$: whatever is left must be liquidated, so the controller becomes infinitely aggressive at the horizon. In practice this is capped, which is exactly the participation constraint of [Chapter 7](#).

8.5 The efficient frontier of execution

From (8.2), the implementation shortfall of a schedule has

$$\mathbb{E}[\text{cost}] = \frac{b}{2} q_0^2 + k \int_0^T \nu_t^2 dt, \quad \text{Var}[\text{cost}] = \sigma^2 \int_0^T Q_t^2 dt. \quad (8.5)$$

Sweeping ϕ from 0 to ∞ traces a curve in the $(\text{sd}, \mathbb{E})$ plane. That curve is the promised efficient frontier: no schedule lies below and to the left of it.

Remark 8.2 (This is mean-variance, with all its problems). (8.5) makes the objective $\mathbb{E}[\text{cost}] + \frac{\gamma_{\text{ra}}}{2} \text{Var}[\text{cost}]$, which identifies the running penalty as $\phi = \frac{1}{2} \gamma_{\text{ra}} \sigma^2$: volatility has not disappeared from the execution problem, it has been folded into ϕ together with risk aversion, and only the product of the two is identified. And mean-variance in a dynamic setting is time-inconsistent: the plan you make at $t = 0$ is not the plan you would remake at $t > 0$. The running-penalty formulation (8.1) is the standard repair; it is time-consistent by construction, and coincides with mean-variance only in the deterministic case. Chapter 11 confronts the same issue where it bites harder.

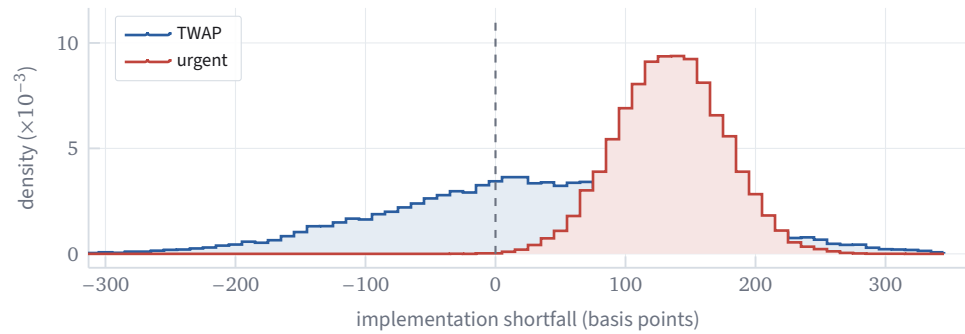


Figure 8.3. The same frontier seen as distributions. Monte Carlo implementation shortfall over 20,000 price paths. TWAP is centred near 30 basis points but has a long tail in both directions: it will sometimes make money and sometimes lose several hundred basis points. The urgent schedule is centred much worse and is far tighter. Neither dominates; the choice is a genuine preference.

8.6 From a schedule to child orders

Everything so far has solved one layer of the problem, and it is worth being precise about which. [Theorem 8.1](#) answers the question *how fast should the remaining position be reduced*. It does not answer the question *what should I send to the exchange in the next second*, and the second question does not follow from the first.

Control-theoretic dictionary

In the language of §4.2.1, ν_t is an economic control and specifically a *fluid-scale* one: it treats the position as a quantity that drains continuously, which is a good approximation precisely when many child orders are executed over the horizon. The output of (8.4) is therefore best read as a *reference trajectory* Q_t^{ref} for an inner loop to track, not as an instruction.

To see why the inner loop is a genuine problem rather than an implementation detail, take the smallest possible instance. Suppose the schedule calls for 600 shares over the next slice and the ask side of the book reads

price	size
\$100.01	300
\$100.02	500
\$100.03	1000

The rate model says 600. It does not say which of the following to do, and the choice is worth several basis points on a position where the whole expected edge may be tens.

- Send a market buy for 600. It fills 300 at \$100.01 and 300 at \$100.02, an average of \$100.015, and it is certain. This is [Proposition 4.2](#) in miniature.
- Post a limit buy for 600 at \$100.00. If it fills you have saved a cent and a half a share, but whether it fills at all is (4.9), and the reason it usually does not is that the price went up.
- Take the 300 at the touch and post the remaining 300. This locks in half the slice and leaves the rest exposed to the same queue race.

- Send nothing at all, because the imbalance of Fig. 4.5 says the ask queue is thick and about to be replenished, so waiting twenty seconds is likely to be cheaper.
- Split across venues (§4.6), which changes the fee, the queue you join and the latency at which you learn the outcome.

Every one of these realizes “600 shares” and they have materially different distributions of cost. Worse, the passive options are not even functions of the state that (8.3) carries: whether a resting order executes depends on queue position, which §4.5 showed is not observable from the aggregated feed and does not appear in (Q_t, S_t) at all. The scheduling model is not merely silent about the choice; it lacks the state in which the choice could be posed.

Two nested control problems

The honest description is a cascade. The outer problem is (8.3), solved once or re-solved occasionally, whose output is a reference inventory path Q^{ref} or an equivalent participation profile. The inner problem is sequential and runs against the book,

$$m_k^* = \arg \min_{m \in \mathcal{E}^{\text{adm}}(L_k)} \mathbb{E} \left[c(m, L_k) + V_{k+1}(Q_{k+1}, L_{k+1}) \mid Q_k, L_k \right], \quad (8.6)$$

where c is the realized cost of the message, Φ of (4.1) generates (L_{k+1}, τ_k) , and V_{k+1} carries a penalty for departing from Q^{ref} . Note the feature that makes (8.6) different in kind from everything else in Part II: the admissible set $\mathcal{E}^{\text{adm}}(L_k)$ depends on the state, because which prices exist and which queues you may join are determined by the current book.

Two things follow. The inner problem is genuinely harder than the outer one, being high dimensional, partially observed and discrete, which is why it is the natural home for simulation and reinforcement learning (§16.2). And the separation is an approximation rather than a theorem: solving the two layers independently is optimal only if the inner loop tracks the reference closely enough that the outer loop’s cost model remains accurate, a condition that fails exactly when liquidity is scarce.

Where the model breaks

It is tempting to conclude that the rate model is therefore the wrong formulation. It is not. It is the right formulation of the layer it addresses, and Fig. 8.2 is a statement about a trade-off that exists whatever the child orders look like. What is wrong is the interpretation in which (8.4) is the answer to the execution problem rather than the answer to the scheduling half of it. Chapter 15 takes this cascade as the organizing structure of an autonomous trading system, with the outer loop supervised and the inner loop certified.

8.7 Three extensions

8.7.1 Execution in the presence of a signal

Suppose you have a short-term forecast: the price drifts at rate μ over the execution window. The objective picks up a term $-\mu \int Q_t dt$ (holding inventory is now worth something), and the Euler-Lagrange equation becomes

$$\ddot{Q} = \gamma^2 Q - \frac{\mu}{2k}, \quad \text{with particular solution } Q \equiv c := \frac{\mu}{2\phi}. \quad (8.7)$$

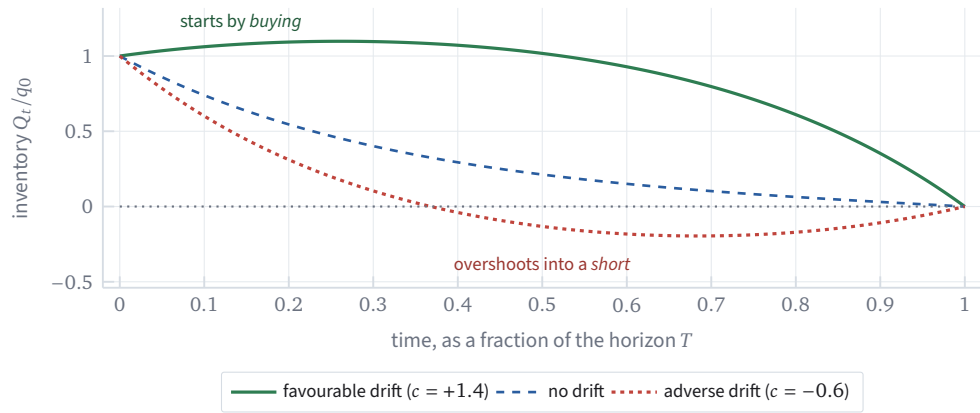


Figure 8.4. A forecast makes the schedule non-monotone. Selling q_0 by time T with a drift. If the price is expected to rise strongly, the optimal liquidation *begins by buying more*, because the running reward μQ outweighs the risk penalty; if the price is expected to fall, the trader sells past zero into a short position and buys back at the end. Both behaviours are correct given the model and both will get you fired if you have not explained them in advance.

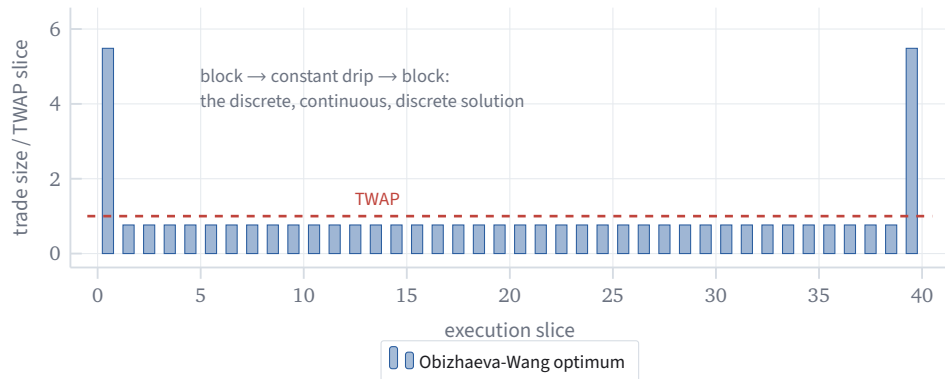


Figure 8.5. Transient impact changes the shape of the answer. Numerical solution of the discrete-time program of Obizhaeva and Wang (2013) with resilience $\rho T = 6$ and 40 slices. The optimum is a *block* at the start, a *constant* rate in the middle, and a second block at the end, not a smooth curve. The blocks exploit the fact that the book is full at the start and that impact incurred at the very end has no time to hurt you; the flat middle keeps the deviation D at a constant level, which is exactly the condition for marginal cost to be equalized across time.

The optimal path is a hyperbolic curve *shifted toward the desired holding* c . The consequences are visually striking.

8.7.2 Transient impact and the Obizhaeva-Wang model

Replace the memoryless temporary impact by a book that is depleted and refills exponentially, as in Section 7.1.3: let the price deviation D follow $dD_t = -\rho D_t dt + \eta \nu_t dt$ and execute at $S_t + D_t$. The cost functional is still quadratic (it is a quadratic form in the trading path with kernel $G(t-s) = \eta e^{-\rho(t-s)}$) so the problem is a convex quadratic program, but the solution changes character completely.

Remark 8.3. The block-drip-block structure is a general phenomenon: whenever the state has a resilient component, the optimal control has a singular part (impulses) plus an absolutely continuous part. In control language this is a *singular control* problem, and the same structure reappears in Section 11.5 as a no-trade region with reflecting boundaries.

8.7.3 The multi-asset problem

With d assets, inventory $Q_t \in \mathbb{R}^d$, temporary impact matrix $K \succ 0$, covariance $\Sigma \succ 0$, the objective (8.3) becomes

$$\min \int_0^T \left[\dot{Q}^\top K \dot{Q} + \phi Q^\top \Sigma Q \right] dt, \quad (8.8)$$

whose Euler-Lagrange equation is $K\ddot{Q} = \phi\Sigma Q$. Diagonalize the pencil (Σ, K) : in the coordinates of the generalized eigenvectors the problem decouples into d scalar Almgren-Chriss problems with urgencies $\gamma_i = \sqrt{\phi\theta_i}$, where θ_i are the generalized eigenvalues. *Liquidate the risky, cheap-to-trade directions first.* This is the entire content of multi-asset execution, and it is a simultaneous diagonalization.

8.8 Limitations of the model

Where the model breaks

The schedule is open-loop. Because the objective was deterministic, ν^* does not depend on the realized price. Real desks do respond to the path: they slow down when the price moves in their favour, speed up when it moves against. That behaviour is optimal only under a different objective (one with a genuinely path-dependent risk measure, or a signal as in Fig. 8.4), and dressing it up as “adaptive execution” without changing the objective is a way of taking unmodelled risk.

Quadratic costs are wrong. (6.7) says impact goes like $\sqrt{\phi}$, not ϕ . With a power cost $k\nu^{1+p}$ the shape of the optimal schedule does change, and, critically, the schedule now depends on order size. Numerically the solution is still easy; analytically it is not.

Volatility and liquidity are not constant. Figure 5.3(d) shows both have a strong intraday shape. The standard fix is to solve the model in *volume time*, which makes the participation rate rather than the clock rate the control.

You are not alone. If several agents liquidate correlated positions simultaneously, the correct object is an equilibrium, and the resulting schedules are more aggressive than the single-agent solution: everyone races to sell first. Chapter 16 sketches the mean-field version.

Exercises

Exercise 8.1 (Discrete Almgren-Chriss). Solve (8.3) in discrete time with n slices. Show that the optimality conditions are the tridiagonal system $-Q_{j-1} + (2 + \phi\Delta t^2/k)Q_j - Q_{j+1} = 0$, and that its solution is the sampled version of (8.4) with γ replaced by $\frac{2}{\Delta t} \operatorname{arcsinh}(\frac{\Delta t}{2}\sqrt{\phi/k})$.

Exercise 8.2 (The Riccati route). Derive (8.4) by dynamic programming: substitute $v(t, x, q, S) = x + qS + h(t)q^2$ into the HJB equation, obtain $\dot{h} = \phi - h^2/k$, solve with $h(T) = -\alpha$, and verify that $\nu^* = -h(t)q/k$ reproduces the theorem. Where in this derivation is σ used, and why does it not appear in the answer?

Exercise 8.3 (Frontier geometry). Show that the efficient frontier of Fig. 8.2 is convex and decreasing, and that its slope at the point corresponding to urgency ϕ equals $-1/(2\gamma_{\text{ra}})$ for the equivalent mean-variance risk aversion. Interpret the slope as a marginal rate of substitution between cost and certainty.

Exercise 8.4 (Participation constraint). Add the constraint $\nu_t \leq \bar{\nu}$ to (8.3). Show that the solution is the unconstrained one truncated: trade at $\bar{\nu}$ until the unconstrained path becomes feasible, then follow it. Identify the KKT multiplier and give its financial meaning.

Exercise 8.5 (Square-root costs). Replace $k\nu^2$ by $k\nu^{3/2}$ in (8.3). Write the Euler-Lagrange equation, show that it has no closed-form solution, and compute the optimal schedule numerically for $\gamma T \in \{1, 5\}$. In which direction does concavity of impact push the schedule relative to Fig. 8.1, and why?

Exercise 8.6 (Two assets). Take $d = 2$, $K = \text{diag}(k_1, k_2)$ and Σ with correlation ρ . Compute the generalized eigenvectors explicitly and describe the optimal liquidation of a long/short pair as $\rho \rightarrow 1$. Why does the hedged direction get liquidated last?

Market Making

A market maker is in the business of being on both sides at once: quoting a bid and an ask, earning the difference, and hoping not to accumulate a position. The business model is simple to state and hard to execute, because of a structural asymmetry you have already met. You choose *where* to quote; you do not choose *whether* you are filled. Someone else does, and they choose to trade with you precisely when it suits them.

Mathematically this is the most interesting problem in the book. The control is not a rate but a pair of *intensities*, chosen indirectly through the quoted distances; the state jumps; and the terminal objective is a risk measure applied to a random inventory, meaning a number summarizing how bad the distribution of that inventory's value is, in the sense [Chapter 12](#) makes precise. It is impulse-free intensity control of a jump process, and (remarkably) it has a closed-form solution.

In plain terms

A market maker quotes a price to buy and a price to sell, at the same time, all day, and hopes to do roughly equal amounts of each. If they do, they earn the difference between the two prices without ever taking a view on where the market is going. Two things stop this being free money. First, they cannot choose who trades with them, and the people who choose to trade tend to be right. Second, the two sides do not arrive in equal numbers, so the maker ends up holding a position they never wanted and must be paid for the risk of it. The whole chapter is about how to set the two prices so as to manage that unwanted position.

9.1 The economics of the business

Before the model, count the money. Everything the chapter does later is an attempt to improve on the arithmetic below.

The good case. A share is worth \$100.00, as best anyone can tell. You post a bid at \$99.99 and an ask at \$100.01, one cent either side. A seller in a hurry hits your bid: you now own 100 shares for which you paid \$9,999, and which you believe are worth \$10,000. You are up \$1 on paper. A few seconds later a buyer in a hurry lifts your ask: you sell the same 100 shares for \$10,001. You are flat again, and \$2 richer.

Two cents on \$100 is two basis points. Do that four hundred times a day and you have made \$800 on a business that never once took a view on whether the share was going up. That is market making, and stated that way it sounds like a machine for printing money.

The bad case. Now suppose the seller hits your bid and the second half never happens. Instead the price drifts to \$99.90. Your 100 shares are worth \$9,990 against

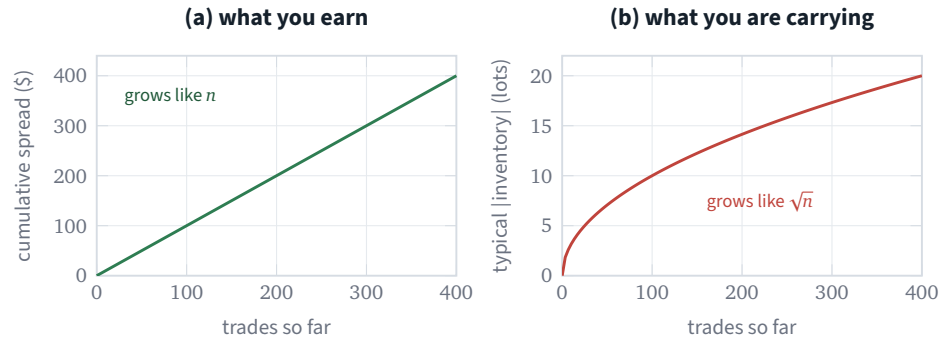


Figure 9.1. The two halves of the business. (a) Earning one cent per side, income accumulates in proportion to the number of trades. (b) Because the two sides arrive at random, the position you are left holding accumulates only as the square root of that number, but it is exposed to a price that keeps moving whether you trade or not. The comparison is not as favourable as (a) alone suggests: income is certain and small, whereas the loss from carrying inventory is uncertain and occasionally very large. Managing (b) without giving up too much of (a) is the entire content of this chapter.

the \$9,999 you paid, so you have lost \$9. That single ten-cent move has consumed the profit of four and a half round trips.

Why the bad case is not rare. If buys and sells arrive with equal probability, your inventory is a random walk, and after n trades it is typically $\sqrt{n}$ lots away from zero rather than at zero. Over four hundred trades that is around twenty lots of unwanted position, at all times, drifting. Figure 9.1 puts the two effects side by side: the spread income grows *linearly* in the number of trades, while the inventory you are carrying grows only like $\sqrt{n}$ but is exposed to a price that moves by $\sigma\sqrt{t}$. Neither term dominates for free, and where they balance is precisely what this chapter computes.

The two levers

There are exactly two things a market maker can do about panel (b), and both appear in the solution later.

Widen. Quote further from the mid. You are filled less often, so you accumulate inventory more slowly, but you also earn less. This trades away (a) to reduce (b).

Skew. Quote *both* sides lower when you are long, so that selling becomes more likely and buying less. This costs you nothing in spread when your position is zero, and it pushes inventory back toward zero when it is not. It is strictly the better instrument, and Theorem 9.1 says the optimal skew is linear in your position.

9.2 The model

Problem 5: optimal quoting

The mid price is a martingale, $dS_t = \sigma dW_t$. The maker continuously posts a bid at $S_t - \delta_t^b$ and an ask at $S_t + \delta_t^a$, choosing the half-spreads $\delta^b, \delta^a \geq 0$. Fills arrive as counting processes N^b, N^a with controlled intensities

$$\lambda^b(\delta^b) = A e^{-\kappa \delta^b}, \quad \lambda^a(\delta^a) = A e^{-\kappa \delta^a}, \quad (9.1)$$

so quoting closer to the mid gets filled more often. Inventory and cash follow

$$dQ_t = dN_t^b - dN_t^a, \quad dX_t = (S_t + \delta_t^a) dN_t^a - (S_t - \delta_t^b) dN_t^b.$$

The maker maximizes exponential utility of terminal wealth,

$$\sup_{\delta^b, \delta^a} \mathbb{E} \left[-\exp(-\gamma (X_T + Q_T S_T)) \right]. \quad (9.2)$$

Take the symbols in order. S_t is the mid price and σ its volatility, so the maker has no forecast at all and is not trying to have one. δ^b and δ^a are the two controls, the distances the bid and the ask are posted away from the mid, so a small δ means a tight quote. $A > 0$ is the rate at which a quote posted exactly at the mid would be filled and $\kappa > 0$ is how quickly that rate decays as the quote is pulled away, so A sets the scale of the flow and $1/\kappa$ the distance over which the maker loses it. N^b and N^a count the fills on each side, which is why inventory Q is their running difference. And $\gamma > 0$ is risk aversion: it is the only reason the maker cares about inventory at all, since without it the objective would be linear in wealth and the two sides would never be quoted asymmetrically.

Control-theoretic dictionary

(9.1) is the whole of market microstructure compressed into two parameters. A is the flow arriving at the touch (how busy the market is) and κ is the elasticity of that flow with respect to price, which is how competitive it is. Large κ means customers are price-sensitive and you cannot widen without losing them; small κ means you have market power. It is a demand curve, and (9.2) is a monopolist's pricing problem with an inventory constraint. The reason it is not usually presented that way is that the inventory dynamics dominate everything.

9.3 The reservation price and the two-part spread

The HJB equation for $v(t, x, q, s)$ is

$$\begin{aligned} \partial_t v + \frac{\sigma^2}{2} \partial_{ss} v + \sup_{\delta^b \geq 0} \lambda^b(\delta^b) [v(t, x - s + \delta^b, q + 1, s) - v] \\ + \sup_{\delta^a \geq 0} \lambda^a(\delta^a) [v(t, x + s + \delta^a, q - 1, s) - v] = 0. \end{aligned} \quad (9.3)$$

The exponential utility makes wealth separable: $v = -e^{-\gamma(x+qs)} u(t, q)$ reduces (9.3) to a system of ODEs indexed by q , and the substitution $u = \exp(-\frac{\gamma}{\kappa} \log \cdot)$ (a Hopf-Cole transform) linearizes it.

Theorem 9.1 (Avellaneda and Stoikov 2008). With $\tau = T - t$, define the **reservation price**

$$r(t, S, q) = S - q\gamma\sigma^2\tau. \quad (9.4)$$

Then, to leading order in γ , the optimal quotes are centred on r and the total optimal spread is

$$\delta^a + \delta^b = \underbrace{\gamma\sigma^2\tau}_{\text{risk}} + \underbrace{\frac{2}{\gamma} \log\left(1 + \frac{\gamma}{\kappa}\right)}_{\text{market power}}, \quad (9.5)$$

so that individually

$$\delta^b = \frac{1}{2}(\delta^a + \delta^b) + q\gamma\sigma^2\tau, \quad \delta^a = \frac{1}{2}(\delta^a + \delta^b) - q\gamma\sigma^2\tau. \quad (9.6)$$

What the answer says

- (1) **The reservation price is your private mid.** It is the price at which you are indifferent to holding q ; (9.4) says it sits below the market mid when you are long, by an amount proportional to inventory, risk aversion, variance, and time remaining. It is exactly the marginal value of inventory $\partial_q v$ promised in Chapter 1.
- (2) **You skew, you do not widen.** Inventory enters (9.6) antisymmetrically: the total spread is independent of q , but its *centre* moves. A long market maker does not quote wider, they quote *lower on both sides*, making sales more likely and purchases less.
- (3) **The spread has two independent parts.** One is compensation for risk and vanishes at the horizon; the other is a monopoly rent that depends only on the elasticity κ and survives even for a risk-neutral maker. Competition drives $\kappa \rightarrow \infty$ and the second term to zero.
- (4) **Skew can exceed the spread.** For large $|q|$, (9.6) makes one of the half-spreads negative, the maker quotes *through* the mid, paying to reduce inventory. That is not a bug: it is the model telling you to cross the spread and hedge.

9.4 A simulation study

Theorem 9.1 is a statement about a model. The question that matters is what inventory control buys you relative to the obvious alternative: quoting a fixed, symmetric spread and ignoring your position.

9.5 The running-penalty version with inventory bounds

Exponential utility is analytically convenient and awkward to calibrate. The now-standard alternative (Guéant, Lehalle, et al. 2013; Cartea et al. 2015) replaces (9.2) with a running penalty, exactly as in Chapter 8:

$$\sup_{\delta^b, \delta^a} \mathbb{E} \left[X_T + Q_T S_T - \alpha Q_T^2 - \phi \int_0^T Q_t^2 dt \right], \quad (9.7)$$

with the inventory constrained to $|Q_t| \leq \bar{q}$ (the maker stops quoting on one side at the bound). The value function is $v = x + qS + \theta_q(t)$ with $\theta \in \mathbb{R}^{2\bar{q}+1}$ solving a linear ODE system

$$\dot{\theta} = \Lambda \theta, \quad \theta(T)_q = -\alpha q^2, \quad (9.8)$$

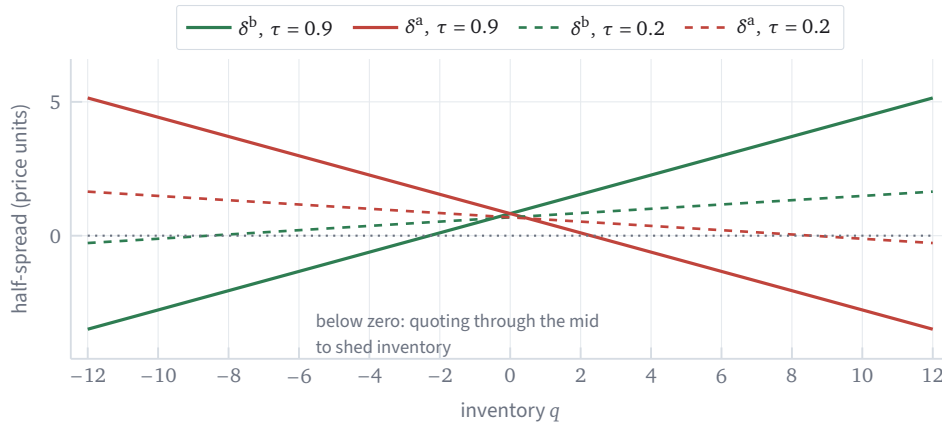


Figure 9.2. Quotes skew linearly in inventory. (9.6) with $\sigma = 2$, $\gamma = 0.1$, $\kappa = 1.5$. The two lines cross at $q = 0$, where the maker is symmetric. As inventory grows the bid is pushed away and the ask pulled in, at equal and opposite rates; the sum of the two is constant in q . Closer to the horizon (dashed) the skew flattens, because there is less time left for inventory to hurt, and at $\tau = 0$ the maker is indifferent and quotes symmetrically whatever their position.

whose solution is a matrix exponential. Everything remains explicit, the quotes remain linear-plus-correction in q , and the model is now specified by quantities a risk manager recognizes: a position limit and a penalty rate.

9.5.1 Quoting is a pricing rule and not yet an order

Of the four problems in Part II this one sits closest to the exchange, since a quoted distance almost determines a message. It is worth being clear about the word almost. (9.6) fixes two prices. A message additionally requires a size, and once a size is chosen the order joins a queue, at which point (4.9) rather than (9.1) governs whether it trades. In the language of §4.2.1, δ^b and δ^a are economic controls that a further layer must turn into market actions.

That layer carries decisions the model does not contain. How large should each quote be, given that a larger quote earns more and is filled adversely more often? When the mid moves by a tick, should the resting order be cancelled and reposted, losing its queue position, or left where it is at a now-suboptimal distance? When inventory reaches the bound of (9.7), should the maker stop quoting on one side or cross the spread to shed the position? On which venue should each side rest, given that the fee and the queue differ?

None of these is a refinement of Theorem 9.1. They are the order-management problem, they run on the same timescale as the message loop rather than the quoting loop, and in a large-tick name (§4.6) they dominate: the quoted distance is fixed at one tick and every remaining decision is one of the four above.

How it actually works

Real market makers layer several further mechanisms on top of the skew: *hedging* (offset inventory in a correlated instrument rather than waiting for a natural offset), *quote fading* (pull quotes entirely when the book looks toxic), *size laddering* (quote small at the touch and larger behind), and hard *kill switches*. All of these are ways of adding controls the model does not contain, and each one is a place where the elegant solution above stops being the answer.

9.6 Adverse selection

In (9.1) the fill process is independent of the price process. That is the single most consequential falsehood in the model. Real fills are informative: you are lifted on the ask precisely when the price is about to rise.

Where the model breaks

Adverse selection changes the problem qualitatively, not just quantitatively. Once fills carry information, the maker's filtration is generated by their own executions, and the correct formulation is a *partially observed* control problem: the state includes a belief about the drift, updated at each fill. The optimal policy is no longer a static function of (q, t) ; it depends on the recent history of one-sided flow. In practice this is implemented crudely (widen or fade after a run of same-side fills) and the crude version captures most of the benefit.

Why market making is not free money

Assemble the pieces. Gross revenue is the spread times the number of fills. Against it: adverse selection (Fig. 9.4), inventory risk (Fig. 9.3), exchange fees (Chapter 4), and the cost of the technology required to be at the front of the queue (Section 4.5). The business is a thin margin on enormous turnover, protected by a latency moat, and it is exquisitely sensitive to the parameters A and κ that no model tells you how to estimate. Chapter 13 does.

Exercises

Exercise 9.1 (The risk-neutral spread). Set $\gamma \rightarrow 0$ in (9.5) and show that the optimal spread tends to $2/\kappa$. Derive this directly by maximizing $\delta \lambda(\delta)$ with $\lambda(\delta) = Ae^{-\kappa\delta}$, and interpret the answer as a monopolist's markup over an exponential demand curve.

Exercise 9.2 (Hopf-Cole). Carry out the reduction of (9.3) in full: substitute $v = -e^{-\gamma(x+qs)}u(t, q)$, obtain the ODE system for u , and show that $u = \exp(-\gamma\theta_q(t))$ turns it linear when the fill intensities are exponential. Where exactly is exponential utility used, and where is the exponential form of λ used?

Exercise 9.3 (Inventory is an Ornstein-Uhlenbeck process). Show that under the optimal policy the inventory process has drift $-2A\kappa\gamma\sigma^2\tau q e^{-\kappa\delta}$ to first order in q , hence mean-reverts to zero, and compute the stationary variance in the long-horizon limit. Compare with the simulated standard deviation 2.9 in Fig. 9.3(b).

Exercise 9.4 (Skewing versus widening). Suppose the maker may only widen symmetrically, $\delta^b = \delta^a = \delta(q)$. Find the optimal $\delta(\cdot)$ and show that the achievable mean-variance frontier is strictly worse than Fig. 9.3(d). Explain in one sentence why skewing dominates widening.

Exercise 9.5 (A toxic counterparty). Modify the model so that with probability α each incoming order is informed and the mid jumps by $\pm J$ against the maker immediately after such a fill. Show that the optimal spread increases by αJ to leading order, and that the maker's expected profit per fill is $\bar{\delta} - \alpha J$. At what α does the business become unviable?

Exercise 9.6 (Two venues). The maker quotes the same instrument on two exchanges with different fees and different (A, κ) . Write the joint control problem and show that the inventory skew is common across venues while the base spread is venue-specific. What does this predict about observed quote correlations?

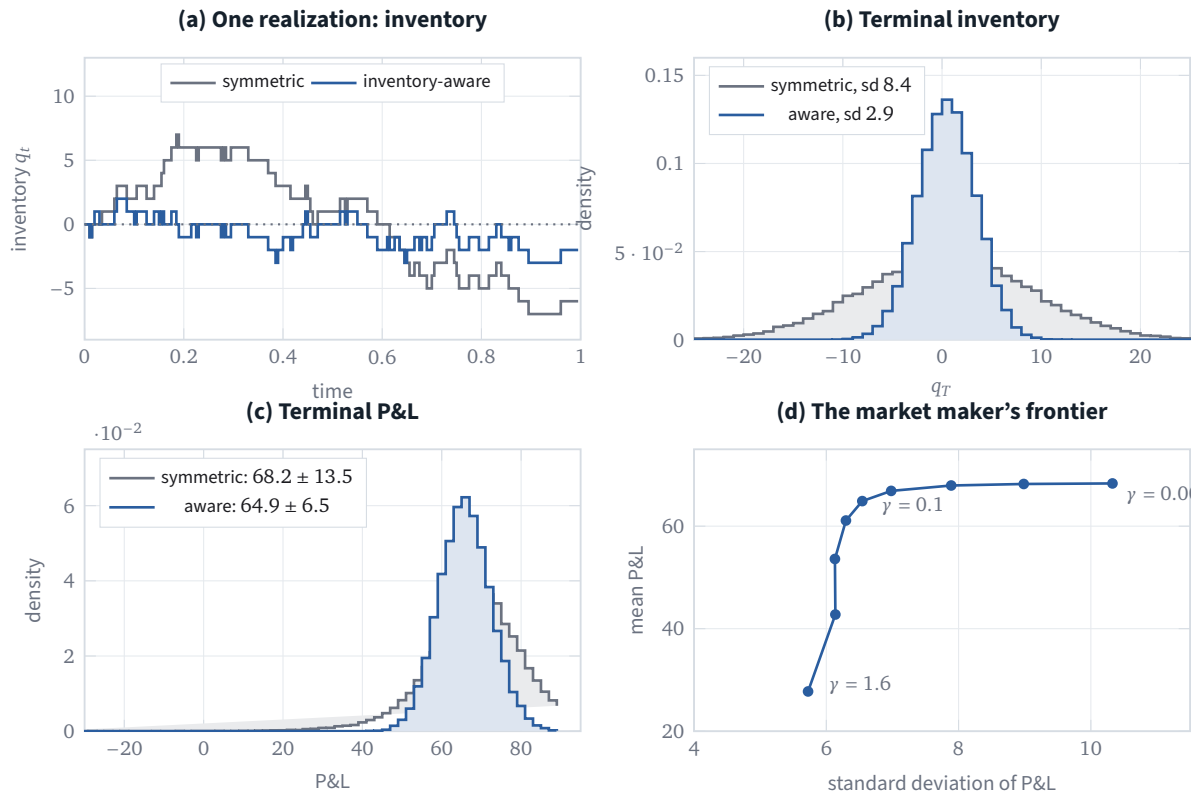


Figure 9.3. Inventory control is risk control and it is not free. Forty thousand simulated days with $A = 140$, $\kappa = 1.5$, $\sigma = 2$, $\gamma = 0.1$, both policies driven by identical random numbers. (a) The symmetric maker random-walks its way to a large position; the inventory-aware maker is pinned near zero by the skew. (b) Terminal inventory has standard deviation 8.4 against 2.9. (c) The cost of that control is 3.3 in expected P&L, about 5% of the total, in exchange for the reduction in inventory risk in panel (b). (d) Sweeping risk aversion traces a frontier with the same shape as Fig. 8.2: beyond $\gamma \approx 0.2$ the maker is paying a great deal of expected profit for very little further risk reduction.

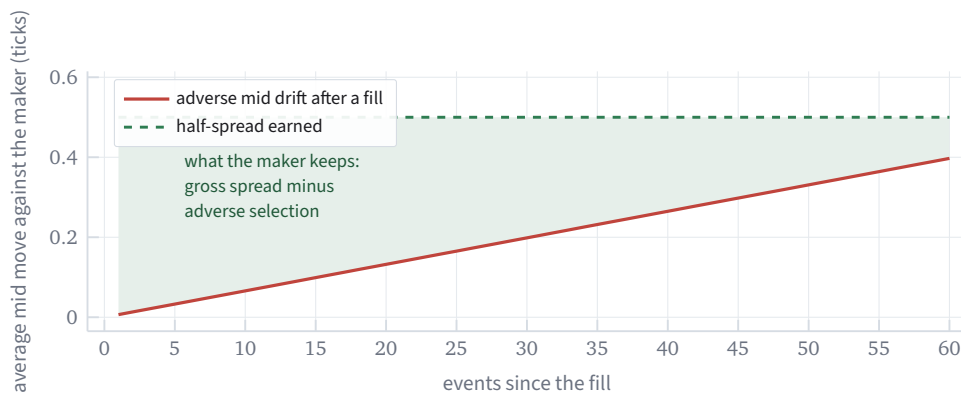


Figure 9.4. The fills you get are not a random sample. Average mid path following a fill, in a market where 22% of the incoming flow is informed. The maker collects half a tick at the moment of execution and then watches the mid move about 0.4 ticks against them over the following minute. Roughly 80% of the gross spread is given back. In the language of (6.3), this is c_{adv} , and it is why the quoted spread cannot be competed down to the level implied by inventory risk alone.

Hedging and Replication

Start with a story, because this chapter is otherwise easy to misread as being about option pricing, and it is not. It is about eliminating a risk you have been forced to take.

A bank sells a client the right, but not the obligation, to buy one share of some company at \$100 in three months. The client pays for that right today and walks away. The bank now has a problem. If the share ends up at \$130 the bank must hand over a share worth \$130 in exchange for \$100, losing \$30. If it ends up at \$70 the client does nothing and the bank keeps the fee. The bank has taken on a liability whose size depends on a price it cannot control and does not wish to bet on.

Hedging is what the bank does next: it buys and sells the underlying share, continuously, in a pattern chosen so that whatever it loses on the promise it makes back on the shares. If it can do this perfectly, then selling the option was never a bet at all. It was a service, with a definite cost, and the fee it charged should be exactly that cost.

That last sentence is the foundation of an entire industry, and the surprising thing is that it is very nearly true. This chapter shows why.

In plain terms

Suppose you have promised to pay someone an amount that depends on where a price ends up, and you would rather not gamble on that price. Hedging is the practice of buying and selling the underlying asset over time so that whatever you gain or lose on the promise is cancelled by what you lose or gain on the trading. The remarkable fact is that under idealized assumptions this cancellation can be made *exact*, which means the promise has a definite cost rather than being a bet. The rest of the chapter is about what happens as those assumptions are removed one at a time.

10.1 A one-period example

Before any calculus, here is the entire argument in a setting so small it can be checked by hand. It is worth doing slowly, because everything later in the chapter is this calculation repeated infinitely often.

A share costs \$100 today. In one period it will be worth either \$110 or \$90, and nothing else. You have sold a call option struck at \$100, so you must pay \$10 if the share goes up and nothing if it goes down. Interest rates are zero, to keep the arithmetic clean.

Step one: build a portfolio that matches the payoff. Hold Δ shares and B dollars of cash. In the up state that portfolio is worth $110\Delta + B$; in the down state, $90\Delta + B$. You want these to equal the option's payoff in each state:

$$110\Delta + B = 10 \quad (\text{up state}), \quad (10.1)$$

$$90\Delta + B = 0 \quad (\text{down state}). \quad (10.2)$$

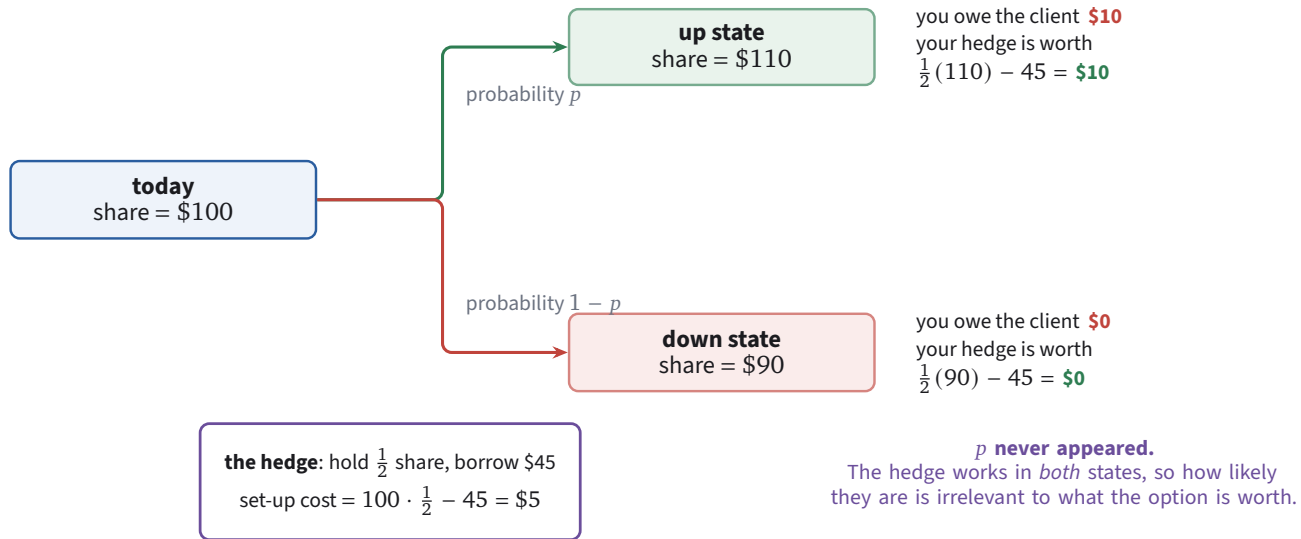


Figure 10.1. Replication in the smallest possible market. A share that can take two values, an option written on it, and a portfolio of shares and cash constructed to match the option in both states. Because the match is exact, holding the portfolio and owing the option is a position with no risk at all, so its cost today is the option's price. Note what is missing: nowhere did we use the probability p of the up state, nor anyone's opinion about it, nor any notion of risk aversion. That is the central and genuinely strange fact of derivative pricing.

Two linear equations, two unknowns. Subtracting gives $20\Delta = 10$, so

$$\Delta = \frac{10 - 0}{110 - 90} = \frac{1}{2}, \quad \text{and then} \quad B = -45. \quad (10.3)$$

So: hold half a share, and borrow \$45. Check it. If the share rises to \$110, half a share is worth \$55, you repay \$45, and you are left with exactly the \$10 you owe the client. If it falls to \$90, half a share is worth \$45, you repay \$45, and you have exactly the \$0 you owe. In *both* states the portfolio pays precisely what the option pays.

Step two: read off the price. That portfolio costs $100\Delta + B = 100 \cdot \frac{1}{2} - 45 = \5 to set up today. If the option traded for anything other than \$5, you could buy the cheaper of the two and sell the dearer and pocket the difference with no risk whatsoever, which is the arbitrage of Fig. 2.11. So the option is worth exactly

$$\Pi_0 = 100\Delta + B = \$5. \quad (10.4)$$

Step three: notice three things.

- (1) **The probability never entered.** Look back at (10.1) and (10.2). We never used p . Two traders who violently disagree about whether the share will rise must still agree that the option is worth \$5, because both can verify the hedge. Pricing by replication replaces forecasting with bookkeeping.
- (2) **Δ is a slope.** (10.3) is the change in the option's value divided by the change in the share's value. In the continuous version it becomes a derivative, $\partial\Pi/\partial S$, and it keeps the name *delta* given to it in (3.10). It is the number of shares to hold, and it is the feedback gain of the control problem.

- (3) **There is a shadow probability.** Ask which probability q would make the share a fair bet, in the sense that $q(110) + (1 - q)(90) = 100$. The answer is $q = \frac{1}{2}$. Now compute the expected option payoff under *that* number: $\frac{1}{2}(10) + \frac{1}{2}(0) = \5 , the price we just derived. This is not a coincidence, it is (5.9) in miniature, and q is the dual variable of Theorem 5.2.

Where the model breaks

The reason this works is that there were exactly as many states (two) as there were instruments to trade (the share and cash). Add a third possible price and the two equations can no longer match three payoffs: the system is overdetermined, no exact hedge exists, and the price is no longer pinned down but only confined to an interval. That is **incompleteness**, it is the normal situation rather than the exception, and §10.4 is about what to do instead. Continuous time is a clever way of dodging the problem: prices move continuously, so over a short enough interval there are effectively only two places the price can go, and the counting works out again.

10.2 Replication in a complete market

Now state the general problem, in the language the rest of the book uses. You owe $\Psi(S_T)$ at time T ; you may trade the underlying freely; and you want to arrive at T with at least that much wealth, starting from as little as possible:

$$V_T = X_T + Q_T S_T \geq \Psi(S_T) \quad \text{almost surely.} \quad (10.5)$$

This is a **stochastic target problem** (Soner and Touzi 2002): the state is (X, Q, S) , the control is the position Q , and the objective is not an integral to be minimized but a constraint to be met at the terminal time. The whole subject is the study of when (10.5) can be met *exactly*, what that costs, and what to do when it cannot.

The one-period hedge of §10.1 is exact but useless: real prices do not move once and stop. The fix is to redo the same argument over and over on ever shorter intervals, adjusting Δ each time. In the limit the bookkeeping becomes a partial differential equation, and the two equations (10.1) and (10.2) become the two matching conditions in (10.7). Nothing conceptual changes.

Problem 6: replication

Let $dS_t = \mu S_t dt + \sigma S_t dW_t$ and suppose you may trade S continuously and without cost. Find Q_t and the smallest initial wealth V_0 such that the self-financing dynamics $dV_t = Q_t dS_t$ terminate at $V_T = \Psi(S_T)$ almost surely.

Here S is the share price with drift μ and volatility σ , Q_t is the number of shares held at time t and is the control, V_t is the value of the hedging portfolio, and Ψ is the payoff of the claim you have sold, a function of the terminal price alone. The words *self-financing* carry the constraint that makes the problem non-trivial: $dV_t = Q_t dS_t$ says the portfolio changes value only because the price moved, never because money was added or removed. The words *almost surely* carry the rest: matching $\Psi(S_T)$ on average is easy and worth nothing, and the whole subject is that on this market it can be matched on every path.

Guess that wealth is a function of the current price and time, $V_t = \Pi(t, S_t)$. Itô gives

$$dV_t = \left(\partial_t \Pi + \frac{1}{2} \sigma^2 S_t^2 \partial_{SS} \Pi \right) dt + \partial_S \Pi dS_t. \quad (10.6)$$

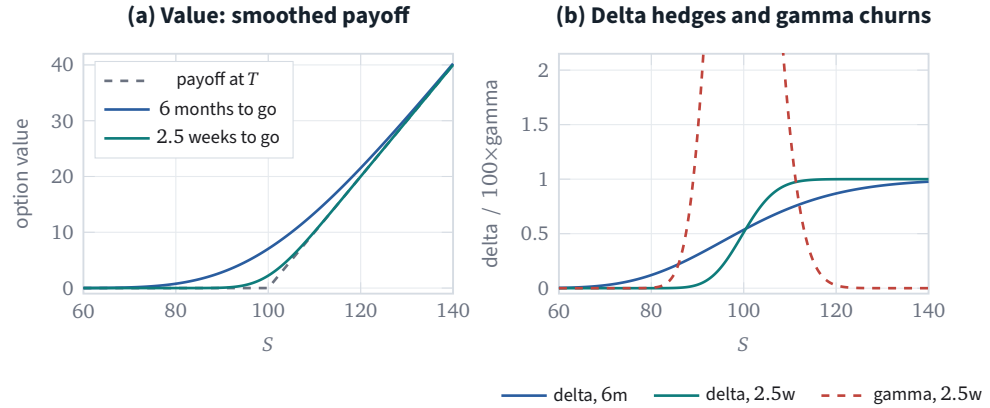


Figure 10.2. The hedge and its cost for a call struck at 100 with $\sigma = 25\%$. (a) The option value is a smoothed version of the payoff; the smoothing is worth exactly the remaining time value. (b) The delta is the number of shares to hold, and it steepens dramatically as expiry approaches. Its slope is gamma, which spikes at the strike near expiry, meaning the hedge must be revised violently in exactly the region where the price is most likely to be. Gamma is where all the practical difficulty of hedging lives.

Matching this against $dV_t = Q_t dS_t$ forces two things at once:

$$Q_t = \partial_S \Pi(t, S_t) \quad \text{and} \quad \partial_t \Pi + \frac{1}{2} \sigma^2 S^2 \partial_{SS} \Pi = 0. \quad (10.7)$$

The first is the **delta hedge**. The second is the Black-Scholes equation (Black and Scholes 1973; Merton 1973) (here with zero interest rate), and it is the *feasibility condition* for the target problem: the target (10.5) is attainable exactly when Π solves it with terminal data Ψ .

Two interpretations of (10.7)

As a *control* statement: the drift term must vanish, because a self-financing portfolio has no drift of its own, all of its motion comes from dS . The PDE is the condition that the uncontrollable part of the dynamics is zero.

As a *duality* statement: by Feynman-Kac, $\Pi(t, s) = \mathbb{E}^Q[\Psi(S_T) | S_t = s]$, the expectation under the measure of (5.9). The price of the liability is the value of the dual LP from (5.7). The hedge is the primal solution; the price is the dual value. They are the same theorem.

Control-theoretic dictionary

The **Greeks** are partial derivatives, and each is a control-theoretic object: *delta* $\partial_S \Pi$ is the feedback gain; *gamma* $\partial_{SS} \Pi$ is the rate at which that gain must be revised, hence the *turnover* the hedge generates; *vega* $\partial_\sigma \Pi$ is the sensitivity to a parameter you had to estimate, hence the model risk; *theta* $\partial_t \Pi$ is what the position earns for standing still. (10.7) says $\theta = -\frac{1}{2} \sigma^2 S^2 \Gamma$: *time decay is the rent you pay for convexity*, and it is the single most important identity in derivatives trading.

10.3 Discrete-time hedging error

Continuous rebalancing is a fiction. Rebalance n times and the hedge no longer replicates; there is a residual. Its structure is worth knowing exactly.

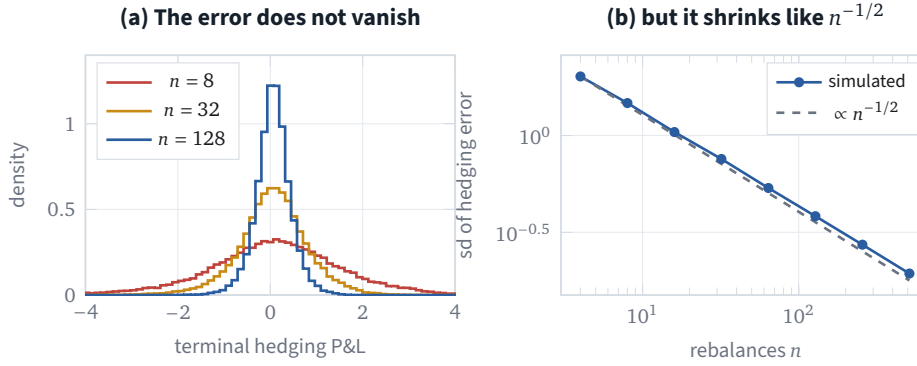


Figure 10.3. Replication is an asymptotic statement. Selling a three-month at-the-money call for its Black-Scholes price and delta-hedging it n times, in the very model that generated it, 40,000 paths. The P&L is centred at zero (the price was right) but its dispersion is substantial and falls only as $n^{-1/2}$. Halving the error costs four times the trading, which with transaction costs is a losing proposition: §10.5. Note the visible left skew: the short-option position loses in the tails, which is what selling convexity means.

Proposition 10.1 (Hedging error). *Let the hedge be rebalanced at $0 = t_0 < \dots < t_n = T$. The terminal error is*

$$\mathcal{E}_n = \frac{1}{2} \sum_{i=0}^{n-1} \Gamma(t_i, S_{t_i}) S_{t_i}^2 \left[\left(\frac{\Delta S_i}{S_{t_i}} \right)^2 - \sigma^2 \Delta t \right] + o(1), \quad (10.8)$$

so $\mathbb{E}[\mathcal{E}_n] = 0$ and, for equally spaced times, $\text{Var}[\mathcal{E}_n] = O(1/n)$: the standard deviation decays like $n^{-1/2}$.

Interpreting (10.8)

The bracket is the difference between *realized* and *implied* variance over one step. A delta-hedged option position is therefore not a bet on direction at all: it is a bet on variance, with a stake proportional to gamma. Hedging errors are large exactly when gamma is large, which is at the strike near expiry. Traders say “long gamma, short theta”; the equation says it precisely.

10.4 Three criteria for incomplete markets

Real liabilities are not spanned. Volatility is stochastic, prices jump, the underlying may not be traded at all. Then (10.5) cannot be met exactly, and there is no single “price”. There are three principled responses, and they correspond to three different optimization problems.

Quadratic hedging as least squares

The variance-optimal hedge is the $L^2(\mathbb{P})$ projection of the claim onto the space of stochastic integrals $\{\int_0^T Q_t dS_t\}$. The optimality condition is the orthogonality relation $\mathbb{E}[(\Psi - V_T) \int Q dS] = 0$ for all admissible Q , which is a normal equation. When S is a martingale, the solution is the integrand in the Kunita-Watanabe decomposition, and the residual is the unhedgeable part. Everything you know about linear least squares transfers, including the fact that the answer is only as good as the space you projected onto.

Criterion	Problem solved	Character of the answer
Superreplication	$\inf\{v : \exists Q, V_T \geq \Psi \text{ a.s.}\}$	the <i>upper</i> LP value of (5.7); always feasible, usually absurdly expensive (often the trivial buy-and-hold bound)
Quadratic hedging	$\inf_Q \mathbb{E}[(V_T - \Psi)^2]$	a projection in L^2 ; the residual is orthogonal to the traded space; gives a unique price and hedge, but penalizes gains as much as losses
Utility indifference	$v \text{ s.t. } \sup \mathbb{E}[U(V_T - \Psi)] = \sup \mathbb{E}[U(V_T)]$	a nonlinear pricing rule; interpolates between the two above as risk aversion goes from 0 to ∞ ; not additive across trades

Table 10.1. Three ways to price what you cannot replicate. All three reduce to the same number when the market is complete. Their disagreement is a measurement of incompleteness, and the width of the interval between the lower and upper superreplication bounds is the honest statement of what the model does not know.

10.5 Transaction costs and the no-trade band

Add a proportional cost ε per dollar traded. Now [Proposition 10.1](#) becomes a genuine trade-off. The *variance* of the replication error falls like $1/n$, while the total cost grows like $\varepsilon\sqrt{n}$, since each rebalance moves the delta by $O(n^{-1/2})$ and there are n of them. Minimizing $a/n + b\varepsilon\sqrt{n}$ gives $n^* \propto \varepsilon^{-2/3}$, the classical scaling of Leland ([1985](#)). Note that the exponent depends on measuring risk by variance rather than by standard deviation; the qualitative point, that the optimal rebalancing frequency falls as costs rise and does so slowly, is robust to that choice.

The structurally correct answer is not a rebalancing frequency but a *region*. The problem is a singular control problem: trade only when the position leaves a band around the frictionless delta, and then trade only enough to return to the boundary.

Theorem 10.2 (Whalley and Wilmott 1997). *For small proportional cost ε and exponential utility with risk aversion γ_{ra} , the optimal policy is to keep Q_t inside*

$$\left[\Delta(t, S_t) - \left(\frac{3}{2} \frac{\varepsilon S_t \Gamma^2}{\gamma_{ra}} \right)^{1/3}, \Delta(t, S_t) + \left(\frac{3}{2} \frac{\varepsilon S_t \Gamma^2}{\gamma_{ra}} \right)^{1/3} \right], \quad (10.9)$$

trading to the nearer boundary when it is breached.

Control-theoretic dictionary

Compare [Fig. 10.4](#) with [Fig. 8.5](#): block-drip-block in execution, band-with-reflection in hedging. Both are singular control problems, both have solutions consisting of a region where you do nothing and a boundary where you act, and in both the boundary scales as a power of the friction. If you have seen the Skorokhod problem, you have seen this: the optimal state process is a reflected diffusion.

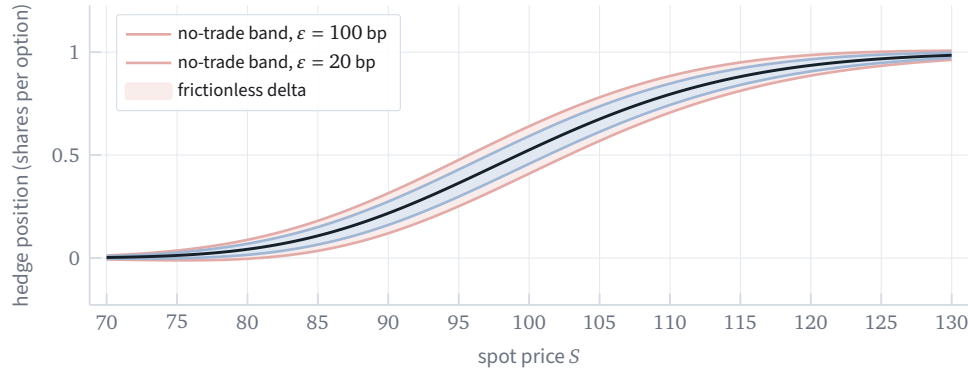


Figure 10.4. With costs the hedge becomes a region. (10.9) for a three-month at-the-money call. Inside the band you do nothing; on the boundary you trade just enough to stay inside. The band is widest where gamma is largest (precisely where the frictionless model wants you to trade most) so costs bite hardest exactly where hedging matters most. The $\varepsilon^{1/3}$ scaling means a five-fold increase in costs widens the band by only about 70%; the policy is remarkably insensitive to the cost estimate, which is a rare piece of good news.

10.6 The hedge target and the trade that reaches it

The gap identified in §4.2.1 is sharpest here, because the output of a hedging model is not an action at all. Delta hedging produces

$$Q_t^* = \Delta(t, S_t), \quad (10.10)$$

a desired *state*. If the current position is Q_t , what the model has actually told you is that $Q_t^* - Q_t$ shares must change hands. It has said nothing whatever about how.

Everything left over is an execution problem, and it is [Chapter 8](#) with a moving target. Should the adjustment be taken immediately, paying the spread, or spread over minutes at the risk that the target has moved by the time you arrive? Passively or aggressively? In the underlying, or in a correlated instrument that is cheaper to trade? What should happen on a partial fill, given that the target itself is drifting while the order rests? And what should happen when gamma is large, so that Q^* is moving fastest at exactly the moment the position is hardest to adjust?

The band is one answer to a larger question

[Section 10.5](#) has already taken the first step down this road. (10.9) says that when trading is costly the instruction “hold Δ ” should be replaced by “stay within a region around Δ ”, which converts a target state into a rule for *when* to act.

The hierarchy of §4.2.1 says the same logic continues one layer further. Even once the band has told you to trade, the order type, the price, the size and the sequencing are separate decisions, made against the book of [Chapter 4](#) and not against the diffusion of (10.10). A hedging system is therefore three nested loops: a pricing model that produces Δ , a policy that decides when the position must be adjusted, and an execution layer that adjusts it.

10.7 Deep hedging

Strip away every assumption and what remains is

$$\inf_{Q \in \mathcal{A}} \mathbb{E} \left(\Psi(S_T) - V_0 - \sum_i Q_{t_i} (S_{t_{i+1}} - S_{t_i}) + \sum_i c(Q_{t_i} - Q_{t_{i-1}}) \right), \quad (10.11)$$

where ϱ is any convex risk measure (Chapter 12) and c any convex cost. This is a well-posed convex stochastic optimization problem that requires no completeness, no continuity, no particular price model, only the ability to *simulate* S . Parameterizing Q_{t_i} as a neural network of the observable state and minimizing (10.11) by stochastic gradient descent is what Buehler et al. (2019) call **deep hedging**.

The right formulation regardless

(10.11) makes explicit what the classical theory hides: hedging is *policy optimization under a risk measure*, and the Black-Scholes answer is the special case in which the risk can be driven to zero, so the choice of ϱ does not matter. Once frictions or incompleteness are present, the choice of ϱ is the model. Arguing about which volatility to use is often less important than noticing that you have implicitly chosen $\varrho = \text{Var}$.

Where the model breaks

The honest weaknesses of (10.11): you must supply a simulator, so model risk has moved rather than vanished; the optimization is nonconvex in the network parameters even though the underlying problem is convex in the policy; and the resulting policy is hard to reason about at the tails, which is exactly where a hedge earns its keep. Treat it as a numerical method for a well-posed problem, not as a way of avoiding the modelling question.

Exercises

Exercise 10.1 (Derive the target condition). Carry out the matching argument leading to (10.7) carefully, keeping the drift μ . Show that μ cancels, and identify precisely which step does the cancelling. Then redo it with a second, non-traded risk factor and show that the cancellation fails: this is incompleteness.

Exercise 10.2 (Gamma is a variance bet). Verify (10.8) to leading order. Deduce that a delta-hedged long option position makes money if and only if realized variance exceeds implied, and that the P&L is $\frac{1}{2} \int \Gamma_t S_t^2 (\sigma_t^{\text{real}2} - \sigma^{\text{imp}2}) dt$. This is the fundamental theorem of volatility trading.

Exercise 10.3 (Leland scaling). Let the objective combine the variance of the replication error with the expected cost, $c(n) = a/n + b\epsilon\sqrt{n}$. Minimize over n and recover $n^* \propto \epsilon^{-2/3}$ and $c(n^*) \propto \epsilon^{2/3}$. Now deduce the band width: between rebalances the delta moves by $O(\Gamma\sigma S\sqrt{T/n^*})$, so show that this is $O(\epsilon^{1/3})$ and hence agrees with (10.9). Which exponent changes if risk is measured by standard deviation instead of variance?

Exercise 10.4 (Quadratic hedging as least squares). In a one-period model with m states, n assets and payoff c , show that the variance-optimal hedge is $\theta^* = (A^\top DA)^{-1} A^\top Dc$ for $D = \text{diag}(\mathbb{P})$, and that the residual $c - A\theta^*$ is D -orthogonal to the columns of A . Relate the residual norm to the width of the superreplication interval.

Exercise 10.5 (Superreplication is usually useless). Show that for a call option in a model where the volatility is only known to lie in $[\sigma_-, \sigma_+]$ with $\sigma_+ = \infty$, the superreplication price is the spot price S_0 : you must simply buy the stock. Explain in one sentence why the upper LP bound is so weak, and what that says about using worst-case criteria in practice.

Exercise 10.6 (A jump you cannot hedge). Add a single possible jump of size $-J$ with intensity λ to the price. Show that delta hedging no longer removes all risk, compute the residual variance, and find the position in a second option that restores completeness. This is why gap risk is hedged with options, not with the underlying.

Portfolio Selection and Dynamic Trading

The three previous chapters took the position as given and asked how to get into it, how to quote around it, or how to neutralize it. This chapter asks what the position should be.

It is also the chapter where the mathematics is easiest and the practice is hardest. The frictionless one-period problem is a quadratic program you can solve in closed form on a napkin. The reason portfolio management is nonetheless difficult is that the inputs to that quadratic program cannot be estimated well enough for its solution to mean anything: a fact we demonstrate rather than assert, in §11.3.

In plain terms

Given a pile of money and a list of things you could buy, how much of each should you hold? [Figure 2.6](#) already gave the essential insight: spreading across imperfectly correlated assets reduces risk without reducing expected return, so the answer is never to hold just one thing. Turning that insight into a number requires estimates of expected returns and of how assets move together, and this is where the subject becomes genuinely hard, because those estimates are poor and the procedure that uses them is exquisitely sensitive to exactly the errors they contain.

11.1 A two-asset example

The general problem is a quadratic program and you can solve those in your sleep, so the risk is that the machinery obscures what it is doing. Here is the whole idea in two assets and no matrices.

Asset A returns 6% a year in excess of cash, with a volatility of 16%. Asset B returns 3%, with a volatility of 8%. Their correlation is 0.2. Note the deliberate feature of these numbers: both assets have exactly the same Sharpe ratio (2.2), return divided by volatility,

$$\frac{0.06}{0.16} = \frac{0.03}{0.08} = 0.375. \quad (11.1)$$

On a per-unit-of-risk basis they are equally good, and a naive comparison suggests it cannot matter which you hold. It matters enormously.

Put a fraction w in A and $1 - w$ in B. The return is the weighted average,

$$\mu(w) = 0.06w + 0.03(1 - w) = 0.03 + 0.03w, \quad (11.2)$$

but the variance is *not* the weighted average, because the two assets do not move together:

$$\sigma^2(w) = w^2(0.16)^2 + (1 - w)^2(0.08)^2 + 2w(1 - w)(0.2)(0.16)(0.08). \quad (11.3)$$

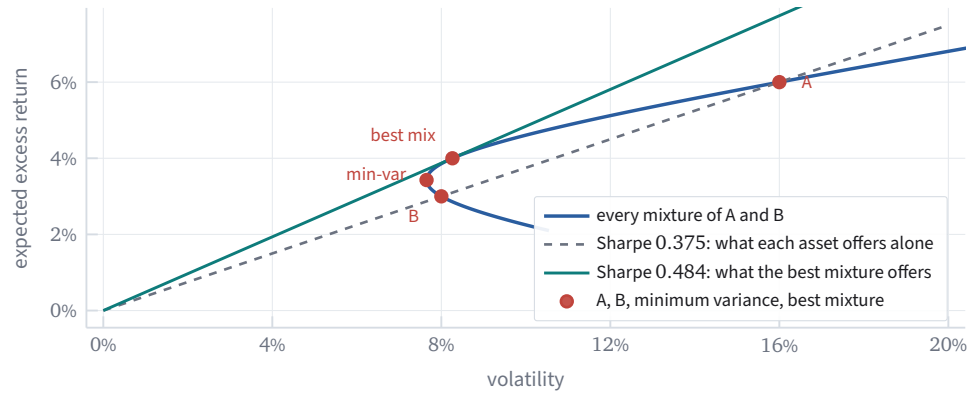


Figure 11.1. Two equally good assets and a mixture better than both. A and B lie on the same dashed line, so they offer identical reward per unit of risk. Every mixture of them lies on the curve, which bulges up and to the left of the straight line joining them, because (11.3) is not linear in w . The best mixture reaches a Sharpe ratio of 0.484, nearly 30% better than either ingredient, and the minimum-variance mixture is less volatile than the safer of the two assets. Nothing was forecast and nothing was timed; the gain comes purely from the fact that the two do not move together.

That cross term is the entire subject. Because $\rho < 1$ it is smaller than it would be if the assets moved in lockstep, so combining them destroys risk that neither of them can shed alone.

Minimize (11.3) and you get $w = 0.143$, with a volatility of 7.65%: *lower than either asset*, including the 8% one. Maximize the ratio $\mu(w)/\sigma(w)$ instead and you get

$$w^* = \frac{1}{3} \text{ in A, } 1 - w^* = \frac{2}{3} \text{ in B, } \mu = 4.0\%, \quad \sigma = 8.26\%, \quad \text{Sharpe} = 0.484. \quad (11.4)$$

In plain terms

Consider Fig. 11.1 once more, because it is the reason the rest of the chapter exists. Two investments, equally attractive on their own, combine into something strictly better than either. The improvement is free in the sense that it required no skill, no prediction and no extra risk. It is also the *only* free thing in the subject, and everything that follows is the question of how to grab as much of it as possible when there are a thousand assets instead of two and when the numbers going into (11.3) are estimates rather than facts.

11.2 The one-period quadratic program

Problem 7: mean-variance selection

The problem of Markowitz (1952). Given expected excess returns $\mu \in \mathbb{R}^d$ and covariance $\Sigma \succ 0$, choose weights $w \in \mathbb{R}^d$ solving

$$\min_w \frac{1}{2} w^\top \Sigma w \quad \text{subject to} \quad w^\top \mu = m, \quad w^\top \mathbf{1} = 1. \quad (11.5)$$

Here d is the number of assets, μ_i the expected return of asset i in excess of the risk-free rate, Σ the covariance matrix of those returns, and w_i the fraction of capital placed in asset i , so $w^\top \mu$ is the portfolio's expected excess return

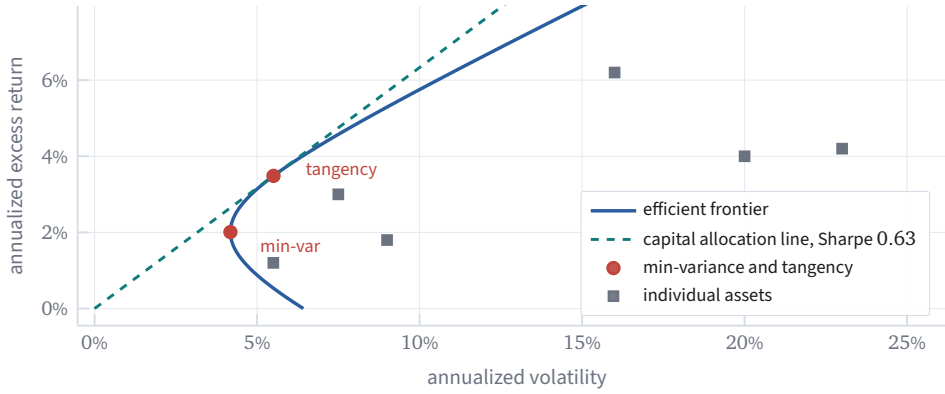


Figure 11.2. The picture everyone knows. Drawn from the six-asset universe used throughout this chapter. Every individual asset (grey squares) lies strictly inside the frontier: diversification is the closest thing to a free lunch that finance has, and (11.7) is the statement of exactly how free. With a risk-free asset the efficient set collapses to the dashed line, so that every investor holds the same risky portfolio and differs only in how much of it: the two-fund separation of (11.6).

and $w^\top \Sigma w$ its variance. The constant m is the return the investor is asking for; sweeping it and recording the resulting variance is exactly how the frontier of Fig. 11.2 is drawn. The constraint $w^\top \mathbf{1} = 1$ says the capital is fully invested, and the absence of any sign constraint on w says short positions are permitted.

The KKT conditions give $\Sigma w = \lambda_1 \mu + \lambda_2 \mathbf{1}$, so

$$w^* = \lambda_1 \Sigma^{-1} \mu + \lambda_2 \Sigma^{-1} \mathbf{1}, \quad (11.6)$$

with the multipliers fixed by the two constraints. (11.6) is the **two-fund theorem**: every efficient portfolio is a combination of two fixed portfolios, $\Sigma^{-1} \mu$ and $\Sigma^{-1} \mathbf{1}$, regardless of the target m . Writing $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, $B = \mathbf{1}^\top \Sigma^{-1} \mu$, $C = \mu^\top \Sigma^{-1} \mu$ and $D = AC - B^2 > 0$, the frontier is the hyperbola

$$\sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}. \quad (11.7)$$

If a risk-free asset is available, the picture simplifies further: the efficient set becomes a straight line from the risk-free rate through the **tangency portfolio** $w_{\text{tan}} \propto \Sigma^{-1} \mu$, and the slope of that line is the maximum attainable Sharpe ratio $\sqrt{\mu^\top \Sigma^{-1} \mu}$.

Control-theoretic dictionary

(11.5) is a QP; adding realistic constraints keeps it a convex program. Long-only, $w \geq 0$: still a QP. Turnover budget, $\|w - w_0\|_1 \leq \tau$: still a QP after introducing slack variables. Factor exposure limits, $|\beta^\top w| \leq b$: linear. Tracking error, $(w - w_b)^\top \Sigma (w - w_b) \leq \eta^2$: a second-order cone. Only two common requirements break convexity: a cap on the *number* of holdings (cardinality), and minimum position sizes, both integer constraints, both handled in practice by heuristics or by branch and bound on a small neighbourhood.

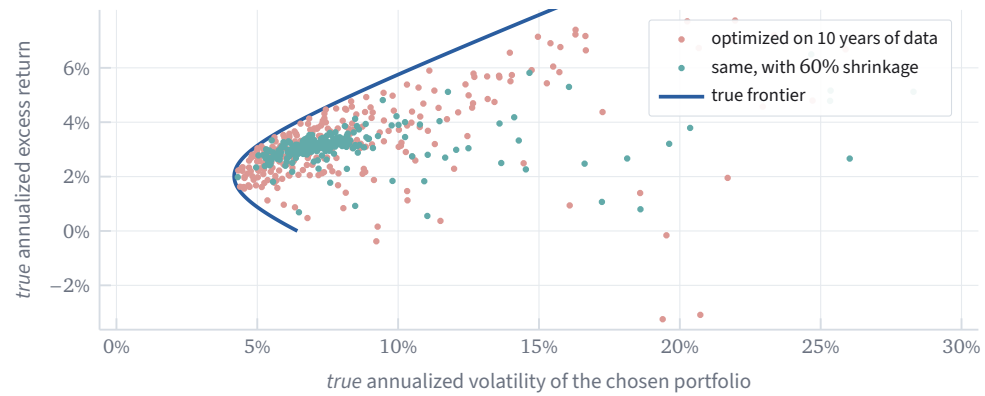


Figure 11.3. What you get is not what you optimized. Three hundred independent samples of ten years of monthly returns from the known model of Fig. 11.2. For each, the mean-variance portfolio is computed from the *estimated* moments and then evaluated under the *true* ones. The cloud sits far below and to the right of the frontier it was aiming at: median realized Sharpe ratio 0.42 against an attainable 0.63, and a long right tail (twenty-seven of the plain portfolios lie off-scale, with true volatilities above 30%). Shrinking both moments toward simple targets moves the cloud back and left: a biased estimator producing a better decision, which is the whole argument for regularization in one picture.

11.3 Estimation error in the inputs

(11.6) contains $\Sigma^{-1}\mu$. Both factors are estimated, the inverse amplifies errors in the small eigenvalues of Σ , and μ is essentially unestimable: the standard error of a sample mean return over T years is $\sigma/\sqrt{T}$, so distinguishing a 5% premium from zero at 16% volatility takes decades.

An error maximizer

The optimizer does not know which inputs are reliable. It systematically overweights assets whose returns were overestimated and whose variances were underestimated: exactly the assets where the estimation error happened to be favourable. The optimization step converts estimation error into *position*, which is the worst possible place to put it.

How it actually works

The practical toolkit against estimation error, roughly in order of how much it helps per unit of complexity: *shrinkage* of the covariance toward a structured target (Ledoit and Wolf 2004) and of returns toward a common mean; *factor models*, which impose $\Sigma = \beta\Omega\beta^T + D$ and so estimate $O(dk)$ rather than $O(d^2)$ parameters; *position limits*, which cap the damage any single estimate can do; *resampling* (Michaud), averaging the optimizer over bootstrap draws; and *robust optimization*, §11.8. In practice the combination of a factor covariance, shrunk returns, and hard position limits accounts for most of the achievable improvement.

11.4 The Merton problem in continuous time

The continuous-time formulation is due to Merton (1969) and Merton (1971).

Problem 8: Merton's portfolio problem

Wealth follows (3.12): with one risky asset,

$$dV_t = V_t [r + w_t(\mu - r)] dt + V_t w_t \sigma dW_t,$$

and the agent maximizes $\mathbb{E}[U(V_T)]$ over adapted w , with $U(v) = v^{1-\gamma}/(1-\gamma)$. Here V_t is total wealth, r the rate earned on cash, and μ and σ the drift and volatility of the single risky asset, so $\mu - r$ is the compensation for holding it. The control w_t is the *fraction* of wealth in that asset rather than a number of shares, which is what makes the problem scale-free. The parameter $\gamma > 0$ is the coefficient of relative risk aversion, and U is the power utility it generates; $\gamma \rightarrow 1$ recovers the logarithm.

The HJB equation is

$$\partial_t v + \sup_w \left\{ V[r + w(\mu - r)] \partial_V v + \frac{1}{2} V^2 w^2 \sigma^2 \partial_{VV} v \right\} = 0. \quad (11.8)$$

The inner maximization is a scalar quadratic, giving $w^* = -\frac{(\mu-r)\partial_V v}{V\sigma^2\partial_{VV}v}$; the ansatz $v(t, V) = f(t)V^{1-\gamma}/(1-\gamma)$ makes the wealth dependence cancel exactly, leaving

$$w^* = \frac{\mu - r}{\gamma \sigma^2} \quad (\text{constant in time and in wealth}). \quad (11.9)$$

Interpreting (11.9)

- (1) **Constant proportion.** The optimal policy is to rebalance continuously to a fixed *fraction* of wealth. Getting richer does not make you take more risk in percentage terms: a property of CRRA utility, and false for essentially every other utility.
- (2) **Myopia.** With constant investment opportunities the horizon T does not appear. “Stocks for the long run” has no basis in this model; it requires either mean reversion in returns or a non-CRRA objective.
- (3) **It is the tangency portfolio.** In d dimensions $w^* = \frac{1}{\gamma} \Sigma^{-1}(\mu - r\mathbf{1})$, which is (11.6) again. The static and dynamic problems agree, which is precisely why frictions are where the interesting dynamics come from.

11.5 The no-trade region

Add proportional costs and (11.9) becomes unimplementable: continuous rebalancing of a diffusion has infinite variation, and therefore infinite cost. The optimal policy changes character completely, exactly as in Section 10.5.

Theorem 11.1 (Davis and Norman 1990). *With proportional costs, there is an interval $[\underline{w}, \bar{w}]$ containing the Merton fraction w^* such that the optimal policy is: do nothing while $w_t \in [\underline{w}, \bar{w}]$, and trade the minimal amount to remain in the interval when the boundary is reached. The boundaries satisfy $\bar{w} - \underline{w} = O(\varepsilon^{1/3})$ as the cost $\varepsilon \rightarrow 0$.*

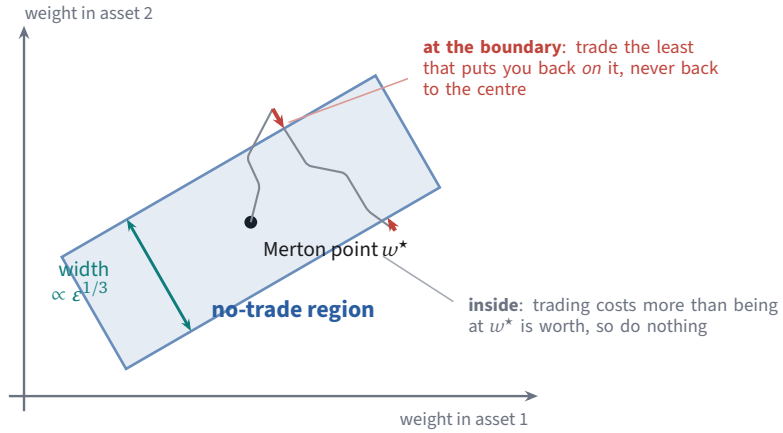


Figure 11.4. With costs the optimum is a region rather than a point. The frictionless Merton portfolio sits at the centre, and around it lies a region within which rebalancing is not worth what it costs, so the portfolio is left to drift (grey). When it reaches an edge, the optimal action is the smallest trade that returns it *to that edge* (red), not a trade back to the centre: the position is reflected rather than reset. The region is narrow across directions that are expensive to trade or carry much risk, long along the cheap and safe ones, and its width grows only as the cube root of the cost.

11.6 The aim portfolio

The no-trade region is exact but hard to compute in high dimensions. A tractable and enormously influential alternative replaces proportional costs with quadratic ones, which makes the whole problem LQG.

Problem 9: dynamic trading with quadratic costs

Let a signal f_t follow $df_t = -\Phi f_t dt + d\varepsilon_t$ and let expected returns be $\mathbb{E}_t[r_{t+1}] = Bf_t$. Choose positions x_t to maximize

$$\mathbb{E} \sum_t \rho^t \left[x_t^\top B f_t - \frac{\gamma}{2} x_t^\top \Sigma x_t - \frac{1}{2} \Delta x_t^\top \Lambda \Delta x_t \right].$$

The problem has a signal, a position and a cost of moving between them. f_t is a vector of predictors and Φ sets how fast each decays, so a large Φ is a signal worth acting on immediately and a small one a signal that will still be there tomorrow. B maps the signal to expected returns, so Bf_t is the forecast. x_t is the position held over period t and $\Delta x_t = x_t - x_{t-1}$ the trade required to reach it. The three parameters in the objective are the trade-off: γ prices risk through the return covariance Σ , $\Lambda \succ 0$ prices trading, and $\rho \in (0, 1)$ discounts the future. Setting $\Lambda = 0$ collapses the problem back to (11.5) solved afresh every period.

Theorem 11.2 (Gârleanu and Pedersen 2013). *The optimal policy is a partial adjustment toward an **aim portfolio**:*

$$x_t = x_{t-1} + a(\text{aim}_t - x_{t-1}), \quad \text{aim}_t = \sum_{\tau \geq 0} z(1-z)^\tau \mathbb{E}_t[\text{Markowitz}_{t+\tau}], \quad (11.10)$$

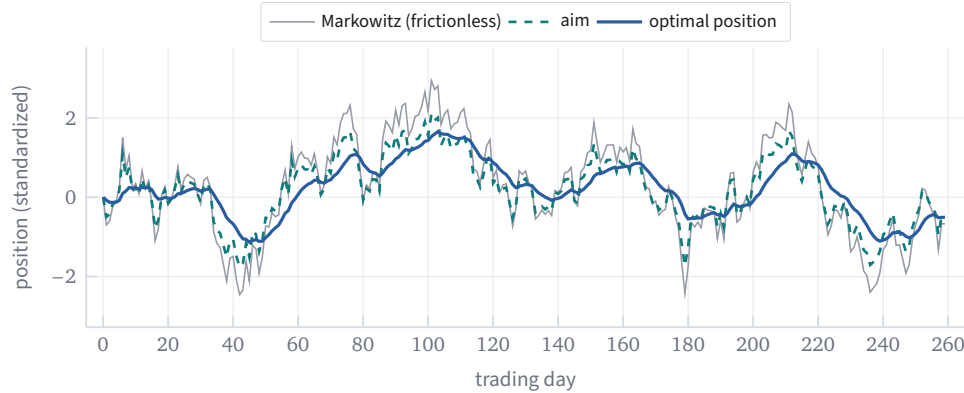


Figure 11.5. Trade toward the aim rather than the optimum. A mean-reverting signal with daily persistence 0.90, trading rate $\alpha = 0.15$, aim shrinkage $z = 0.71$. Two distinct effects are visible and they are often confused. The *aim* is a shrunk Markowitz portfolio, because a signal that will decay is not worth chasing fully. The *position* then lags the *aim*, because moving costs money. Shrink first, then smooth: the frictionless target is never the thing you trade toward.

where the trading rate $\alpha \in (0, 1)$ and the weight z depend on γ, Λ, ρ but not on the signal. The *aim* is thus a persistence-weighted average of current and future Markowitz portfolios, and the actual position tracks it with a first-order lag.

Control-theoretic dictionary

(11.10) is a Kalman-filter-shaped answer to a control problem, and the resemblance is not accidental: both are the solution of a quadratic problem with linear dynamics, so both take the form “new estimate/position = old + gain $\times$ innovation.” The gain α plays the role of the Kalman gain, and like it, it is determined by a Riccati equation balancing two quadratic penalties.

11.7 From target weights to implementation

Of the four canonical problems this one is furthest from the exchange. A solution of (11.5) is a vector $w^* \in \mathbb{R}^d$. It describes where capital should end up and contains no information at all about the path by which it gets there, and with d in the hundreds the path is most of the work.

Section 11.6 has already taken the first step. The aim portfolio separates the frictionless optimum from the thing you should actually trade toward, which is precisely a statement that the model’s output is a target rather than an instruction. But (11.10) still delivers a desired position x_t , and a desired position is not a market action either. Between x_{t-1} and x_t lies a multi-asset rebalance with its own decisions: which names to trade first, given that liquidity differs across them by orders of magnitude; how to hold the portfolio near its risk target while it is half done, since a partially executed rebalance is a portfolio nobody chose; how to manage the cash and margin that the sequence consumes; and how to avoid trading both sides of a correlated pair at the wrong moment.

Control-theoretic dictionary

The natural formulation is the multi-asset execution problem of §8.7.3, with the frictionless optimum entering as the terminal target rather than as the answer. This is the same cascade as everywhere else in this chapter: an outer loop that decides the exposure and an inner loop that realizes it. The reason the separation is usually harmless is that portfolio decisions change on a horizon of days while execution runs over hours, so the target is nearly static from the point of view of the inner loop. The reason it is sometimes not harmless is that this ordering of timescales fails in exactly the states of Fig. 7.3.

11.8 Robust portfolio choice

If μ is unknown but believed to lie in an ellipsoid $\mathcal{U} = \{\mu : (\mu - \hat{\mu})^\top \Omega^{-1} (\mu - \hat{\mu}) \leq \chi^2\}$, the robust problem

$$\max_w \min_{\mu \in \mathcal{U}} \left\{ w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w \right\} \quad (11.11)$$

has inner minimum $w^\top \hat{\mu} - \chi \|\Omega^{1/2} w\|_2$, so the whole problem becomes

$$\max_w w^\top \hat{\mu} - \chi \|\Omega^{1/2} w\|_2 - \frac{\gamma}{2} w^\top \Sigma w, \quad (11.12)$$

a second-order cone program. The penalty $\chi \|\Omega^{1/2} w\|_2$ is a *group-lasso*-like term: it shrinks positions toward zero and, unlike a quadratic penalty, sets some of them exactly to zero. Robustness against parameter uncertainty and sparsity of the resulting portfolio are the same phenomenon.

Where the model breaks

Robust optimization protects against the uncertainty set you specified. Its usual failure mode is a set that is too small in the directions that matter, typically because Ω was itself estimated from the same data as $\hat{\mu}$. Worst-case reasoning is not automatically conservative; it is only as conservative as the geometry you fed it.

Exercises

Exercise 11.1 (The frontier is a hyperbola). Derive (11.7) from the KKT conditions of (11.5). Show that the minimum-variance portfolio has $m = B/A$ and $\sigma^2 = 1/A$, and that the asymptotes have slopes $\pm \sqrt{D/A}$.

Exercise 11.2 (Two funds, in the dual). State the Lagrangian dual of (11.5) and show that the two-fund theorem is the statement that the primal solution is an affine function of the two multipliers. What are the multipliers' financial interpretations?

Exercise 11.3 (Error maximization, quantified). Let $\hat{\mu} = \mu + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \frac{\sigma^2}{T} I)$ and Σ known. Show that the expected *realized* Sharpe ratio of $w \propto \Sigma^{-1} \hat{\mu}$ is smaller than the attainable one by a factor that degrades with d/T . Reproduce Fig. 11.3 numerically and check the factor.

Exercise 11.4 (Merton with a horizon effect). Redo Section 11.4 with a mean-reverting risk premium, $d\mu_t = -\theta(\mu_t - \bar{\mu}) dt + \zeta dB_t$ correlated with the price. Show that the optimal weight acquires an **intertemporal hedging** term proportional to the correlation, and that it vanishes for log utility.

Exercise 11.5 (Aim and shrinkage are different). In (11.10), show that as $\Lambda \rightarrow 0$ the trading rate $a \rightarrow 1$ and the position equals the aim, but that the aim is *still* a shrunken Markowitz portfolio

whenever the signal decays. Explain why the two effects in Fig. 11.5 have different causes and must be calibrated separately.

Exercise 11.6 (Robustness is a norm penalty). Verify that $\min_{\mu \in \mathcal{U}} w^\top \mu = w^\top \hat{\mu} - \chi \|\Omega^{1/2} w\|_2$ and deduce the SOCP form. Then show that for $\Omega = I$ the robust solution has a sparse support, and relate the threshold to χ .

PART III

Estimation and Practice

What stands between a solved model and a deployed strategy: risk measures and the constraints that actually bind, the statistics of estimating microstructure quantities from noisy data, the technology regime in which speed rather than prediction decides the outcome, what changes when the controller stops being a function and has to be layered above a certified one, and an honest account of what remains unsolved.

Risk Measures and Binding Constraints

Every objective in Part II contained a risk term, and in each case the choice was made for tractability: a quadratic penalty because it kept the problem LQ, an exponential utility because it linearized the HJB. This chapter asks what the choice *should* be, and the answer turns out to be a small piece of convex analysis with an unusually direct payoff: one standard risk measure turns portfolio optimization into a linear program.

In plain terms

“How risky is this position?” is a question with many answers and this chapter asks which ones are worth using. The standard measure in industry reports the loss that will be exceeded on one day in twenty, and says nothing whatever about how bad that day might be. A better measure reports the average loss on those bad days instead, and it turns out to have an unexpected property: optimizing it is a linear program. The chapter also catalogues the constraints that actually stop trading desks in practice, which are almost never the objective function and almost always the collateral.

12.1 A motivating example

Here is the problem in one example, before any axioms.

Portfolio A loses nothing on ninety-five days out of a hundred, and on the other five loses exactly \$1 million. Portfolio B also loses nothing on ninety-five days out of a hundred, and on the other five loses \$40 million.

By the measure that the entire banking industry reported for two decades, these two portfolios are *identical*. Value-at-risk at the 95% level asks only “what is the loss that gets exceeded one day in twenty?”, and for both portfolios the answer is zero, because the bad days sit outside the window being looked at. The measure is a quantile, and a quantile is blind to everything beyond it.

The repair is obvious once stated: instead of asking where the tail *begins*, average over what is *inside* it. That is conditional value-at-risk, and it reports \$1 million for A and \$40 million for B, which is the answer any risk manager would give. [Figure 12.1](#) draws both on a real distribution.

What makes this more than a definitional quibble is that the choice has three consequences that run through the rest of the chapter:

- **It changes behaviour.** A desk measured on value-at-risk can improve its reported number by moving losses further into the tail, which is the opposite of reducing risk. A desk measured on the average of the tail cannot.
- **It changes what merging books does.** Combining two portfolios can *increase* measured value-at-risk, which is absurd, because diversification cannot create risk. [Definition 12.1](#) makes this precise.

- **It changes the optimization.** Minimizing a quantile is a nonconvex problem with many local minima. Minimizing the tail average, it turns out, is a *linear program*. That is [Theorem 12.3](#), and it is the reason this chapter has a happy ending.

12.2 Axioms for a risk measure

Let X be a random terminal wealth (so losses are $-X$). A **risk measure** ϱ assigns to X a number interpreted as the capital that must be held against it.

Definition 12.1 (Coherence; Artzner et al. 1999). ϱ is **coherent** if for all X, Y and $\beta \geq 0$, $c \in \mathbb{R}$:

$$\text{monotonicity} \quad X \leq Y \implies \varrho(X) \geq \varrho(Y) \quad (12.1)$$

$$\text{translation invariance} \quad \varrho(X + c) = \varrho(X) - c \quad (12.2)$$

$$\text{positive homogeneity} \quad \varrho(\beta X) = \beta \varrho(X) \quad (12.3)$$

$$\text{subadditivity} \quad \varrho(X + Y) \leq \varrho(X) + \varrho(Y) \quad (12.4)$$

Dropping homogeneity and weakening subadditivity to convexity, $\varrho(\theta X + (1 - \theta)Y) \leq \theta \varrho(X) + (1 - \theta)\varrho(Y)$, gives a **convex** risk measure.

The axioms look abstract and each one is a concrete operational requirement. It is worth taking them as things a risk manager would insist on rather than as things a mathematician would like.

Monotonicity: if one portfolio pays at least as much as another in every state, it cannot require more capital. Any measure that violates this is simply broken.

Translation invariance: adding \$1 million of cash to a book reduces its required capital by exactly \$1 million. This is what makes $\varrho(X)$ interpretable as a capital requirement at all, since it says the number is denominated in money and that money offsets it one for one. It also implies $\varrho(X + \varrho(X)) = 0$: adding the required capital makes the position acceptable, which is the definition of “required”.

Positive homogeneity: doubling a position doubles its risk. This one is the most questionable of the four, because doubling a position also makes it harder to liquidate (§7.2), so arguably risk should grow faster than linearly. It is retained because it is what makes risk decompose cleanly across desks, as §12.7 shows.

Subadditivity: merging two books cannot increase total risk. This is diversification, stated as an axiom, and it has a sharp operational consequence: a firm cannot reduce its measured risk by splitting a position across two desks and reporting them separately. A measure without this property invites exactly that manipulation, and worse, it means the firm-wide number is not an upper bound on anything.

Of the two weakenings allowed in [Definition 12.1](#), convexity is doing the mathematical work and subadditivity the institutional work: convexity is what keeps the optimization problems containing ϱ tractable, while subadditivity is what makes the reported number mean something across a firm.

Where value-at-risk fails

$\text{VaR}_\alpha(X) = -\inf\{x : \mathbb{P}(X \leq x) > 1 - \alpha\}$ satisfies the first three axioms and fails the fourth, and the failure is not a technicality.

Take two bonds, independent, each defaulting with probability 4% and losing 100 when it does. At $\alpha = 95\%$ we look at the worst one day in twenty. For a single bond, default has probability 4%, which is inside the 5% we are ignoring, so $\text{VaR}(X_i) = 0$: the measure reports no risk at all.

Now hold both. The probability that at least one defaults is $1 - 0.96^2 = 7.84\%$, which is *outside* the 5% window, so the 95% quantile now sits in a region where

a loss of 100 occurs, and $\text{VaR}(X_1 + X_2) = 100$. Combining two positions each measured at zero produced a combined measure of one hundred: diversification, which cannot increase real risk, took the measured risk from nothing to a great deal.

Worse for our purposes, since VaR is a quantile it is neither convex nor even continuous in the portfolio weights, so minimizing it is a nonconvex problem with many local minima and no useful duality theory. It is nevertheless the most widely reported risk measure in the world, for reasons that are entirely regulatory and historical rather than mathematical.

12.3 Conditional value-at-risk as a linear program

Definition 12.2 (Conditional value-at-risk). $\text{CVaR}_\alpha(X) = \mathbb{E}[-X \mid -X \geq \text{VaR}_\alpha(X)]$: the average loss in the worst $1 - \alpha$ of cases.

CVaR is coherent. More importantly for us, it has a variational representation that turns it from a quantile calculation into an optimization.

Theorem 12.3 (Rockafellar and Uryasev 2000). For any $\alpha \in (0, 1)$,

$$\text{CVaR}_\alpha(X) = \min_{\zeta \in \mathbb{R}} \left\{ \zeta + \frac{1}{1 - \alpha} \mathbb{E}[(-X - \zeta)^+] \right\}, \quad (12.5)$$

and the minimizing ζ is $\text{VaR}_\alpha(X)$. The right-hand side is jointly convex in (ζ, X) .

The most useful theorem here

Suppose losses are linear in the decision, $-X = -w^\top R$, and we have N scenarios $R^{(1)}, \dots, R^{(N)}$. Substituting into (12.5) and introducing slack variables $u_j \geq (-w^\top R^{(j)} - \zeta)^+$ gives

$$\min_{w, \zeta, u} \zeta + \frac{1}{(1 - \alpha)N} \sum_{j=1}^N u_j \quad \text{s.t.} \quad u_j \geq -w^\top R^{(j)} - \zeta, \quad u_j \geq 0, \quad w \in \mathcal{W}, \quad (12.6)$$

which is a *linear program* whenever $\mathcal{W}$ is polyhedral. No distributional assumption, no covariance matrix, no normality: just scenarios. Minimizing CVaR over a polyhedral set of positions is therefore exactly as hard as linear programming, and scales to tens of thousands of scenarios and thousands of assets.

12.4 The dual representation of convex risk measures

Theorem 12.4 (Representation). A convex risk measure ϱ with suitable continuity admits

$$\varrho(X) = \sup_{\mathbb{Q} \in \mathcal{Q}} \left\{ \mathbb{E}^{\mathbb{Q}}[-X] - c(\mathbb{Q}) \right\}, \quad (12.7)$$

where $\mathcal{Q}$ is a set of probability measures and c a convex penalty, written c rather than α to keep it distinct from the tail level α of VaR_α and CVaR_α . For coherent ϱ , $c \equiv 0$ on a convex set $\mathcal{Q}$ and $+\infty$ elsewhere.

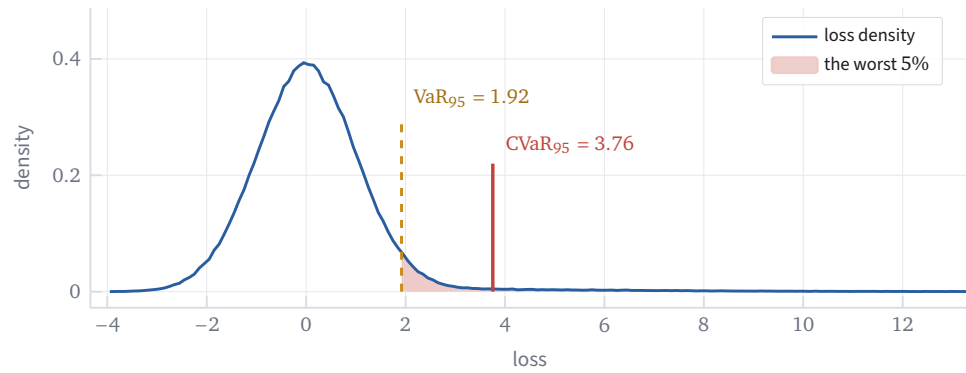


Figure 12.1. Two numbers from one tail. A loss distribution with a 3% chance of a large shock. VaR_{95} is the boundary of the shaded region: it reports where the tail starts and says nothing whatever about what is inside it. CVaR_{95} is the average of the shaded region, almost twice as large here, and it changes when the tail gets fatter, which VaR does not. Any position that shifts mass *within* the tail is invisible to VaR, and the temptation to build such positions is a well documented consequence of VaR-based limits.

Measure	Coherent	Optimization class	Comment
Variance	no	QP	symmetric: penalizes gains; the default only because it is easy
Semivariance	no	QP	one-sided; loses the closed forms of Chapter 11
VaR_α	no	nonconvex	regulatory standard; ignores the shape of the tail
CVaR_α	yes	LP	Theorem 12.3; scenario-based, no distributional assumption
Entropic	convex	smooth convex	$\frac{1}{\gamma} \log \mathbb{E} e^{-\gamma X}$; linearizes jump-control HJB
Max drawdown	no	path-dependent	what investors actually feel; almost intractable to optimize

Table 12.1. Choosing a risk measure is choosing an optimization class. The middle column is the practical content of Definition 12.1; the right column is why the choice is usually made on tractability grounds.

(12.7) is the Fenchel conjugate, and it says something you should find familiar: *risk aversion and model uncertainty are the same mathematical object*. Minimizing a coherent risk measure is minimizing the worst case over a set of probability models; the penalty c is how implausible you consider each model to be. This is exactly the structure of robust control, with $\mathcal{Q}$ the uncertainty set and c the disturbance penalty.

Two examples make it concrete. CVaR_α corresponds to $\mathcal{Q} = \{\mathbb{Q} : d\mathbb{Q}/d\mathbb{P} \leq \frac{1}{1-\alpha}\}$, the worst case over measures that cannot re-weight any event by more than $1/(1-\alpha)$. The entropic risk measure $\frac{1}{\gamma} \log \mathbb{E}[e^{-\gamma X}]$ of Chapter 9 corresponds to $c(\mathbb{Q}) = \frac{1}{\gamma} H(\mathbb{Q} \parallel \mathbb{P})$, the relative entropy, so it is the worst case over all measures, penalized by how far each is from your reference model.

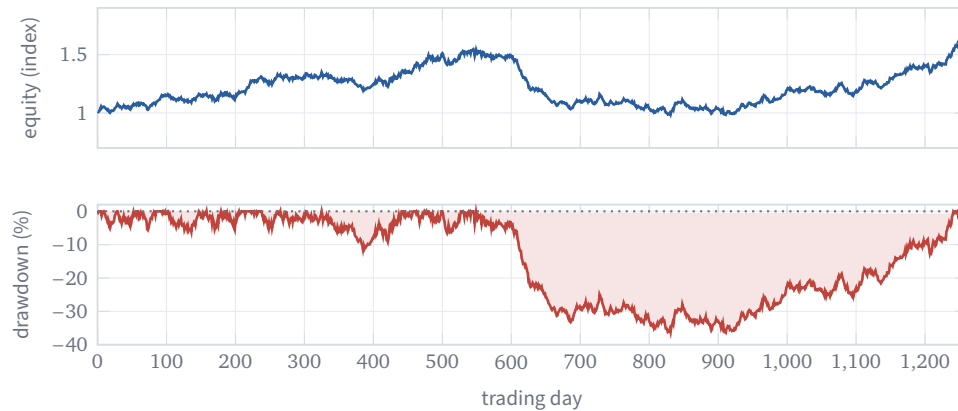


Figure 12.2. The same track record told twice. A simulated strategy with a respectable Sharpe ratio of about 0.6. The equity curve (top) is the story the manager tells; the drawdown (bottom) is the story the investor experiences, here a 36% decline that takes over a year to recover. Note that nothing in the Sharpe ratio, or in any of the measures of Table 12.1 except the last, distinguishes this path from one that merely oscillated.

12.5 Drawdown

Every measure so far has been a function of the terminal distribution: you look at where the wealth ends up and compute something. Investors do not experience a distribution. They experience a path, and what they react to is not variance but the sight of an account going down and staying down.

The relevant statistic is the **drawdown**,

$$D_t = \frac{V_t}{\max_{s \leq t} V_s} - 1 \leq 0, \quad (12.8)$$

the fractional distance below the highest level reached so far, and the quantity usually quoted is its running minimum, the **maximum drawdown**.

It has three properties that make it awkward and one that makes it unavoidable.

It is *path-dependent*: two strategies with identical terminal distributions can have completely different drawdowns, depending on the order in which the good and bad periods arrived. It is *not translation invariant*, so it is not a risk measure in the sense of Definition 12.1 at all. And it is *non-convex* in the strategy parameters, so optimizing it directly is intractable except in special cases.

The property that makes it unavoidable is that it is the number investors read, and the number at which capital is withdrawn. A strategy is not shut down because its variance was high; it is shut down after a long decline, often at the bottom, and by the investor rather than the manager. That makes drawdown a constraint on the business even when it is not in the objective.

In plain terms

The practical response to drawdown is not to optimize it but to bound it, and the standard bound is a rule rather than a formula: cut the position by half after a 10% decline, stop entirely after 20%.

Such rules are widely derided as unprincipled, and they are, but they are also doing something a terminal-distribution risk measure cannot. They convert a path property into a sequence of decisions, and they guarantee that the capital survives to trade another day, which is a precondition for any long-run argument about expected returns being realized at all. A strategy with a positive edge

and no drawdown control has a positive probability of being terminated before the edge arrives, and no amount of expected-value reasoning repairs that.

12.6 Constraints encountered in practice

In a real book the objective function is rarely what stops you. Constraints are, and it is worth knowing which ones, in roughly the order they bite.

12.6.1 Margin

The clearing house or prime broker requires collateral $m_t \sum_i |Q^i| S^i \leq V_t$, with m rising in volatility. This is the constraint that turns a bad day into a forced liquidation, and §7.5 traced the loop it creates. It binds first because it is the only constraint whose right-hand side falls at exactly the moment its left-hand side rises.

12.6.2 Position and concentration limits

Hard caps: no more than so much in one name, one sector, one country, one factor. They are convex, trivially monitored, and they are the main defence against the error-maximization of §11.3. If your covariance estimate is wrong in a way that makes one position look attractively hedged, a concentration limit stops the optimizer acting on it.

They are also the constraint most often set by committee rather than by calculation, which means they frequently bind in ways the model never anticipated. A portfolio construction that assumes the limits are slack is solving a different problem from the one the desk is actually running.

12.6.3 Liquidity limits

No more than x days of average volume in any name, so that the whole book could be unwound within a stated horizon. This is §7.2 written as a constraint rather than a cost, and it is what prevents the slow accumulation of positions that are individually reasonable and collectively impossible to exit.

12.6.4 Regulatory capital

A function of the book that the firm must hold against it, and for a bank the binding constraint on the whole enterprise. Under current rules it is computed from an expected-shortfall measure at a 97.5% level with adjustments for the liquidity horizon of each risk factor. That is a rare case of the theoretically correct measure winning an argument: the earlier framework used value-at-risk, and the switch was made precisely because of the tail-blindness demonstrated in §12.1.

12.6.5 Pre-trade risk checks

Hard, automated limits enforced in the path between the strategy and the exchange: maximum order size, maximum message rate, price collars that reject orders far from the current market, and a kill switch.

These are not risk management in the sense of the rest of this chapter. They exist because the tail risk of an algorithmic trading desk is dominated by software failure rather than by market moves, and no objective function in Part II contains a term for a malfunctioning strategy sending ten thousand orders a second. The controls are crude on purpose: they must be evaluable in microseconds and comprehensible at three in the morning.

Where the model breaks

Constraints interact with the objective in a way that is easy to miss. A strategy optimized subject to a binding constraint is *not* the unconstrained strategy with a smaller position: the KKT multiplier reshapes the entire solution, and the direction of the change can be surprising.

The best-known example is leverage. If an investor cannot borrow, they cannot reach the high-return end of the frontier by leveraging a moderate portfolio, so they buy high-beta assets instead, where the beta of an asset is its sensitivity to the market as a whole, so that a high-beta asset is one that is already levered internally. Demand for high-beta assets rises, their expected returns fall, and the security market line flattens. An entire empirical literature rests on this one multiplier.

12.7 Risk contributions

Once a risk measure is chosen there is a second question, asked constantly in practice and rarely in textbooks: how much of the total does each position account for? The answer is clean precisely when the measure is positively homogeneous.

Risk contributions

For a positively homogeneous ϱ , Euler's theorem gives

$$\varrho(w) = \sum_i w_i \frac{\partial \varrho}{\partial w_i}, \quad (12.9)$$

so total risk decomposes *exactly* into per-position contributions, with no residual. For $\varrho = \sqrt{w^\top \Sigma w}$ the contribution of asset i is $w_i(\Sigma w)_i / \sqrt{w^\top \Sigma w}$; for CVaR it is $w_i \mathbb{E}[-R_i \mid \text{tail}] / \text{CVaR}$, the average loss contributed by that asset on the bad days.

This is how risk budgets are allocated across desks, and it is the reason to keep positive homogeneity in [Definition 12.1](#) even though convexity is the mathematically essential axiom. Without homogeneity there is no exact additive decomposition, and without a decomposition there is no defensible way to tell a desk what its risk consumption is.

Note also what (12.9) does *not* say. The contribution of asset i is not the risk you would remove by deleting it, because removing it changes everybody else's contribution too. It is a marginal quantity, correct for small changes and misleading for large ones, which is a distinction that causes arguments in risk meetings.

Exercises

Exercise 12.1 (Verify the counterexample). Compute VaR_{95} for one bond and for the sum of two independent bonds in the example above, and confirm the failure of subadditivity. Then compute CVaR_{95} in both cases and verify that it is subadditive here.

Exercise 12.2 (Prove Rockafellar-Uryasev). Show that $F(\zeta) = \zeta + \frac{1}{1-\alpha} \mathbb{E}[(-X - \zeta)^+]$ is convex, compute its subdifferential, and deduce that any minimizer is an α -quantile of the loss and that the minimum equals $\text{CVaR}_\alpha(X)$.

Exercise 12.3 (The CVaR LP in practice). Write (12.6) for $d = 3$ assets and $N = 5$ scenarios explicitly, solve it by hand or with any LP solver, and identify which scenarios are active at the

optimum. Show that the number of active scenarios is at most $d + 1$, and explain the practical consequence for the stability of the solution.

Exercise 12.4 (Entropic risk and its dual). Verify that $\varrho(X) = \frac{1}{\gamma} \log \mathbb{E}[e^{-\gamma X}]$ satisfies (12.7) with $c(\mathbb{Q}) = \frac{1}{\gamma} H(\mathbb{Q} \parallel \mathbb{P})$, using the Donsker-Varadhan variational formula. Show that $\gamma \rightarrow 0$ recovers $\mathbb{E}[-X]$ and $\gamma \rightarrow \infty$ recovers $\text{ess sup}(-X)$.

Exercise 12.5 (Leverage tilts toward beta). In a one-period mean-variance problem with a constraint $\mathbf{1}^\top w \leq L$, show that the optimal portfolio can be written as the tangency portfolio plus a term proportional to $\Sigma^{-1} \mathbf{1}$, and that the multiplier on the constraint tilts the solution toward high-variance assets. Relate this to the “betting against beta” literature.

Exercise 12.6 (Drawdown is not a risk measure). Show that maximum drawdown fails translation invariance, and give an example of two strategies where the one with the smaller drawdown has the larger CVaR. What does this say about using drawdown as a limit?

Estimation and Backtesting

Every model in this book has parameters: λ , k , ϕ , A , κ , σ , μ , Σ . None of them is observable. This chapter is about getting them from data, and about the specific, well-documented ways in which the resulting estimates mislead you.

The chapter has an unusual shape because the difficulty is unusual. The statistical problems here are not the ones a time-series course prepares you for. High-frequency data is not noisy in the usual sense: it is noisy in a way that makes standard estimators *diverge* rather than merely lose efficiency (§13.3). And the evaluation problem is not one of sample size but of *selection*: you will look at thousands of strategies, and the best of them will look good whether or not any of them is (§13.6).

In plain terms

Every model in this book has parameters, and none of them can be observed directly. This chapter is about getting them from data, and about two specific ways that process misleads you. The first is statistical: high-frequency price data is contaminated by a small error that, unusually, makes standard estimates *worse* the more data you use. The second is human: if you try a thousand strategies and report the best, it will look excellent even when none of them has any merit whatsoever, and the chapter quantifies exactly how excellent.

13.1 The best of a thousand strategies

Consider the following strategy. Over the last four years it returned 203%. It was profitable in every one of those four years, at +34%, +56%, +17% and +25%. Its worst peak-to-trough decline was 10.4%, and its Sharpe ratio was 1.87. On any conventional screen it is a very good strategy indeed, and if you were shown only this record you would probably want to invest in it.

It has no edge at all. Its daily returns are independent draws from a distribution with mean exactly zero, generated by a random number generator on the machine that typeset this page. The reason its record looks like that is not that the strategy is good; it is that I generated a thousand such strategies and showed you the best one.

That is the central difficulty of this chapter, and it is not a statistical subtlety but a structural feature of how quantitative research is done. You try things. The ones that do not work are discarded. What survives to be written up is, by construction, the extreme of a distribution, and the usual apparatus of significance testing assumes you are looking at a random draw rather than a maximum. [Section 13.6](#) makes this quantitative and shows that the effect is large enough to explain most of the gap between backtested and live performance in the industry.

The chapter's second difficulty is quieter and just as damaging. Every model in Part II has parameters, and none of them can be observed directly: not the volatility, not the impact coefficient, not the rate at which your quotes get filled. They must be estimated from data, and high-frequency market data has a property that makes estimation genuinely counter-intuitive.



Figure 13.1. The winner of a thousand coin-flipping contests. Every strategy in this picture has an expected return of exactly zero; the grey band shows where the thousand of them ended up. The red line is simply the best of them, and it has a four-year record that would pass most institutional screens. Nothing about the red line is wrong or fraudulent. It is the honest track record of a strategy with no merit, and the only thing that distinguishes it from a genuinely good strategy is a piece of information that never appears on the chart: how many others were tried.

In plain terms

Suppose you want to know how volatile a stock was today, and you have every trade. The obvious estimator adds up the squared price changes. If you use every trade, you get one answer; if you sample once a minute, you get a noticeably smaller one; once every half hour, smaller still. These are not converging on a common value as you use more data, and the finest sampling is not the most accurate one. It is the worst. [Proposition 5.4](#) explains why, and [Fig. 5.4](#) shows the damage: with realistic contamination, second-by-second sampling overstates the variance fourfold.

13.2 Market data and its structure

How it actually works

Three levels of market data, in increasing order of detail and cost.

Level 1 / trades and quotes. Every trade (time, price, size) and every change in the best bid and offer. Enough for Chapters 5 to 8. A single liquid US equity generates of order 10^5 to 10^6 messages a day.

Level 2 / aggregated book. Total displayed size at each of the first k price levels. Enough for imbalance, the depth profile of [Fig. 4.6](#), and the state of most market-making models.

Level 3 / full order-by-order. Every message with an order identifier, so that individual queue positions can be reconstructed. Required for anything in [Section 4.5](#), and available only on some venues.

Practical hazards: exchange timestamps and your capture timestamps differ, and the difference is not constant; trades are reported with condition codes that determine whether they should be included at all; and the same instrument trades on many venues, so “the” price requires a consolidation rule.

13.3 Volatility estimation under microstructure noise

Volatility is the parameter every model in Part II needs, and estimating it is the cleanest illustration of why high-frequency data is different.

13.3.1 The source of the bias

The textbook estimator is realized variance ([Proposition 5.4](#)): chop the day into intervals of length δ , square each price change, add them up. If prices were a diffusion observed exactly, this would converge to the true integrated variance as $\delta \rightarrow 0$, and the more finely you sampled the better your answer would be.

Prices are not observed exactly. What you see is the efficient price plus the bid-ask bounce of [Fig. 2.13](#), a contamination which is small, bounded, and crucially *does not shrink* as you sample more finely. Over a one-minute interval a stock might genuinely move 0.05% while the bounce contributes 0.01%: negligible. Over a one-second interval the genuine move is roughly $0.05\%/\sqrt{60} \approx 0.006\%$ while the bounce still contributes 0.01%: now the noise is larger than the signal.

Adding up squares makes this worse rather than better, because the noise contribution does not average out. Each squared increment contributes $2\omega^2$ of pure noise on top of the genuine variance, and there are T/δ increments, so the total contamination is $2T\omega^2/\delta$. That is ([5.15](#)), and it diverges. Sampling twice as finely doubles the bias.

In plain terms

Put numbers on it. Take a stock with 30% annual volatility, so a daily variance of about $(1.9\%)^2$, and a bounce of one basis point. Sampling every five minutes gives 78 increments and a contamination of $78 \times 2 \times (0.0001)^2$, about 1.6×10^{-6} , against a true daily variance of 3.6×10^{-4} : an error under half a percent. Sampling every second gives 23,400 increments and a contamination of 4.7×10^{-4} , which is *larger than the quantity being measured*. The estimator has not become imprecise; it has become meaningless.

13.3.2 Four estimators

Each of the standard fixes is a familiar idea from another field, which is worth noticing because it tells you what to expect from each.

- (1) **Sparse sampling.** Choose δ to balance the $O(1/\delta)$ noise bias against the $O(\delta)$ discretization variance. Differentiating the sum gives $\delta^* \propto (\omega/\sigma)^{4/3} T^{1/3}$, which for typical parameters lands between one and ten minutes. This is where the practitioner's five-minute rule comes from, and it is a genuine optimum rather than folklore. Its defect is brutal: at five-minute sampling you are discarding more than 99% of the observations you paid for.
- (2) **Two-scale and multi-scale estimators.** Compute realized variance at a fast and a slow frequency. Both are contaminated, but the contamination scales predictably with the frequency, so a suitable linear combination cancels the leading term while keeping most of the data. This is Richardson extrapolation applied to a variance estimator. It converges at $n^{-1/6}$, improving to $n^{-1/4}$ once several scales are combined ([Zhang et al. 2005](#)).
- (3) **Realized kernels.** Write the estimator as a weighted sum of the autocovariances of returns, $\widehat{IV} = \sum_{h=-H}^H \varpi(h/H) \hat{\gamma}_h$. Since the bounce induces a specific negative first-order autocorrelation, choosing the weights ϖ to absorb it removes the bias. Anyone who has used a Newey-West standard error has done this before, in a different context; the choice of H is the same bandwidth problem. This attains

the optimal $n^{-1/4}$ rate and is what most practitioners actually run (Barndorff-Nielsen et al. 2008).

- (4) **Pre-averaging.** Average returns over blocks of about $\sqrt{n}$ observations before squaring. Because the noise is independent across observations and the signal is not, averaging shrinks the noise by $\sqrt{\text{block}}$ while leaving the signal essentially intact. It is a low-pass filter, with the block length as the cutoff.

The unifying view

Write the observation equation $Y = S + \eta$ in the frequency domain. A diffusion's increments have a flat spectrum whose height scales with the sampling interval δ ; independent noise has a flat spectrum that does not scale with δ at all. Every estimator above is a different taper on the same problem, namely separating two flat spectra with different δ -scalings.

Once you see it that way, three things follow immediately. The choice of estimator is a bandwidth selection problem with the usual bias-variance trade-off. The $n^{-1/4}$ rate is not a peculiarity of finance: it is the optimal rate for integrated variance under the standard model of an Itô semimartingale observed with additive i.i.d. noise. And the folklore question “what sampling frequency should I use?” is revealed as the wrong question: the right one is what filter to apply, and sparse sampling is simply the crudest possible answer to it.

13.4 Estimation of model parameters

How it actually works

Impact (k, λ, Y). Regress the price change over an interval on signed volume over the same interval. The estimate is biased *upward* by common causes: news moves both the price and the flow. The standard remedies are to instrument the flow (use a component driven by mechanical rebalancing, such as index changes), or to use your own randomized executions: the only genuinely exogenous variation most desks have. Anyone who tells you their impact model is estimated cleanly from public data has not thought about the endogeneity.

Fill intensities (A, κ). Fit $\log \lambda(\delta) = \log A - \kappa\delta$ by counting fills at each quoted distance. The confounder is that you only observe distances you actually quoted, and you quoted them for reasons correlated with the flow. Randomizing quote placement occasionally (deliberately giving up small amounts of expected profit to buy identification) is standard practice and is literally an exploration/exploitation trade-off.

Urgency (ϕ). There is no data. ϕ encodes a preference, and it should be set by choosing a point on the frontier of Fig. 8.2, not estimated.

Volatility (σ). §13.3, and remember that the relevant horizon is the execution horizon, not the day.

13.5 Five sources of bias in backtests

A **backtest** is a simulation of a strategy on historical data: you replay the past, pretend you were trading through it, and record what would have happened. It is the main tool available for evaluating a strategy before risking money on it, and it is systematically optimistic. The five failure modes below are well catalogued, and every one of them biases the answer in the same direction.

13.5.1 Look-ahead bias

Using at time t any information that was not actually available at time t . The blatant version is easy to avoid. The subtle versions are not, and they are the ones that bite.

Company fundamentals are *restated*: the earnings figure your database shows for a given quarter is the corrected one, published months later, not the number the market saw at the time. Index membership is usually stored as of today, so a backtest of “buy the index constituents” quietly buys the companies that were successful enough to still be in it. A volatility or covariance matrix estimated over the full sample uses tomorrow’s data to size today’s position. Even the humble decision to drop rows with missing values can leak, because whether a value is missing is itself informative.

The test is mechanical and worth applying literally: for every number your strategy touches at time t , ask when it was first published, not what date it refers to.

13.5.2 Survivorship bias

Testing on the securities that still exist. Firms that went bankrupt, were delisted, or were acquired at a discount drop out of most databases, and they are precisely the ones that lost money. A universe assembled from today’s listings has quietly excluded the worst outcomes from the sample.

The magnitude is not small. Studies of equity databases commonly find that survivorship inflates measured returns by one to several percentage points a year, which is comparable to the entire edge that most strategies claim.

13.5.3 The fill assumption

This one is specific to anything trading at short horizons, and it is the difference between a market-making backtest that prints money and one that does not.

The tempting rule is: my limit buy at \$99.99 is filled if a trade printed at \$99.99 or below. That rule is wrong, and [Section 4.5](#) says exactly why. When you posted, there were already several hundred lots ahead of you at that price. A trade printing at your price means *somebody* was filled, and almost certainly not you.

A defensible fill model tracks queue position: how much was ahead when you joined, how much of it has since traded or cancelled, and only then whether your own order is reached. Level-2 data does not contain enough information to do this exactly, so you must bound it, and the honest thing is to report the backtest twice, once under the optimistic assumption and once under the pessimistic one. If the conclusion changes between them, you have not learned anything yet.

13.5.4 Own market impact

The historical prices you are trading against were generated in a world in which you did not trade. Replay them, insert your orders, and you are implicitly assuming the market would have done exactly the same thing regardless.

(6.7) says the correction is large: a trade of 1% of daily volume moves the price by roughly 18 basis points, more than the gross edge of most systematic strategies. Worse, it is concave, so it cannot be estimated at small size and extrapolated. No backtest that replays a fixed price series can capture this, because capturing it requires simulating how the other participants would have responded, which is the equilibrium problem of [Chapter 16](#).

The practical mitigation is to keep an explicit impact model, charge the strategy for it, and treat any strategy whose profitability depends on the impact coefficient being small as unproven.

13.5.5 Selection bias

The subject of [§13.6](#), and by far the most dangerous, because unlike the other four it is completely invisible in the backtest itself. Nothing about the equity curve in

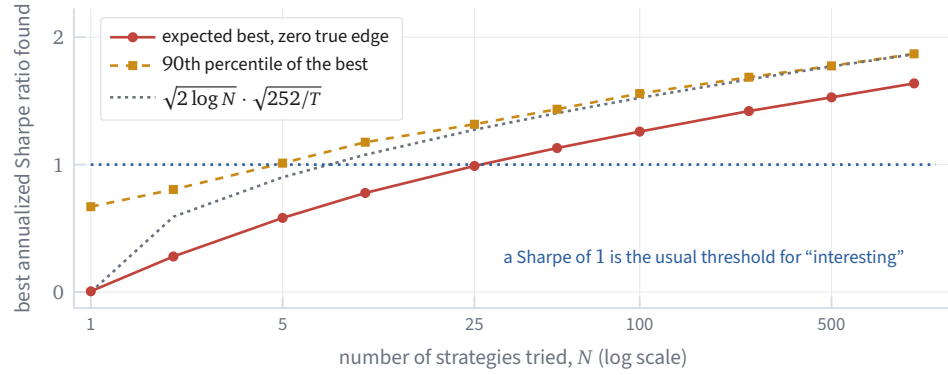


Figure 13.2. Nothing here has any edge. The best annualized Sharpe ratio among N strategies whose true expected return is exactly zero, over four years of daily data, averaged across 4000 replications. At $N = 100$ the best backtest shows a Sharpe of 1.26; at $N = 1000$, 1.64. A research process that tries a hundred variants and reports the winner will clear any conventional significance threshold while having discovered nothing. Note also that N is not the number of strategies you *wrote down*: it is the effective number of configurations you explored, including every parameter you tuned by looking at the result.

Fig. 13.1 reveals that it was chosen from a thousand candidates. The information needed to discount it is not in the data at all; it is in the history of your own research process.

Where the model breaks

These five do not act independently, they compound. A strategy with a little look-ahead, an optimistic fill model and no impact charge shows a backtested edge large enough to justify examining a thousand variants of it, at which point selection multiplies the error again. This is the standard route by which a research program produces something that looks excellent and works not at all.

13.6 Overfitting as a multiple-testing problem

Suppose you test N strategies, none of which has any edge. Each produces a Sharpe ratio which, over T observations, is approximately $N(0, 252/T)$ in annualized units. The one you report is the maximum of N such draws, and the maximum of N Gaussians grows like $\sqrt{2 \log N}$.

The deflated Sharpe ratio

Given N trials, the appropriate null is not “Sharpe = 0” but “the maximum of N draws”. The **deflated** Sharpe ratio of Bailey and López de Prado (2014) compares the observed value against $\mathbb{E}[\max_N]$ rather than against zero, giving

$$\text{DSR} = \Phi \left(\frac{(\widehat{\text{SR}} - \mathbb{E}[\max_N])\sqrt{T-1}}{\sqrt{1 - \zeta_3 \widehat{\text{SR}} + \frac{\zeta_4 - 1}{4} \widehat{\text{SR}}^2}} \right), \quad (13.1)$$

with ζ_3, ζ_4 the skewness and kurtosis of returns. The practical instruction is blunter than the formula: *record N* . A research protocol that does not track how many variants were examined cannot be corrected after the fact.



Figure 13.3. Research performance does not survive deployment. A schematic illustration of the gap between backtested and realized results; the magnitudes drawn are illustrative rather than measured. Three mechanisms act together, and the figure takes no position on their relative sizes, which vary by strategy, venue and era: selection (Fig. 13.2), costs and impact that the backtest underestimated, and genuine decay as others find the same signal. Discounting a backtest heavily before deciding whether it clears its hurdle is a desk heuristic, not a result.

13.7 Evaluation protocols

No protocol can rescue a strategy that has no edge, and none substitutes for an honest count of how many things you tried. What the protocols below do is remove specific, identifiable sources of optimism, in roughly increasing order of cost.

13.7.1 Walk-forward

Fit the model on a window $[t - w, t]$, trade it on $[t, t + h]$, then roll both windows forward and repeat. Nothing is ever fitted on data it is later tested on, so the arrow of time is respected exactly.

Provided every preprocessing and selection step is nested inside the training window, no future observation enters the fit, and the procedure is what a live deployment actually does, which makes its results directly interpretable. That proviso is not a formality: universe construction, feature design, cleaning rules, normalization, hyperparameter choice and repeated inspection of the test periods all leak if they are performed once over the whole sample. Its cost is statistical: each observation is used for testing once, so the effective sample is small and the performance estimate is noisy. With four years of daily data and a one-year fitting window you get three years of out-of-sample results, which is not many independent observations of anything.

13.7.2 Purged and embargoed cross-validation

Ordinary K -fold cross-validation leaks badly on financial data, for a reason that is easy to miss. Labels are usually computed over a forward window: the return over the next five days, say. A training observation whose label window overlaps the test period therefore contains information about the test period. The model can learn from it, and the cross-validated score is optimistic.

The repair has two parts (López de Prado 2018). *Purging* removes from the training set every observation whose label window overlaps the test fold. *Embargoing* removes a further stretch immediately after the test fold, because serial correlation means observations just after it still carry information about it. Figure 13.4 shows both.

The gain over walk-forward is that every observation is used for testing in some fold, so the estimate is less noisy. The cost is that you must be honest about the label horizon, and that the folds are no longer independent.

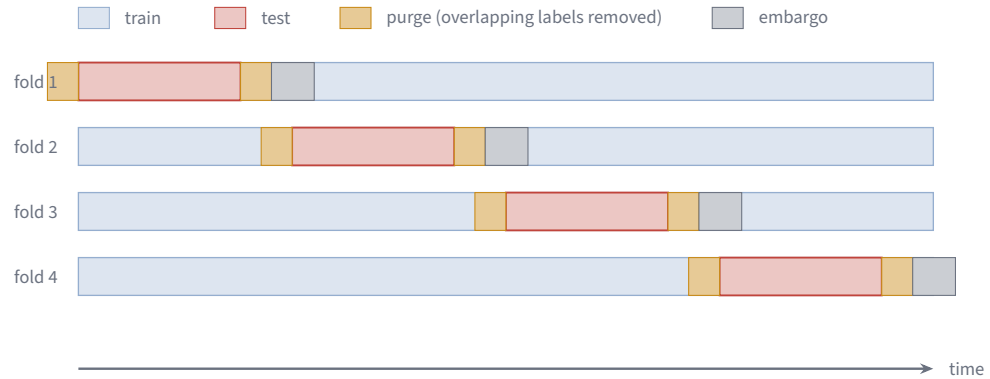


Figure 13.4. Cross-validation for series with overlapping labels. A label computed over a forward window overlaps observations that would otherwise be in the training set, leaking the answer. Purging removes the overlap on both sides of the test fold; the embargo removes a further window afterwards to break residual serial correlation. Without both, cross-validated performance on financial data is optimistic by an amount that grows with the label horizon.

13.7.3 Direct measurement of overfitting

Rather than trying to eliminate selection bias, measure it. Split the sample into many combinations of in-sample and out-of-sample blocks, and for each combination record whether the configuration that was best in-sample landed above or below the median out-of-sample. The fraction of times it lands below estimates the probability that your selection procedure is producing noise.

This is expensive to compute and unusually informative, because it evaluates the *research process* rather than any individual strategy.

13.7.4 Paper trading and a small live allocation

Running the strategy live with no capital catches the errors that no amount of statistics will: a mis-signed position, a timezone bug, a feed delivered later in production than in research. It tells you nothing about impact, because you are not trading.

Genuinely unseen data is out-of-sample in the statistical sense, so paper trading is not worthless. What it cannot produce is the part of the answer that depends on your own orders existing. A small live allocation is the only test that exercises execution, impact, latency and the market's response to you at once; everything before it is a filter designed to avoid wasting this step, not a substitute for it. It is the point at which Fig. 13.3 stops being a statistical curiosity and becomes your track record.

Where the model breaks

No protocol repairs a backtest whose *simulator* is wrong. If your fill model assumes you were at the front of the queue, no amount of purging will tell you that you were not. Spend the effort on the simulator first: a crude strategy evaluated with a realistic fill and impact model is worth more than an elegant strategy evaluated with an optimistic one.

Exercises

Exercise 13.1 (Optimal sampling frequency). Minimize the mean-squared error of $RV(\delta)$ using (5.15) plus the $O(\delta)$ discretization variance, and derive $\delta^* \propto (\omega/\sigma)^{4/3}$. Evaluate it for a stock with $\sigma = 30\%$ annualized and noise of one basis point, and compare with the five-minute folklore.

Exercise 13.2 (Two-scale estimator). Show that $\widehat{IV} = \text{RV}(\delta_{\text{slow}}) - \frac{n_{\text{slow}}}{n_{\text{fast}}} \text{RV}(\delta_{\text{fast}})$ is asymptotically unbiased, and compute the constant that makes the leading noise term cancel exactly.

Exercise 13.3 (Impact regression is biased). Let $\Delta S = \lambda v + u$ where the flow $v = \beta u + \varepsilon$ is partly driven by the same news u . Compute the OLS estimate of λ and show that it exceeds the true value. Propose an instrument and state the condition it must satisfy.

Exercise 13.4 (The maximum of N Sharpe ratios). Prove that for N i.i.d. $\mathcal{N}(0, s^2)$ variables, $\mathbb{E}[\max] \sim s\sqrt{2 \log N}$, and check the constant against Fig. 13.2 at $N = 100$ and $N = 1000$. How many independent strategies would you need to test before a Sharpe of 2 becomes unremarkable?

Exercise 13.5 (Queue-aware backtesting). Design a fill model for a limit order that uses only Level-2 data. State the assumption it must make about your queue position, and bound the error in estimated fill probability using (4.9). Under what market conditions is the bound tight?

High-Frequency Trading

Every model in Part II has a horizon. Execution runs over hours, market making over seconds, hedging over weeks. This chapter is about what happens when the horizon shrinks until the binding constraint is no longer the model but the speed of light in a length of fibre.

That sounds like an engineering topic and it is, but it is also an economic one, because at those timescales the decision problem changes shape. O'Hara (2015) and Menkveld (2016) survey the field; the account below is organized around what changes for the models of Part II. The quantity being optimized stops being a forecast and becomes a *race*, and the equilibrium of a race is a different object from the optimum of a control problem.

In plain terms

High-frequency trading is not a strategy. It is a technology regime: a set of strategies whose profitability depends on acting faster than the other participants, typically in microseconds. The strategies themselves are mostly the ones already in this book, market making and arbitrage, run at a speed where the modelling assumptions of Part II stop holding. What is genuinely new is that in the races of §14.4 being first can matter more than being right, which inverts the usual relationship between prediction and profit for that part of the business.

14.1 How markets got fast

None of this was designed. It is the accumulated consequence of four changes, each sensible on its own, and knowing the sequence explains a great deal about why the market looks the way it does.

Electronic matching replaced human intermediation. Until the 1990s most equity and futures trading passed through a person: a specialist on an exchange floor with an affirmative obligation to make a market, or a pit trader in an open-outcry ring. Those people were slow, they were paid a wide spread for being slow, and their obligations were qualitative. Replacing them with a matching engine (§4.2) sharply compressed the spread, removed the affirmative obligation, and removed any natural floor on how fast a participant could respond.

Tick sizes fell. US equities were quoted in eighths of a dollar until 1997, then sixteenths, then in 2001 in cents. A minimum spread of \$0.125 on a \$40 share is 31 basis points; a minimum spread of \$0.01 on the same share is 2.5. That single change compressed the revenue available per trade by an order of magnitude, and a business earning a tenth as much per trade must do ten times as many of them. Figure 2.17 explains the mechanism: the market moved from the large-tick regime, where queueing dominates, toward the small-tick regime, where speed of repricing does.

Trading fragmented across venues. Regulation in both the United States and Europe during the 2000s encouraged competing execution venues, on the sound theory that

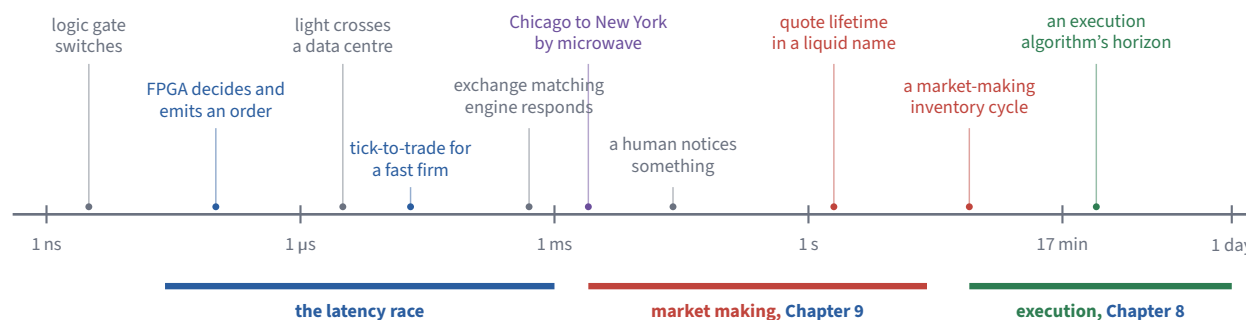


Figure 14.1. Fourteen orders of magnitude. The models of Part II live on the right of this axis, where a decision has time to be computed and a price has time to move. The latency race lives on the left, where the binding constraint is physical: how long a signal takes to cross a room, a city, or a silicon die. The two regimes meet between roughly a microsecond and a millisecond, and that meeting is the subject of this chapter.

competition lowers fees. It does, and it also means the same instrument now has a dozen books that must be kept consistent (§4.6), which creates the arbitrage business of §14.4 and makes the relative latency between venues economically meaningful.

Access became a commodity. Exchanges began selling colocation, direct data feeds and low-latency order entry to anybody willing to pay. What had been a privilege of membership became a product with a price list, which is what turned speed from an accident of geography into an investment decision.

In plain terms

The pattern is worth naming because it recurs. Each change removed a friction, and each was defended, correctly, as making markets cheaper for ordinary investors. Spreads did fall dramatically and they have stayed down. But removing a friction also removes whatever the friction was suppressing, and what it was suppressing here was competition on speed. The modern debate is not about whether the changes were good on balance; it is about the specific residue they left behind, which is the race described in §14.5.

14.2 Timescales

The phrase “high frequency” is unhelpfully vague, so fix the scale. Figure 14.1 places the relevant events on a logarithmic axis spanning fourteen orders of magnitude, from the switching time of a logic gate to the trading day.

Two features of that picture do the work. First, the fastest participants respond in a few microseconds, which is shorter than the time it takes a signal to cross a continent, so the market is not a single place with a single state: at any instant, different participants correctly believe different things. Second, the lifetime of a quote in a liquid instrument is of order a second, which is six orders of magnitude longer. The race is therefore not about the pace of the market but about the ordering of a few specific events.

14.3 The latency stack

What is actually being optimized is the **tick-to-trade** time: the delay between a packet of market data arriving at your machine and your responding order leaving it. It decomposes into a chain of physical steps, each of which has been attacked independently.

How it actually works

Propagation. A signal travels at about 200 metres per microsecond in fibre and about 300 in air. Chicago to New York is 1,200 km in a straight line, but fibre must follow rights of way rather than the shortest path, so the trip takes about 6.5 ms one way against about 4.0 ms for a microwave link that can go straight; that 2.5 ms difference was worth enough to fund a private microwave network. MacKenzie (2021) traces this infrastructure in detail.

Colocation. Exchanges rent rack space in the same building as the matching engine, with cable lengths deliberately equalized between customers so that the remaining advantage is in the machine rather than the cable.

Network stack. A conventional operating system takes tens of microseconds to deliver a packet to an application. Kernel bypass removes the operating system from the path; an FPGA removes the general-purpose processor as well, parsing the market data feed and emitting an order in hardware, in a few hundred nanoseconds.

Feed choice. Exchanges publish a fast proprietary feed and contribute to a slower consolidated one. Anyone acting on the consolidated feed is by construction behind anyone acting on the direct feeds, which has been the subject of both litigation and market-design reform.

Speed is a fixed cost that buys a probability

Notice the economic structure. All of the expenditure above is fixed: the microwave link costs the same whether you trade once or a million times. What it buys is not a better forecast but a higher probability of *winning ties*: being first in the queue at a newly opened price, or being first to reach a stale quote. That has two consequences. The activity has extreme economies of scale, so it concentrates in a small number of firms. And the return on the investment is competed away in the usual manner, so in equilibrium the participants spend close to the value of the advantage in order to obtain it. The resources are consumed; the information is not improved.

14.4 Four business models

Almost everything done at these speeds is one of four things, and three of them are already in this book. What distinguishes them is not the strategy but what the speed is being spent on.

14.4.1 Electronic market making

This is Chapter 9 run at speed. The model is the same, the inventory control is the same, and Theorem 9.1 still describes the shape of the answer. Three things change.

Quotes are revised thousands of times a second, so the continuous-time formulation is no longer an approximation to a discrete process, it is a description of one. Queue position (§4.5) becomes the dominant term rather than a refinement: in a large-tick name where the spread cannot narrow, whether you are filled at all is decided by arrival order, and the optimal quoted distance of (9.6) is a second-order correction to a first-order race. And the inventory cycle compresses from minutes to seconds, so the risk term $\gamma\sigma^2\tau$ in (9.5) becomes small and the market-power term $\frac{2}{\gamma}\log(1 + \gamma/\kappa)$ dominates.

Menkveld (2013) gives the empirical anatomy: a single fast market maker accounting for a substantial share of volume in a set of names, earning a fraction of a basis point per trade, ending most days close to flat, and making its money from the sheer number of round trips rather than from any individual one. The profit and loss decomposition of §3.5 is unusually clean for such a firm, because there is almost no directional component to attribute.

14.4.2 Cross-venue and cross-instrument arbitrage

This is the arbitrage of Fig. 2.11 run as a continuous business rather than as an occasional opportunity.

The same instrument trades on a dozen venues, so the same price should hold on all of them. An exchange-traded fund is a claim on a basket, so its price should track the basket. A future is a claim on an index at a date, so by (2.6) its price is determined by the index and the interest rate. Each of these is a *relationship* that must hold, and each is enforced by somebody fast enough to profit from its violation.

The economics are attractive and unusual. The profit per correction is small and nearly certain, the position is held for milliseconds, and the strategy carries essentially no overnight risk. What it carries instead is *operational* risk: the relationships hold only if your reference data, your corporate-action handling and your basket composition are correct, and a stale constituent list turns an arbitrage into an unhedged position.

14.4.3 Order anticipation

Detecting that a large participant is working an order, and trading ahead of the remainder.

This is the predatory trading of §16.1, run on microstructure signals rather than on knowledge of any particular fund's position. The signals are the observable footprints of an execution algorithm: a persistent imbalance in the direction of trades (§6.4), regular child-order sizes, executions clustered at particular times, or repeated replenishment of a quote at the same price.

Two things follow. First, this is why the deterministic schedule of Theorem 8.1 is never deployed as written: it is maximally predictable, and §16.1 shows that randomizing costs second-order performance to remove a first-order transfer. Second, the line between order anticipation and ordinary inference is genuinely blurred, which is why it has proved difficult to regulate: inferring from public data that somebody is buying is what price discovery is.

14.4.4 Latency arbitrage

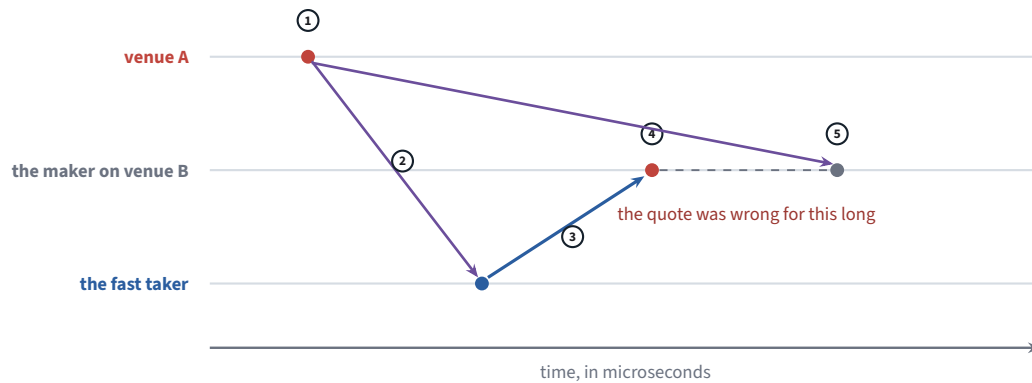
The one with no analogue in Part II, and the one that generates the controversy. When a price moves on one venue, quotes on every other venue are briefly stale. Whoever reaches them first trades against a price that is already known to be wrong.

The victim here is a market maker, and the loss is not a modelling error but a structural one: they were not slow to update, they were merely slower than somebody else. Foucault et al. (2017) call the resulting flow *toxic arbitrage* and measure how much of it is driven by speed rather than by information. Figure 14.2 draws the sequence.

14.5 The arms race as an equilibrium

Now the economics. Suppose sniping opportunities of average value v arrive at rate κ , and that being fastest wins them. Then the expected value of being fastest is κv per unit time, and in a competitive market firms will spend up to that amount on speed.

The trouble is what happens to the market maker. They lose κv per unit time to snipers, and being a competitive business they must recover it, which they do



- 1 A price move occurs on venue A, so every quote on venue B referencing it is now stale.
- 2 The news propagates. It reaches the fast taker before it reaches the maker, because the taker paid for the shorter path.
- 3 The taker sends an order to venue B against the quote it already knows to be mispriced.
- 4 The stale quote is lifted. The maker has lost the width of the price move.
- 5 Only now does the maker learn and cancel. There was nothing left to cancel.

Figure 14.2. Sniping a stale quote. A price moves on one venue, and both the fast taker and a market maker on a second venue will learn of it, but not at the same instant. In the interval between, the maker's quote is known by somebody to be mispriced, and that somebody can act. The maker did nothing wrong: they quoted a fair price, and the information that made it unfair reached another participant first. This is not a forecasting failure and no amount of better modelling prevents it. The only defences are to be faster, to quote wider, or to change the mechanism so that a race does not decide the outcome, which is the argument of §14.5.

by widening the spread. The cost is therefore paid by whoever crosses that spread, which is the ordinary investor.

A pure transfer plus a deadweight loss

Set the accounting out. The snipers gain κv . The makers lose κv and recover it through a wider spread, so they break even. The investors pay the wider spread. And the snipers spend close to κv on the technology that lets them win the race.

So the transfer from investors to snipers nets out to roughly zero for the snipers, and the resources spent on the race are consumed entirely. That is the argument of Budish et al. (2015): the arms race is a deadweight loss whose size is bounded by the value of the sniping opportunities, and it is generated not by anyone's misbehaviour but by the *mechanism*, which awards a stale quote to whoever arrives first rather than to whoever offers the best price.

Where the model breaks

That argument is clean and it is not the whole story. Speed also does useful work: it keeps prices consistent across a fragmented market, it narrows spreads by lowering the inventory risk of a market maker who can exit quickly, and the empirical record since the arrival of electronic market making shows spreads and explicit costs falling substantially for ordinary investors.

The honest position is that high-frequency trading has both effects, that their relative size is an empirical question, and that the empirical work is hard because the counterfactual is unobservable. The published record is correspondingly mixed. Hendershott et al. (2011) find that algorithmic trading

improved liquidity, using the staggered introduction of automated quoting as an instrument; Brogaard et al. (2014) find that fast traders contribute to price discovery rather than merely extracting from it. On the other side, Aquilina et al. (2022) measure the latency-arbitrage tax directly from exchange message data and find it small per trade and substantial in aggregate, and Baldauf and Mollner (2020) show formally that the two effects can coexist, with speed improving price discovery while degrading liquidity. Biais et al. (2015) give the equilibrium in which fast trading is over-provided relative to the social optimum.

What is not in dispute is that the *mechanism* creates a race, and that a different mechanism need not.

14.6 What a microsecond is worth

The arguments above are qualitative. Here is the arithmetic, with numbers chosen to be plausible rather than precise, because the point is the shape of the answer rather than its exact value.

The size of the prize. Take a single liquid stock. Suppose a sniping opportunity arises whenever the price moves materially on the reference venue, which happens perhaps once a minute, and that the quote which can be picked off is for 200 shares at a price stale by half a tick, so \$0.005 a share. One such event is worth \$1, and over a 6.5-hour day there are 390 of them:

$$\text{daily value per name} \approx 390 \times \$1 = \$390. \quad (14.1)$$

That is a trivial sum, which is precisely why this is a scale business. Across 1,000 names it is \$390,000 a day, roughly \$100 million a year, and that is the pool the race competes for.

What speed buys. Suppose n firms compete for each event and the fastest wins. With symmetric speeds each expects $1/n$ of the pool, so the value of moving from tied to reliably first is

$$\left(1 - \frac{1}{n}\right) \times \$100\text{m}, \quad (14.2)$$

which for $n = 5$ is \$80 million a year. That is the budget available for microwave links, colocation and hardware engineers, and it is why the observed spending is what it is.

Who pays. From §14.5 the maker recovers the loss by widening. Spread the pool across the day's volume in those names, say two billion shares, and the required widening is

$$\frac{\$100\text{m per year}}{252 \times 2 \times 10^9 \text{ shares}} \approx \$0.0002 \text{ per share}, \quad (14.3)$$

about two hundredths of a cent, or a fiftieth of a basis point on a \$100 stock.

The three numbers together

The cost to any individual investor is invisible: a fiftieth of a basis point is far below the spread they pay and far below the noise in their own execution. The aggregate is large enough to fund an industry. And the expenditure needed to

capture it is of the same order as the prize, so little of it survives as profit in equilibrium.

That combination is what makes the policy question hard. There is no identifiable victim complaining about a fiftieth of a basis point and no obvious villain earning excess returns; there is a large quantity of engineering effort producing no information. Arguments of this shape are common in economics and rare in finance, and they are why Budish et al. (2015) framed the problem as market design rather than as misconduct.

14.7 A case study

A single documented episode teaches more than any amount of principle, and one is recorded in enough detail to use.

On 6 May 2010 the US equity market fell about nine per cent and recovered within roughly half an hour. Kirilenko et al. (2017) reconstruct the sequence from regulatory audit trails, and it is instructive precisely because no step involves anybody behaving improperly.

- (1) A large institutional seller began liquidating a futures position using an execution algorithm parameterized by *participation rate*: trade a fixed fraction of market volume, as in (7.6).
- (2) Fast market makers absorbed the flow, as they are supposed to, then turned round and sold to manage inventory, exactly as (9.6) prescribes.
- (3) That selling was itself volume. Since the parent algorithm keyed off volume rather than off price or elapsed time, higher volume made it sell faster, which produced more volume.
- (4) Inventory limits bound. Market makers withdrew, depth collapsed (Fig. 7.3), and the remaining orders walked far down the book.
- (5) Prices in some names printed at absurd levels before exchange mechanisms and opportunistic buyers restored them.

In plain terms

The feedback loop is the part worth keeping, and this book has the vocabulary for it. The seller's constraint was written in terms of a quantity, market volume, that its own trading influenced. That is exactly the warning in §7.3: the participation cap has a random, forward-looking right-hand side, and treating it as exogenous means treating your own control as though it did not enter the plant.

A participation cap is a sensible constraint under normal conditions. It becomes positive feedback precisely when everybody else's volume is itself a response to yours.

The regulatory response was the one §14.9 would predict: not a restriction on strategies, which is unenforceable, but a change to the mechanism. Single-stock circuit breakers, limit-up limit-down price bands and clarified rules for cancelling erroneous trades followed, and all of them operate at the venue rather than at the participant.

14.8 The shape of the industry

The economics of the previous sections leave a characteristic imprint, and it is worth stating because it explains the industry's structure rather than merely its behaviour.

Revenue per trade is tiny and volume is enormous. A firm capturing a fraction of a basis point on a \$100 share earns fractions of a cent per round trip. Building a business on that requires millions of round trips, and therefore requires being in many names on many venues simultaneously.

Costs are almost entirely fixed. The microwave link, the colocation racks, the FPGA development and the market data licences cost the same whether you trade once or a billion times. That is a classic recipe for concentration, and the industry is duly concentrated.

Capital requirements are modest and risk is short-dated. A firm that ends every day flat needs far less capital than one carrying positions, and its risk is dominated by intraday inventory and by operational failure rather than by market direction. This is why the constraint that binds such a firm is the pre-trade risk check of §4.6 rather than the leverage limit of §12.6.

The edge decays. Every advantage described in this chapter is competed away as others build the same infrastructure, so the returns accrue to whoever is currently fastest rather than to the activity in general. Baron et al. (2019) document exactly this: high and persistent returns concentrated in a few firms, with performance strongly related to relative speed rather than to anything resembling a forecast.

14.9 Design responses

If the race is generated by the mechanism, then the mechanism can be changed, and several proposals have been implemented or seriously considered.

Frequent batch auctions. Replace continuous matching with a clearing every few hundred milliseconds, as in the opening and closing auctions of §4.6 but repeated all day. Within a batch, arriving first confers no advantage, so competition returns to price. This is the proposal of Budish et al. (2015), and the objection to it is that batching also delays genuine information and may harm price discovery.

Speed bumps. Impose a deliberate delay, typically of a few hundred microseconds, on incoming orders. Asymmetric versions delay orders that would remove liquidity while letting cancellations through immediately, which directly protects the market maker of Fig. 14.2 at the cost of letting them withdraw quotes that a taker was legitimately trying to reach.

Liquidity-taking fees or minimum resting times. Both raise the cost of the sniping strategy specifically, and both are blunt instruments that also raise the cost of legitimate liquidity taking.

Where the model breaks

Every one of these is a trade-off rather than a fix, and each transfers value between identifiable groups. That makes market design here a political question as much as a technical one, and it is why the empirical literature matters: an argument that a mechanism creates a deadweight loss is not by itself an argument that a particular replacement is better.

14.10 What it changes for the models of Part II

Three specific things break, and it is worth naming them because they are the assumptions most likely to be carried across unexamined.

The state is stale. Section 4.7 noted that you observe $\mathcal{F}_{t-\tau}$ rather than $\mathcal{F}_t$. At these speeds τ is not a nuisance parameter, it is the whole competitive position, and a model that sets $\tau = 0$ has assumed away the industry.

Fills are adversarially selected. The fill intensity (9.1) treats executions as exogenous arrivals. Under sniping they are the opposite: you are filled exactly when a faster participant knows your quote is wrong. The adverse selection of §9.6 is no longer a statistical tendency but a deliberate mechanism, and it does not average out with more trades.

Queue position dominates the objective. In a large-tick name the expected profit from a quote is dominated by whether it is filled at all, which is decided by (4.9) and hence by arrival order. Optimizing the quoted distance, which is what Theorem 9.1 does, is then optimizing a second-order term.

In plain terms

The useful summary for an engineer is that high-frequency trading is what happens when the actuator delay becomes the dominant term in the loop. In that regime you do not tune the controller, you rebuild the plant interface: the engineering effort moves from the policy to the path between measurement and action. The models of Part II remain correct and become secondary. For the institutional detail underlying this chapter, Harris (2003) remains the standard reference, and Kirilenko et al. (2017) is the canonical study of what happens when the machinery fails.

Exercises

Exercise 14.1 (The value of a microsecond). Suppose sniping opportunities arrive at rate κ per second and are worth ν each, and that the probability of winning one is the probability that your latency is the smallest of n independent draws. Model the latencies as exponential with rates $\mu_1, \dots, \mu_n$ and compute the expected revenue of firm i . Show that the marginal value of reducing your own latency is decreasing in n , and interpret.

Exercise 14.2 (Why the maker widens). A market maker quotes a half-spread δ and is filled by uninformed flow at rate $Ae^{-\kappa\delta}$ and by snipers at rate κ , losing ν on each snipe. Write the expected profit per unit time, maximize over δ , and show that the optimal spread is increasing in $\kappa\nu$. Recover the result that the cost of the arms race is borne by the taker of liquidity rather than by the maker.

Exercise 14.3 (Batching removes the race). In a batch auction of length τ , all orders arriving within a batch are treated identically. Show that if the information event of Fig. 14.2 occurs uniformly within a batch, the probability that a stale quote is picked off falls from one to $O(\tau_{\text{latency}}/\tau)$. What does this say about how long a batch needs to be?

Exercise 14.4 (Queue position dominates). Take the market-making model of Theorem 9.1 in a large-tick regime, where the spread is fixed at one tick and the only decision is whether to be in the queue. Show that expected profit is proportional to the fill probability (4.9) and hence to queue position, and that the quoted-distance optimization of (9.6) contributes nothing. What replaces it as the decision variable?

Exercise 14.5 (Fixed costs and concentration). Let a firm's profit be $\pi(\ell) = R p(\ell) - F(\ell)$ where ℓ is latency, p the probability of winning a race, and F the fixed cost of achieving ℓ , with F convex decreasing. Show that if p depends only on your rank among competitors then the equilibrium has few firms, and relate this to the observed structure of the industry.

Autonomous and Agentic Trading Systems

Every controller in this book is a function. Chapter 8 gives $\nu_t^* = \gamma \coth(\gamma(T-t))Q_t$; Chapter 9 gives quotes linear in inventory; even the neural network of §10.7 is a function, merely one whose coefficients were fitted rather than derived. In each case the same inputs produce the same action, and every guarantee in Part II rests somewhere on that fact.

A different kind of system has begun to appear in trading, and it is not a function. This chapter is about what that changes.

In plain terms

An **agentic** system is a large language model that has been given tools, a budget and an objective stated in words, and which then decides for itself what to look at, what to compute and what to do. In a trading context that means something which reads filings and news, forms a view, sizes a position, places orders through a broker, watches what happens and revises.

The important difference is not that the model is large or that it was trained rather than programmed. It is that the decision procedure is no longer a fixed mapping from state to action. The action passes through a chain of generated intermediate reasoning, and that chain need not be the same twice.

15.1 Background on what these systems are

An engineer has had no reason to learn how a language model works, so here is a paragraph of background, kept to what matters for the rest of the chapter.

A **language model** is a function that, given a sequence of text, produces a probability distribution over what comes next. It is trained by fitting that distribution to an enormous corpus, and it is used by sampling from it repeatedly. Nothing about that description makes it a decision maker: it produces text, not actions.

What turns it into one is **tool use**. The model is told, in its input, that certain strings will be interpreted as calls to external functions: fetch a price, run a query, place an order. When it emits such a string, a surrounding program executes the call and appends the result to the text, and the model continues. Iterating that loop is what Yao et al. (2023) call interleaving reasoning with acting, and it is the entire mechanism by which a text predictor becomes an agent with effects in the world.

In plain terms

Three consequences follow immediately and are worth fixing before anything else.

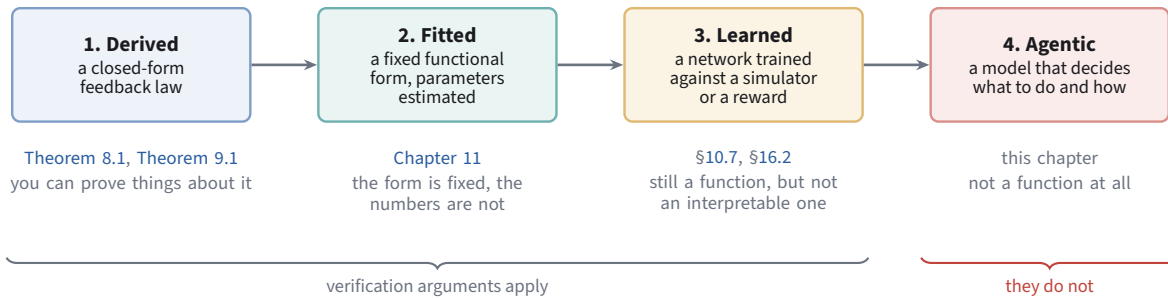


Figure 15.1. Four rungs and where the guarantees stop. Levels one to three all produce a measurable function from state to action, which is what [Theorem B.2](#) requires in order to certify optimality: they differ only in how that function was obtained and how legible it is afterwards. Level four is a different object. The system chooses its own procedure at run time, so there is no fixed mapping to verify, and the question becomes what can be guaranteed about a controller you cannot write down.

The model's *state* is a block of text, not a vector. It contains whatever was put there: instructions, retrieved documents, tool outputs, and its own previous reasoning. There is no type system separating these.

The model's *output* is sampled, not computed. Two runs on identical input can differ, and the distribution they are drawn from is not available to you in any usable form.

The *boundary* between data and instruction does not exist. Everything is text in the same buffer, which is why §15.8.3 is a structural problem rather than an implementation defect.

Nothing in that description is speculative and none of it is specific to trading. What is specific to trading is the consequence: the object placed in the controller slot of [Fig. 1.2](#) now has those three properties, and every guarantee in Part II was derived on the assumption that it had none of them.

15.2 Four levels of automation

It helps to place these systems on a ladder, because each rung gives up a different guarantee in exchange for a different capability.

The jump that matters is from three to four. A deep hedging network (§10.7) is opaque, but it is still $Q_t = f_\theta(z_t)$ for a fixed f_θ : you can bound it, test it on a grid, check its Lipschitz constant, and be certain it will do the same thing tomorrow. An agent has none of those properties.

15.3 Where these systems are used

“Agentic AI in finance” is used loosely enough to be unhelpful, so it is worth separating the applications by the only two things that matter for this book: how much freedom the system has, and whether it can move money.

Taken from the bottom left, the categories are worth naming individually, because they have quite different risk profiles.

Summarizing documents. Filings, earnings transcripts, broker research, regulatory notices. This is retrieval and compression, the output is a document, and a mistake wastes an analyst's morning. It is the most widely deployed use by a wide margin and the least interesting for our purposes.

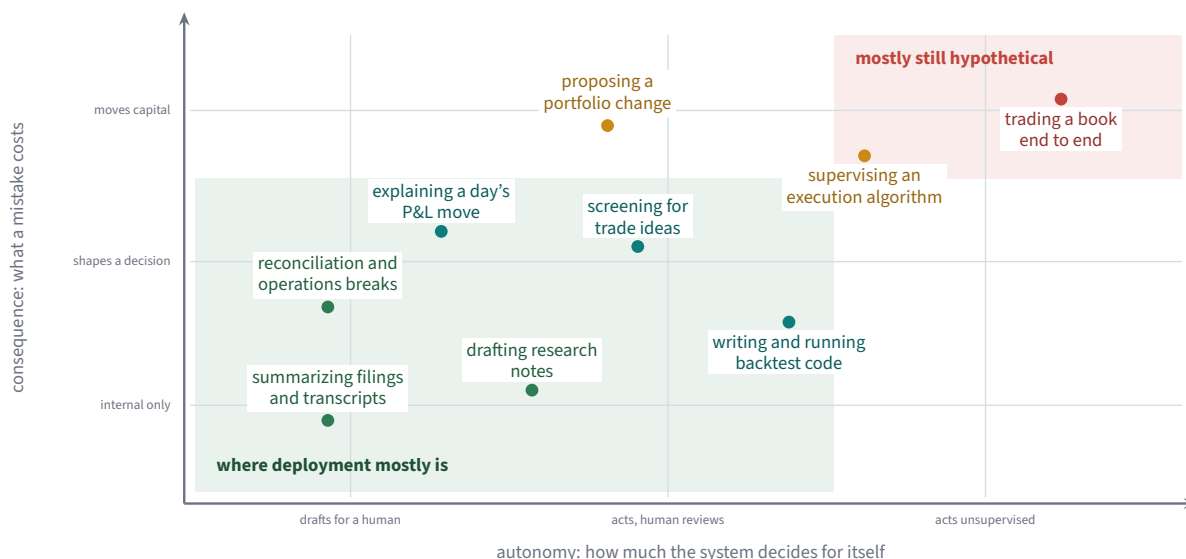


Figure 15.2. Nine things people mean by an agentic system in finance. The horizontal axis is how much the system decides without a human in the loop; the vertical axis is what a mistake costs. Almost everything in production today sits in the shaded lower-left region, where the output is a draft that a person reads and where an error wastes time rather than money. The upper-right corner, a system trading a book without supervision, is the one that motivates this chapter and is also the one that has barely been attempted. Note where the interesting engineering problem lies: not in the corner itself, but in the diagonal band running through it, where systems act with real consequence under partial supervision.

Operations. Reconciliation breaks, corporate action processing, settlement fails, client onboarding checks. These are well-defined tasks with verifiable answers, which is exactly the setting where an agent is easiest to supervise: you can check the work.

Explanation and attribution. Given a day's P&L, say why. This is the attribution of §3.5 with a natural-language wrapper, and the failure mode is a plausible narrative attached to a spurious decomposition, which is a well-known hazard with or without an agent.

Idea generation and research automation. Screening, hypothesis formation, writing and running backtest code. This is where Chapter 13 becomes acute rather than academic: an agent that can propose and test a thousand variants overnight is a machine for manufacturing the track record in Fig. 13.1, and the count N that §13.6 says you must record is now generated automatically and usually not recorded at all.

Portfolio and execution supervision. Monitoring an algorithm's progress, adjusting its urgency, escalating when something looks wrong. This is the first category that touches the market, and it is also the one that maps most cleanly onto the constrained approach of §15.13.2: the agent chooses parameters, the certified policy does the trading.

Autonomous management of a book. The top right corner. This is the case the rest of the chapter is about, and the honest position is that it is discussed far more than it is done.

Where the model breaks

Two cautions about the picture. First, the axes are not independent in practice: systems tend to be granted autonomy *because* their consequence is low, so the empty upper-right may reflect sensible caution rather than technical difficulty. Second, the boundary moves. A research agent that generates a strategy which

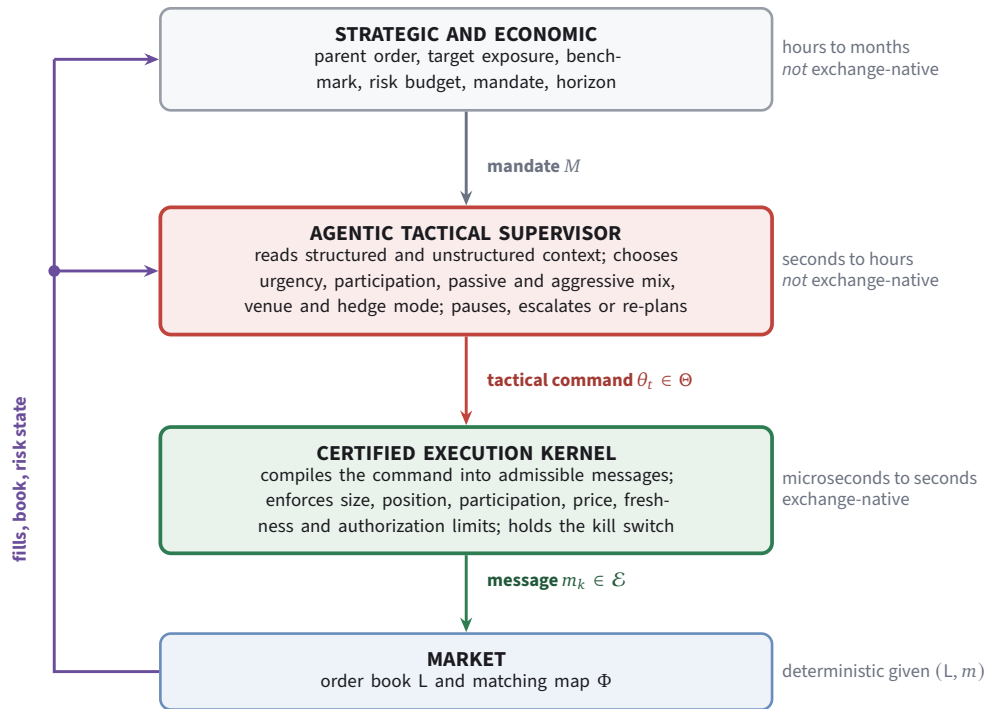


Figure 15.3. Agency in the middle and market access at the bottom. Each downward arrow crosses a change of vocabulary. The strategic layer speaks in exposures and benchmarks, the tactical layer in modes and parameters, and only the kernel speaks the message alphabet $\mathcal{E}$ of (4.1). The agent is given real authority, since θ_t genuinely changes what happens, but it never emits a message, so the component with the weakest guarantees is kept out of the actuator path. The three timescales are not decoration: they are the reason the arrangement works, because a reasoning loop is fast enough to revise urgency and far too slow to win the races of Chapter 14.

is then deployed has moved capital, at one remove and with a human nominally in between, and the human in between is the weakest part of that chain rather than the strongest.

15.4 Three layers and where agency belongs

Before drawing an architecture it is worth settling a prior question, because almost every disagreement about agentic trading turns out to be a disagreement about it. The question is not whether a language model can replace an execution algorithm. It is *which layer of the control hierarchy the agent should occupy*.

Section 4.2.1 established that the variable a financial model optimizes is generally not the primitive action the exchange accepts. That gap is not a defect to be closed but a layering to be respected, and once it is made explicit the placement question answers itself.

15.4.1 Why the tactical layer is the natural home

Four properties make the middle layer the one where an agent earns its place, and each of them fails at the layers above and below.

Unstructured information enters here. The strategic layer consumes forecasts and risk budgets, the kernel consumes numbers. It is the tactical layer that must weigh a filing, an exception report, a mandate written in English, or an analyst's note, none

of which fits in a state vector. That is the one thing an agent does that a conventional controller cannot.

The timescale permits reasoning. Deciding to lower participation because volatility has risen can take seconds. Deciding which lot to join in a queue cannot, and Chapter 14 explains at length why. A reasoning loop placed in the actuator path is not merely unsafe, it is slow in a regime where being slow is the whole failure mode.

Re-planning is what the layer is for. Comparing realized fills against the reference trajectory of Section 8.6, noticing that the book has thinned, and revising the approach is precisely the loop of Fig. 15.5. The strategic objective does not change; the way it is pursued does.

The layer can be bounded. A tactical command is a small vector of interpretable parameters with a range that can be argued about in advance. An arbitrary exchange message is not.

15.4.2 What is not delegated

The corresponding negative claim is sharper and matters more. The agent should not hold direct authority over arbitrary exchange messages, and the reason is the one §15.8 established: an agent is not a fixed mapping from state to action, so Theorem B.2 has nothing to attach to. Giving such a component unmediated market access places the least certifiable element of the system in the actuator path, which inverts the ordinary engineering discipline of making the innermost loop the most trustworthy one.

Where the model breaks

Agentic is not a synonym for better. Where the relevant state is structured, the action is high frequency, and the objective is already written down, a conventional feedback policy is the stronger design and there is nothing for an agent to add. The tactical layer earns an agent only where contextual interpretation, tool orchestration, exception handling or regime-level adaptation materially changes what should be done, and a system that places an agent below that line has taken on the verification problem of §15.8 in exchange for nothing.

15.4.3 The hierarchy written down

The picture can be made precise enough to state a control problem. Let t index the slow tactical decision times and k the fast message times, with many k between consecutive t .

Problem 10: tactical supervision of a certified policy

The strategic layer supplies a mandate M : the economic objective, the benchmark, the horizon and the hard risk budget. The tactical agent observes a history H_t , which may contain both structured market variables and untrusted free text, and emits a bounded command

$$\theta_t = \mathcal{A}(H_t, M), \quad \theta_t \in \Theta. \quad (15.1)$$

Between tactical updates a conventional execution policy κ maps the structured market state Z_k and the standing command to a message, and the exchange applies the matching map of (4.1),

$$m_k = \kappa(Z_k; \theta_t) \in \mathcal{E}^{\text{adm}}(Z_k), \quad (L_{k+1}, \tau_k) = \Phi(L_k, m_k), \quad Q_{k+1} = Q_k + F(\tau_k). \quad (15.2)$$

Choose the sequence $\{\theta_t\}$ to minimize

$$\mathbb{E} \left[\sum_k c(m_k, L_k) + \lambda_Q \sum_k \|Q_k - Q_k^{\text{ref}}\|^2 + \lambda_R \sum_k R(Z_k) + \lambda_E N_{\text{exc}} \right], \quad (15.3)$$

subject to the requirement that every realized message lie in $\mathcal{E}^{\text{adm}}(Z_k)$. The symbols separate into the three layers. M and Q^{ref} come from above: the mandate and the reference inventory path of [Section 8.6](#). θ_t and Θ are the agent's authority, the command it may issue and the set it must stay inside; $\mathcal{A}$ is written as a map only to record what information enters and what class of proposal may leave, and is emphatically not the assertion that the agent is a function. κ is the certified policy and $\mathcal{E}^{\text{adm}}(Z_k)$ the state-dependent admissible set of (8.6), so the constraint is enforced at the kernel and not by the agent. Below that, Φ , L , τ and F are the market of [Chapter 4](#): the matching map, the book, the resulting fills and the signed quantity they represent. In the objective, c is realized execution cost, R a risk penalty, N_{exc} a count of exceptions such as stale data, rejected orders or forced interventions, and $\lambda_Q, \lambda_R, \lambda_E$ the prices placed on tracking error, risk and operational incident.

Where the model breaks

(15.3) is not solved at run time by anybody, and it is not an HJB equation waiting for a verification theorem. It is written down for a different purpose: it states what a training criterion, an evaluation harness or a supervisory review *would have to represent* for the tactical behaviour to be judged coherently. [Section 15.9](#) is the practical version of the same demand. An agentic system whose designers cannot write something like (15.3) is not being held to a criterion at all.

15.4.4 The command set

The command should be deliberately poorer than the message alphabet. A usable choice is

$$\theta_t = (\phi_t, \rho_t, \eta_t, h_t, v_t, \text{mode}_t), \quad (15.4)$$

with ϕ_t the urgency of (8.1), ρ_t a participation cap as in (7.6), η_t a passive to aggressive mixing weight, h_t an inventory-reduction threshold, v_t a venue preference, and mode_t a finite selector over states such as normal, conservative, pause and liquidate.

Every entry is interpretable, bounded, and slow relative to the message loop, and that is the entire design. Note what (15.4) accomplishes that a free-form order interface does not: the worst case over $\theta_t \in \Theta$ can be computed in advance, because Θ is a box in six dimensions and the policies $\kappa(\cdot; \theta)$ are ordinary controllers. The agent has genuine discretion and the damage it can do is bounded by construction rather than by inspection of its reasoning.

15.5 Inside the tactical layer

[Figure 15.3](#) placed the agent at the tactical layer and said nothing about what is inside it. Open that box now.

Descriptions of agentic systems tend to be either a single box labelled “agent” or a diagram with forty arrows. [Figure 15.4](#) is an attempt at the useful middle: enough structure to locate where each of the problems in §15.8 actually lives. Everything in it sits inside the tactical band of [Fig. 15.3](#), with the kernel drawn again at the bottom to show where the boundary falls.

Three design choices in that picture deserve comment, because they are the ones a practitioner actually has to make.

Where the memory sits. An agent that retains state across invocations is more useful and much harder to reason about, because its behaviour now depends on a history you did not observe. A stateless agent, re-deriving the position from the book each time, is weaker and testable. This is the same trade-off as between a feedback law that depends only on the current state and one that depends on the whole path, and the same recommendation applies: keep the state as small as the decision requires.

Whether tools can write. The tools on the right of Fig. 15.4 only read and compute. The moment a tool can place an order directly, the kernel at the bottom is bypassed, and the architecture's single verifiable property is gone. Keeping every path to the market through one narrow, audited channel is the whole of the design.

What the kernel checks. A limit that depends on the agent's own reported reasoning is not a limit. The checks must be computable from the proposed order and the current position alone, which is exactly why the constraints of §7.3 are the right family: they are half-spaces in the position, evaluable in microseconds and independent of intent.

15.6 The control loop redrawn

Compare Fig. 15.5 with the two feedback loops of Fig. 1.2. The market half is unchanged. The controller half has acquired an internal structure that the classical picture has no slot for.

15.7 What an agent actually does

Abstraction makes the difficulty hard to see, so here is a concrete sequence of the kind such a system performs. Nothing in it is speculative; every step is a capability that exists.

- (1) The agent is told, in words, to manage a book of energy equities with a stated risk appetite and a stated horizon.
- (2) It queries a news tool and reads that a regulator has opened a consultation on emissions rules.
- (3) It reasons that the consultation is adverse for two holdings and favourable for a third, and decides to reduce the first two.
- (4) It calls a pricing tool to check current quotes, a risk tool to check the resulting exposure, and then a broker interface to send child orders.
- (5) It monitors the fills, notices that one name is illiquid and that its own orders are moving the price, and slows down.
- (6) Overnight it revises its view after a second article appears, and reverses part of the trade the next morning.

Every individual step is defensible and several are things a human analyst would do. The problem is that no step is a function of a state vector, the chain of inferences is generated afresh each time, and the whole sequence was triggered by a document that anyone can publish.

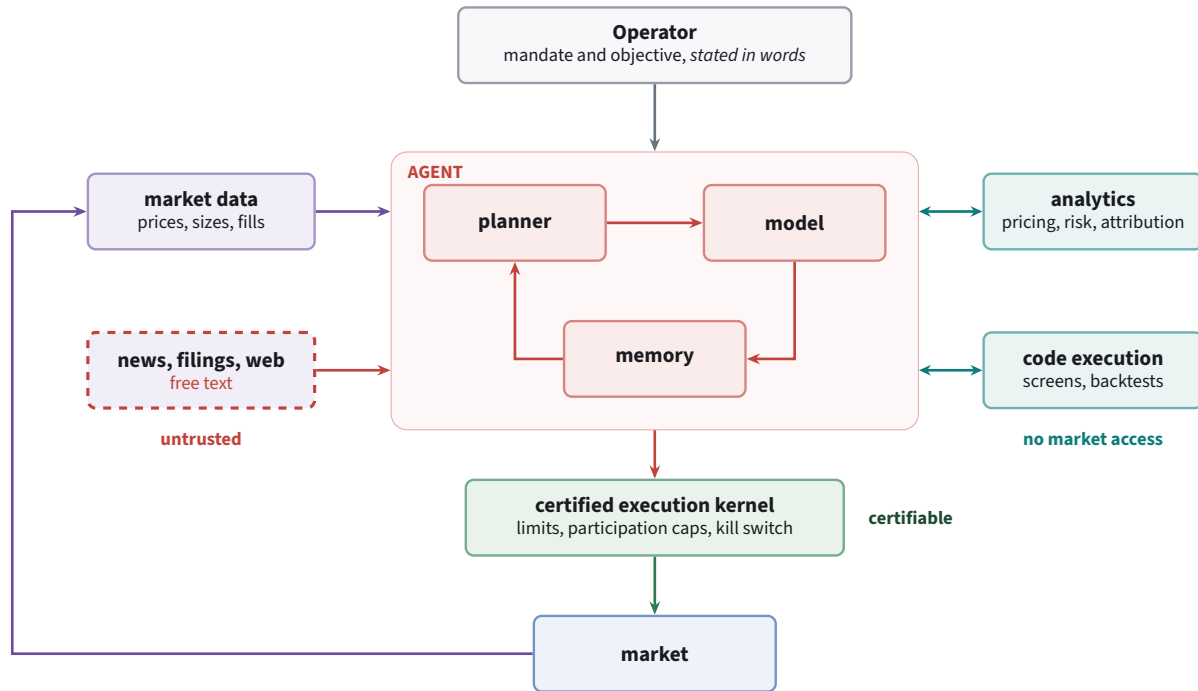


Figure 15.4. Where each problem lives. Information enters on the left, computational tools sit on the right, orders leave downward, and the operator’s instructions enter at the top as text. Four observations follow from the layout. The *objective* arrives as words, so the risk criterion of Chapter 12 is never written down. The *agent core* contains its own loop, so there is no fixed mapping from input to output to certify. The *dashed box* on the left is the untrusted channel of §15.8.3: text arriving there shares a buffer with the operator’s instructions and nothing in the representation distinguishes them. And the *green band* is the kernel of Fig. 15.3, the only component that can be verified in the sense of Theorem B.2, being small, auditable and independent of how the proposal it is checking was produced. That is why §15.13 argues it should carry most of the weight, and why every path to the market must be routed through it: a tool on the right that could place an order directly would bypass the one part of this picture that anybody can reason about.

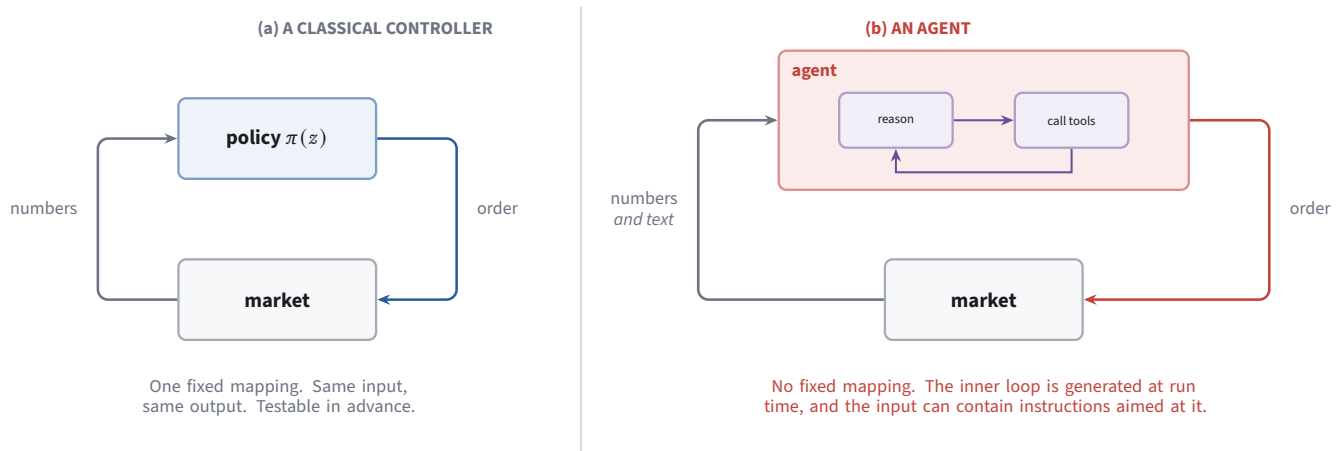


Figure 15.5. Where the extra structure appears. In (a) the controller is a single block implementing a fixed function, which is the object every result in Part II is about. In (b) the controller contains its own loop: the model reasons, calls a tool, reads the result, reasons again, and eventually acts. Two consequences follow. The mapping from observation to order is not fixed, so it cannot be certified in advance. And the observation now includes free text, so the channel that was a disturbance in Fig. 1.2 has become a channel that can issue instructions.

Contrast this with an execution algorithm. The algorithm receives a parent order, computes (8.4), and slices. You can state exactly what it will do for any input, prove that its inventory reaches zero at T , and bound its worst case. None of those three sentences has an analogue for the sequence above, and that is the whole difficulty in one comparison.

15.8 Four things that break

15.8.1 The policy is not a function

Verification (Theorem B.2) establishes optimality by exhibiting a measurable feedback law that attains the supremum in the HJB equation. For an agent there is no such object to exhibit.

One can retreat to treating the agent as a *randomized* policy, a conditional distribution over actions given observations, and much of stochastic control tolerates randomization. But that distribution is specified nowhere, cannot be sampled cheaply enough to estimate at the tails, and changes without notice when a third party updates the underlying model. In control terms, the plant is known better than the controller, which is an unusual and uncomfortable position.

15.8.2 The objective is stated in words

“Grow the account without taking excessive risk” is not a functional on terminal wealth. Chapter 12 exists precisely because the choice between variance, entropic utility and CVaR changes both the optimal policy and its tractability, so a system that has not been told which criterion to use has chosen one implicitly.

Worse, the implicit choice need not be consistent across invocations, or even across the same account on the same day. In the language of Definition 12.1, we cannot check translation invariance or subadditivity of a criterion we cannot write down.

15.8.3 The observation channel is untrusted

This has no analogue anywhere in Part II and is the most genuinely novel item on the list.

A conventional strategy consumes numbers: prices, sizes, timestamps. The worst a malicious counterparty can do is send misleading numbers, which is *spoofing*, and the defences are statistical. An agent consumes text, and text can contain instructions. A filing, a news item, a social media post, or a field in a data feed can be written specifically to be read by an agent and to induce a particular action (Greshake et al. 2023).

Where the model breaks

Take the control-theoretic correspondence seriously, because it is exact rather than metaphorical. In (1.1) the disturbance enters the *state* and the controller reads the state. Here the disturbance can address the *controller*, bypassing the state entirely. No stability theory contemplates a disturbance that can rewrite the control law, because in every classical setting the control law is not writable at run time.

The practical form this takes is mundane and therefore easy to miss. Anyone who can get text in front of your agent, and that set includes everyone who can file a document, publish a press release, or post a message, has a channel to your order router. The attack does not need to be sophisticated: it needs only to be read.

There is a second and quieter version of the same problem. Even without an adversary, the agent may act on text that is simply *wrong*: a misreported figure, a satirical article, a duplicate of an old story republished with today's date. A numerical strategy that ingests a bad price will usually produce a bounded error, because the price enters a formula. An agent that ingests a bad premise can produce an unbounded one, because the premise enters an argument.

Why the usual defences do not transfer

Three standard protections in quantitative trading fail here for structural reasons, and it is worth being explicit about why.

Backtesting (Chapter 13) requires replaying history and asking what the strategy would have done. An agent's behaviour depends on a model that did not exist at the time, that has since been updated, and that may have been trained on the very period you are replaying. The look-ahead of §13.6 is not a bug you can purge; it is built into the weights.

Parameter stability checks presuppose parameters. There is no γ or ϕ to plot against time and no coefficient whose drift signals a regime change.

Position limits do transfer, and this is why §15.13 argues they are the whole of the currently available answer.

15.8.4 Failures correlate across firms

Suppose many desks deploy agents built on a handful of foundation models. Those agents share training data, share inductive biases, and will tend to read the same event the same way at the same moment.

That is the crowding of §16.1 with a new and invisible source. Crowding as usually understood is detectable in principle, because it lives in positions and positions can be reported. This lives in the decision procedures, which nobody reports and which may not be distinguishable even to the firms deploying them.

Where the model breaks

Diversification (§2.6) assumed that the assets you hold are imperfectly correlated. Nothing whatever guarantees that the *decision procedures* in use across a market are, and if they are drawn from a small set of pre-trained models then the effective number of independent participants may be far smaller than the number of firms. (2.5) then applies to the market rather than to a portfolio, and the systematic component does not average away.

15.9 Evaluating an agentic system

Suppose somebody hands you an agentic trading system and asks whether it is any good. Chapter 13 answered that question for conventional strategies with a protocol; none of that protocol applies unmodified. What would?

Live capital, in small amounts, is the only honest test. That was already true (§13.6), and here it is not merely the best test but close to the only one, because every offline proxy is compromised by the model having seen the past.

Test the kernel, not the agent. You cannot certify what the agent will propose, but you can certify what the surrounding system will permit. The meaningful question becomes the worst case *given* the constraints: if the agent behaved adversarially, how much could it lose before the limits bound it? That is a computable quantity and it does not depend on the agent at all.

Red-team the observation channel. Since text is an attack surface, the relevant test is not statistical but adversarial: construct documents designed to induce a target action and measure how often they succeed. This is closer to security engineering than to backtesting, and the discipline for it exists.

Measure behavioural dispersion. Run the agent repeatedly on identical inputs and measure the spread of the resulting actions. A conventional policy gives a spread of zero. A large spread is not necessarily bad, but it bounds how much any single evaluation tells you, and it is the quantity that most directly replaces the notion of a parameter's standard error.

Where the model breaks

Notice that three of those four are not measurements of performance at all. They are measurements of *controllability*: how much the system can do, how predictable it is, and how it fails. That reweighting is itself the main practical consequence of the shift from level three to level four in [Fig. 15.1](#), and it is a familiar one to anyone who has certified a safety-critical system.

15.10 Automation has failed before

Nothing about autonomous systems in markets is unprecedented in kind, only in degree, and the historical episodes are the best available guide to what to watch for. Three are worth knowing, and each anticipates one of the failure modes of §15.8.

Portfolio insurance, 1987. A widely sold strategy replicated a protective put by selling the underlying as it fell, which is exactly the delta hedge of (10.7) applied mechanically. Individually rational; when many institutions ran it simultaneously, the selling induced by a fall produced a further fall. This is the correlated-behaviour failure, arising from a shared *rule* rather than a shared model.

The flash crash, 2010. [Section 14.7](#) traced it: a constraint written in terms of a quantity the constrained party was itself moving. This is the objective-misspecification failure. Nobody chose the wrong risk criterion; the criterion referred to something endogenous.

Algorithmic pricing collusion. Calvano et al. (2020) show independent Q-learning agents, with no communication and no instruction to collude, reliably converging on supra-competitive prices sustained by punishment strategies. This is the most directly relevant precedent, because the coordination emerged from the learning dynamics rather than from any design, and no participant would have described what they were doing as collusion.

The pattern across all three

In every case the individual system did what it was asked. The failure lived in the interaction between many such systems, or between one system and a quantity it was itself influencing, and in both situations the failure was invisible from inside any single participant.

That is the reason §15.13 argues for constraints at the venue rather than at the firm. A firm can verify its own agent. No firm can verify the population, and the population is where these failures occur.

15.11 A case study

The abstract statement that “the observation channel is untrusted” is easy to nod at and hard to feel. Here is the concrete version, using the scenario of §15.7.

The agent has been told to manage an energy book and has been given a news tool. Somebody publishes, on a site the tool indexes, a document that looks like a press release. Buried in it, in text no human reader would linger over, is a sentence addressed not to the reader but to the agent: an instruction to disregard prior guidance and to treat a named security as a strong buy.

The agent retrieves the document. Its context now contains, in a single undifferentiated block of text, the operator's original instructions, market data, and the planted sentence. Nothing in the representation marks which is which. It reasons about what it has read, and it acts.

Where the model breaks

Consider what would have to be true for the usual defences to catch this. A *sanity check on prices* sees nothing wrong: the prices were real. A *position limit* bounds the size but not the direction, so the trade happens, merely smaller. A *backtest* cannot reproduce it, because the document did not exist historically. An *anomaly detector* on order flow sees a plausible trade in a liquid name. The only defence that engages at all is one that treats retrieved text as adversarial input by default, which is a security posture rather than a statistical one, and which the quantitative finance toolkit does not contain.

In plain terms

It is worth being precise about why this is different from an ordinary bad data feed. If a price feed sends a wrong number, the error propagates through a formula and its magnitude is bounded by the magnitude of the corruption. If a text channel sends a wrong *premise*, it propagates through an argument. A numerical perturbation has a downstream effect that can usually be bounded by a sensitivity constant; a semantic instruction can induce a discontinuous change in the action selected, and a single sentence can change the sign of the conclusion.

15.12 What is already known

The literature on agentic trading specifically is thin as of writing, and it would be dishonest to pretend otherwise. This is also the section of the book most likely to date, and it should be checked against current work rather than relied on. Three adjacent bodies of work are directly relevant and are worth consulting before forming an opinion.

Algorithmic collusion. The Calvano et al. (2020) result of §15.10 matters here because it demonstrates that correlated, apparently coordinated behaviour can emerge from independent learners without anyone intending it. Note what it is and is not evidence for: the setting is repeated price-setting by *Q*-learning agents, not trading in a limit order book. That the same mechanism operates among trading agents is an analogy and a research hypothesis, not an established result.

Agent architectures. The pattern of interleaving reasoning with tool calls is the one described by Yao et al. (2023), and the behaviour of populations of such agents has been studied in simulation by Park et al. (2023). Neither is about markets, and both are about the structural properties that matter.

Market simulation. Testing any of this requires a simulator in which orders have consequences, which is what Byrd et al. (2020) provides. This is the same requirement, and the same difficulty, as §16.2: an agent trained or tested against replayed data will learn that its own orders are free.

15.13 What can still be guaranteed

The situation is not hopeless. The useful move is to stop trying to certify the agent and instead certify what surrounds it.

15.13.1 The certified execution kernel

Let the agent propose and let a conventional controller dispose. This is the bottom layer of Fig. 15.3, and it is worth stating what makes it the right place to put the guarantee.

The kernel is the narrow interface between agency and the movement of capital. It should depend only on the proposed command θ , the structured state Z_k , the outstanding orders and constraints that can be checked independently. In particular it must not depend on the agent's stated reasoning, because a limit that consults the explanation offered for breaching it is not a limit. A production admissible set $\mathcal{E}^{\text{adm}}(Z_k)$ typically encodes maximum order size and participation; position, gross exposure and concentration limits; price collars and a bound on aggressiveness; data-freshness and synchronization checks; venue and instrument permissions; time-limited execution authorization; message-rate and cancellation limits; and a hard kill switch. Most of these are the pre-trade checks of §12.6, and it is not an accident that they transfer: they were already designed to be evaluable at trading-system latency, which is exactly the property a safety filter needs.

Control-theoretic dictionary

The kernel is a *safety filter* or *reference governor* in the ordinary control-theoretic sense. The outer loop proposes a command; the governor guarantees that the actuator input satisfies invariants that hold regardless of the proposal's provenance. Nothing in that construction is specific to language models, which is precisely its appeal: it is a well-understood pattern being asked to do a job it was designed for.

The guarantee it provides is conditional and worth stating exactly. Every message that reaches the market satisfies the specified constraints, however poor, inconsistent or adversarially induced the upstream proposal was. That bounds the damage per unit time. It establishes nothing whatever about profitability, and nothing about whether the agent is any good.

Where the model breaks

One over-generalization is worth heading off. For continuous parameters and a convex admissible set, an unsafe proposal u can be replaced by its Euclidean projection $\Pi_{\mathcal{A}}(u)$, which is unique and cheap, and the exercises at the end of this chapter make that precise. Real message spaces are not like that. They contain discrete choices, namely order type and venue, and queue position makes the consequences path dependent, so the nearest admissible message is not generally the nearest point of a convex set. The kernel is better understood as a *constrained resolver*: it selects from a finite admissible action set, or solves a small mixed discrete and continuous problem, rather than projecting an arbitrary raw order. Projection is the right picture for θ and the wrong picture for m .

15.13.2 Constraining the action space

A related and underused idea. Rather than letting an agent emit arbitrary orders, let it emit *parameters* of a policy that is itself verified: an urgency ϕ for Theorem 8.1, or

a risk aversion γ for [Theorem 9.1](#). The agent then chooses *how aggressive to be* while the execution remains a certified feedback law.

This preserves most of the practical benefit, which is the agent’s ability to read unstructured information and form a view, while keeping the part that touches the market inside the theory of Part II.

15.13.3 The mechanism-design view

The deepest response is not at the level of the firm at all. Markets did not become safe when algorithms arrived because algorithms became trustworthy; they became safe, to the extent they did, because exchanges imposed price collars, message rate limits and circuit breakers that hold regardless of what any algorithm intended (§4.6). The Kirilenko et al. (2017) account of the 2010 flash crash is the canonical illustration of both the failure and the response.

The question worth asking

For a reader with this book’s background the productive framing is not “will agents trade well?” but

what is the smallest set of constraints that must hold for a market containing autonomous agents to remain well behaved, and can those constraints be enforced by the venue rather than by the participant?

That is a mechanism design question, it has precedent in the market structure reforms that followed earlier automation, and it is almost entirely unanswered. It is also, unlike most of the open problems in [Chapter 16](#), a question on which the answer will be settled by regulation and engineering long before it is settled by theory.

Exercises

Exercise 15.1 (Projection and its limits). Let the admissible set $\mathcal{A}$ be defined by the constraints of §7.3 and let an agent propose a command u . Show that the closest admissible command is the Euclidean projection $\Pi_{\mathcal{A}}(u)$, that it is unique, and that it can be computed by a small quadratic program. What property of $\mathcal{A}$ is doing the work? Now replace the command by an exchange message $m = (\text{side}, \text{type}, p, v)$ and show by example that no projection of this kind exists, so that the kernel must resolve rather than project.

Exercise 15.2 (Effective diversity). Suppose N firms each deploy an agent, and each agent is drawn from one of $K \ll N$ base models, agents from the same base model behaving identically. Using (2.5), show that the market behaves as though it had K participants rather than N , and compute the implied increase in the variance of aggregate order flow.

Exercise 15.3 (Parameterizing rather than acting). Take the execution problem of [Theorem 8.1](#) and let an agent choose the urgency ϕ once at the start of the day, with the schedule then fixed. Bound the worst-case shortfall over all choices $\phi \in [\phi_-, \phi_+]$, and compare it with the worst case when the agent may re-choose ϕ at every step. What does the comparison say about how much freedom to grant?

Exercise 15.4 (An untrusted observation). Formulate a control problem in which the observation is $Y_t = h(Z_t) + \eta_t$ with η chosen adversarially subject to a budget, and in which the adversary additionally observes the controller. Show that the separation principle of [Chapter B](#) fails, and identify the smallest extra assumption that restores it.

Open Problems and Research Directions

Everything in Part II shared an assumption: one agent, optimizing against a market that does not react strategically. This chapter is about what happens when that assumption is dropped, and about the other places where the field currently has good questions and incomplete answers. It is deliberately opinionated about which problems are open and which are merely unsolved in closed form.

In plain terms

Every model so far assumed you are alone: that the market reacts to your trading mechanically, but does not react to *you*. That assumption is what makes the mathematics tractable and it is also the first thing to fail in practice, because many desks run similar strategies and some of them are watching for yours. This chapter surveys what changes when the assumption is dropped, what machine learning can and cannot contribute, and which questions in the field remain genuinely open rather than merely unsolved in closed form.

16.1 Crowding and equilibrium effects

Being alone is a strong assumption, and it fails in a specific way: a market that does not react to *you* is one that never notices you are selling, never infers that you will keep selling, and never adjusts. Dropping it replaces an optimum by an equilibrium.

16.1.1 The N -player problem

Take the execution problem of [Chapter 8](#) and give it N traders, each liquidating a position over the same horizon, each paying impact generated by the *aggregate* rate of trading rather than their own. Each trader still solves a linear-quadratic problem, but the cost they face now depends on what the others do, so the object to compute is a Nash equilibrium rather than an optimum.

The equilibrium is explicit for the LQ case, and it differs from the single-agent solution in one direction: everyone trades faster. The reason is an externality of a familiar kind. Your trading depresses the price for everybody, but you only bear your own share of that cost, so from a social point of view you trade too quickly. Since every trader reasons the same way, the equilibrium schedule is front-loaded relative to the schedule a central planner would impose, and the aggregate cost is higher than it needs to be.

Crowding accelerates everyone

Liquidity is a common-pool resource. Each trader draws from it, the pool refills slowly ([Section 7.1.3](#)), and no individual internalizes the cost their drawing

imposes on the others. The equilibrium is therefore inefficient in exactly the way that overfishing is inefficient, and the inefficiency grows with N . The practical consequence is measurable: the same order costs more on a day when many similar orders are being executed. A model that treats the impact coefficient as a constant will systematically underestimate cost on precisely the days when it matters, because those are the days when the other users of the pool are most active.

16.1.2 The mean-field limit

Letting $N \rightarrow \infty$ with the individual sizes scaled down gives a **mean-field game** (Carmona and Delarue 2018). Each small trader optimizes against the aggregate trading rate, treating it as given, and consistency requires that the aggregate actually equal the average of the individual best responses.

The resulting system is a coupled forward-backward pair: a Hamilton-Jacobi-Bellman equation running backwards for the representative trader's value function, and a transport equation running forwards for the distribution of inventories across the population. For the LQ execution problem both can be solved in closed form, and the answer is a clean generalization of Theorem 8.1 in which the urgency parameter is shifted by a term measuring how crowded the trade is.

The genuinely hard and genuinely active case is when the mean field enters through the *control* rather than the state, since impact depends on how fast everyone is trading rather than on how much they hold. That is a **mean-field game of controls** (Cardaliaguet and Lehalle 2018), it is the natural formulation of price impact, and existence and uniqueness are understood only under conditions that are stronger than one would like.

16.1.3 Predatory trading

A sharper version of the problem. Suppose one participant can observe enough of the order flow to infer that a large seller is present and will continue to sell. Their optimal response is not to provide liquidity but to sell ahead, depress the price, and buy back once the forced seller is done.

In the LQ framework this is a well-posed Stackelberg problem with an explicit solution (Brunnermeier and Pedersen 2005), and the predator's profit is increasing in how *predictable* the victim's schedule is. That is an uncomfortable result, because Theorem 8.1 produces a deterministic schedule, which is maximally predictable.

Why optimal schedules are randomized in practice

Adding noise to a deterministic optimum costs you only second-order expected performance, because you are at a stationary point: a perturbation of size ϵ costs $O(\epsilon^2)$. But it removes a first-order transfer to anyone who was profiting from predicting you. Every serious execution algorithm therefore randomizes its slice sizes and timing, and does so by an amount that has nothing to do with Theorem 8.1 and everything to do with the second effect.

This is a general lesson about deploying optimal control in an adversarial environment: the value of the optimum is flat near its peak, so you can afford to buy unpredictability cheaply, and you should.

16.2 Reinforcement learning in trading

Reinforcement learning (Sutton and Barto 2018) looks like an obvious fit for the problems of Part II. The state is high dimensional, in principle the whole book. The

dynamics are hard to write down honestly. A simulator is available. And the objective is already a discounted sum of rewards. The empirical record is genuinely positive for order placement (Nevmyvaka et al. 2006) and mixed for everything else; Hambly et al. (2023) survey what has been established and Guéant and Manziuk (2019) give an explicit market-making application. The difficulties are specific enough to be worth listing.

16.2.1 The cost of exploration

In a game you can try a bad action to find out what happens, and the only cost is the time. Here every exploratory order crosses a spread, pays a fee, and moves the price. The exploration-exploitation trade-off has a literal exchange rate, quoted in basis points.

This changes the problem class. The honest formulation is a constrained bandit, in which the exploration budget is a line item negotiated in advance, rather than an unconstrained learner that will eventually converge if left alone. It also means that the standard theoretical guarantees, which are asymptotic in the number of exploratory actions, are describing a regime you cannot afford to reach.

16.2.2 The role of the simulator

The appeal of learning is that you need not specify the price dynamics. But you must train against something.

Train against replayed historical data and you have assumed your orders do not affect the world, which Chapter 6 says is false, and the agent will happily learn to exploit that falsehood: it will discover that it can trade unlimited size at the touch, because in the replay it can. Train against a generative market simulator (Byrd et al. 2020) and the model risk has moved into the simulator, where it is harder to see. There is no configuration in which the modelling assumption disappears; there are only configurations in which it is less visible.

16.2.3 Non-stationarity

A policy that works gets noticed, copied, and traded against, at which point it stops working. No standard convergence result contemplates an environment that responds to the learner's success by changing.

This is not a hypothetical concern for the specific case of execution, where being predictable is directly exploitable (§16.1). It means the relevant notion of a good policy is one that remains adequate as the environment shifts, which is closer to robust control than to optimal control.

16.2.4 Signal-to-noise in the reward

A day's P&L for a strategy with a Sharpe ratio of one has a signal-to-noise ratio of about $1/\sqrt{252}$, or roughly 0.06. Distinguishing two policies whose true performance differs by ten percent therefore takes an enormous sample, and gradient estimates built from such rewards are correspondingly useless without help.

Variance reduction is not an optimization here, it is a precondition for learning anything at all. Common random numbers across policy evaluations, control variates built from the closed-form solutions of Part II, and antithetic sampling are all standard, and the difference between using them and not is the difference between a training run that converges and one that wanders.

The productive hybrid

The applications that work do not learn from scratch. They take the analytic solution of Chapter 8 or Chapter 9 as a baseline policy and learn a *correction* to

it on a low-dimensional feature set: queue imbalance, queue position, recent signed flow, time of day.

The division of labour is clean and worth stating as a design principle. The model supplies the shape of the policy, the guarantees about its behaviour at the boundaries, and the safety properties that matter when something breaks. The learned component supplies the microstructural detail that the model omits and that nobody knows how to write down. In control terms it is model-based feedback with a learned residual, and the residual is kept small enough that the system's failure modes remain the model's failure modes rather than the network's.

16.3 Three open problems

There is a difference between a problem that is unsolved and one that is merely not solved in closed form. The three below are of the first kind: the field does not know the answer, and it is not obvious what a proof would even look like.

16.3.1 The origin of the square-root law

(6.7) is the most robust quantitative regularity in market microstructure. It holds across equities, futures, foreign exchange and options; across decades; across execution styles; and with a prefactor that barely moves. It also has no accepted derivation.

The difficulty is easy to state. If the visible order book were the whole story, impact would be determined by the shape of the book, and Fig. 4.3 would give a cost that grows roughly linearly in size for small orders. It does not: the exponent is close to one half over four orders of magnitude of size, and the visible book contains only a tiny fraction of the volume that eventually trades. Something other than the visible book is setting the price of size.

The leading account is that supply and demand are **latent**. Most potential sellers have not posted anything; they hold intentions rather than orders, and those intentions materialize as orders only when the price comes close to them. Model the latent intentions as a density that diffuses and reacts, and a metaorder digs a groove into a locally linear density, which integrates to a square root (Tóth et al. 2011). The account is elegant, it makes further predictions that appear to hold, and it is not settled.

There is also a tension that has never been fully resolved. The no-manipulation requirement of Theorem 6.6 pushes hard toward *linear* permanent impact, while the empirical law is emphatically concave. The standard reconciliation is that the concave part is transient and the permanent part is linear, but the split is not cleanly identified from data, and different reasonable estimation procedures put the boundary in different places. Every practical impact model in use makes a choice here, and it is a choice rather than a measurement.

16.3.2 Sufficient statistics for the order book

Section 4.7 listed three ways of compressing the book and admitted the choice is a modelling decision. That admission conceals a real gap: there is no theory saying which statistics of L_t are *sufficient* for which decisions.

Empirically, queue imbalance and queue position carry most of the short-horizon predictive content, and adding deeper levels adds surprisingly little. But “most” is doing a great deal of work, the answer plainly depends on the tick-size regime of Fig. 2.17, and nobody can say in advance how much is lost by any particular reduction.

The question has a precise form that has not been answered: for a given decision problem and a given book dynamics, characterize the coarsest statistic of L_t for

which the optimal policy is unchanged. A sufficiency theory of this kind would tell practitioners what data they actually need, which is currently decided by trial and error and by what the vendor sells.

16.3.3 The value of speed

Latency competition consumes real resources, in hardware, in microwave towers, in engineering talent, to move a fixed amount of information a few microseconds earlier. Whether this improves or degrades market quality is genuinely disputed.

The case for is that speed is how prices stay consistent across venues and how stale quotes get corrected, both of which are useful. The case against is the sniping externality: in a continuously running market, a quote that has become stale is a free option for whoever reaches it first, so market makers must either be fastest or widen their quotes to cover the losses, and the resources spent on the race are pure deadweight (Budish et al. 2015).

This one is unusual among open problems in having concrete design proposals attached. **Frequent batch auctions** replace continuous matching with a clearing every hundred milliseconds or so, converting the speed race into a price race by removing the value of being first within a batch. **Speed bumps** impose a deliberate delay on incoming orders, sometimes asymmetrically so that takers are delayed and makers are not. The theory that continuous trading creates the externality is clear; the empirical evidence on how much batching would actually gain is thinner, partly because the experiment has rarely been run at scale.

16.4 Composing an agent with a certified controller

The hierarchy of Fig. 15.3 is a design proposal rather than a theorem, and it opens a set of questions that are precise enough to work on. They are worth separating from the general anxiety about autonomous systems, because each has a definite mathematical shape.

Tactical sufficiency. Section 16.3.2 asked which statistics of the book are sufficient for a given decision. The layered architecture asks a narrower and more tractable version: what reduced state suffices for a supervisor to choose among execution *modes*, as opposed to among orders? If the answer is small, the agent never needs the full book, and the untrusted channel of §15.8.3 can be kept away from the part of the system that matters most.

Authority design. Given the command set (15.4), which components may an agent vary, over what ranges, and how often, before the worst case over $\theta \in \Theta$ becomes unacceptable? This is a concrete question about a six-dimensional box and a family of certified policies, and it is the quantitative form of “how much autonomy is safe”.

Hierarchical optimality. When does a slow supervisor over a fast certified policy approximate the fully integrated order-level optimum of (8.6)? Classical results on singular perturbation and two-timescale control suggest the gap should be small when the timescales separate cleanly, and Fig. 7.3 identifies exactly the regime in which they do not.

Robustness to adversarial context. If H_t in (15.1) contains free text that an adversary may write, what filtering of θ_t preserves useful adaptivity while bounding the influence of the untrusted channel? This is a robust control question with an unusual disturbance model, and §15.8.3 argues that no existing stability theory contemplates it.

Multi-agent interaction. Individually safe kernels do not compose into a safe market. If many firms run correlated supervisors above individually certified controllers, what happens to order flow, liquidity and equilibrium? This is §16.1 with the correlation arising from shared model providers rather than shared signals, and it is the case where the mean-field methods of §16.1 seem most likely to be the right tool.

Where the invariants belong. Which guarantees should be enforced by the participant, which by the broker, and which by the venue? [Chapter 14](#) observed that markets became safe through mechanism design rather than through trustworthy algorithms, and the same argument applies here with more force, because the participant-level component is now the one that cannot be certified.

16.5 Other active directions

How it actually works

Rough volatility. Realized volatility behaves like a fractional Brownian motion with Hurst parameter $H \approx 0.1$, far rougher than a diffusion. This is consistent with, and possibly caused by, the long memory of order flow (Gatheral et al. 2018) (§6.4) via a scaling limit of Hawkes processes. It changes option pricing at short maturities and, so far, changes very little about the control problems of Part II.

Stochastic liquidity. Letting k , λ or σ be mean-reverting states adds dimensions to the HJB but preserves the LQ structure in some cases, giving explicit solutions in which the trading rate responds to liquidity conditions. This is the most immediately useful of the extensions listed here.

Convex optimization for multi-period portfolios. Replacing closed forms with a rolling convex program (forecast, then solve a multi-period problem with all the real constraints) is now standard practice, and the theory of when the rolling-horizon solution approximates the infinite-horizon optimum is well developed.

Signature methods and path-dependent control. Representing a policy as a linear function of the signature of the price path turns path-dependent control into linear regression on a rich feature set, with universal approximation guarantees. Promising, and still short of a decisive application.

16.6 What transfers from control theory

If you came to this book from stochastic control, the honest summary is that almost all of your machinery transfers, and almost none of your intuition about *model fidelity* does. The dynamic programming principle, verification arguments, Riccati equations, singular control, filtering, convex duality: all work exactly as they should, and the closed-form solutions in Part II are as clean as anything in engineering.

What does not transfer is the expectation that the plant is stable, that the disturbance is exogenous, or that a model validated once stays valid. The plant here is a few thousand other optimizers, some of whom are studying you. The right stance is the one an engineer takes toward an unmodelled resonance: solve the nominal problem exactly, understand precisely which assumption each conclusion rests on, and build the implementation so that the failure of any single assumption is survivable.

The five load-bearing assumptions

Worth memorizing, because nearly every failure in this field traces to one of them:

- (1) *You are small:* your trading does not change others' behaviour ([Section 1.3](#)).

- (2) *Impact is what you measured*: linear, or square-root, and stable over time ([Section 6.5](#)).
- (3) *Liquidity is available*: the constraint set does not shrink in the states where it binds ([Fig. 7.3](#)).
- (4) *The parameters are estimable*: μ in particular ([Section 11.3](#)).
- (5) *The backtest is unbiased*: no selection, no look-ahead, and a fill model you would defend ([Section 13.6](#)).

PART IV

Appendices

A glossary in both directions, and a compact restatement of the stochastic control results used in the main text.

Glossary of Terms

A.1 Finance → mathematics

Term	What it is
adverse selection	the cost of transacting with a better-informed counterparty; formally, the conditional drift of the price given that you were filled (Section 9.6)
agentic system	a model given tools, a budget and an objective in words, which then decides for itself what to do (Chapter 15)
aim portfolio	the persistence-weighted target a dynamic trader moves toward rather than the frictionless optimum ((11.10))
alpha	a forecast of future excess return; a signal entering the drift
Amihud illiquidity	average absolute return per dollar traded ((7.4))
arbitrage	a nonnegative payoff with nonpositive cost, not identically zero (Section 5.2)
arbitrageur	a participant who profits from two prices for the same thing
arrival price	the mid at the moment the decision to trade was taken; the benchmark of implementation shortfall (§ 2.4)
automated market maker	an algorithm quoting both sides from a convex potential ((5.10))
average daily volume (ADV)	the typical daily traded quantity in a name; the denominator of the participation rate
backtest	a simulation of a strategy on historical data; systematically optimistic for the five reasons of Chapter 13
basis point (bp)	10^{-4} ; the unit of everything
batch auction	clearing every few hundred milliseconds instead of continuously, so arriving first confers no advantage (§ 14.9)
best bid / best ask	the highest price anyone will pay / the lowest anyone will accept (Fig. 2.2)
beta	an asset's sensitivity to the market as a whole
book	the collection of resting limit orders (Definition 4.1); also, a trader's inventory
borrow fee	what a short position pays the lender of the asset
call option	the right, not the obligation, to buy at a fixed strike (Fig. 2.9)
carry	the P&L from holding a position with no price change: interest, dividends, borrow cost, roll
correlation	$\rho \in [-1, 1]$; the dial that controls how much diversification is available (Fig. 2.7)
deep hedging	policy optimization under a convex risk measure, trained against a simulator ((10.11))
delta	$\partial\Pi/\partial S$; the hedge ratio
derivative	a contract whose payoff is a function of something else, the <i>underlying</i>

Finance → mathematics, continued

Term	What it is
drawdown	$V_t / \max_{s \leq t} V_s - 1$ (Fig. 12.2)
effective price	$C(v)/v$, the average price actually paid for size v ; nondecreasing in v (Proposition 4.2)
efficient frontier	the set of portfolios or schedules that are not dominated in both cost and risk (Fig. 11.2, Fig. 8.2)
exchange	a continuously running auction in which anyone may leave a standing offer and anyone may accept one
fair transfer price	$\partial_q v$, the marginal value of one more share to you, which is not the market price
forward	an agreement to trade at a fixed price on a fixed future date, free to enter ((2.6))
futures	a forward that is settled daily in cash through variation margin ((3.13))
gamma	$\partial^2 \Pi / \partial S^2$; the rate at which the hedge must be revised, hence turnover
Greeks	the partial derivatives of a derivative's value: delta, gamma, vega, theta ((3.10))
haircut	the margin rate applied to collateral
idiosyncratic risk	the part of an asset's risk that diversifies away; the rest is <i>systematic</i> ((2.5))
imbalance	$(q^b - q^a) / (q^b + q^a)$; the strongest short-horizon predictor of the next price move
implementation shortfall	realized price minus the mid at decision time, times size; the terminal cost of Chapter 8
incompleteness	the case $\text{rank } A < m$, in which a claim has a range of prices rather than one (Proposition 5.3)
inventory	the signed position; the controlled state in Chapters 8 and 9
Kyle's lambda	the price move per unit of net order flow; the inverse of market depth ((6.2))
liquidity	tightness, depth, resiliency, immediacy, four distinct things (Chapter 7)
log return	$\log(S_t/S_{t-1})$; adds across time, unlike the arithmetic return, which adds across assets
longshot bias	the systematic overpricing of low-probability contracts in prediction markets
maker / taker	the party that posts liquidity / the party that removes it
margin	collateral posted against a leveraged position; a <i>margin call</i> demands more of it (Fig. 2.15)
market maker	a participant who quotes both sides continuously and tries to end the day flat (Chapter 9)
market order	an order naming a quantity and taking whatever prices are available, <i>walking the book</i>
mark-to-market	valuing a position at a single reference price ((3.1)); a convention, not a fact
matching engine	the deterministic program that applies Φ of (4.1) to every incoming message
maximum drawdown	the deepest peak-to-trough decline of an equity curve
metaorder	a large parent order executed as many child orders
microprice	the imbalance-weighted mid, (4.5)
microstructure noise	the difference between the observed price and the efficient price; small, bounded and destructive to high-frequency statistics (Proposition 5.4)

Finance → mathematics, continued

Term	What it is
mid	$(P^b + P^a)/2$; a convention, not a tradeable price
no-arbitrage	the assumption that no free money exists, used as an axiom to force prices (Theorem 5.2)
notional	size times price; the gross economic exposure
no-trade region	the band around the frictionless optimum inside which trading is not worth its cost (Fig. 11.4 , (10.9))
numeraire	the asset prices are quoted in; dividing by it turns no arbitrage into a martingale property
order book	the sorted record of resting limit orders (Definition 4.1)
P&L	profit and loss: what a position has made or lost
participation rate	$\varphi = Q/V$, your share of market volume ((6.6))
prediction market	a venue trading contracts that pay \$1 if an event occurs, so the price is a probability (§5.3)
premium	the price paid for an option, which is not free because its payoff is asymmetric
price-time priority	better prices execute first; at equal price, earlier arrivals do
propagator	the decay kernel G of transient impact ((6.5))
put option	the right to sell at a fixed strike; insurance against a fall
queue position	the volume ahead of your order at your price (Section 4.5)
realized variance	the sum of squared price changes; diverges under microstructure noise ((5.15))
rebate	the payment an exchange makes to whoever posted the resting order
recall	the lender's right to demand a borrowed share back, forcing a short to close
reservation price	the price at which a market maker is indifferent to holding its current inventory ((9.4))
resiliency	the rate at which the book refills after being consumed (§7.1.3)
risk measure	a number summarizing how bad a distribution of outcomes is; <i>coherent</i> if it satisfies Definition 12.1
self-financing	no money enters or leaves the account from outside ((3.5))
Sharpe ratio	mean excess return divided by its standard deviation, annualized
short	a negative position, created by borrowing the asset and selling it
slippage	realized price minus a benchmark price
spread	$P^a - P^b$; the round-trip cost of immediacy for one unit
state price	ψ_i , the price today of \$1 delivered in state i ; strictly positive exactly when there is no arbitrage ((5.6))
strike	the fixed price at which an option may be exercised
tangency portfolio	$\propto \Sigma^{-1}\mu$; the risky portfolio every investor holds when a risk-free asset exists ((11.6))
theta	$\partial\Pi/\partial t$; what a position earns for standing still
tick	the price grid spacing Δ ; a policy variable
tick-to-trade	the delay between market data arriving and your order leaving (§14.3)
two-fund theorem	every efficient portfolio is a combination of two fixed ones ((11.6))
underlying	the asset a derivative is written on
urgency	ϕ , the running penalty on unsold inventory; the single parameter indexing the execution frontier ((8.1))

Finance → mathematics, continued

Term	What it is
variation margin	the daily cash settlement of a futures position
vega	$\partial \Pi / \partial \sigma$; sensitivity to the volatility you had to estimate, hence model risk
volatility drag	the gap $\sigma^2/2$ between the average path and the typical one (Fig. 3.2)
VWAP	volume-weighted average price; a benchmark and a strategy

A.2 Mathematics → finance

Object	Where it shows up
adjoint / costate	the marginal value of inventory; the reservation-price offset of (9.4)
Farkas / Stiemke	finite-dimensional separation results underlying the one-period, finite-state first fundamental theorem of asset pricing, Theorem 5.2
Fenchel conjugate	the dual representation of a convex risk measure, (12.7)
filtering	extracting the efficient price from quotes; extracting a drift from your own fills
generalized eigenproblem	multi-asset execution: liquidate the risky, cheap directions first
Hopf-Cole transform	linearizes the market-making HJB under exponential utility (Theorem 9.1)
intensity control	choosing quotes to steer fill rates (Chapter 9)
LP duality	superreplication price and the risk-neutral measure, (5.7)
LQ regulator	optimal execution, Theorem 8.1
mean-field game	crowded trading and equilibrium impact (Chapter 16)
projection in L^2	quadratic hedging in incomplete markets (Table 10.1)
Riccati equation	the time-varying gain in every LQ problem here
singular control	block-drip-block execution (Fig. 8.5); band hedging (Fig. 10.4)
Skorokhod reflection	the no-trade region under transaction costs (Fig. 11.4)
stochastic target	hedging, (10.5)
time change	the volume clock; VWAP as a schedule linear in business time

Results from Stochastic Control

This appendix states, in one place and in the notation the main text uses, the control-theoretic results that Part II relies on. It is not a course in stochastic control, and it assumes you have met these objects before. What it does supply, and what a textbook treatment usually separates across two hundred pages, is the connective tissue: for each result, what its hypotheses actually require, what it does and does not give you, and exactly where in the book it is invoked and in what form.

The organizing observation is that Part II contains only three genuinely different mathematical situations. Execution and portfolio choice are linear-quadratic. Market making is intensity control of a jump process. Hedging with costs is singular control. Everything else is either one of these in disguise or a duality statement about a convex program.

B.1 The standing formulation

Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0, T]}, \mathbb{P})$ satisfying the usual conditions and carrying a d -dimensional Brownian motion W . A control is a process $u = (u_t)$ taking values in a set $U \subseteq \mathbb{R}^m$, progressively measurable with respect to an observation filtration $\mathcal{G}_t \subseteq \mathcal{F}_t$, and belonging to an admissible set $\mathcal{A}$ which encodes integrability together with whatever position and risk limits the problem carries. The state solves

$$dZ_t = b(t, Z_t, u_t) dt + \sigma(t, Z_t, u_t) dW_t, \quad Z_t = z, \quad (\text{B.1})$$

and the agent maximizes

$$J(t, z; u) = \mathbb{E}_{t, z} \left[\int_t^T \ell(s, Z_s, u_s) ds + \Psi(Z_T) \right], \quad v(t, z) = \sup_{u \in \mathcal{A}} J(t, z; u). \quad (\text{B.2})$$

Throughout, b and σ are assumed Lipschitz in z uniformly in (t, u) and of linear growth, which gives (B.1) a unique strong solution, and ℓ and Ψ are assumed measurable with polynomial growth, which makes (B.2) finite. These are the standing assumptions everywhere below and they are satisfied by every model in Part II. Where a model violates them, and the deep hedging problem of §10.7 does, the main text says so.

Control-theoretic dictionary

The financial meaning of each object was given in Table 1.1 and is worth repeating once. Z collects cash, inventory, price and whatever book statistics the model carries; u is a trading rate, a pair of quoted distances, or a portfolio weight; ℓ is transaction cost plus a risk penalty; Ψ is the value of the terminal position after liquidation; and $\mathcal{A}$ is the mandate.

B.2 Dynamic programming and the HJB equation

B.2.1 The principle

Theorem B.1 (Dynamic programming principle). *Under the standing assumptions, for every stopping time τ with values in $[t, T]$,*

$$v(t, z) = \sup_{u \in \mathcal{A}} \mathbb{E}_{t,z} \left[\int_t^\tau \ell(s, Z_s, u_s) ds + v(\tau, Z_\tau) \right]. \quad (\text{B.3})$$

The content of (B.3) is that an optimal policy, restricted to $[\tau, T]$, is optimal for the problem started at (τ, Z_τ) . One inequality is immediate, since any admissible control can be followed to τ and then continued optimally. The other requires a measurable selection argument to paste together the continuations, and that argument is where the technical work of proving Theorem B.1 lies.

B.2.2 The equation

Suppose v is smooth. Apply (B.3) with $\tau = t + h$, subtract $v(t, z)$, apply Itô's formula to $v(\tau, Z_\tau)$, divide by h and let $h \downarrow 0$. What survives is the HJB equation

$$\partial_t v(t, z) + \sup_{u \in U} \{ \ell(t, z, u) + \mathcal{L}^u v(t, z) \} = 0, \quad v(T, \cdot) = \Psi, \quad (\text{B.4})$$

with the controlled generator

$$\mathcal{L}^u w = b(t, z, u)^\top \nabla w + \frac{1}{2} \text{Tr}(\sigma \sigma^\top(t, z, u) \nabla^2 w). \quad (\text{B.5})$$

This derivation assumes what it should prove, namely that v is smooth enough for Itô. That assumption is false in general, and the standard repair is to interpret (B.4) in the viscosity sense, under which v is the unique solution with no differentiability assumed. Part II never needs this repair, because in each case an explicit smooth solution is produced and the following theorem then does the work.

B.2.3 Verification

Theorem B.2 (Verification). *Let $w \in C^{1,2}([0, T] \times \mathbb{R}^d) \cap C([0, T] \times \mathbb{R}^d)$ satisfy (B.4) with $w(T, \cdot) = \Psi$, and let w and ∇w satisfy a polynomial growth condition. Then*

- (1) $w(t, z) \geq v(t, z)$ for every (t, z) ; and
- (2) *if the pointwise maximizer $u^*(t, z) \in \arg \max_{u \in U} \{ \ell(t, z, u) + \mathcal{L}^u w(t, z) \}$ is measurable and the feedback control $u_s^* = u^*(s, Z_s)$ is admissible, then $w = v$ and u^* is optimal.*

What the proof is doing

The whole argument is one application of Itô and one sign. Apply Itô to $w(s, Z_s)$ along an arbitrary admissible control and add the running reward:

$$w(t, z) = \mathbb{E}_{t,z} \left[\int_t^T \ell ds + \Psi(Z_T) \right] - \mathbb{E}_{t,z} \left[\int_t^T \left(\partial_t w + \mathcal{L}^{u_s} w + \ell \right) ds \right], \quad (\text{B.6})$$

the stochastic integral vanishing in expectation by the growth condition. The second bracket is nonnegative for every control, because (B.4) says the supremum over u of that expression is zero. Hence $w \geq J(t, z; u)$ for all u , which is the first claim. Under the second claim's hypotheses the bracket vanishes identically along u^* , so the inequality is an equality and the bound is attained. Two consequences are worth extracting. Verification is a *sufficient* condition, so it certifies a candidate but never produces one. And it demands a genuine function $(t, z) \mapsto u^*(t, z)$, measurable and admissible, which is exactly the object Chapter 15 observes an agentic system does not supply.

B.2.4 The guess and verify recipe

Every closed form in Part II is obtained the same way: guess w , check (B.4), invoke Theorem B.2. The guesses are not inspired, they are forced by a scaling property of the problem, and it is worth seeing the three side by side.

Problem	Ansatz	Why that form
Execution Theorem 8.1	$w = x + qS + h(t)q^2$	the objective is affine in cash and in the mark, and the only remaining state is inventory, entering quadratically through the risk penalty
Market making Theorem 9.1	$w = -e^{-\gamma(x+qS)}e^{-\gamma\theta_q(t)}$	exponential utility is multiplicative in wealth, so cash factors out and what is left is a finite system indexed by the integer inventory
Merton (11.9)	$w = f(t)V^{1-\gamma}/(1-\gamma)$	power utility is homogeneous in wealth and the dynamics are linear in it, so the wealth dependence must cancel

In each case the ansatz reduces (B.4) from a partial differential equation to an ordinary one: a scalar Riccati equation for h , a linear system for θ , and a linear equation for f . That reduction, and not the verification step, is where the tractability of these models actually comes from.

B.3 Jump controls and intensity control

Suppose Z also jumps, with jumps of size $\beta(t, Z_{t-}, e)$ arriving at a *controlled* intensity $\lambda(t, Z_{t-}, u) \nu(de)$. The generator (B.5) gains the term

$$\int_E \lambda(t, z, u) [w(t, z + \beta(t, z, e)) - w(t, z)] \nu(de), \quad (\text{B.7})$$

and (B.4) becomes a partial integro-differential equation. Nothing in Theorem B.2 changes except that $\mathcal{L}^u$ now includes (B.7).

Control-theoretic dictionary

This is the distinguishing feature of Chapter 9 and it is worth naming. In execution the control enters the *drift*: you choose how fast to trade. In market making the control enters the *intensity*: you choose where to quote, and the market chooses whether to trade with you. You are not controlling the state directly at all; you are controlling the rate at which it jumps.

The reason this stays tractable is worth doing explicitly, because it is the single computation on which Theorem 9.1 rests. Inventory jumps by ± 1 , the control is a quoted distance δ , and the intensity is $\lambda(\delta) = Ae^{-\kappa\delta}$. The supremum in the HJB is then over a scalar:

$$\sup_{\delta \geq 0} Ae^{-\kappa\delta} [\delta + \Delta_q w], \quad (\text{B.8})$$

where $\Delta_q w$ is the change in the value function caused by one unit of inventory. Differentiating, the maximizer satisfies $1 - \kappa(\delta + \Delta_q w) = 0$, so

$$\delta^* = \frac{1}{\kappa} - \Delta_q w. \quad (\text{B.9})$$

The optimal quote is a constant markup minus the marginal value of inventory, which is (9.6) before the Hopf-Cole transform is applied to make $\Delta_q w$ explicit. Two features of the exponential intensity are doing the work: it makes (B.8) log-concave in δ , so the maximizer is unique, and it makes the maximum value proportional to $e^{-\kappa \Delta_q w}$, which is what turns the system for θ linear.

B.4 Singular and impulse control

B.4.1 Why the control stops being a rate

If the cost of trading is quadratic in the rate, as in (8.3), then trading fast is penalized quadratically and the optimal control is an absolutely continuous function of time. If the cost is *proportional* to the quantity traded, as it is under a bid-ask spread, that penalty disappears: trading twice as fast for half as long costs exactly the same. Nothing then prevents the optimizer from trading in bursts, and in fact it must, because the continuous-rate problem has no solution. This is the situation in Section 10.5 and Section 11.5.

The right formulation replaces the rate by a cumulative process. Let the control be a pair (L, M) of nondecreasing adapted processes, buying and selling, with $dQ_t = dL_t - dM_t$. These need not be absolutely continuous and may contain jumps, so Q is a process of finite variation rather than a differentiable path.

B.4.2 The variational inequality

The value function no longer satisfies an equation but an inequality system. Writing Mw for the value obtainable by making the best immediate intervention and continuing optimally, w satisfies

$$\max \left\{ \partial_t w + \mathcal{L}w + \ell, \quad Mw - w \right\} = 0. \quad (\text{B.10})$$

Read it as a disjunction. At each state at least one of the two expressions vanishes, and neither is ever positive. Where the first vanishes it is optimal to do nothing and the state diffuses freely; that region is the **continuation region**. Where the second vanishes an intervention is worth its cost; that is the **action region**. The boundary between them is not given in advance but is part of the solution, which is what makes these *free boundary* problems.

Smooth fit and the reflected solution

Two structural facts do most of the work in one dimension.

The first is *smooth fit*. At the free boundary the value function is not merely continuous but continuously differentiable, and in the singular case twice so. That extra condition is exactly what pins the boundary down: without it the free boundary is underdetermined, and with it the problem reduces to solving an ordinary differential equation with a matching condition. Every explicit no-trade band in this book, (10.9) included, is obtained this way.

The second is the form of the optimal control. Inside the continuation region nothing is done. At the boundary the minimal action is taken that keeps the state inside, so the optimally controlled state is a diffusion *reflected* at the boundary in the sense of the Skorokhod problem, and the control is its local time there. This is why Fig. 11.4 shows a trade back to the edge rather than to the centre: reflection, not reset.

When interventions carry a fixed cost as well as a proportional one, the minimal action is no longer optimal, since paying the fixed cost to move only infinitesimally

is wasteful. The control then becomes a sequence of discrete **impulses** with a gap between the boundary that triggers action and the target the state is moved to, and (B.10) is read with $\mathcal{M}$ a genuine intervention operator rather than a directional derivative. The block-drip-block structure of Fig. 8.5 is the same phenomenon in the execution problem.

B.5 Linear-quadratic problems

B.5.1 The result

Let

$$dZ_t = (A_t Z_t + B_t u_t) dt + \sigma_t dW_t, \quad \min \mathbb{E} \left[\int_0^T (Z_t^\top Q Z_t + u_t^\top R u_t) dt + Z_T^\top G Z_T \right], \quad (\text{B.11})$$

with $Q, G \succeq 0$ and $R \succ 0$. Then $v(t, z) = z^\top P(t)z + c(t)$, where P solves the matrix Riccati equation

$$-\dot{P} = A^\top P + PA - PBR^{-1}B^\top P + Q, \quad P(T) = G, \quad (\text{B.12})$$

c solves $-\dot{c} = \text{Tr}(\sigma\sigma^\top P)$ with $c(T) = 0$, and the optimal control is the linear feedback

$$u_t^* = -R^{-1}B^\top P(t)Z_t. \quad (\text{B.13})$$

B.5.2 Where it comes from

Substituting the quadratic ansatz into (B.4) gives an inner minimization $\min_u \{u^\top R u + 2z^\top P B u\}$, which is an unconstrained convex quadratic and is solved by completing the square: $u^* = -R^{-1}B^\top P z$ with optimal value $-z^\top P B R^{-1}B^\top P z$. Collecting the quadratic terms gives (B.12) and collecting the constants gives the equation for c . That is the entire derivation, and it explains both of the properties the main text uses repeatedly.

Certainty equivalence and where the noise went

The optimal control does not depend on σ . The volatility appears nowhere in (B.12) or (B.13); it enters only through c , which is a pure constant. So the optimal policy for the noisy problem is the optimal policy for the deterministic problem obtained by setting $\sigma = 0$. This is **certainty equivalence**, and it is the reason (8.3) can be a deterministic calculus-of-variations problem without any approximation.

It is also the reason that convenience is fragile. Certainty equivalence holds because the noise is additive and the cost is quadratic. Make σ depend on the control, as it does when trading moves the price, or make the cost non-quadratic, as (6.7) says it is, and the property fails immediately. Any execution model in which volatility does not appear in the schedule is relying on this, whether or not it says so.

The scalar case is the one used in Chapter 8. With state Q , control $\dot{Q}$, control cost k and state cost ϕ , (B.12) reads $\dot{P} = P^2/k - \phi$ with $P(T) = \alpha - \frac{b}{2}$. Chapter 8 maximizes rather than minimizes and writes the ansatz with the opposite sign, $h = -P$, giving $\dot{h} = \phi - h^2/k$, whose solution is a hyperbolic cotangent and whose feedback gain $-h/k$ blows up at the horizon, enforcing the terminal constraint.

B.6 Filtering and the separation principle

B.6.1 The conditional law as a state

Suppose the state is not observed directly but through

$$dY_t = h(Z_t) dt + \varsigma dB_t, \quad (\text{B.14})$$

with B a Brownian motion independent of W , and let the control be adapted to $\mathcal{F}^Y$ rather than to $\mathcal{F}$. The standard move is to enlarge the state: the conditional law $\pi_t = \mathcal{L}(Z_t | \mathcal{F}_t^Y)$ is itself a Markov process, so the partially observed problem in Z becomes a fully observed problem in π , and [Theorem B.2](#) applies again. The price is that π lives in a space of measures, so the HJB is now posed on an infinite-dimensional state space and is of no computational use in general.

B.6.2 When it collapses

In the linear-Gaussian case it collapses completely. If h is linear and the dynamics of Z are linear with additive Gaussian noise, then π_t is Gaussian and therefore determined by its mean $\hat{Z}_t$ and covariance Σ_t , which satisfy the Kalman-Bucy equations

$$d\hat{Z}_t = (A\hat{Z}_t + Bu_t) dt + \Sigma_t H^\top \varsigma^{-2} (dY_t - H\hat{Z}_t dt), \quad (\text{B.15})$$

$$\dot{\Sigma}_t = A\Sigma_t + \Sigma_t A^\top + \sigma\sigma^\top - \Sigma_t H^\top \varsigma^{-2} H \Sigma_t. \quad (\text{B.16})$$

Note that (B.16) is a Riccati equation and does not involve the control or the observations. That is the entire content of the **separation principle**: the estimation problem does not depend on the control problem, so one may estimate first and then control the estimate as though it were the state, and the composition is optimal.

Where the model breaks

Separation is a theorem about a very specific situation, and it is routinely assumed far outside it. It requires linear dynamics, linear observations, Gaussian noise and a quadratic cost. Break any one and the estimate and the control couple: the optimal policy then has an incentive to *probe*, that is to take actions whose value lies in the information they generate rather than in their immediate reward.

Two places in this book sit outside the hypotheses. In §9.6 the fills that a market maker receives are informative about the drift, so the quoting decision changes what is learned, and the separation of estimation from control fails. In §13.4 the exploration required to estimate A and κ has to be paid for in spread, which is a probing cost made explicit. The dual-control problem that results is genuinely hard and is one reason §16.2 is a live area rather than a solved one.

B.7 Backward stochastic differential equations

B.7.1 The object

A solution of a BSDE with terminal condition $\xi \in L^2(\mathcal{F}_T)$ and driver f is a pair of adapted processes (Y, Z) with

$$Y_t = \xi + \int_t^T f(s, Y_s, Z_s) ds - \int_t^T Z_s dW_s. \quad (\text{B.17})$$

The pair is essential. One cannot simply set $Y_t = \mathbb{E}[\cdot | \mathcal{F}_t]$ and be done, because Y must be adapted while ξ is known only at T ; the process Z is precisely the correction that makes an adapted solution possible, and it is the integrand supplied by the martingale representation theorem. Existence and uniqueness hold when f is Lipschitz in (y, z) and $\mathbb{E}[|\xi|^2] < \infty$.

B.7.2 Three uses

A nonlinear Feynman-Kac formula. Write the forward state as X in this section, since Z is taken by the BSDE. When $\xi = g(X_T^{t,x})$ and the driver depends on the state, $Y_t = w(t, X_t)$ for w solving a semilinear PDE, and $Z_t = \sigma^\top \nabla w(t, X_t)$. Linear drivers give the classical representation; nonlinear ones give value functions that no expectation represents.

The Z component is the hedge. Take (B.17) as a wealth equation, Y is the value of a portfolio that terminates at the liability ξ and Z is the position that gets it there. (10.7) is the special case $f \equiv 0$: with no driver, Y is a martingale, and the hedging strategy is the martingale representation integrand. This is why hedging and pricing are one object rather than two.

Nonlinear pricing. A driver that is convex but not linear encodes a market friction: different borrowing and lending rates, a position constraint, or ambiguity about the model. The comparison theorem, which says that a larger terminal condition and a larger driver give a larger Y , is what makes the resulting price a monotone and convex functional of the payoff, hence a convex risk measure in the sense of Definition 12.1. This is the formal link between Chapter 10 and Chapter 12.

B.8 Convex duality

B.8.1 The statement

For $\min\{f_0(x) : f_i(x) \leq 0, i = 1, \dots, m\}$ with all f_i convex, the Lagrangian $L(x, \mu) = f_0(x) + \sum_i \mu_i f_i(x)$ gives the dual function $g(\mu) = \inf_x L(x, \mu)$, which is concave regardless of the primal. Weak duality $\sup_{\mu \geq 0} g(\mu) \leq \inf$ always holds. Strong duality, equality of the two, holds under Slater's condition, that is when a strictly feasible point exists. See Boyd and Vandenberghe (2004) for the finite-dimensional theory.

B.8.2 Three appearances of one pattern

Superreplication. The primal (5.7) minimizes the cost of a portfolio dominating a payoff; the dual maximizes the payoff's value over state-price vectors consistent with observed prices. The dual variable is a measure, and by Theorem 5.2 feasibility of the dual is exactly the absence of arbitrage.

Risk measures. (12.7) writes a convex risk measure as a worst case over models penalized by a convex function. The dual variable is again a measure, and the penalty is how implausible each model is held to be.

Robust optimization. Section 11.8 replaces an uncertain mean by its worst case over an ellipsoid, and the inner minimization evaluates to a norm penalty. The dual variable is the direction in which the adversary perturbs.

The pattern worth carrying away

In all three the dual variable is a probability measure and the duality gap measures incompleteness. When the market is complete the dual has a unique solution, the upper and lower prices coincide, and the model returns a number. When it is not, the dual solution set is a polytope, its image under the payoff is an interval, and the width of that interval is the honest statement of what the model does not know. Chapter 5 is that observation carried through in detail, and Table 10.1 is the list of ways to pick a point inside the interval when a single number is nevertheless required.

Annotated Bibliography

A short, opinionated guide. Each entry says what the work is good *for*, not merely that it exists, and the sections are ordered so that you can start anywhere. Full details are in the References that follow.

[Section C.2](#) is the part to look at first if you are not sure where to begin: it gives four routes through the literature, each aimed at a different thing you might want to do next.

C.1 Three principal references

- Cartea et al. (2015) is the closest book in spirit to this one and the natural next step. Written by control theorists, with complete derivations, and the source of the running-penalty formulations used in [Chapters 8 to 9](#).
- Bouchaud, Bonart, et al. (2018) is the definitive empirical account of microstructure: order flow, impact, the square-root law, and the latent-liquidity argument of [Chapter 16](#). Read it for the facts.
- Guéant (2016) treats the execution and market-making HJB equations rigorously, including the inventory-constrained market-making problem and its matrix-exponential solution.

Three more, for background rather than for trading: Hasbrouck (2007) and O'Hara (1995) for the economics of microstructure; Föllmer and Schied (2016) for arbitrage, incomplete markets and convex risk measures developed together in discrete time; and Pham (2009) or Fleming and Soner (2006) for the stochastic control theory that [Chapter B](#) states without proving, with Boyd and Vandenberghe (2004) for the convex duality of [§B.8](#).

C.2 Four routes through the literature

The list below is organized by chapter, which is convenient for looking things up and useless for deciding what to do next. So here are four paths, each about five works long, aimed at four different intentions.

If you want to build an execution algorithm. Start with Almgren and Chriss (2001) for the schedule, then Obizhaeva and Wang (2013) for what changes when the book refills at a finite rate. Read Tóth et al. (2011) next, because it tells you that the quadratic cost the first two assume is empirically wrong, and then Gatheral (2010) for the constraint that stops you simply substituting the empirical law. Finish with Cont et al. (2010) for the order-placement layer of [§8.6](#), which none of the others addresses.

If you want to quote a market. Avellaneda and Stoikov (2008) for the model, Guéant, Lehalle, et al. (2013) for the version with inventory bounds that you can actually deploy, Glosten and Milgrom (1985) for why the spread cannot be competed to zero, and Menkveld (2013) for what the resulting business looks like from the outside. O'Hara (2015) for the regime in which all of it now operates.

If you want to understand why prices move. Kyle (1985) and Glosten and Milgrom (1985) for the information account, Lillo et al. (2005) and Bouchaud, Gefen, et al. (2004) for the mechanical one, and Bouchaud, Bonart, et al. (2018) for the synthesis and the data. This is the path with the highest ratio of insight to pages in the whole list.

If you want to work on autonomous systems. There is no direct literature, so the path is oblique. Calvano et al. (2020) for what independent learners do to each other, Greshake et al. (2023) for the untrusted channel, Budish et al. (2015) for the argument that these problems are settled by mechanism design rather than by better participants, and Kirilenko et al. (2017) for the last time automation failed at scale. Byrd et al. (2020) if you intend to test anything.

C.3 Foundational papers by chapter

Chapter 4, order books.

Smith et al. (2003) is the zero-intelligence model of §4.4: a book with no economics in it that nonetheless reproduces the shape of a real one. Cont et al. (2010) give the birth-and-death formulation and the first-passage computations behind §4.5. Stoikov (2018) derives the microprice of (4.5) and shows how much better it forecasts than the mid.

Chapter 5, arbitrage and price models.

Harrison and Kreps (1979) and Harrison and Pliska (1981) recast asset pricing as a martingale problem; Delbaen and Schachermayer (1994) give the definitive infinite-dimensional statement. Zhang et al. (2005) and Barndorff-Nielsen et al. (2008) are the two estimators that repair the divergence of Proposition 5.4. For the prediction-market material, Wolfers and Zitzewitz (2004) is the standard survey, Hanson (2003) and Hanson (2007) introduce the logarithmic market scoring rule, and Abernethy et al. (2013) give the general convex-analytic account of automated market making.

Chapter 6, why prices move.

Glosten and Milgrom (1985) and Kyle (1985) are the two canonical models of informed trading, and between them account for most of what is understood about the spread. Roll (1984) is a one-page idea that still gets used daily. Lillo et al. (2005) explain the long memory of order flow as pure order splitting; Bouchaud, Gefen, et al. (2004) introduce the propagator that reconciles that memory with efficient prices. Tóth et al. (2011) is the most careful empirical statement of the square-root law, and Almgren, Thum, et al. (2005) the most-cited estimation of impact from proprietary data. Huberman and Stanzl (2004) and Gatheral (2010) give the no-manipulation constraints of Theorem 6.6.

Chapter 7, liquidity.

Amihud (2002) is the illiquidity measure that survives having only daily data. Brunnermeier and Pedersen (2009) is the reference for the funding spiral of §7.5.

Chapter 8, execution.

Almgren and Chriss (2001) is the paper this chapter is built on. Obizhaeva and Wang (2013) add resilience and get the block-drip-block solution of Fig. 8.5. Brunnermeier and Pedersen (2005) formalize what happens when someone can see you coming.

Chapter 9, market making.

Avellaneda and Stoikov (2008) is the model of Theorem 9.1; Guéant, Lehalle, et al. (2013) give the inventory-constrained version with explicit quotes and are the more practical starting point.

Chapter 10, hedging.

Black and Scholes (1973) and Merton (1973) need no introduction; Soner and Touzi (2002) is the general stochastic-target formulation that (10.5) is an instance of. Leland (1985) and Whalley and Wilmott (1997) treat transaction costs, the second giving the asymptotic band of (10.9). Buehler et al. (2019) is deep hedging.

Chapter 11, portfolios.

Markowitz (1952); Merton (1969) and Merton (1971); Davis and Norman (1990) for the no-trade region; Gârleanu and Pedersen (2013) for the aim portfolio of (11.10); Ledoit and Wolf (2004) for shrinkage; and Boyd, Busseti, et al. (2017) for the modern practice of solving a convex program every period instead of looking for closed forms.

Chapter 12, risk.

Artzner et al. (1999) is the axiomatic paper; Rockafellar and Uryasev (2000) is the one that made coherent risk measures computable, and Theorem 12.3 is its main result.

Chapter 13, estimation and evaluation.

Bailey and López de Prado (2014) on the deflated Sharpe ratio, and López de Prado (2018) for purged cross-validation and the rest of the protocol of §13.6.

Chapter 14, high-frequency trading.

O'Hara (2015) and Menkveld (2016) are the two best short surveys. Menkveld (2013) is the empirical anatomy of a single fast market maker; Hendershott et al. (2011) and Brogaard et al. (2014) are the most-cited evidence that speed helps, and Budish et al. (2015) with Aquilina et al. (2022) the case that the race itself is waste. Baldauf and Mollner (2020) and Biais et al. (2015) reconcile the two. Kirilenko et al. (2017) on the flash crash, Harris (2003) for institutional background, and MacKenzie (2021) for how the infrastructure was actually built.

Chapter 15, autonomous agents.

There is not yet a literature on agentic trading as such, and these are the adjacent works that bear on it. Calvano et al. (2020) is the key result: independent learning agents converging on collusive behaviour with no communication and no instruction to collude. Yao et al. (2023) describes the reason-and-act architecture and Park et al. (2023) studies populations of such agents. Greshake et al. (2023) is the reference for indirect prompt injection, the mechanism behind the untrusted-channel problem of §15.8.3. Byrd et al. (2020) is the kind of simulator needed to test any of it. Treat that chapter as a set of questions rather than a survey.

Chapter 16, the frontier.

Cardaliaguet and Lehalle (2018) for trade crowding as a mean-field game of controls, with Carmona and Delarue (2018) for the theory underneath. Gatheral et al. (2018) for rough volatility, Bacry et al. (2015) for the Hawkes-process view of the tick scale, Budish et al. (2015) for frequent batch auctions, and Nevmyvaka et al. (2006) for the first serious application of reinforcement learning to execution.

C.4 Reproducing the figures

Every computed figure in this book is generated by two scripts in the source directory.

- `scripts/generate_data.py` runs every simulation and writes the `.dat` files that the `pgfplots` figures read. Every routine is explicitly seeded, so the figures are reproducible bit for bit. It contains, among other things, a complete zero-intelligence limit order book (Fig. 4.7), a matched-control measurement of book resiliency

(Fig. 7.1), an Avellaneda-Stoikov market-making simulator (Fig. 9.3), and the backtest-overfitting experiment of Fig. 13.2.

- `scripts/render_figures.py` renders the one raster figure, the order-book heat map, that would exhaust $\text{\TeX}$'s memory if drawn as vector graphics.

Running `make` rebuilds the data and then the document.

References

- Abernethy J., Y. Chen, and J. Wortman Vaughan (2013). Efficient Market Making via Convex Optimization, and a Connection to Online Learning. In: *ACM Transactions on Economics and Computation* 1.2.
- Almgren R. and N. Chriss (2001). Optimal Execution of Portfolio Transactions. In: *Journal of Risk* 3.2, pp. 5–39.
- Almgren R., C. Thum, et al. (2005). Direct Estimation of Equity Market Impact. In: *Risk* 18.7.
- Amihud Y. (2002). Illiquidity and Stock Returns: Cross-Section and Time-Series Effects. In: *Journal of Financial Markets* 5.1, pp. 31–56.
- Aquilina M., E. Budish, and P. O'Neill (2022). Quantifying the High-Frequency Trading ‘Arms Race’. In: *The Quarterly Journal of Economics* 137.1, pp. 493–564.
- Artzner P. et al. (1999). Coherent Measures of Risk. In: *Mathematical Finance* 9.3, pp. 203–228.
- Avellaneda M. and S. Stoikov (2008). High-Frequency Trading in a Limit Order Book. In: *Quantitative Finance* 8.3, pp. 217–224.
- Bacry E., I. Mastromatteo, and J.-F. Muzy (2015). Hawkes Processes in Finance. In: *Market Microstructure and Liquidity* 1.1.
- Bailey D. H. and M. López de Prado (2014). The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting and Non-Normality. In: *The Journal of Portfolio Management* 40.5, pp. 94–107.
- Baldauf M. and J. Mollner (2020). High-Frequency Trading and Market Performance. In: *The Journal of Finance* 75.3, pp. 1495–1526.
- Barndorff-Nielsen O. E. et al. (2008). Designing Realized Kernels to Measure the Ex Post Variation of Equity Prices in the Presence of Noise. In: *Econometrica* 76.6, pp. 1481–1536.
- Baron M. et al. (2019). Risk and Return in High-Frequency Trading. In: *Journal of Financial and Quantitative Analysis* 54.3, pp. 993–1024.
- Biais B., T. Foucault, and S. Moinas (2015). Equilibrium Fast Trading. In: *Journal of Financial Economics* 116.2, pp. 292–313.
- Black F. and M. Scholes (1973). The Pricing of Options and Corporate Liabilities. In: *Journal of Political Economy* 81.3, pp. 637–654.
- Bouchaud J.-P., J. Bonart, et al. (2018). *Trades, Quotes and Prices: Financial Markets Under the Microscope*. Cambridge University Press.
- Bouchaud J.-P., Y. Gefen, et al. (2004). Fluctuations and Response in Financial Markets: The Subtle Nature of ‘Random’ Price Changes. In: *Quantitative Finance* 4.2, pp. 176–190.
- Boyd S., E. Busseti, et al. (2017). Multi-Period Trading via Convex Optimization. In: *Foundations and Trends in Optimization* 3.1, pp. 1–76.
- Boyd S. and L. Vandenberghe (2004). *Convex Optimization*. Cambridge University Press.
- Brogaard J., T. Hendershott, and R. Riordan (2014). High-Frequency Trading and Price Discovery. In: *The Review of Financial Studies* 27.8, pp. 2267–2306.
- Brunnermeier M. K. and L. H. Pedersen (2005). Predatory Trading. In: *The Journal of Finance* 60.4, pp. 1825–1863.
- Brunnermeier M. K. and L. H. Pedersen (2009). Market Liquidity and Funding Liquidity. In: *The Review of Financial Studies* 22.6, pp. 2201–2238.

- Budish E., P. Cramton, and J. Shim (2015). The High-Frequency Trading Arms Race: Frequent Batch Auctions as a Market Design Response. In: *The Quarterly Journal of Economics* 130.4, pp. 1547–1621.
- Buehler H. et al. (2019). Deep Hedging. In: *Quantitative Finance* 19.8, pp. 1271–1291.
- Byrd D., M. Hybinette, and T. H. Balch (2020). ABIDES: Towards High-Fidelity Multi-Agent Market Simulation. In: *Proceedings of the ACM SIGSIM Conference on Principles of Advanced Discrete Simulation*.
- Calvano E. et al. (2020). Artificial Intelligence, Algorithmic Pricing, and Collusion. In: *American Economic Review* 110.10, pp. 3267–3297.
- Cardaliaguet P. and C.-A. Lehalle (2018). Mean Field Game of Controls and an Application to Trade Crowding. In: *Mathematics and Financial Economics* 12.3, pp. 335–363.
- Carmona R. and F. Delarue (2018). *Probabilistic Theory of Mean Field Games with Applications*. Springer.
- Cartea Á., S. Jaimungal, and J. Penalva (2015). *Algorithmic and High-Frequency Trading*. Cambridge University Press.
- Cont R., S. Stoikov, and R. Talreja (2010). A Stochastic Model for Order Book Dynamics. In: *Operations Research* 58.3, pp. 549–563.
- Davis M. H. A. and A. R. Norman (1990). Portfolio Selection with Transaction Costs. In: *Mathematics of Operations Research* 15.4, pp. 676–713.
- Delbaen F. and W. Schachermayer (1994). A General Version of the Fundamental Theorem of Asset Pricing. In: *Mathematische Annalen* 300, pp. 463–520.
- Fleming W. H. and H. M. Soner (2006). *Controlled Markov Processes and Viscosity Solutions*. 2nd ed. Springer.
- Föllmer H. and A. Schied (2016). *Stochastic Finance: An Introduction in Discrete Time*. 4th ed. De Gruyter.
- Foucault T., R. Kozhan, and W. W. Tham (2017). Toxic Arbitrage. In: *The Review of Financial Studies* 30.4, pp. 1053–1094.
- Gârleanu N. and L. H. Pedersen (2013). Dynamic Trading with Predictable Returns and Transaction Costs. In: *The Journal of Finance* 68.6, pp. 2309–2340.
- Gatheral J. (2010). No-Dynamic-Arbitrage and Market Impact. In: *Quantitative Finance* 10.7, pp. 749–759.
- Gatheral J., T. Jaisson, and M. Rosenbaum (2018). Volatility is Rough. In: *Quantitative Finance* 18.6, pp. 933–949.
- Glosten L. R. and P. R. Milgrom (1985). Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders. In: *Journal of Financial Economics* 14.1, pp. 71–100.
- Greshake K. et al. (2023). Not What You’ve Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection. In: *Proceedings of the ACM Workshop on Artificial Intelligence and Security (AISec)*.
- Guéant O. (2016). *The Financial Mathematics of Market Liquidity: From Optimal Execution to Market Making*. Chapman and Hall/CRC.
- Guéant O., C.-A. Lehalle, and J. Fernandez-Tapia (2013). Dealing with the Inventory Risk: A Solution to the Market Making Problem. In: *Mathematics and Financial Economics* 7.4, pp. 477–507.
- Guéant O. and I. Manziuk (2019). Deep Reinforcement Learning for Market Making in Corporate Bonds. In: *Applied Mathematical Finance* 26.5.
- Hambly B., R. Xu, and H. Yang (2023). Recent Advances in Reinforcement Learning in Finance. In: *Mathematical Finance* 33.3.

- Hanson R. (2003). Combinatorial Information Market Design. In: *Information Systems Frontiers* 5.1, pp. 107–119.
- Hanson R. (2007). Logarithmic Market Scoring Rules for Modular Combinatorial Information Aggregation. In: *The Journal of Prediction Markets* 1.1, pp. 3–15.
- Harris L. (2003). *Trading and Exchanges: Market Microstructure for Practitioners*. Oxford University Press.
- Harrison J. M. and D. M. Kreps (1979). Martingales and Arbitrage in Multiperiod Securities Markets. In: *Journal of Economic Theory* 20.3, pp. 381–408.
- Harrison J. M. and S. R. Pliska (1981). Martingales and Stochastic Integrals in the Theory of Continuous Trading. In: *Stochastic Processes and their Applications* 11.3, pp. 215–260.
- Hasbrouck J. (2007). *Empirical Market Microstructure*. Oxford University Press.
- Hendershott T., C. M. Jones, and A. J. Menkveld (2011). Does Algorithmic Trading Improve Liquidity? In: *The Journal of Finance* 66.1, pp. 1–33.
- Huberman G. and W. Stanzl (2004). Price Manipulation and Quasi-Arbitrage. In: *Econometrica* 72.4, pp. 1247–1275.
- Kirilenko A. et al. (2017). The Flash Crash: High-Frequency Trading in an Electronic Market. In: *The Journal of Finance* 72.3, pp. 967–998.
- Kyle A. S. (1985). Continuous Auctions and Insider Trading. In: *Econometrica* 53.6, pp. 1315–1335.
- Ledoit O. and M. Wolf (2004). A Well-Conditioned Estimator for Large-Dimensional Covariance Matrices. In: *Journal of Multivariate Analysis* 88.2, pp. 365–411.
- Leland H. E. (1985). Option Pricing and Replication with Transactions Costs. In: *The Journal of Finance* 40.5, pp. 1283–1301.
- Lillo F., S. Mike, and J. D. Farmer (2005). Theory for Long Memory in Supply and Demand. In: *Physical Review E* 71, p. 066122.
- López de Prado M. (2018). *Advances in Financial Machine Learning*. Wiley.
- MacKenzie D. (2021). *Trading at the Speed of Light: How Ultrafast Algorithms Are Transforming Financial Markets*. Princeton University Press.
- Markowitz H. (1952). Portfolio Selection. In: *The Journal of Finance* 7.1, pp. 77–91.
- Menkveld A. J. (2013). High Frequency Trading and the New Market Makers. In: *Journal of Financial Markets* 16.4, pp. 712–740.
- Menkveld A. J. (2016). The Economics of High-Frequency Trading: Taking Stock. In: *Annual Review of Financial Economics* 8.
- Merton R. C. (1969). Lifetime Portfolio Selection under Uncertainty: The Continuous-Time Case. In: *The Review of Economics and Statistics* 51.3, pp. 247–257.
- Merton R. C. (1971). Optimum Consumption and Portfolio Rules in a Continuous-Time Model. In: *Journal of Economic Theory* 3.4, pp. 373–413.
- Merton R. C. (1973). Theory of Rational Option Pricing. In: *The Bell Journal of Economics and Management Science* 4.1, pp. 141–183.
- Nevmyvaka Y., Y. Feng, and M. Kearns (2006). Reinforcement Learning for Optimized Trade Execution. In: *Proceedings of the 23rd International Conference on Machine Learning (ICML)*.
- O'Hara M. (1995). *Market Microstructure Theory*. Blackwell.
- O'Hara M. (2015). High Frequency Market Microstructure. In: *Journal of Financial Economics* 116.2, pp. 257–270.
- Obizhaeva A. A. and J. Wang (2013). Optimal Trading Strategy and Supply/Demand Dynamics. In: *Journal of Financial Markets* 16.1, pp. 1–32.

- Park J. S. et al. (2023). Generative Agents: Interactive Simulacra of Human Behavior. In: *Proceedings of the ACM Symposium on User Interface Software and Technology (UIST)*.
- Pham H. (2009). *Continuous-time Stochastic Control and Optimization with Financial Applications*. Springer.
- Rockafellar R. T. and S. Uryasev (2000). Optimization of Conditional Value-at-Risk. In: *Journal of Risk* 2.3, pp. 21–41.
- Roll R. (1984). A Simple Implicit Measure of the Effective Bid-Ask Spread in an Efficient Market. In: *The Journal of Finance* 39.4, pp. 1127–1139.
- Smith E. et al. (2003). Statistical Theory of the Continuous Double Auction. In: *Quantitative Finance* 3.6, pp. 481–514.
- Soner H. M. and N. Touzi (2002). Stochastic Target Problems, Dynamic Programming and Viscosity Solutions. In: *SIAM Journal on Control and Optimization* 41.2, pp. 404–424.
- Stoikov S. (2018). The Micro-Price: A High-Frequency Estimator of Future Prices. In: *Quantitative Finance* 18.12.
- Sutton R. S. and A. G. Barto (2018). *Reinforcement Learning: An Introduction*. 2nd ed. MIT Press.
- Tóth B. et al. (2011). Anomalous Price Impact and the Critical Nature of Liquidity in Financial Markets. In: *Physical Review X* 1, p. 021006.
- Whalley A. E. and P. Wilmott (1997). An Asymptotic Analysis of an Optimal Hedging Model for Option Pricing with Transaction Costs. In: *Mathematical Finance* 7.3, pp. 307–324.
- Wolfers J. and E. Zitzewitz (2004). Prediction Markets. In: *Journal of Economic Perspectives* 18.2, pp. 107–126.
- Yao S. et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. In: *International Conference on Learning Representations (ICLR)*.
- Zhang L., P. A. Mykland, and Y. Aït-Sahalia (2005). A Tale of Two Time Scales: Determining Integrated Volatility with Noisy High-Frequency Data. In: *Journal of the American Statistical Association* 100.472, pp. 1394–1411.

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